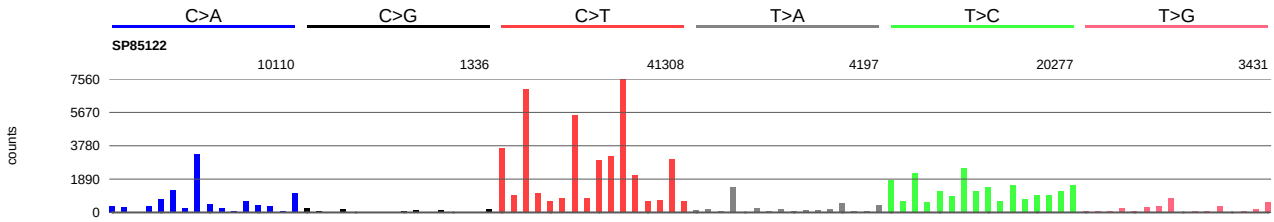
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41308
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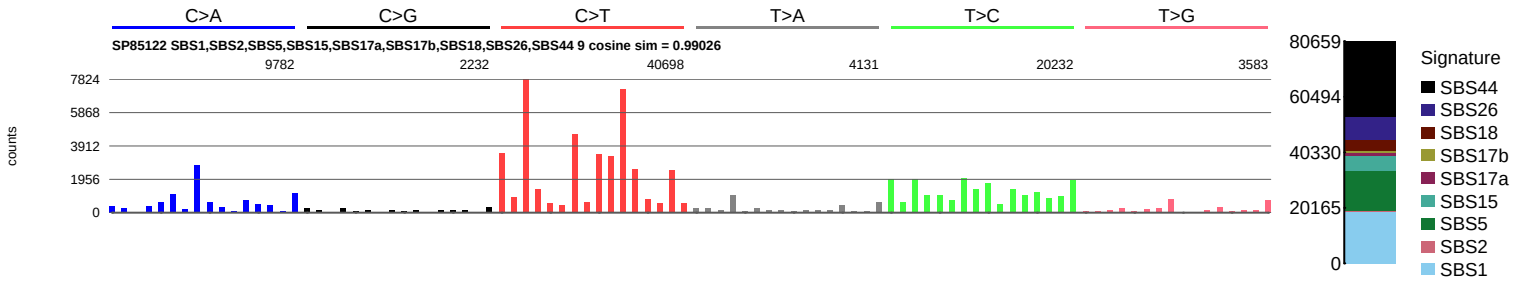
20277
3431
7560
5670
3780
1890
0
counts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SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.99026
9782
2232
40698
4131
20232
3583
7824
5868
3912
1956
0
counts
80659
60494
40330
20165
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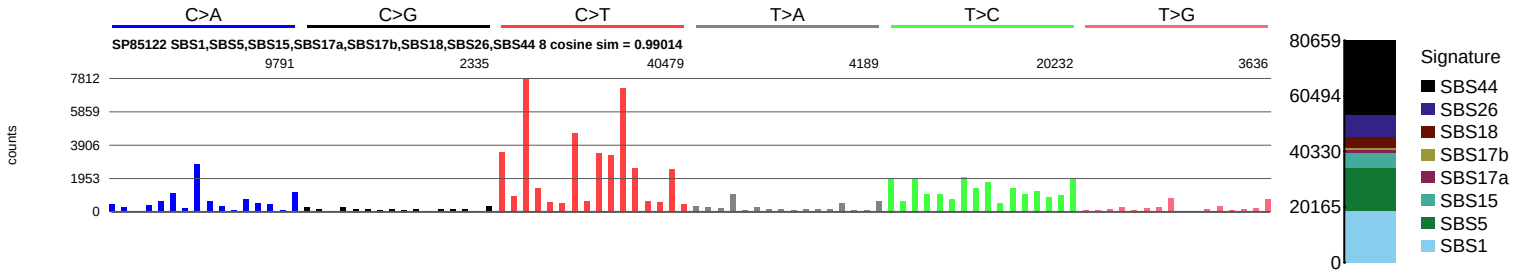
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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9791
2335
40479
4189
20232
3636
7812
5859
3906
1953
0
counts
80659
60494
40330
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SBS44
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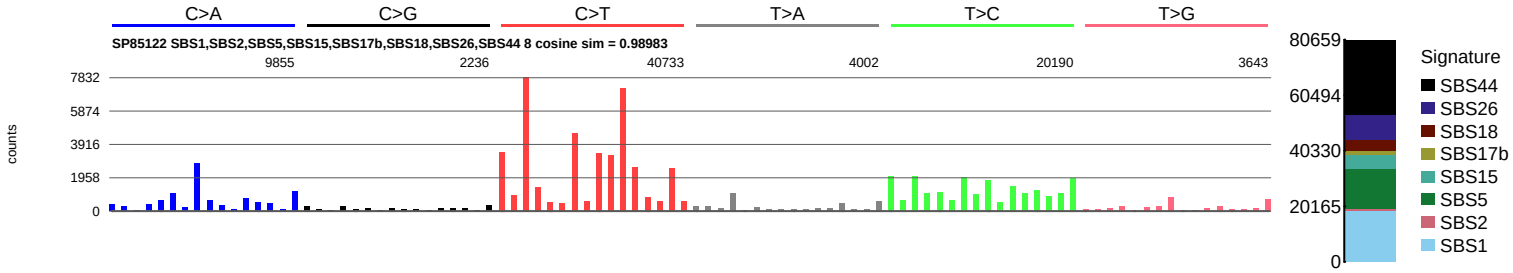
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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9855
2236
40733
4002
20190
3643
7832
5874
3916
1958
0
counts
80659
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40330
20165
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SBS17b
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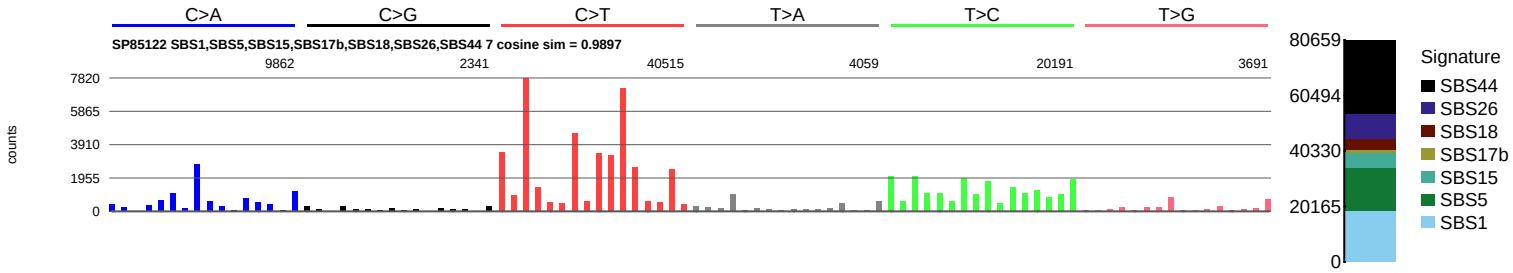
SBS5
SBS2
SBS1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9862
2341
40515
4059
20191
3691
7820
5865
3910
1955
0
counts
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20165
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SBS44
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SBS18
SBS17b
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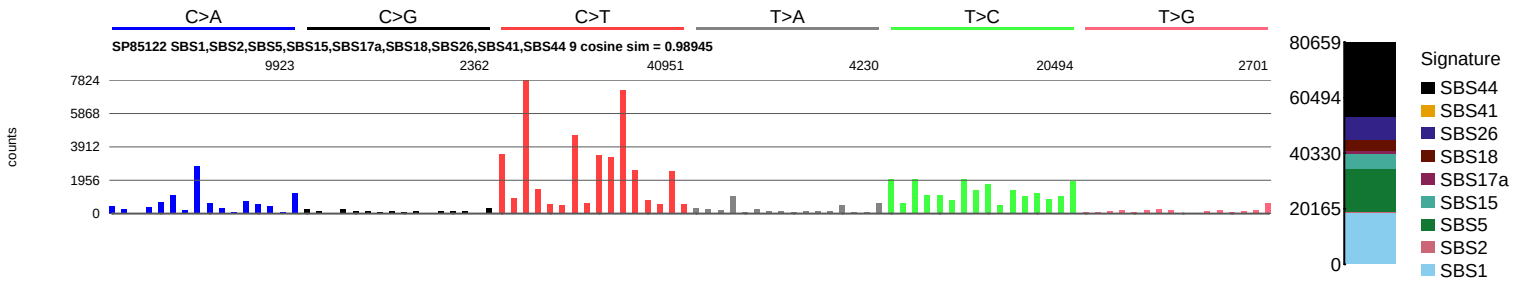
SBS5
SBS1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9923
2362
40951
4230
20494
2701
7824
5868
3912
1956
0
counts
80659
60494
40330
20165
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SBS17a
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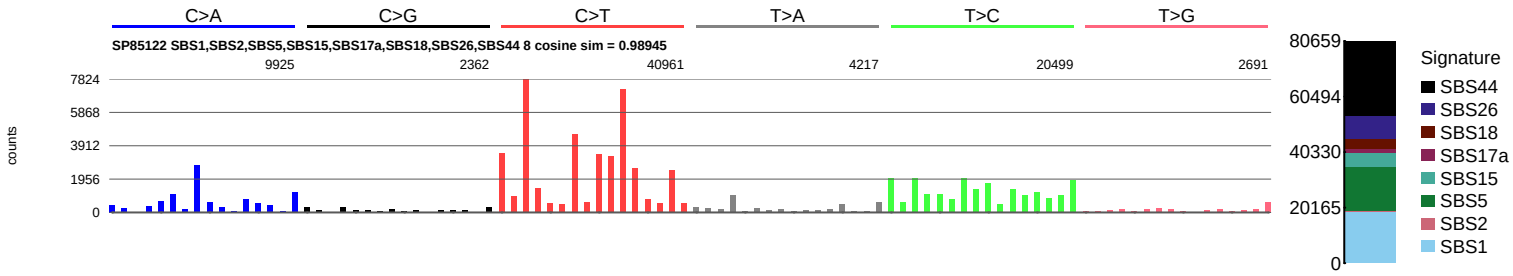
SBS5
SBS2
SBS1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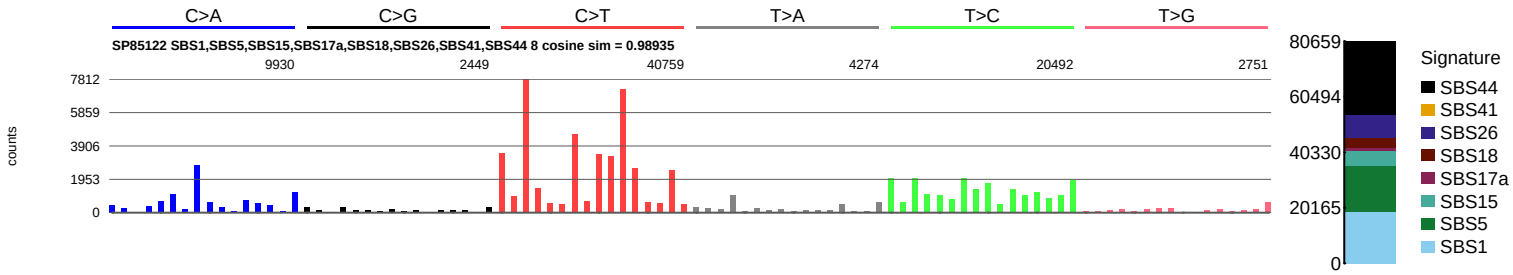
SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS44 8 cosine sim = 0.98945
9925
2362
40961
4217
20499
2691
counts
7824
5868
3912
1956
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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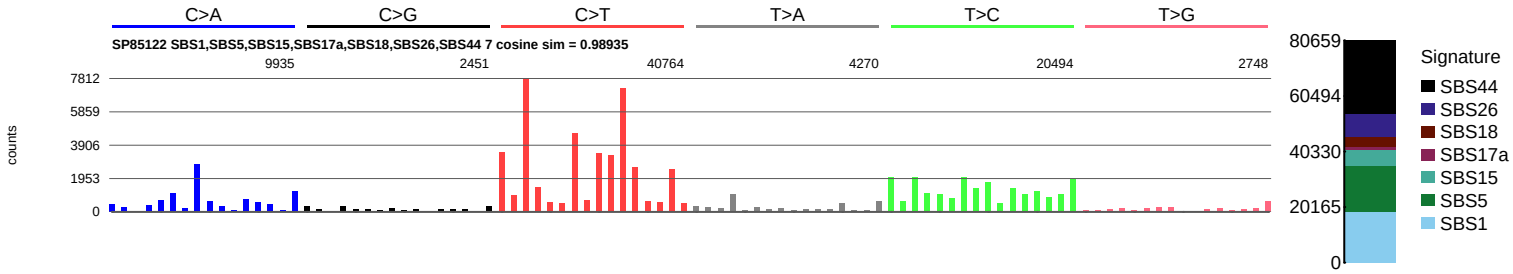
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98935
9930
2449
40759
4274
20492
2751
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1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SP85122 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98935
9935
2451
40764
4270
20494
2748
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
0
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SBS18
SBS17a
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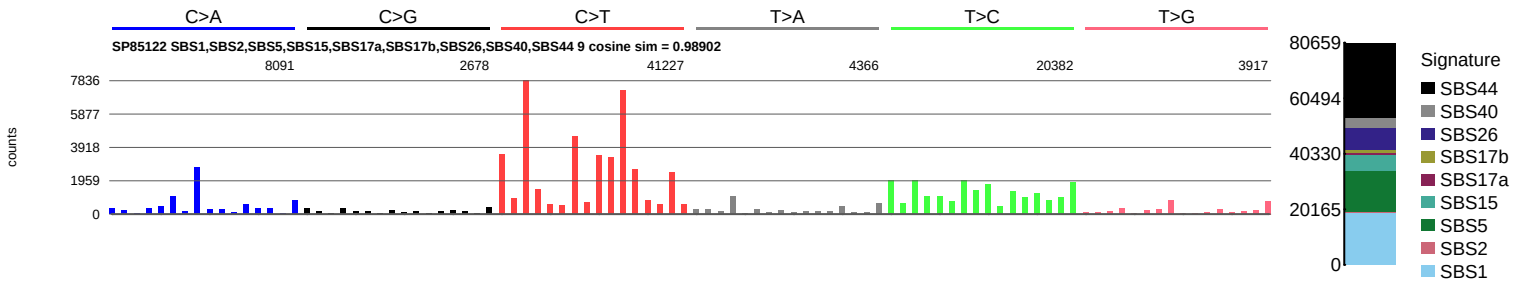
SBS5
SBS1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SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98902
8091
2678
41227
4366
20382
3917
counts
7836
5877
3918
1959
0
80659
60494
40330
20165
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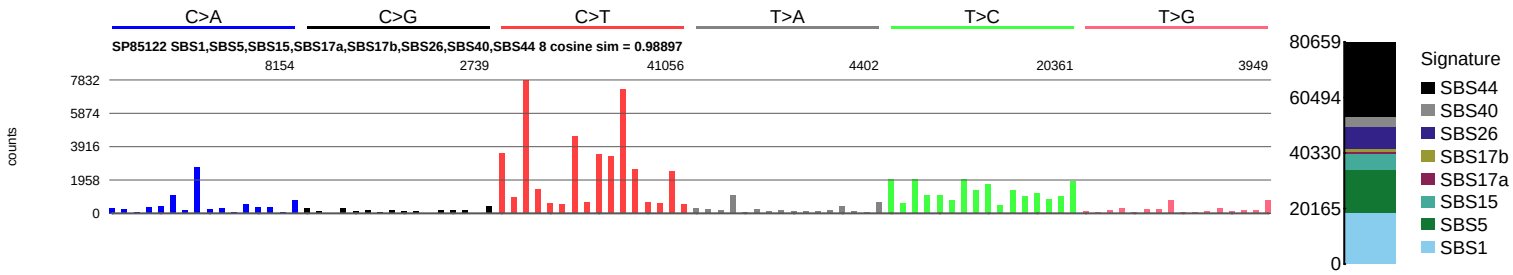
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>T
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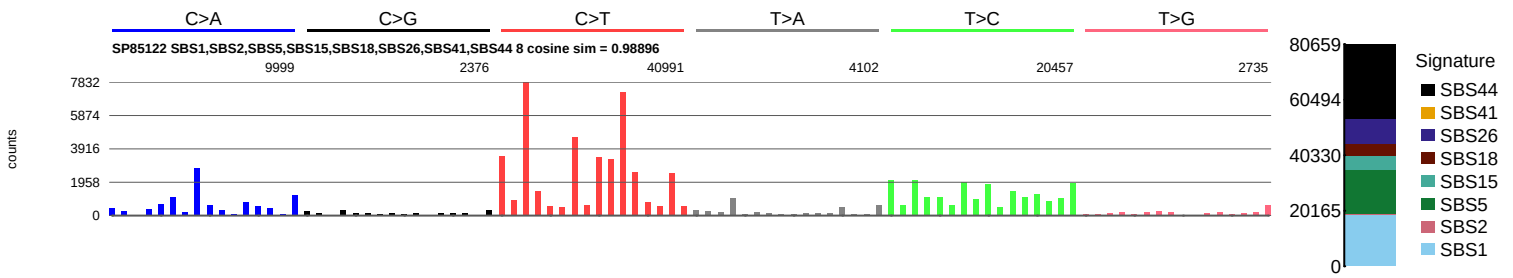
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98897
8154
2739
41056
4402
20361
3949
counts
7832
5874
3916
1958
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98896
9999
2376
40991
4102
20457
2735
counts
7832
5874
3916
1958
0
80659
60494
40330
20165
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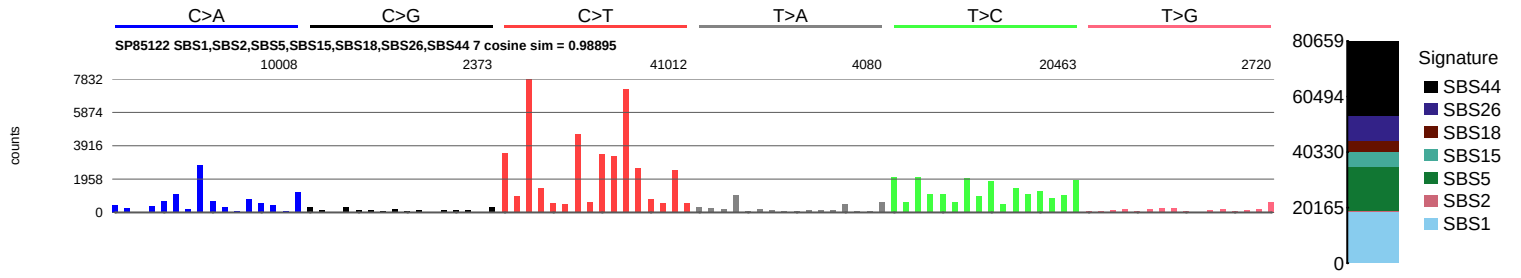
SBS44
SBS41
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1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T>C
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SP85122 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98895
10008
2373
41012
4080
20463
2720
counts
7832
5874
3916
1958
0
80659
60494
40330
20165
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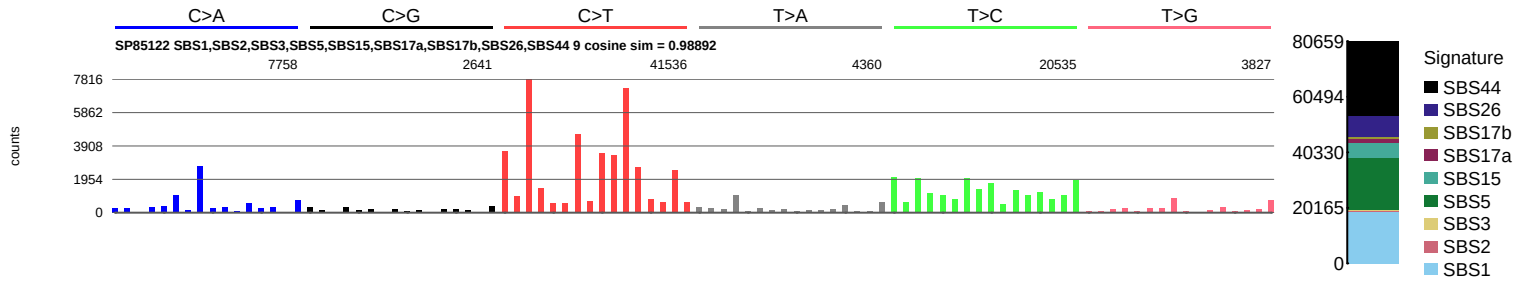
SBS15
SBS5
SBS2
SBS1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T>C
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SP85122 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98892
7758
2641
41536
4360
20535
3827
counts
7816
5862
3908
1954
0
80659
60494
40330
20165
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Signature
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SBS26
SBS17b
SBS17a
SBS15
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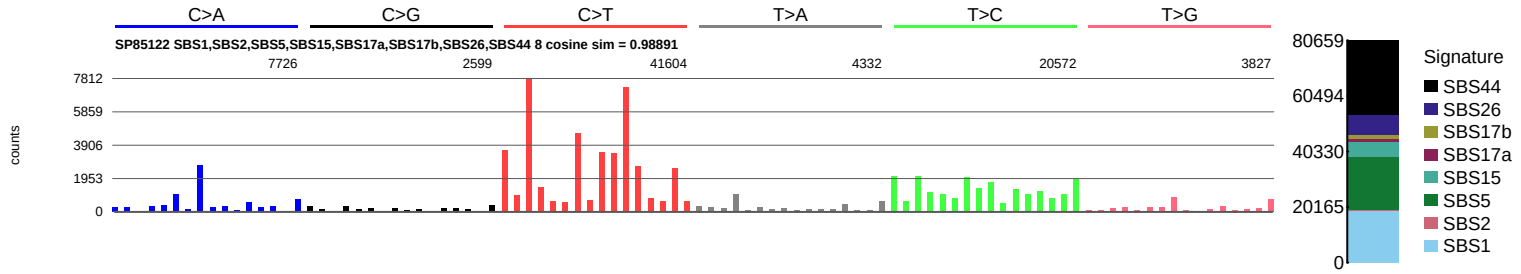
SBS3
SBS2
SBS1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C>T
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T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98891
7726
2599
41604
4332
20572
3827
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
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SBS17a
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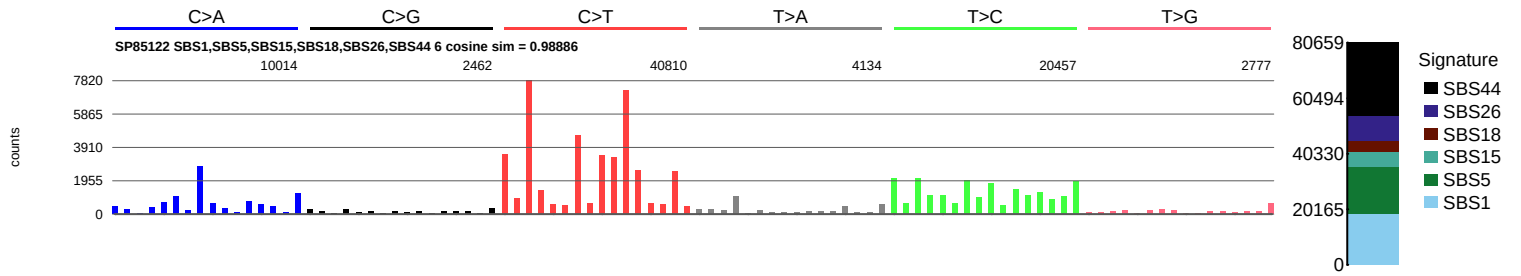
SBS2
SBS1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C>G
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T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98886
10014
2462
40810
4134
20457
2777
counts
7820
5865
3910
1955
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80659
60494
40330
20165
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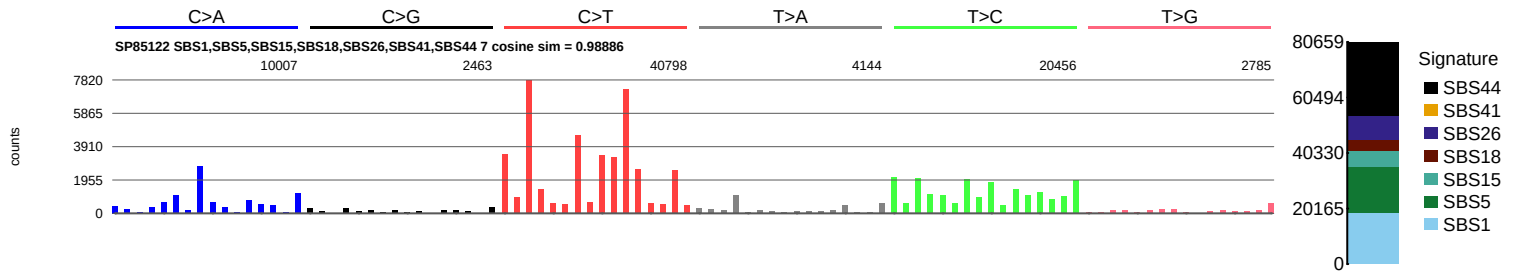
SBS26
SBS18
SBS15
SBS5
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98886
10007
2463
40798
4144
20456
2785
counts
7820
5865
3910
1955
0
80659
60494
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20165
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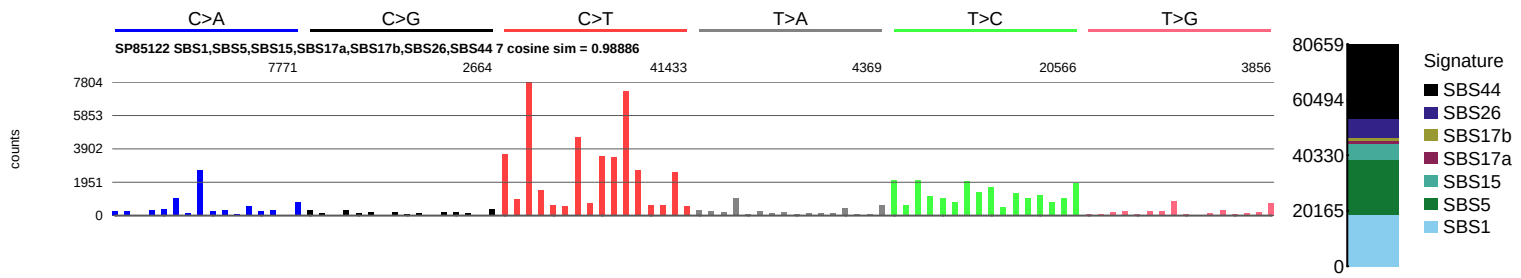
SBS5
SBS1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T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98886
7771
2664
41433
4369
20566
3856
counts
7804
5853
3902
1951
0
80659
60494
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SBS5
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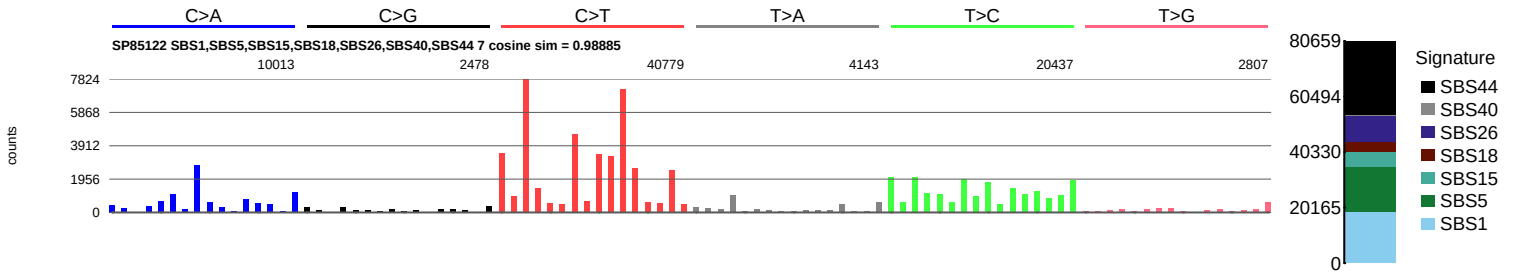
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T>C
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SP85122 SBS1,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98885
10013
2478
40779
4143
20437
2807
counts
7824
5868
3912
1956
0
80659
60494
40330
20165
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SBS26
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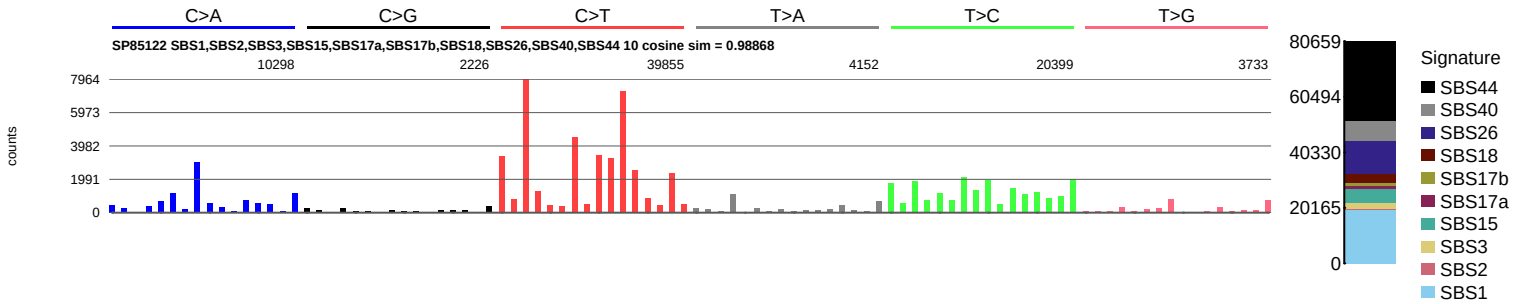
SBS5
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98868
10298
2226
39855
4152
20399
3733
counts
7964
5973
3982
1991
0
80659
60494
40330
20165
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SBS26
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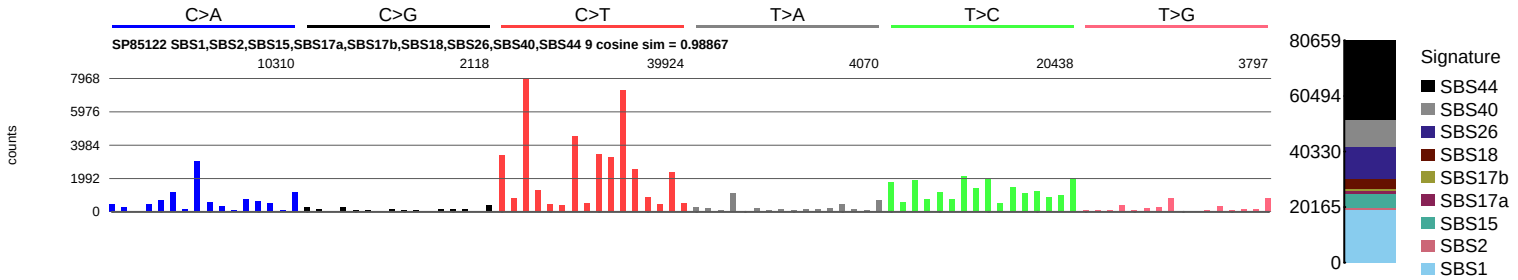
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98867
10310
2118
39924
4070
20438
3797
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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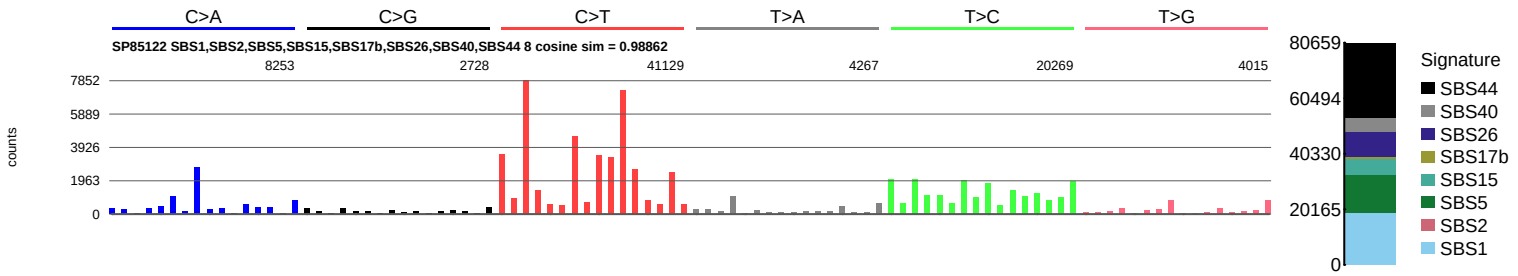
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98862
8253
2728
41129
4267
20269
4015
counts
7852
5889
3926
1963
0
80659
60494
40330
20165
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SBS17b
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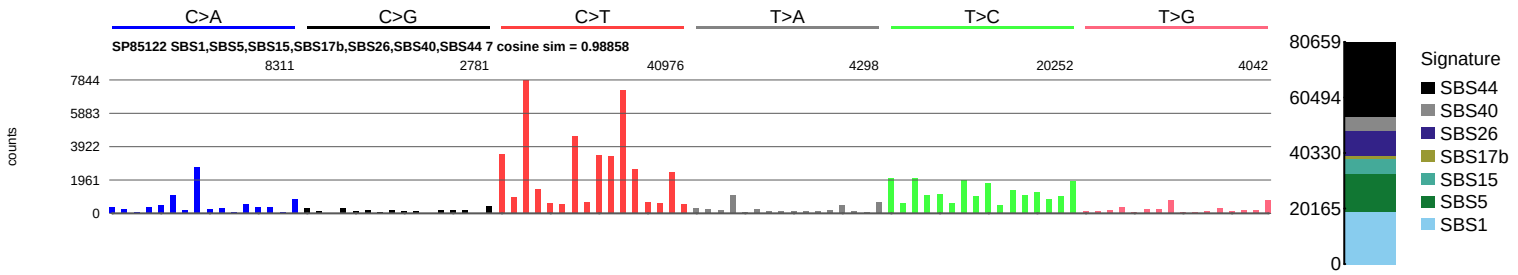
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98858
8311
2781
40976
4298
20252
4042
counts
7844
5883
3922
1961
0
80659
60494
40330
20165
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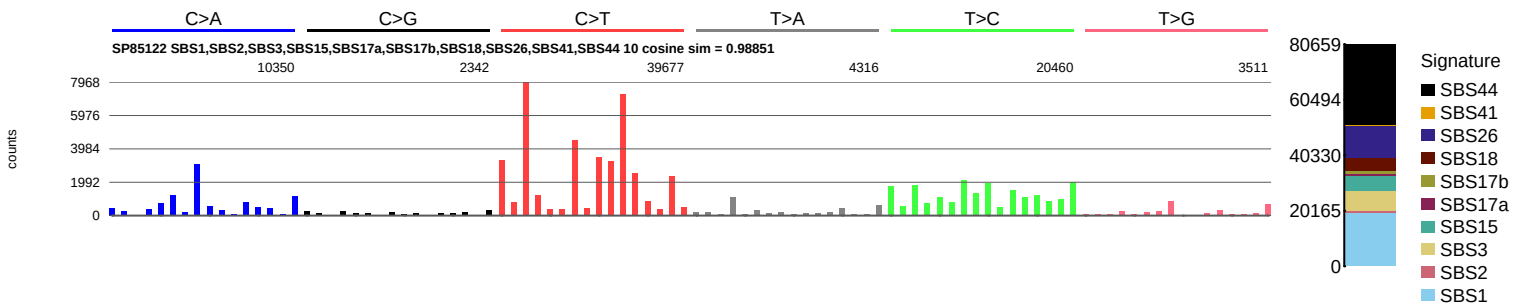
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98851
10350
2342
39677
4316
20460
3511
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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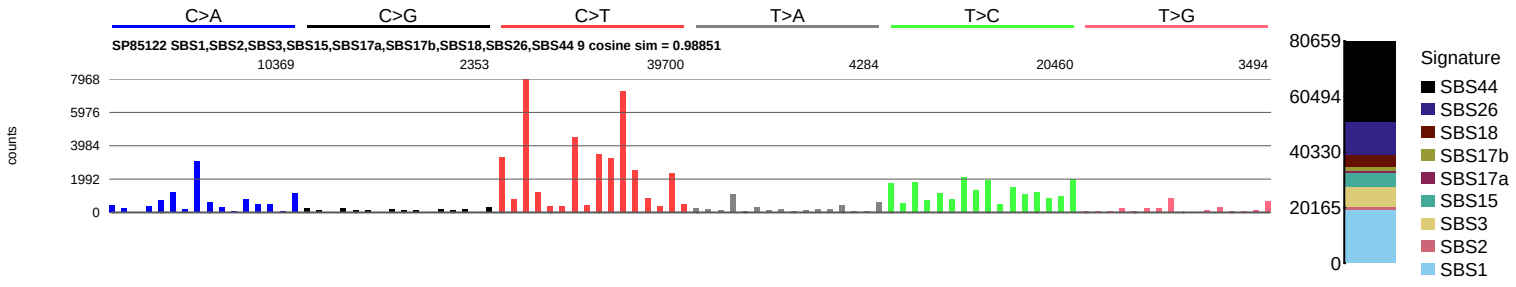
SBS2
SBS1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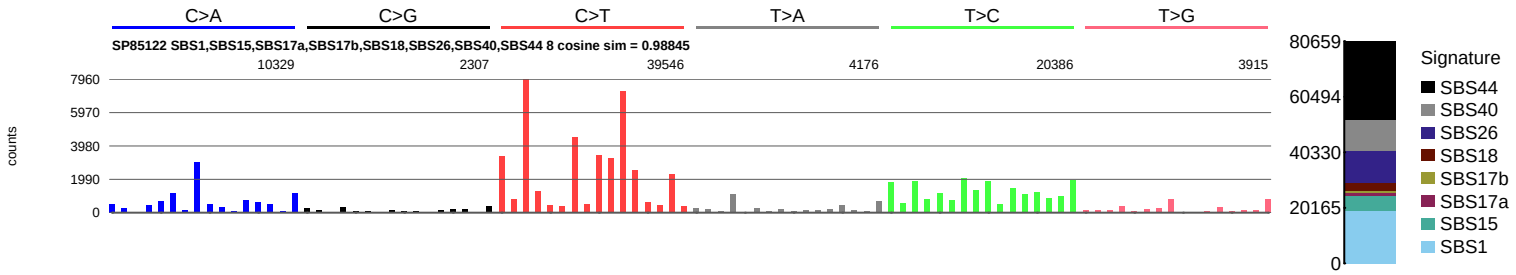
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98851
10369
2353
39700
4284
20460
3494
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98845
10329
2307
39546
4176
20386
3915
counts
7960
5970
3980
1990
0
80659
60494
40330
20165
0
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SBS44
SBS40
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SBS17b
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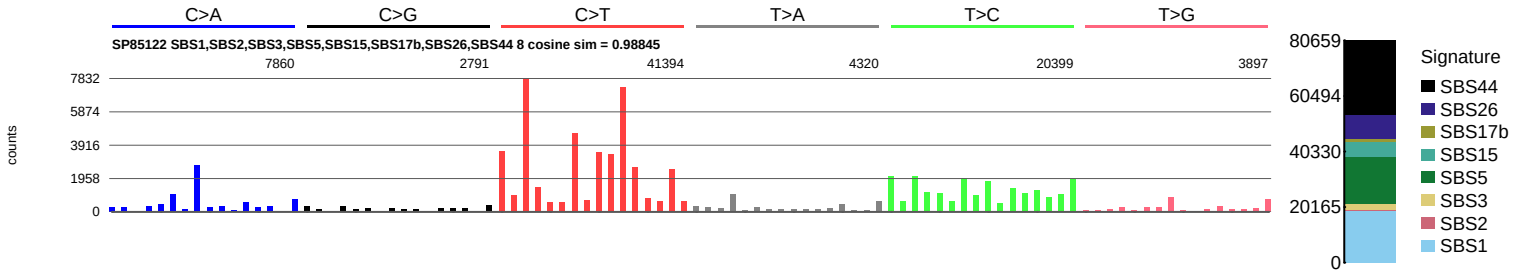
SBS15
SBS1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SP85122 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS44 8 cosine sim = 0.98845
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2791
41394
4320
20399
3897
counts
7832
5874
3916
1958
0
80659
60494
40330
20165
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SBS26
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SBS5
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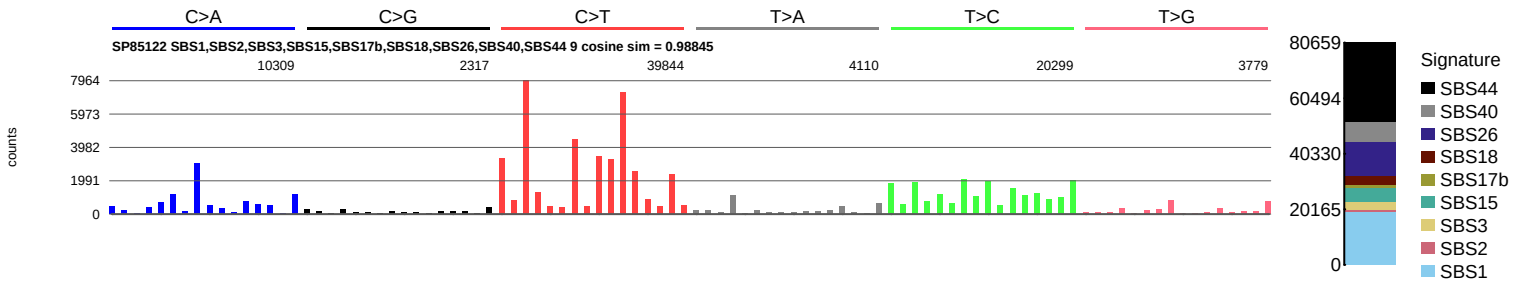
SBS2
SBS1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SP85122 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98845
10309
2317
39844
4110
20299
3779
counts
7964
5973
3982
1991
0
80659
60494
40330
20165
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SBS17b
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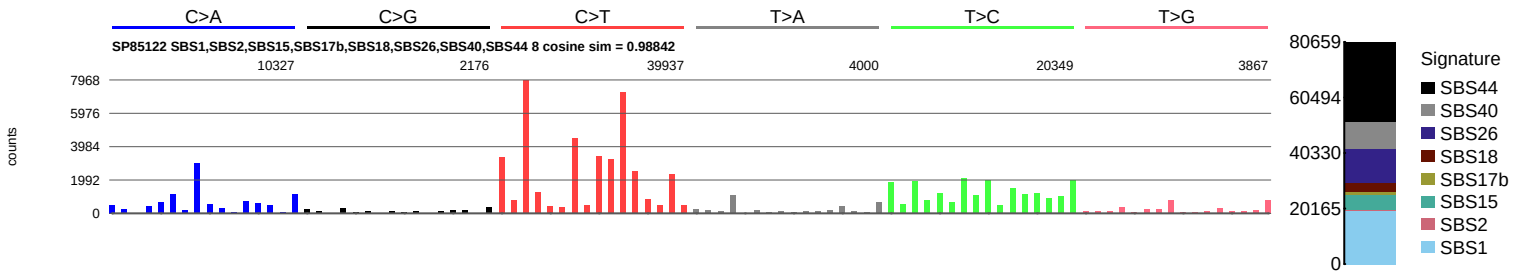
SBS3
SBS2
SBS1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10327
2176
39937
4000
20349
3867
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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SBS15
SBS2
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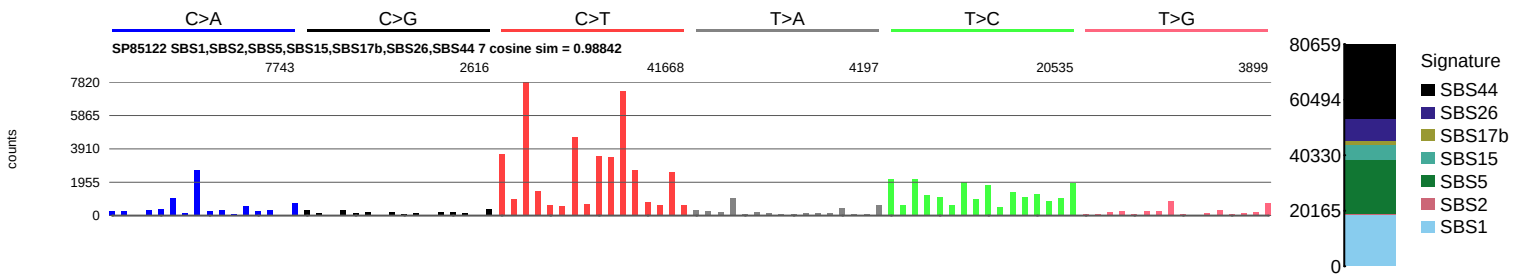
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SP85122 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98842
7743
2616
41668
4197
20535
3899
counts
7820
5865
3910
1955
0
80659
60494
40330
20165
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SBS26
SBS17b
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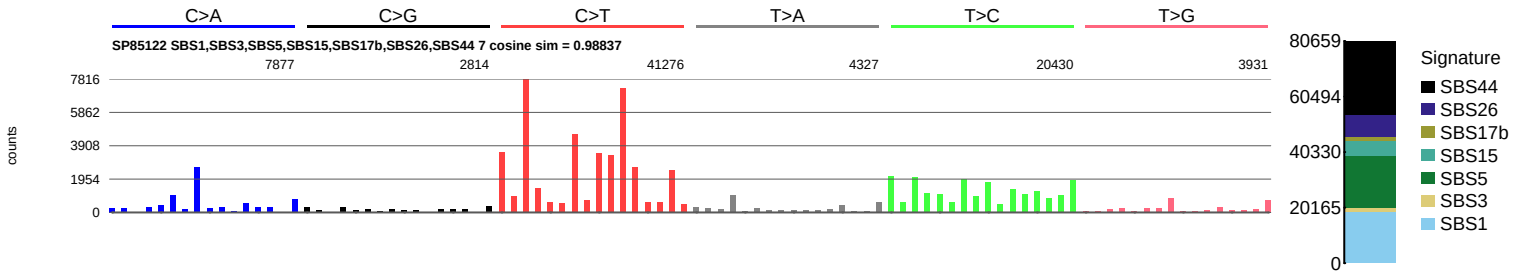
SBS5
SBS2
SBS1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SP85122 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98837
7877
2814
41276
4327
20430
3931
counts
7816
5862
3908
1954
0
80659
60494
40330
20165
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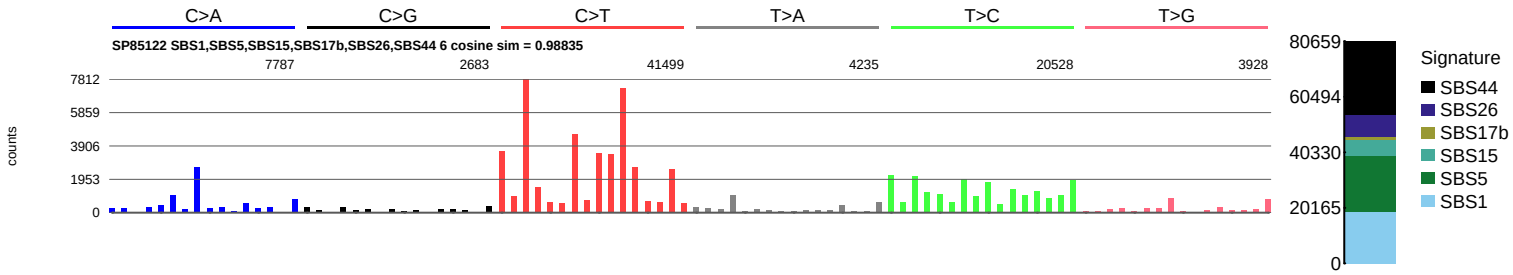
SBS26
SBS17b
SBS15
SBS5
SBS3
SBS1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SP85122 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98835
7787
2683
41499
4235
20528
3928
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
0
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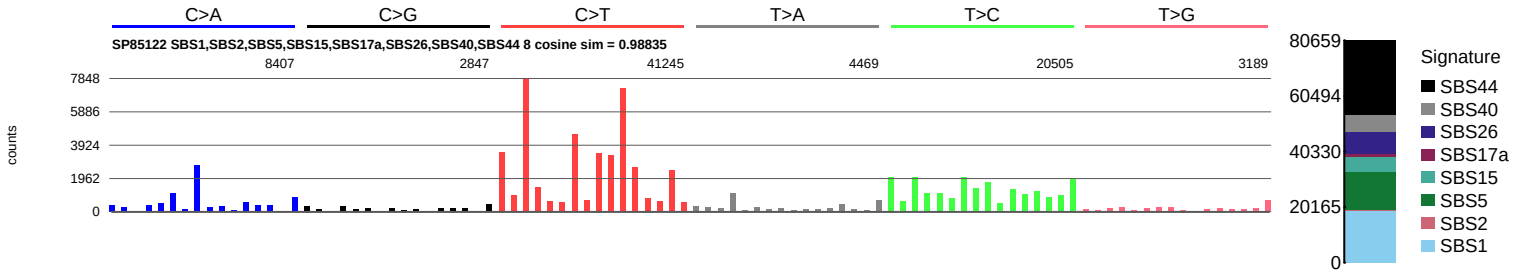
SBS44
SBS26
SBS17b
SBS15
SBS5
SBS1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SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS26,SBS40,SBS44 8 cosine sim = 0.98835
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2847
41245
4469
20505
3189
counts
7848
5886
3924
1962
0
80659
60494
40330
20165
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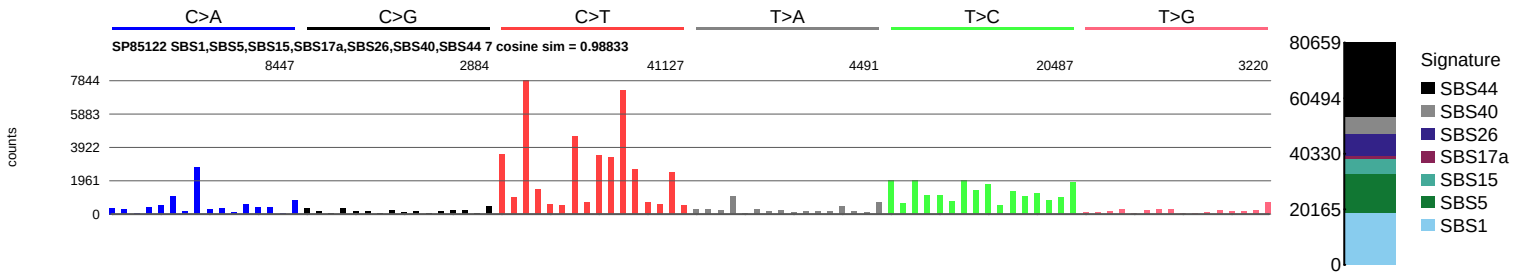
SBS40
SBS26
SBS17a
SBS15
SBS5
SBS2
SBS1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SP85122 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS40,SBS44 7 cosine sim = 0.98833
8447
2884
41127
4491
20487
3220
counts
7844
5883
3922
1961
0
80659
60494
40330
20165
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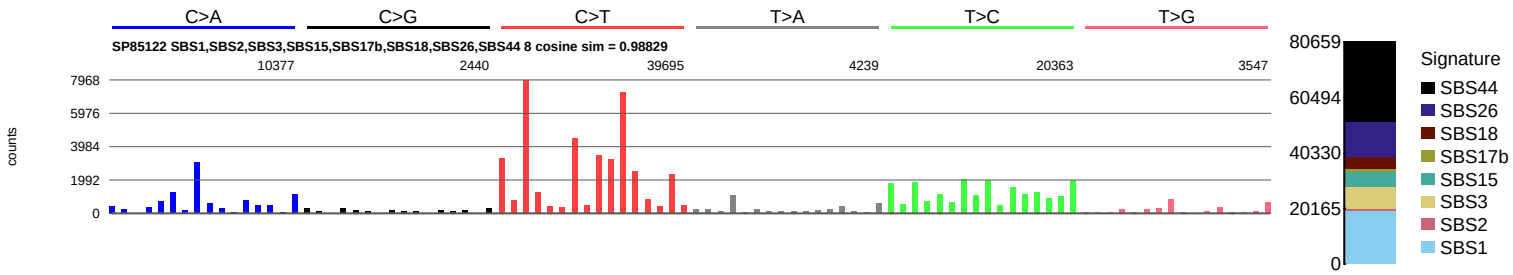
SBS40
SBS26
SBS17a
SBS15
SBS5
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98829
10377
2440
39695
4239
20363
3547
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
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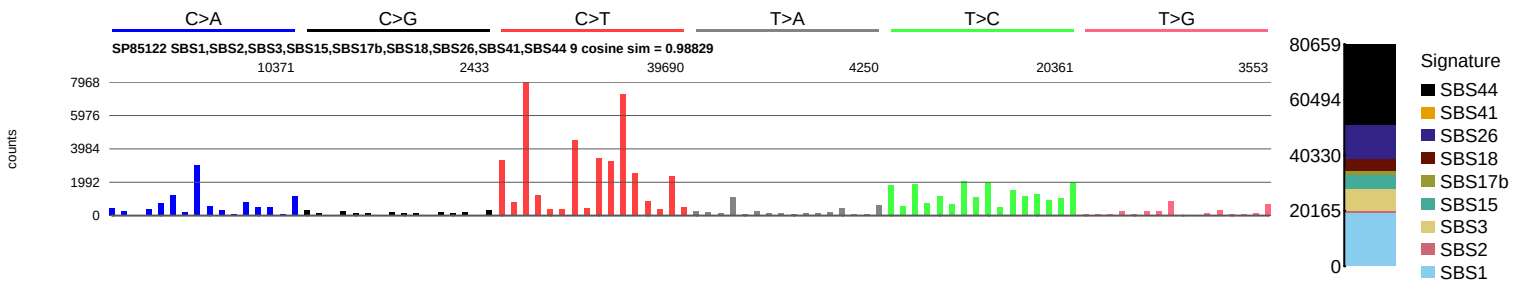
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98829
10371
2433
39690
4250
20361
3553
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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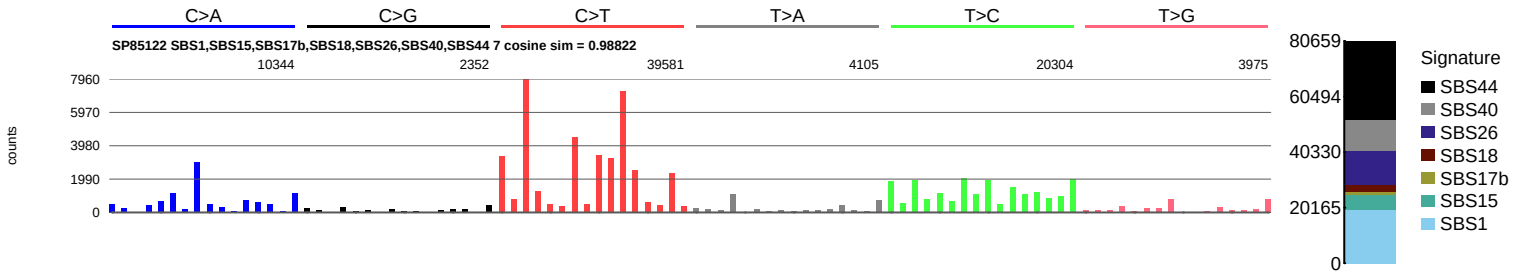
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98822
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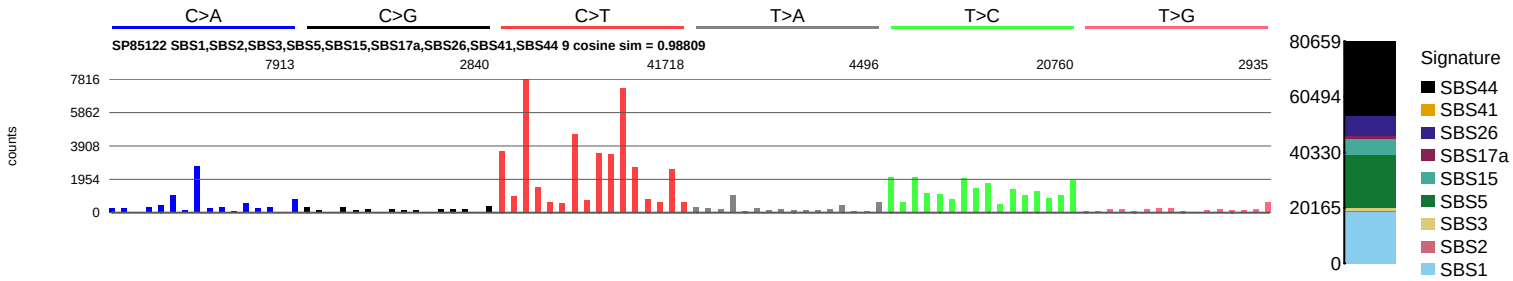
2352
39581
4105
20304
3975
counts
7960
5970
3980
1990
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 9 cosine sim = 0.98809
7913
2840
41718
4496
20760
2935
counts
7816
5862
3908
1954
0
80659
60494
40330
20165
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Signature
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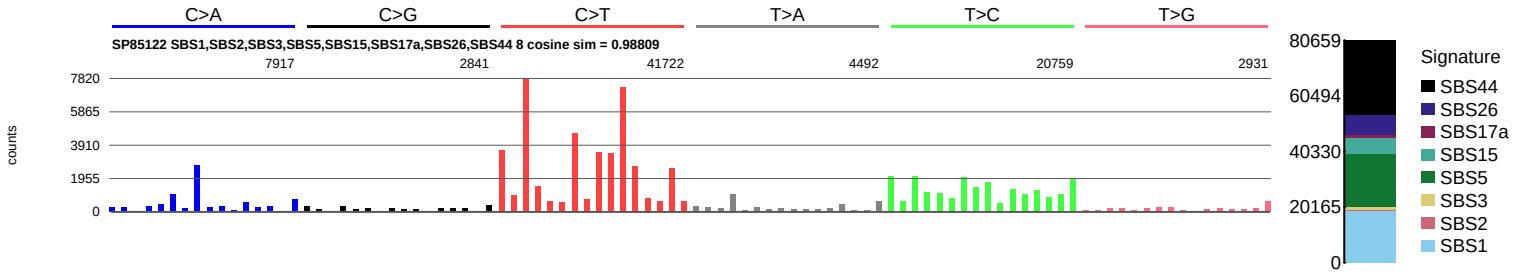
SBS41
SBS26
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS44 8 cosine sim = 0.98809
7917
2841
41722
4492
20759
2931
counts
7820
5865
3910
1955
0
80659
60494
40330
20165
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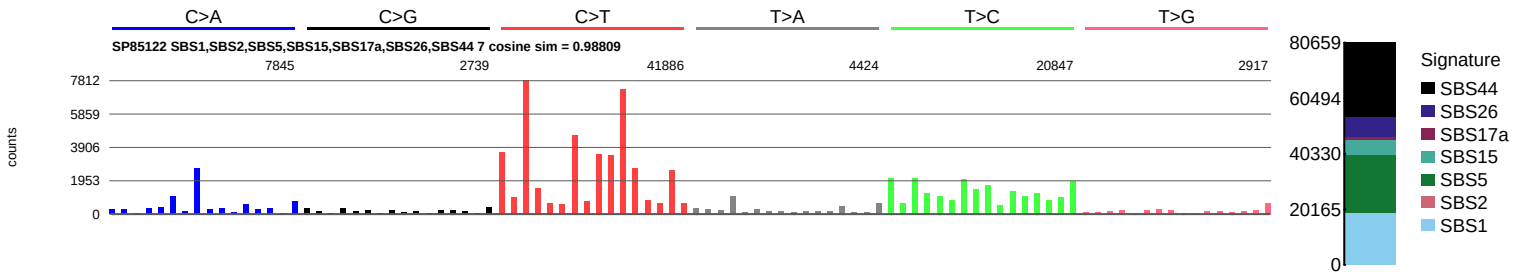
SBS2
SBS1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C>G
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T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98809
7845
2739
41886
4424
20847
2917
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
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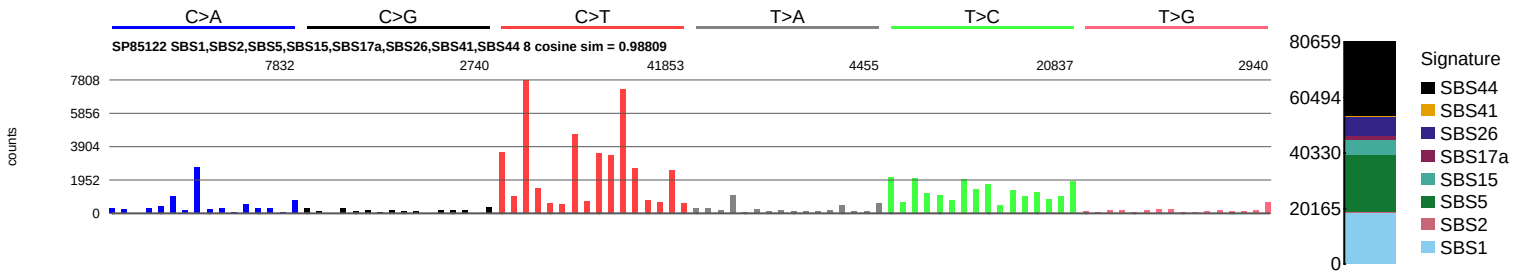
SBS17a
SBS15
SBS5
SBS2
SBS1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SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98809
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2740
41853
4455
20837
2940
counts
7808
5856
3904
1952
0
80659
60494
40330
20165
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SBS41
SBS26
SBS17a
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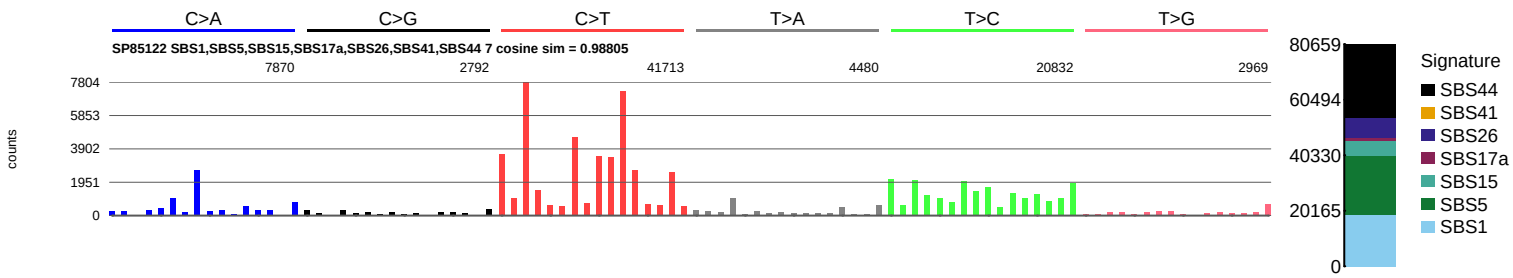
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98805
7870
2792
41713
4480
20832
2969
counts
7804
5853
3902
1951
0
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60494
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20165
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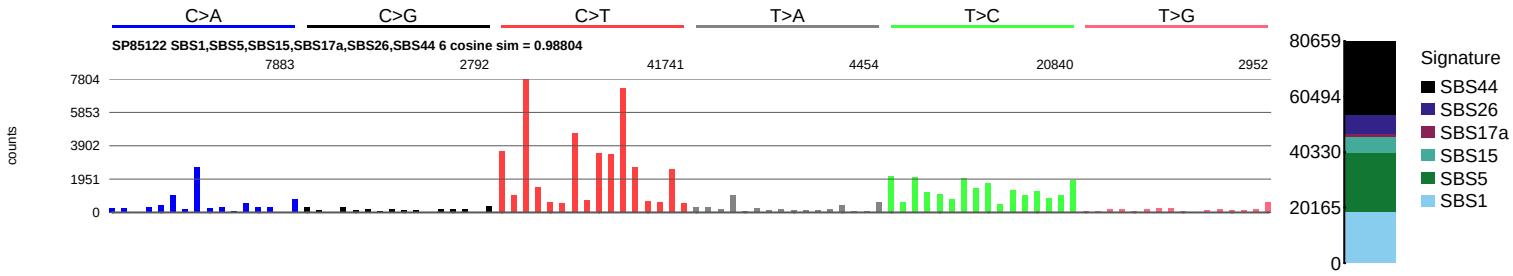
SBS5
SBS1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T>C
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SP85122 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98804
7883
2792
41741
4454
20840
2952
counts
7804
5853
3902
1951
0
80659
60494
40330
20165
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SBS26
SBS17a
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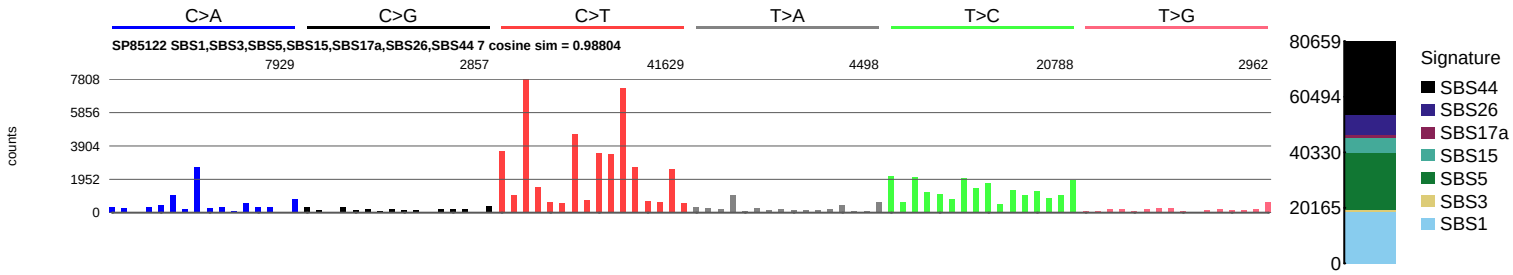
SBS5
SBS1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C>G
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T>C
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SP85122 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98804
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2857
41629
4498
20788
2962
counts
7808
5856
3904
1952
0
80659
60494
40330
20165
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SBS26
SBS17a
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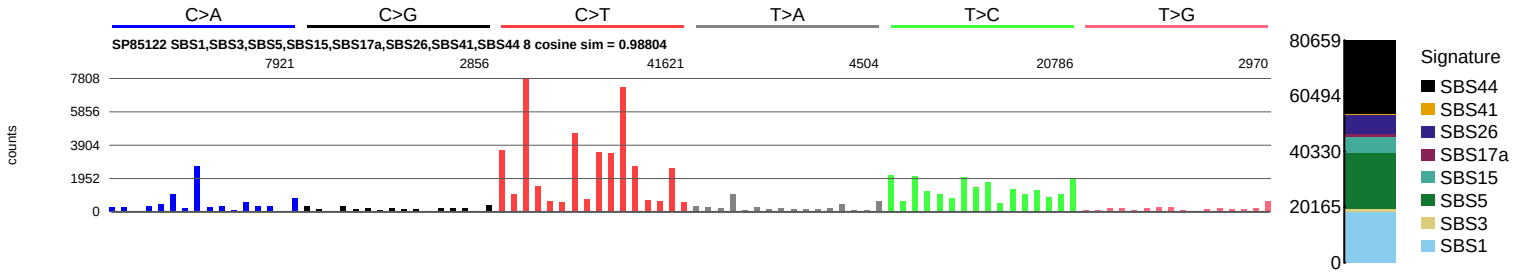
SBS3
SBS1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T>C
T>G
SP85122 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98804
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2856
41621
4504
20786
2970
counts
7808
5856
3904
1952
0
80659
60494
40330
20165
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Signature
SBS44
SBS41
SBS26
SBS17a
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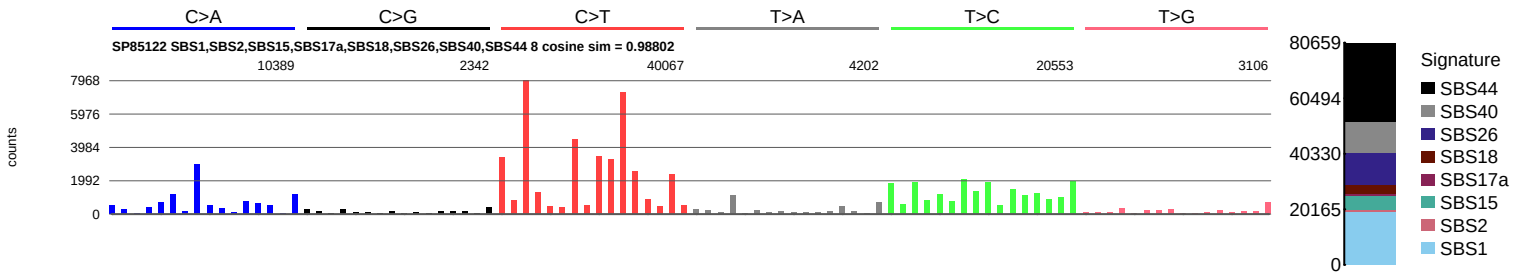
SBS3
SBS1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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98802
10389
2342
40067
4202
20553
3106
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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Signature
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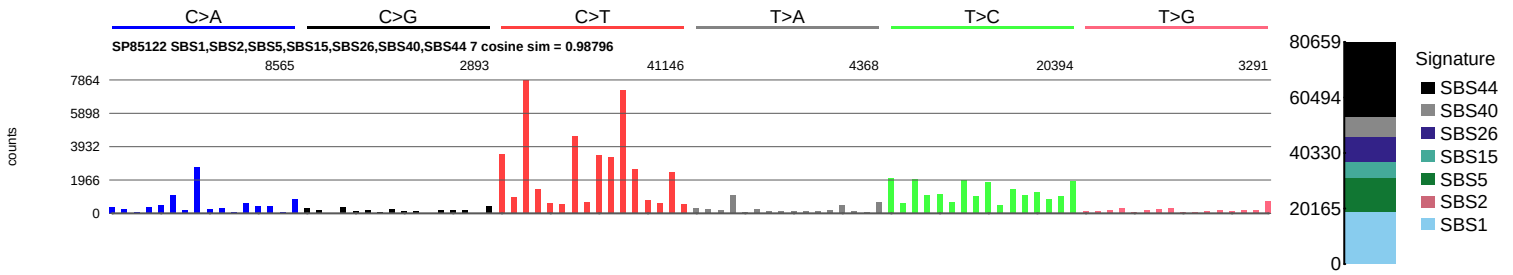
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS26,SBS40,SBS44 7 cosine sim = 0.98796
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2893
41146
4368
20394
3291
counts
7864
5898
3932
1966
0
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60494
40330
20165
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SBS44
SBS40
SBS26
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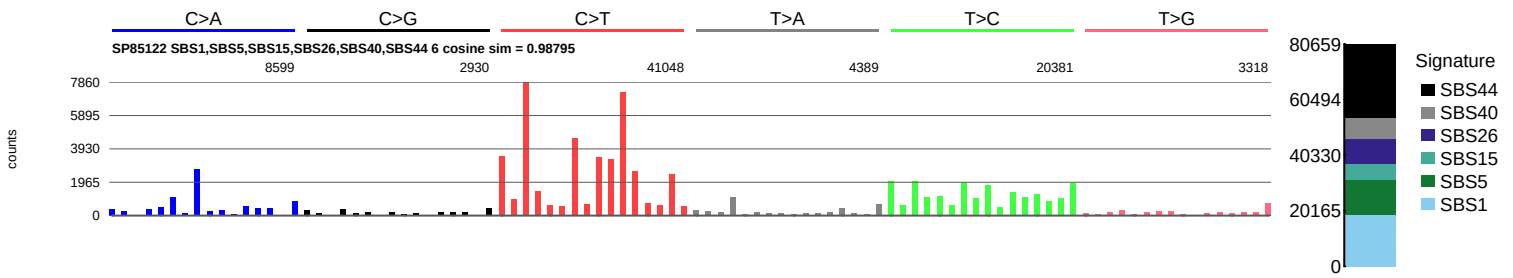
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS26,SBS40,SBS44 6 cosine sim = 0.98795
8599
2930
41048
4389
20381
3318
counts
7860
5895
3930
1965
0
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60494
40330
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SBS40
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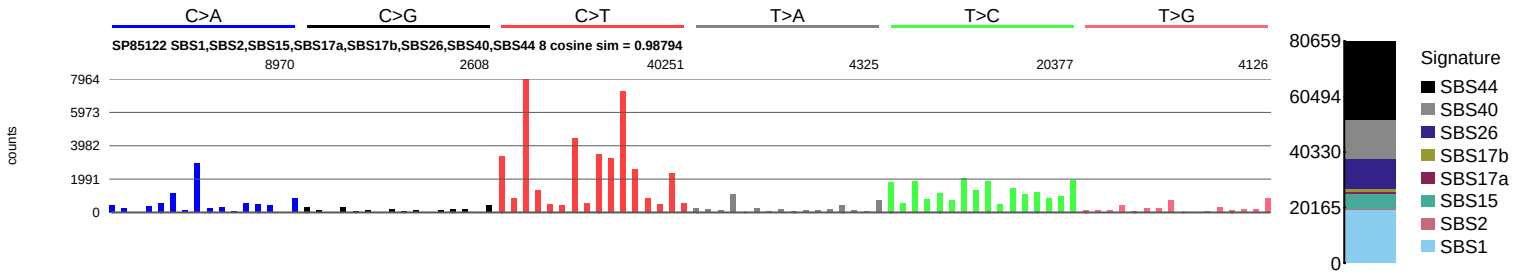
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98794
8970
2608
40251
4325
20377
4126
counts
7964
5973
3982
1991
0
80659
60494
40330
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0
Signature
SBS44
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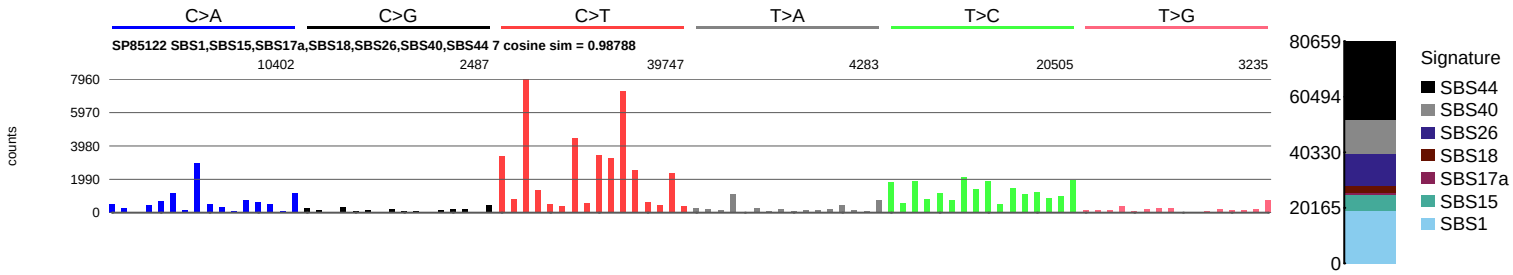
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98788
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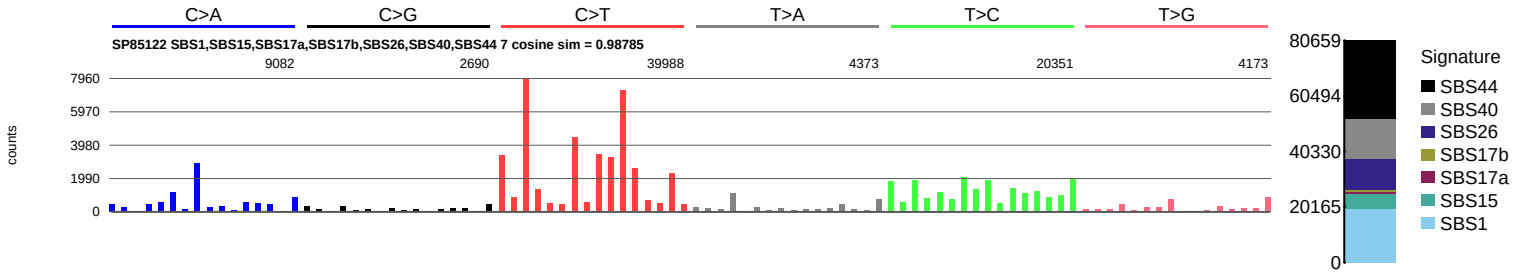
2487
39747
4283
20505
3235
counts
7960
5970
3980
1990
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1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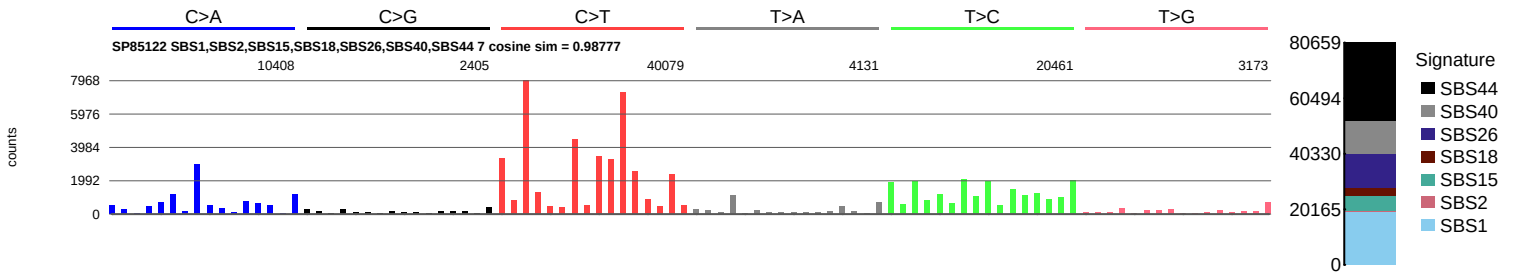
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98785
9082
2690
39988
4373
20351
4173
counts
7960
5970
3980
1990
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1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SP85122 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98777
10408
2405
40079
4131
20461
3173
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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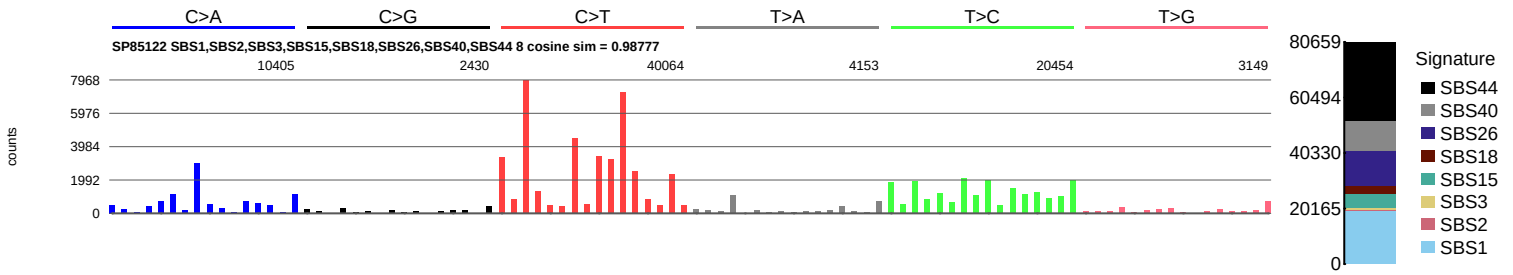
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98777
10405
2430
40064
4153
20454
3149
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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Signature
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SBS40
SBS26
SBS18
SBS15
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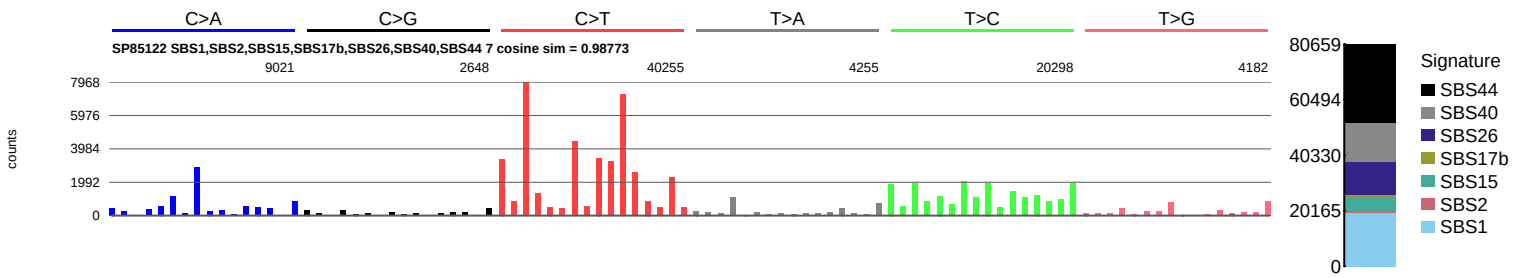
SBS2
SBS1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SP85122 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98773
9021
2648
40255
4255
20298
4182
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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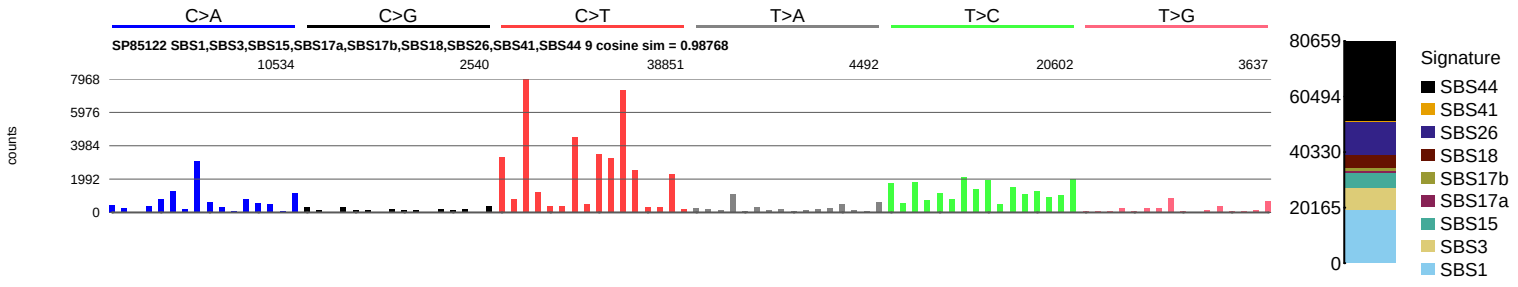
SBS40
SBS26
SBS17b
SBS15
SBS2
SBS1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SP85122 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98768
10534
2540
38851
4492
20602
3637
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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Signature
SBS44
SBS41
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SBS18
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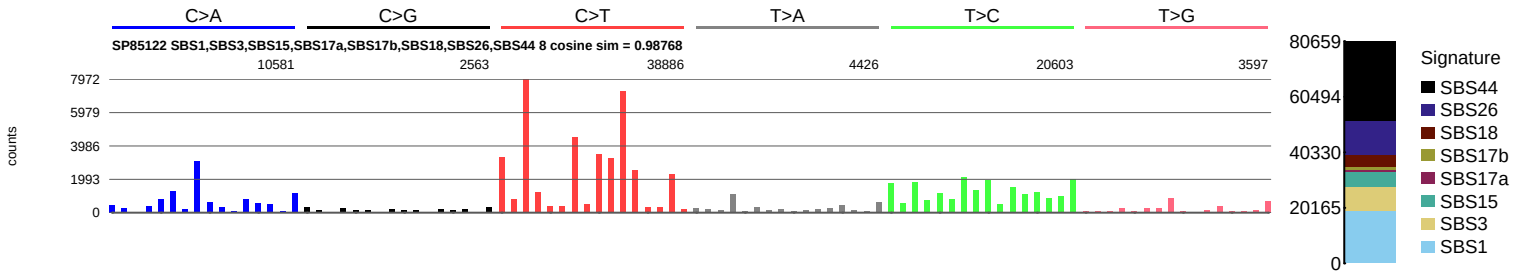
SBS17a
SBS15
SBS3
SBS1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C>T
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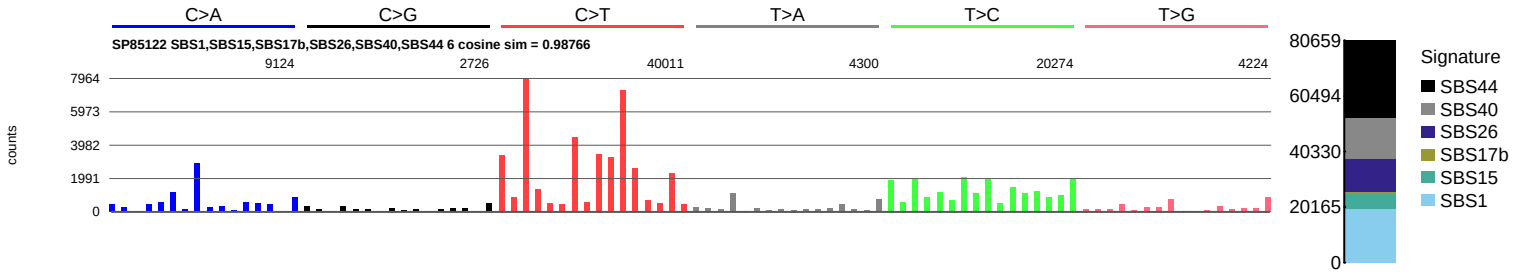
T>G
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98768
10581
2563
38886
4426
20603
3597
counts
7972
5979
3986
1993
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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T>G
SP85122 SBS1,SBS15,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98766
9124
2726
40011
4300
20274
4224
counts
7964
5973
3982
1991
0
80659
60494
40330
20165
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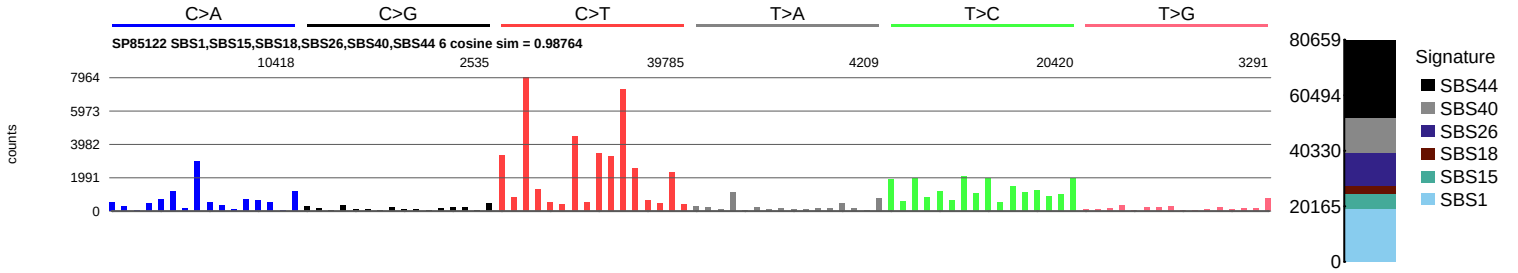
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98764
10418
2535
39785
4209
20420
3291
counts
7964
5973
3982
1991
0
80659
60494
40330
20165
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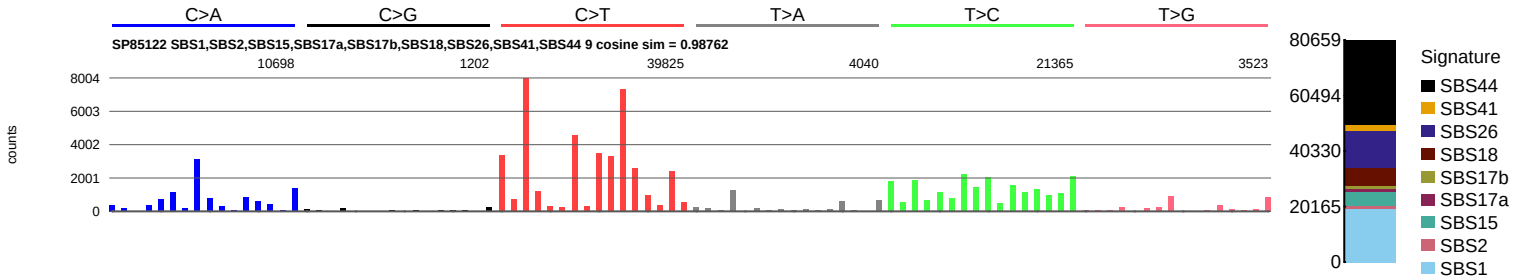
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS1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SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98762
10698
1202
39825
4040
21365
3523
counts
8004
6003
4002
2001
0
80659
60494
40330
20165
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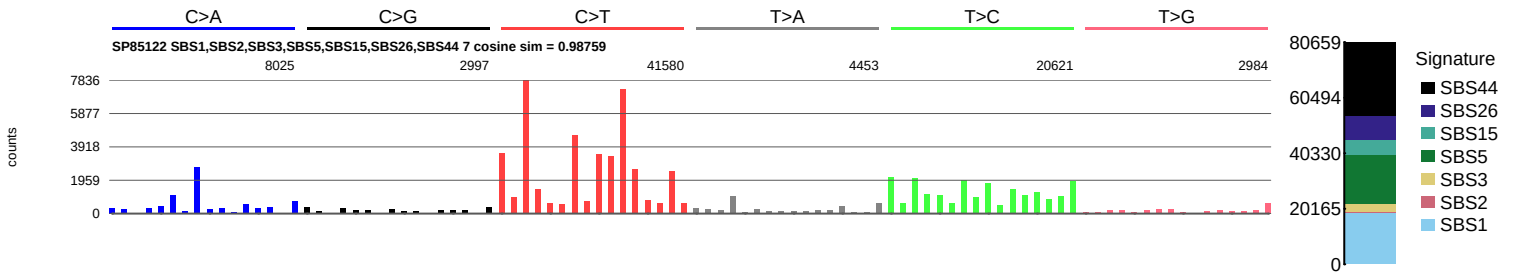
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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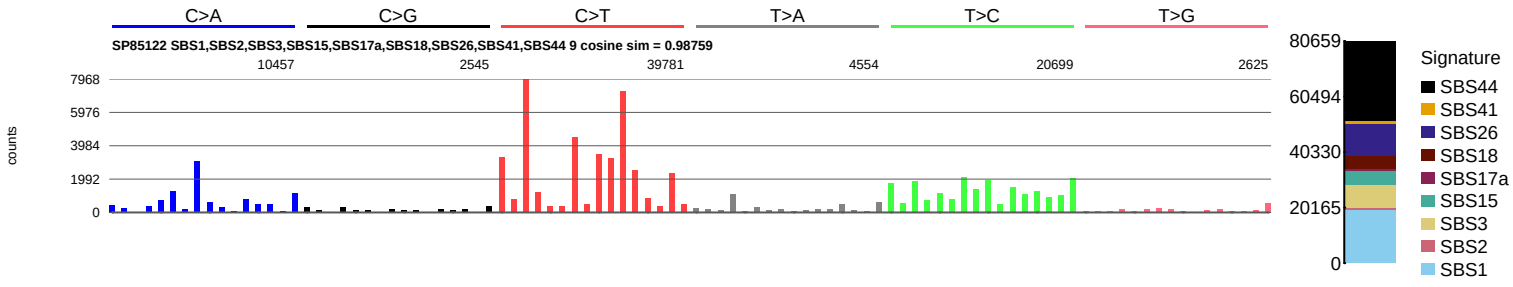
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS15,SBS26,SBS44 7 cosine sim = 0.98759
8025
2997
41580
4453
20621
2984
counts
7836
5877
3918
1959
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS15
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98759
10457
2545
39781
4554
20699
2625
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
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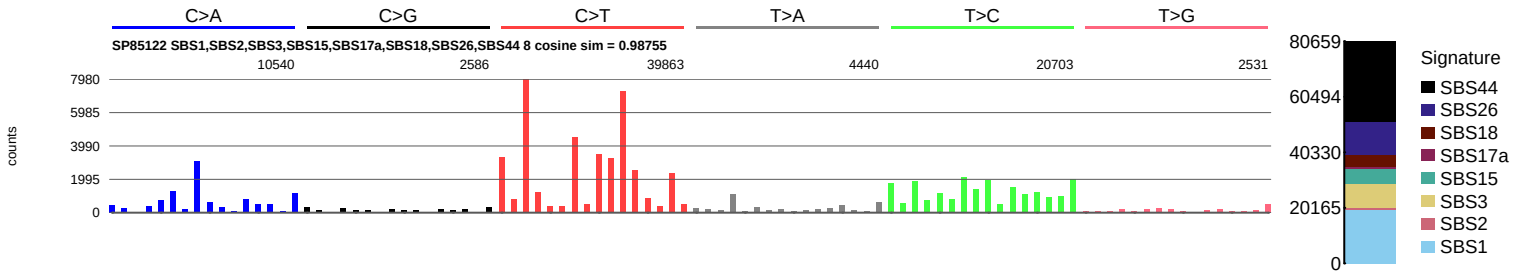
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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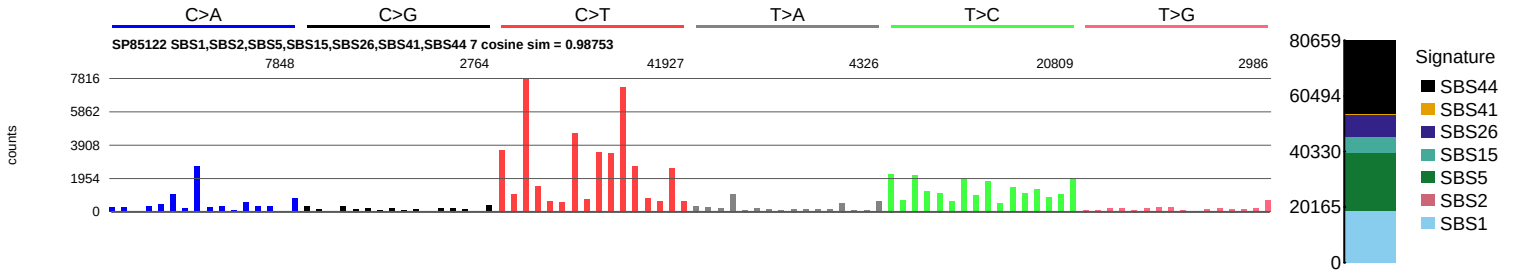
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS44 8 cosine sim = 0.98755
10540
2586
39863
4440
20703
2531
counts
7980
5985
3990
1995
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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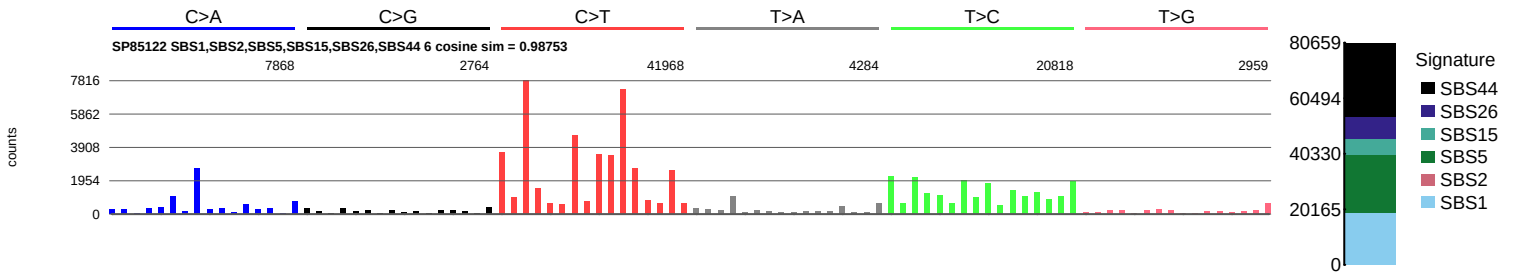
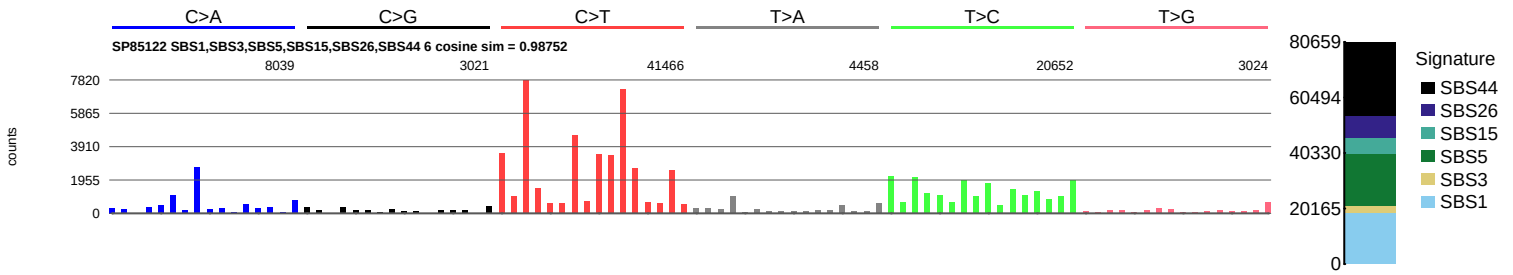
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.98753
7848
2764
41927
4326
20809
2986
counts
7816
5862
3908
1954
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS26,SBS44 6 cosine sim = 0.98753
7868
2764
41968
4284
20818
2959
counts
7816
5862
3908
1954
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS5,SBS15,SBS26,SBS44 6 cosine sim = 0.98752
8039
3021
41466
4458
20652
3024
counts
7820
5865
3910
1955
0
80659
60494
40330
20165
0
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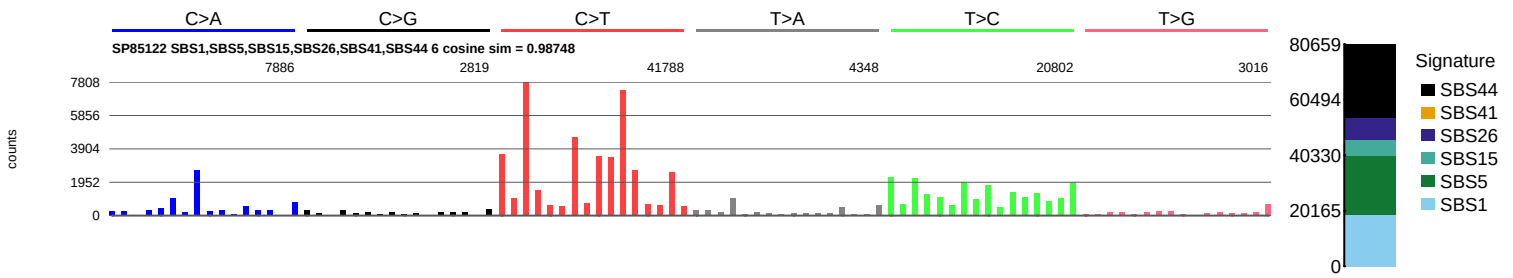
SBS44
SBS26
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98748
7886
2819
41788
4348
20802
3016
counts
7808
5856
3904
1952
0
80659
60494
40330
20165
0
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SBS41
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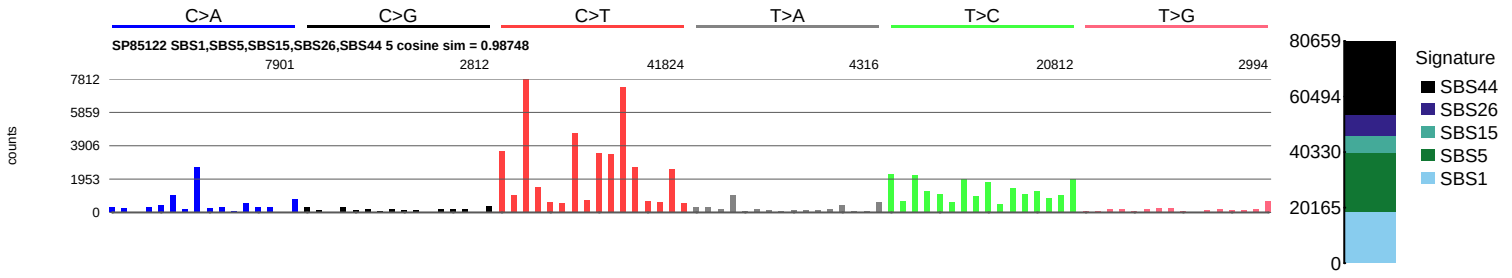
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS26,SBS44 5 cosine sim = 0.98748
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2812
41824
4316
20812
2994
counts
7812
5859
3906
1953
0
80659
60494
40330
20165
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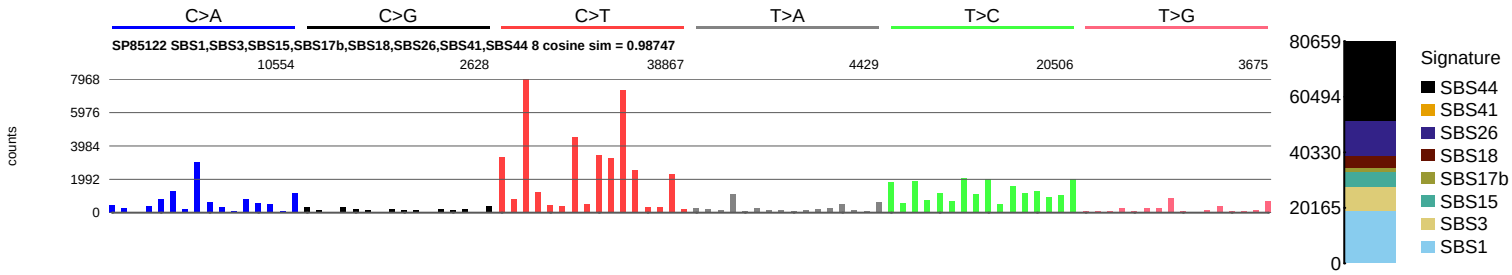
Signature
SBS44
SBS26
SBS15
SBS5
SBS1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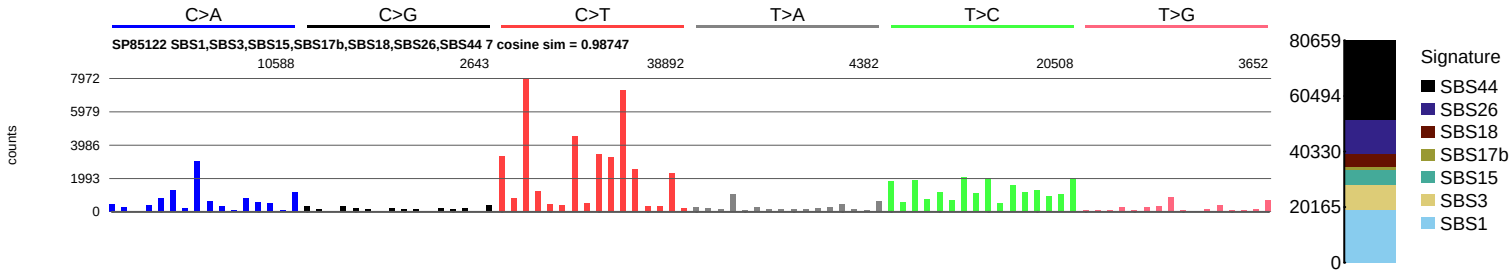
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98747
10554
2628
38867
4429
20506
3675
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS1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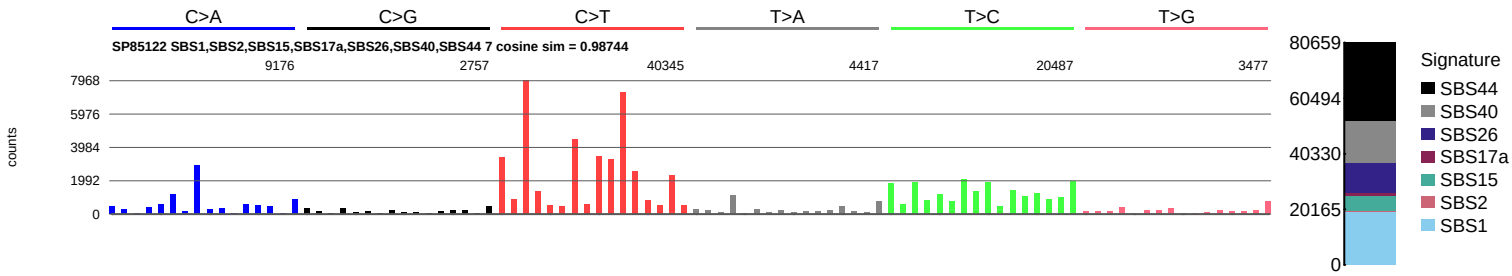
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98747
10588
2643
38892
4382
20508
3652
counts
7972
5979
3986
1993
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS1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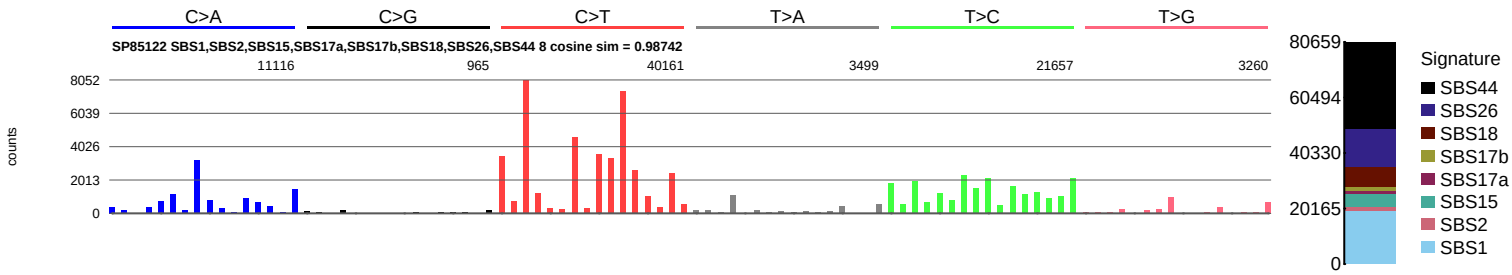
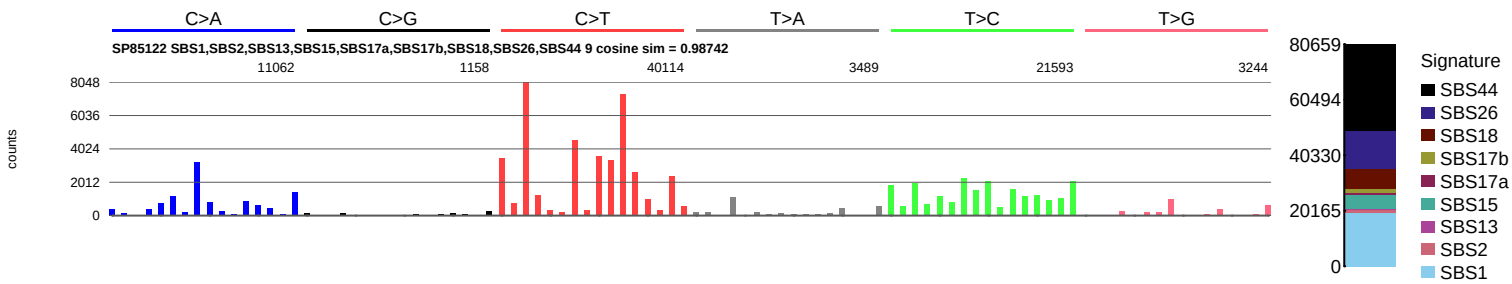
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS26,SBS40,SBS44 7 cosine sim = 0.98744
9176
2757
40345
4417
20487
3477
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98742
11116
965
40161
3499
21657
3260
counts
8052
6039
4026
2013
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98742
11062
1158
40114
3489
21593
3244
counts
8048
6036
4024
2012
0
80659
60494
40330
20165
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Signature
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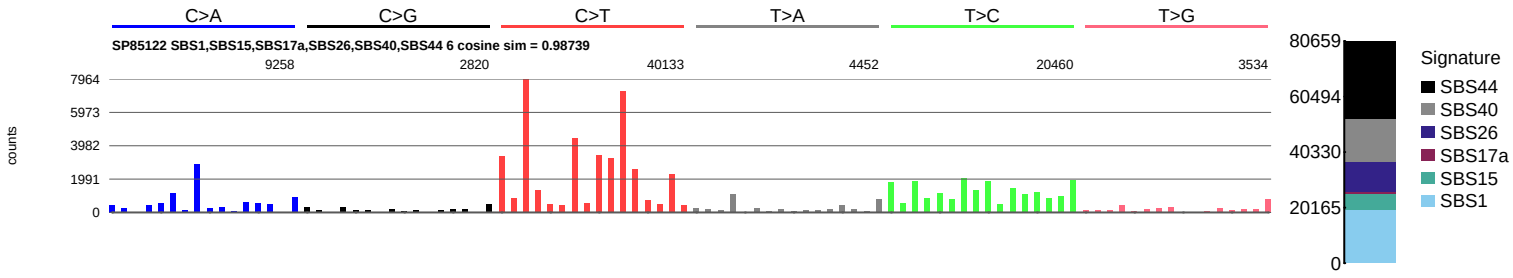
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS26,SBS40,SBS44 6 cosine sim = 0.98739
9258
2820
40133
4452
20460
3534
counts
7964
5973
3982
1991
0
80659
60494
40330
20165
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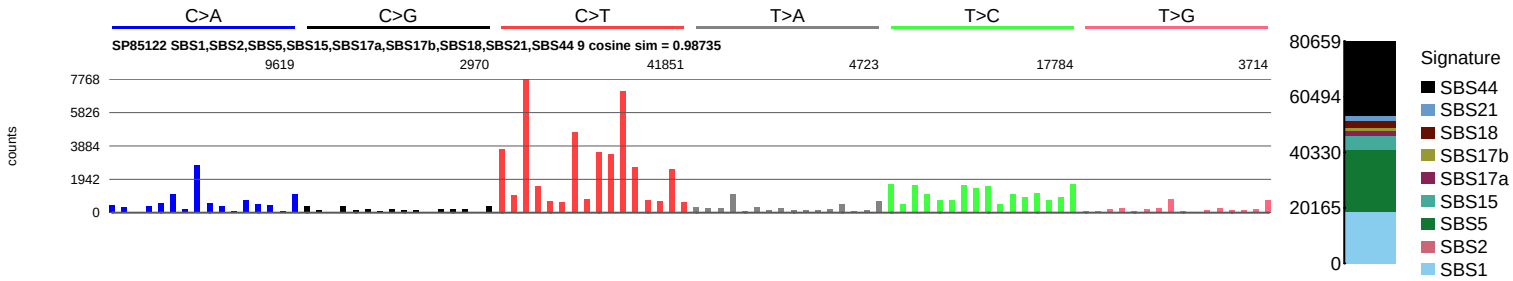
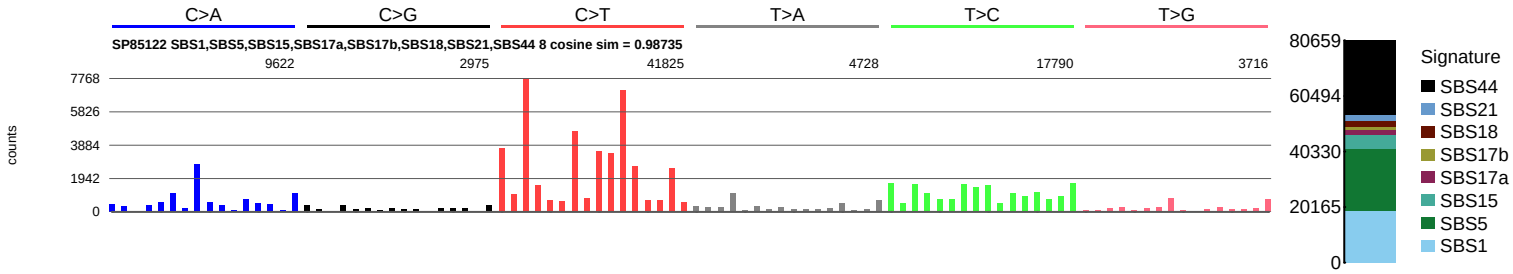
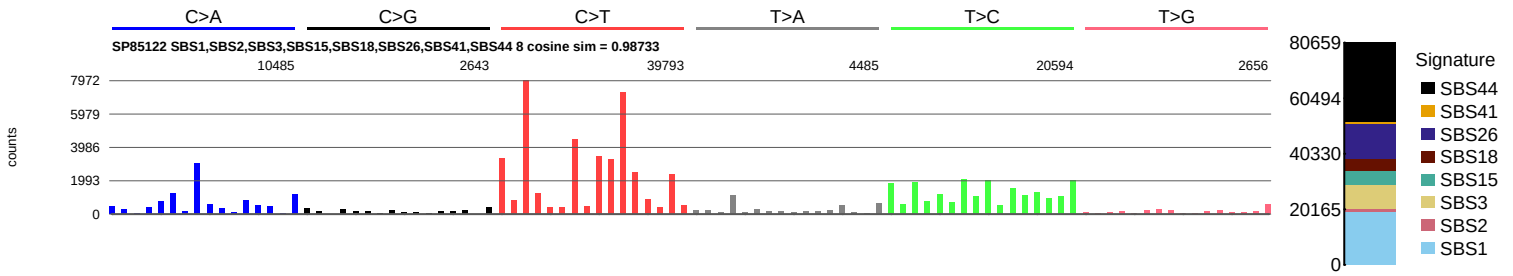
SBS40
SBS26
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98735
9619
2970
41851
4723
17784
3714
counts
7768
5826
3884
1942
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98735
9622
2975
41825
4728
17790
3716
counts
7768
5826
3884
1942
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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SP85122 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98733
10485
2643
39793
4485
20594
2656
counts
7972
5979
3986
1993
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
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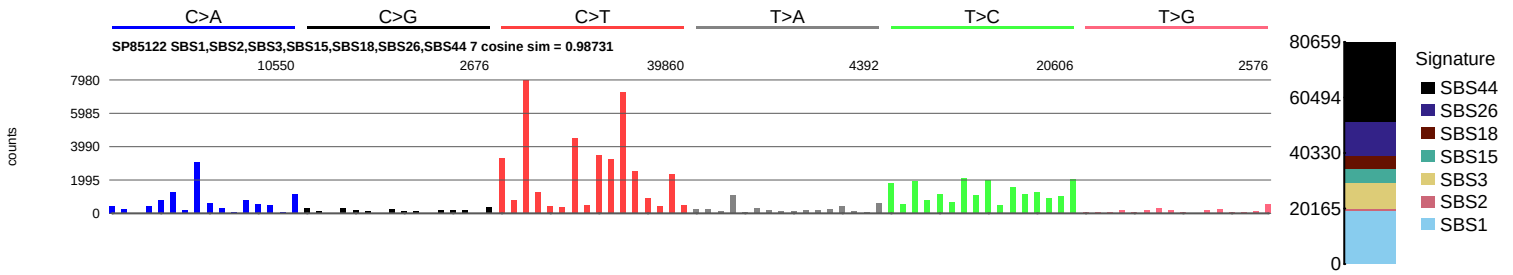
SBS18
SBS15
SBS3
SBS2
SBS1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T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98731
10550
2676
39860
4392
20606
2576
counts
7980
5985
3990
1995
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
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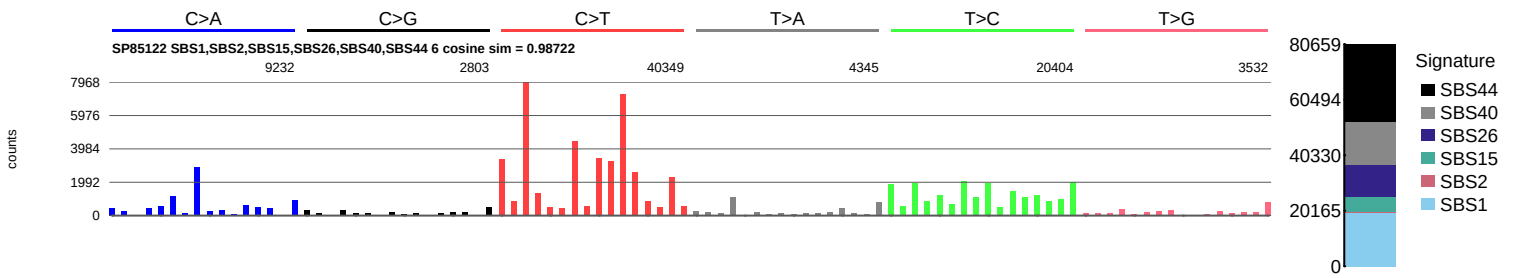
SBS2
SBS1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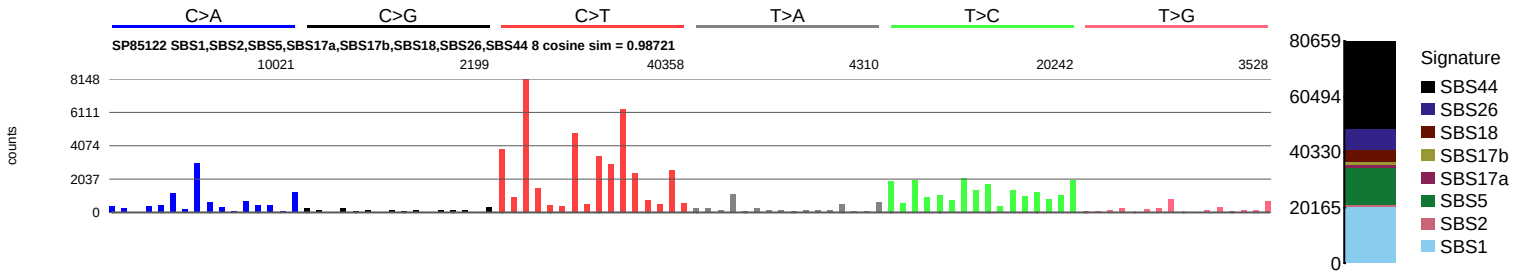
SP85122 SBS1,SBS2,SBS15,SBS26,SBS40,SBS44 6 cosine sim = 0.98722
9232
2803
40349
4345
20404
3532
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS15
SBS2
SBS1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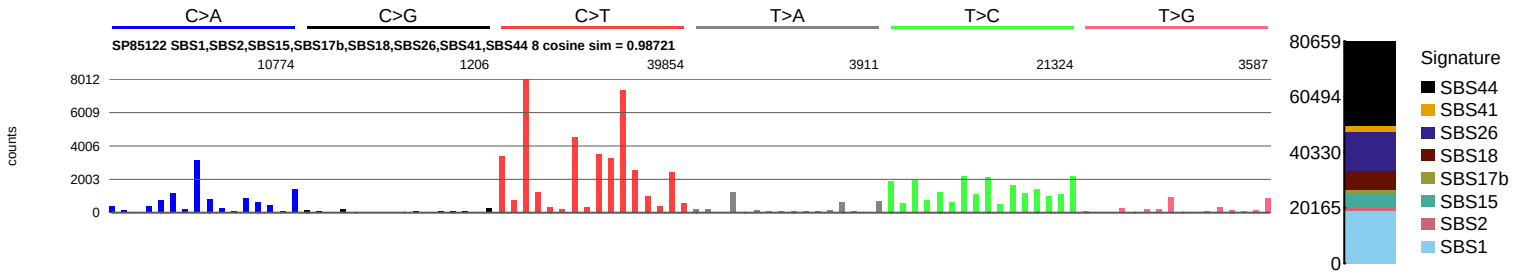
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98721
10021
2199
40358
4310
20242
3528
8148
6111
4074
2037
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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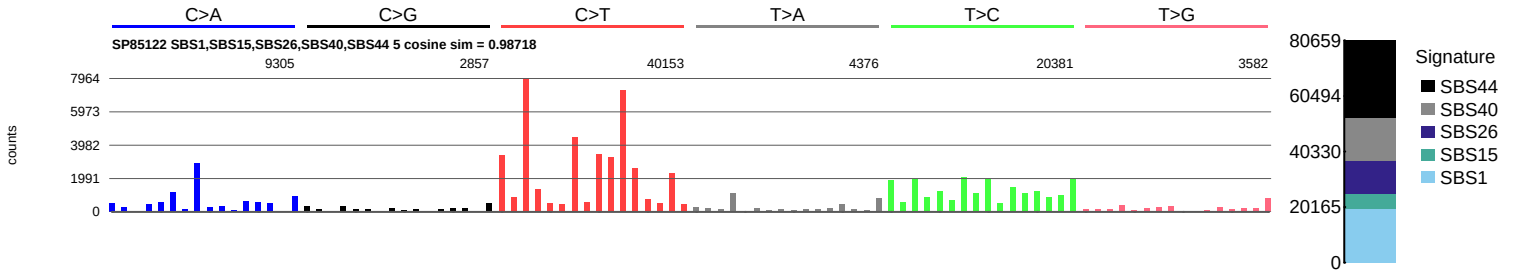
SP85122 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98721
10774
1206
39854
3911
21324
3587
8012
6009
4006
2003
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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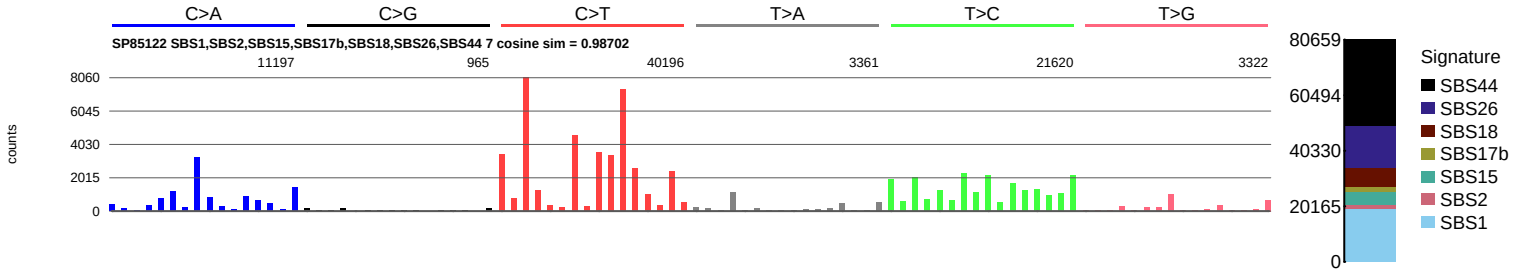
T>C
T>G
SP85122 SBS1,SBS15,SBS26,SBS40,SBS44 5 cosine sim = 0.98718
9305
2857
40153
4376
20381
3582
7964
5973
3982
1991
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS15
SBS1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C>G
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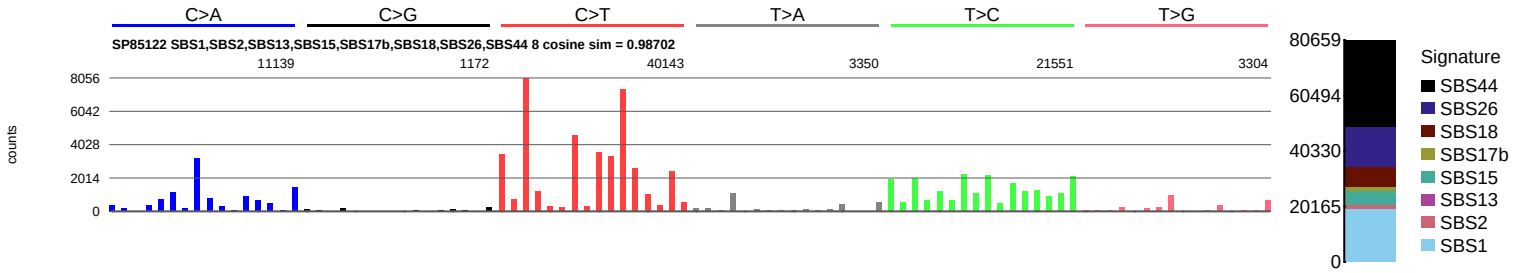
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98702
11197
965
40196
3361
21620
3322
8060
6045
4030
2015
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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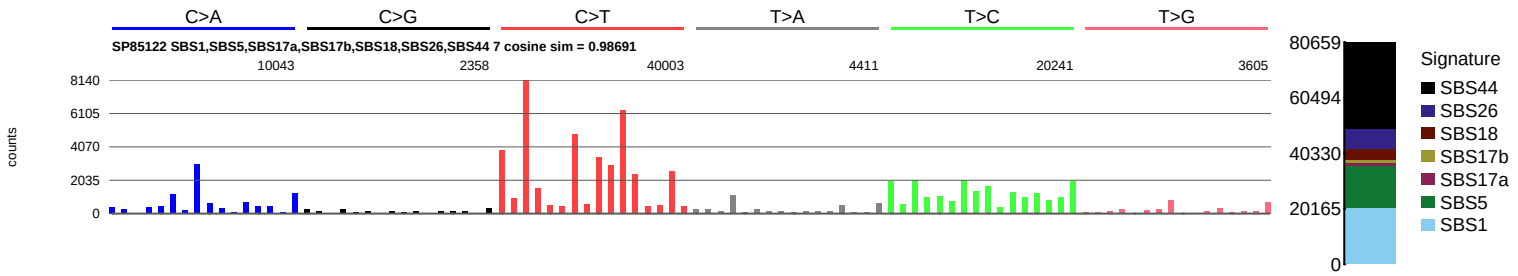
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98702
11139
1172
40143
3350
21551
3304
8056
6042
4028
2014
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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C>G
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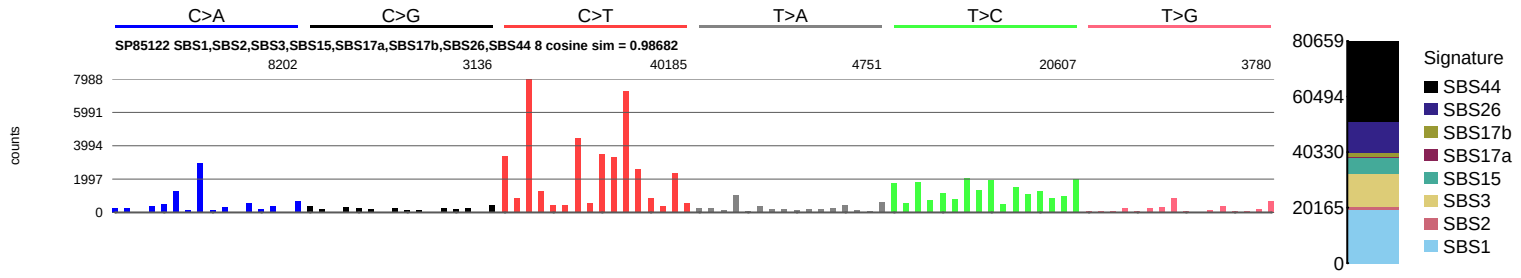
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98691
10043
2358
40003
4411
20241
3605
8140
6105
4070
2035
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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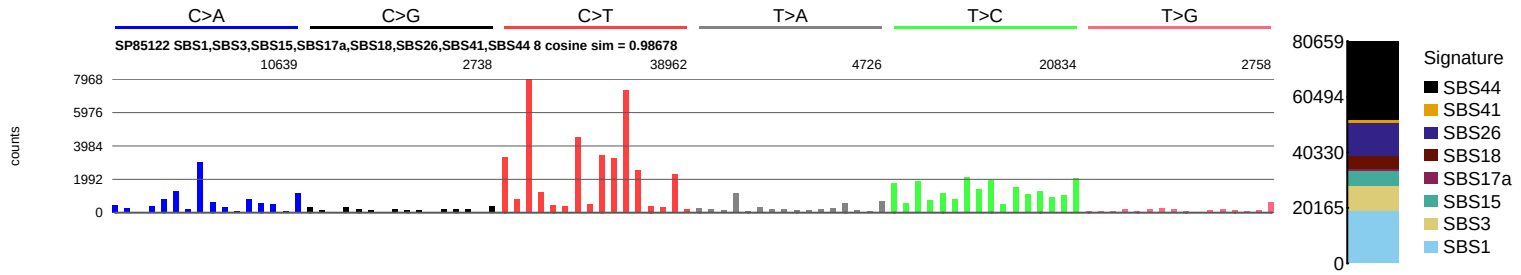
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98682
8202
3136
40185
4751
20607
3780
counts
7988
5991
3994
1997
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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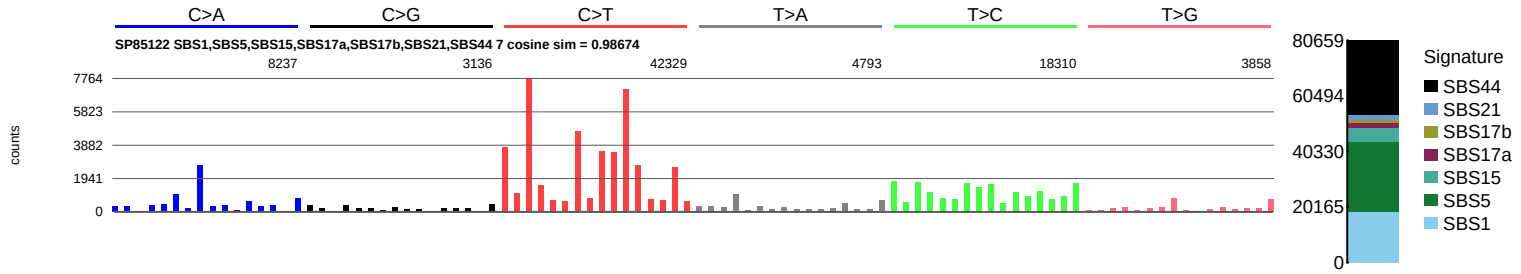
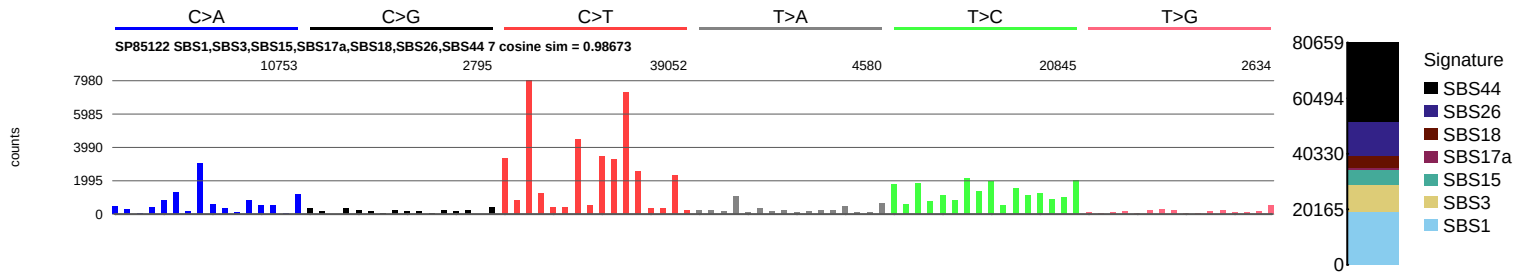
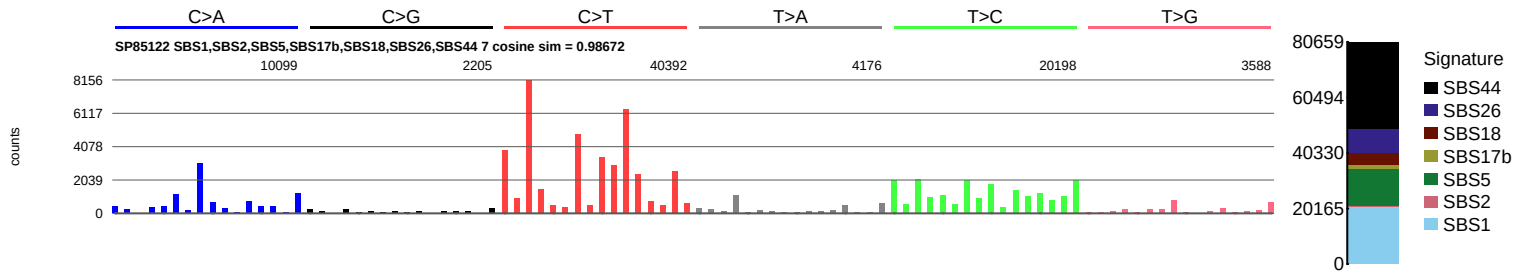
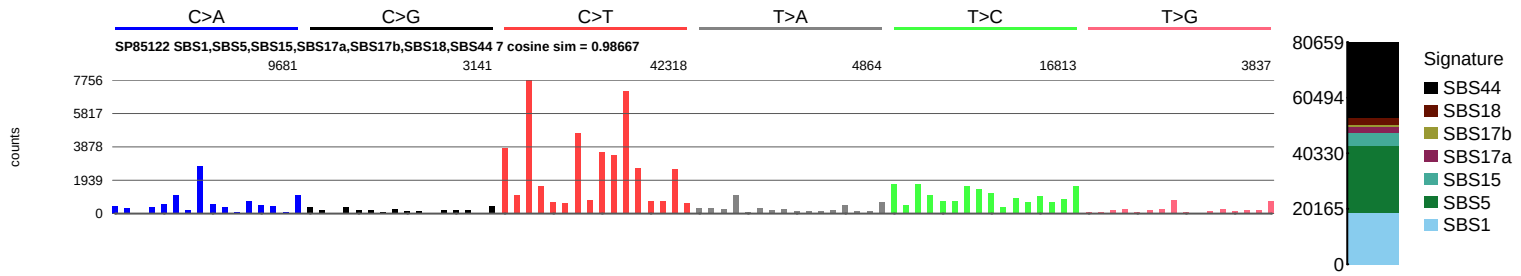
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98678
10639
2738
38962
4726
20834
2758
counts
7968
5976
3984
1992
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98674
8237
3136
42329
4793
18310
3858
counts
7764
5823
3882
1941
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98673
10753
2795
39052
4580
20845
2634
counts
7980
5985
3990
1995
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98672
10099
2205
40392
4176
20198
3588
counts
8156
6117
4078
2039
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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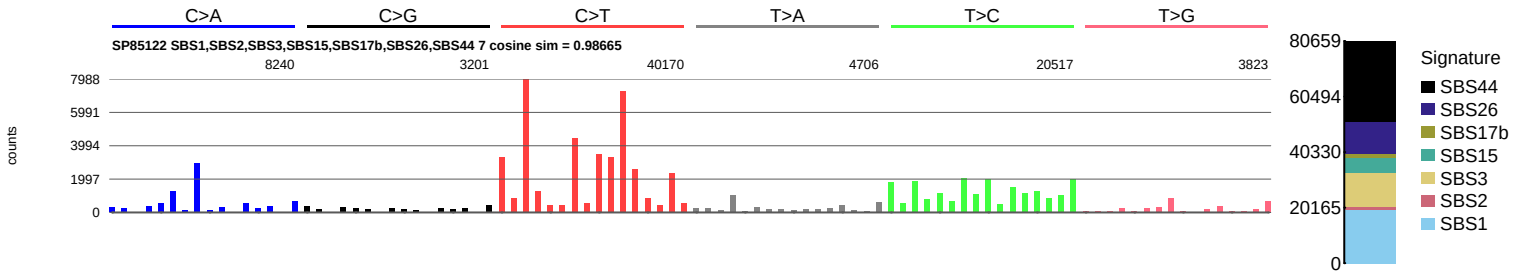
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98667
9681
3141
42318
4864
16813
3837
counts
7756
5817
3878
1939
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98665
8240
3201
40170
4706
20517
3823
counts
7988
5991
3994
1997
0
80659
60494
40330
20165
0
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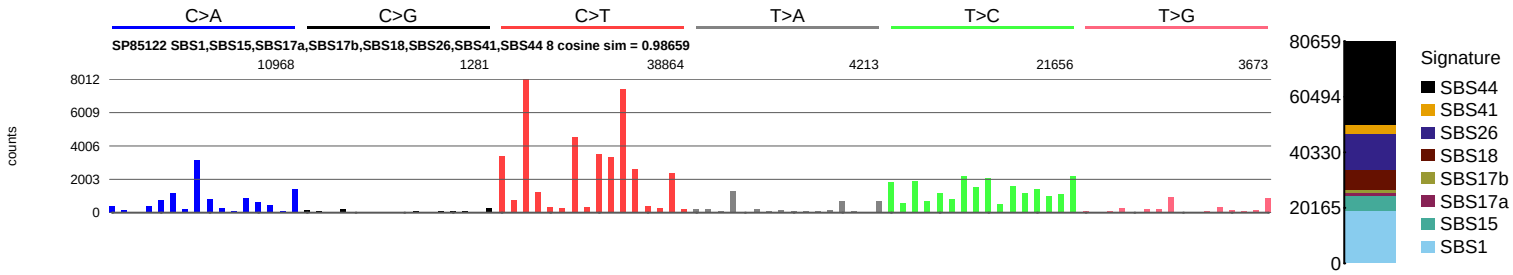
SBS44
SBS26
SBS17b
SBS15
SBS3
SBS2
SBS1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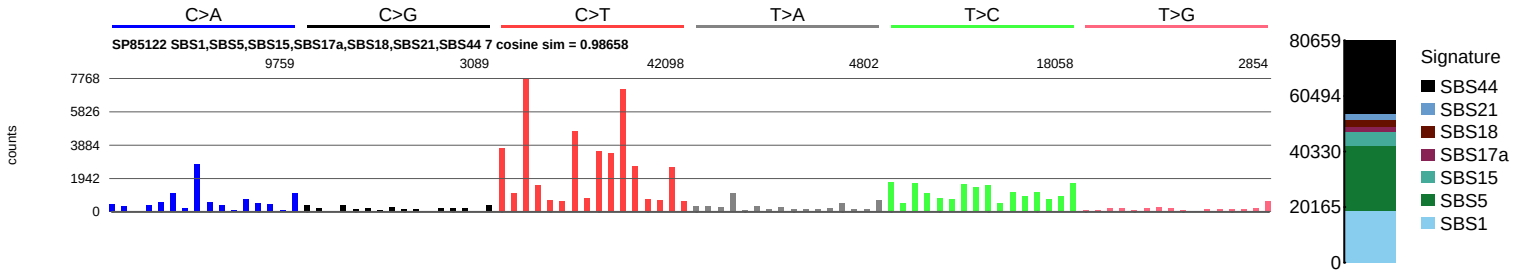
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98659
10968
1281
38864
4213
21656
3673
counts
8012
6009
4006
2003
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS44 7 cosine sim = 0.98658
9759
3089
42098
4802
18058
2854
counts
7768
5826
3884
1942
0
80659
60494
40330
20165
0
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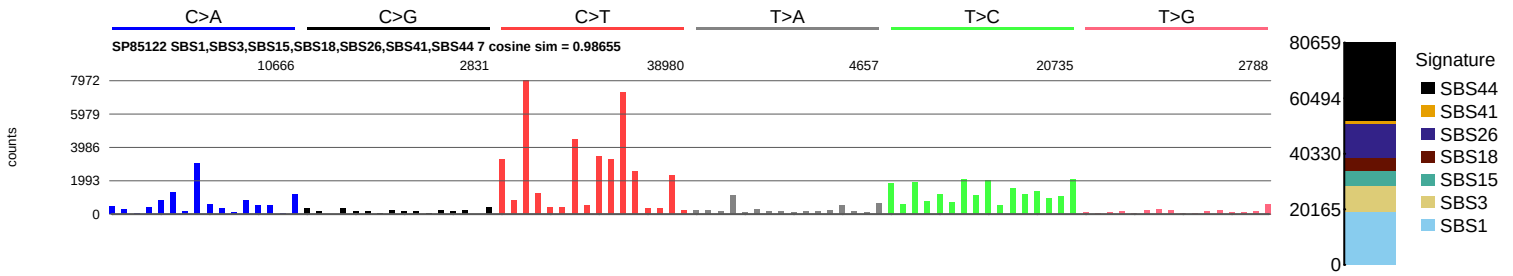
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98655
10666
2831
38980
4657
20735
2788
counts
7972
5979
3986
1993
0
80659
60494
40330
20165
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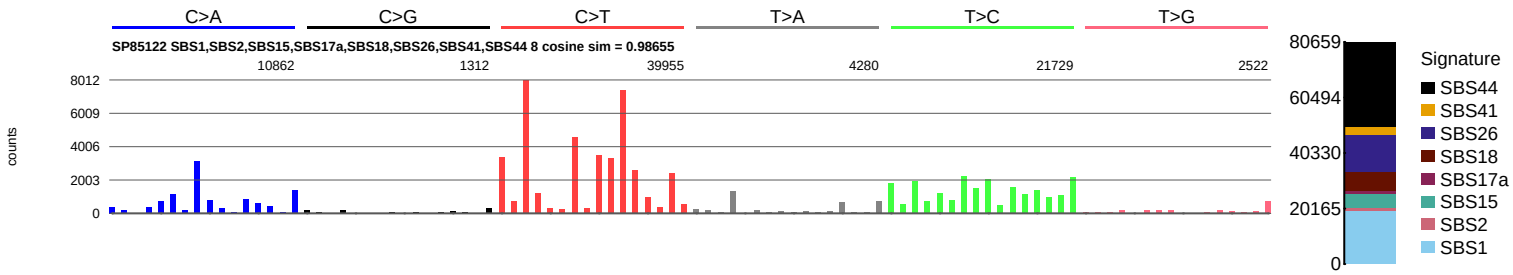
Signature
SBS44
SBS41
SBS26
SBS18
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98655
10862
1312
39955
4280
21729
2522
counts
8012
6009
4006
2003
0
80659
60494
40330
20165
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Signature
SBS44
SBS41
SBS26
SBS18
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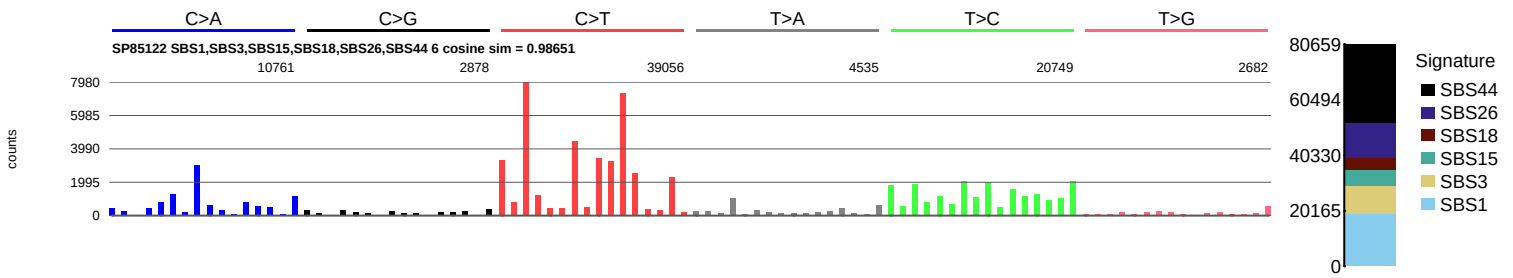
SBS2
SBS1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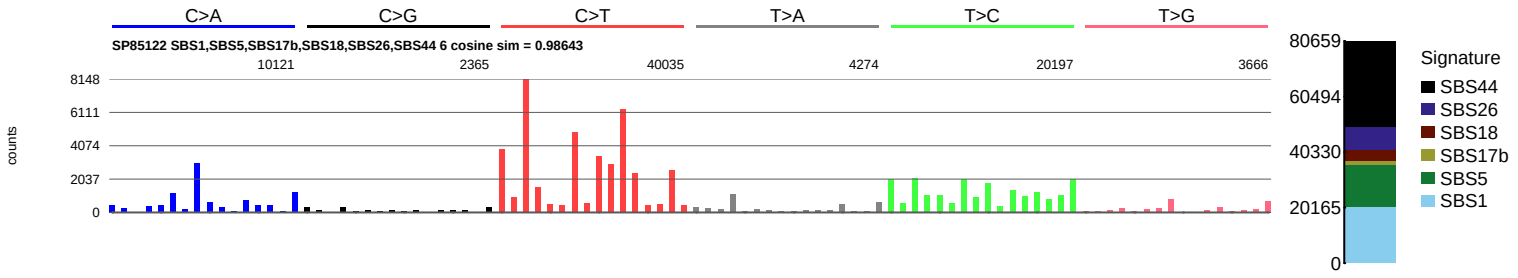
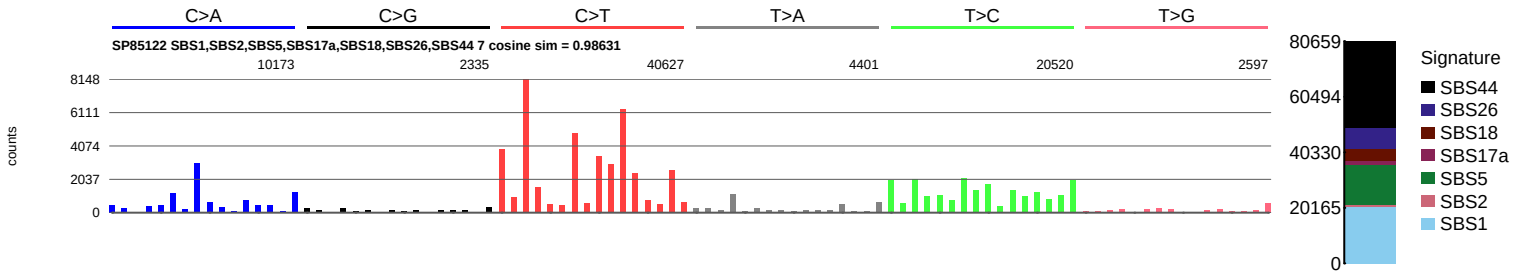
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98651
10761
2878
39056
4535
20749
2682
counts
7980
5985
3990
1995
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98643
10121
2365
40035
4274
20197
3666
counts
8148
6111
4074
2037
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98631
10173
2335
40627
4401
20520
2597
counts
8148
6111
4074
2037
0
80659
60494
40330
20165
0
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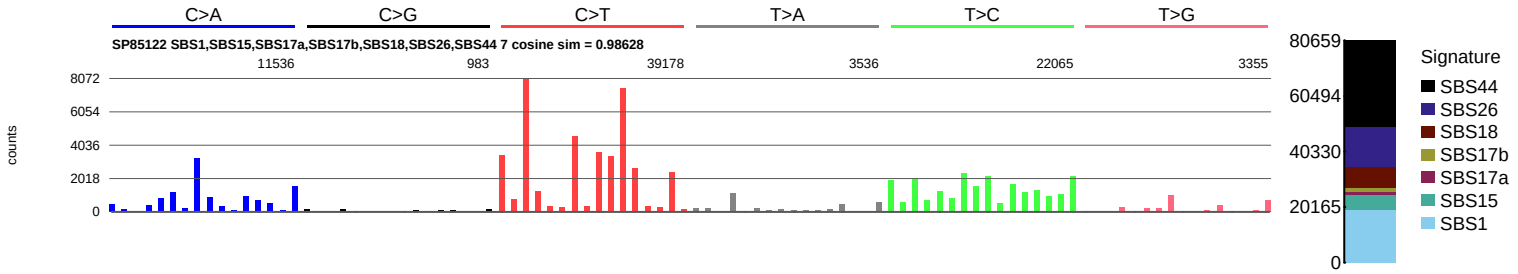
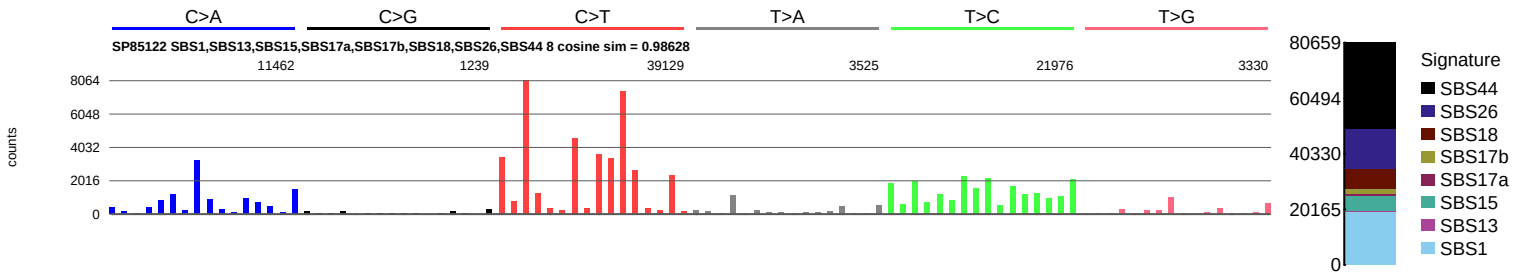
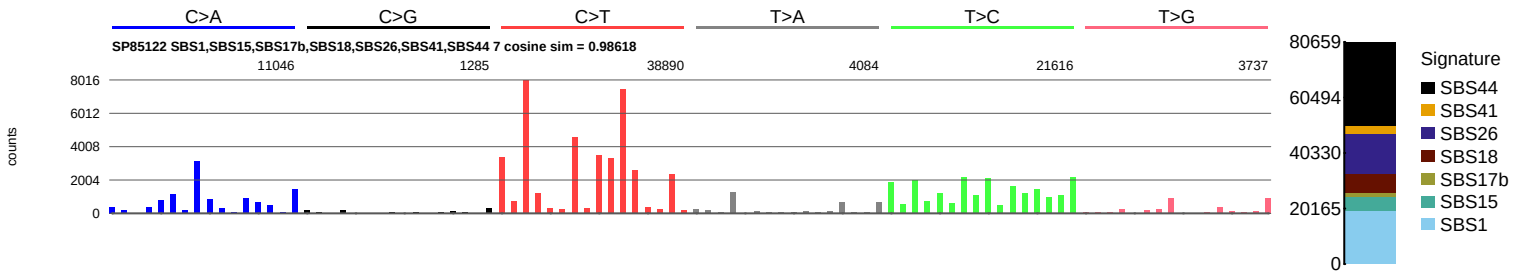
SBS44
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98628
11536
983
39178
3536
22065
3355
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98628
11462
1239
39129
3525
21976
3330
counts
8064
6048
4032
2016
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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SP85122 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98618
11046
1285
38890
4084
21616
3737
counts
8016
6012
4008
2004
0
80659
60494
40330
20165
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Signature
SBS44
SBS41
SBS26
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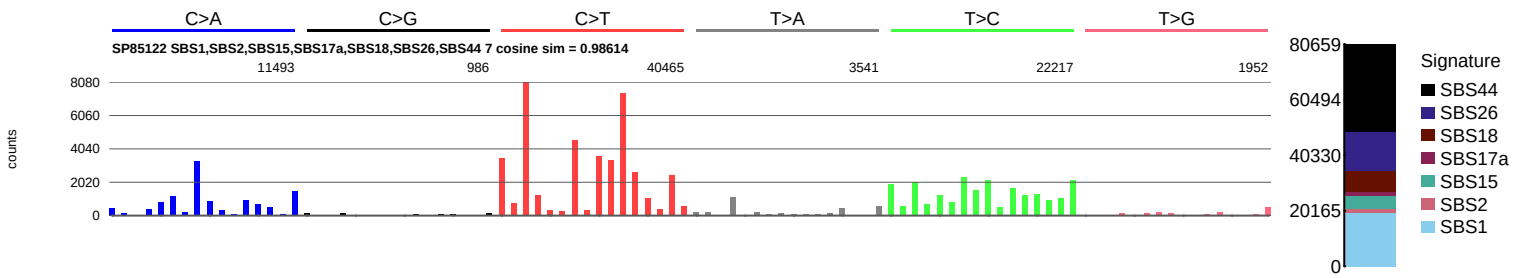
SBS1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SP85122 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98614
11493
986
40465
3541
22217
1952
counts
8080
6060
4040
2020
0
80659
60494
40330
20165
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Signature
SBS44
SBS26
SBS18
SBS17a
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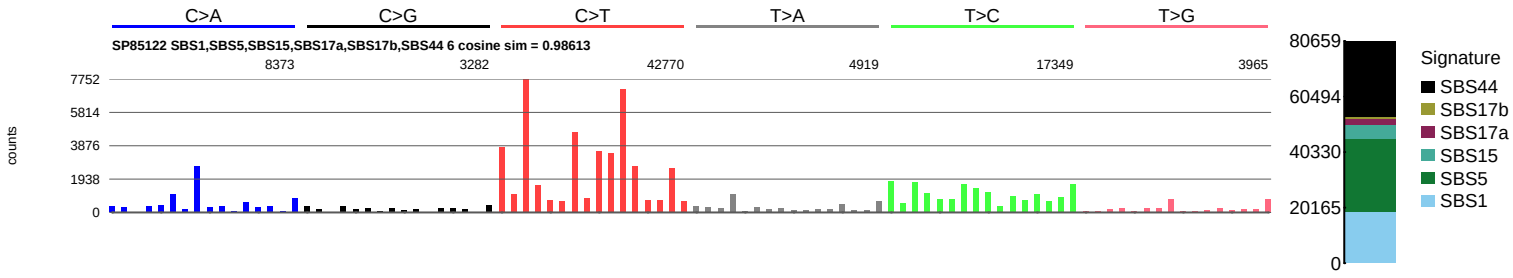
SBS2
SBS1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SP85122 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98613
8373
3282
42770
4919
17349
3965
counts
7752
5814
3876
1938
0
80659
60494
40330
20165
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SBS17b
SBS17a
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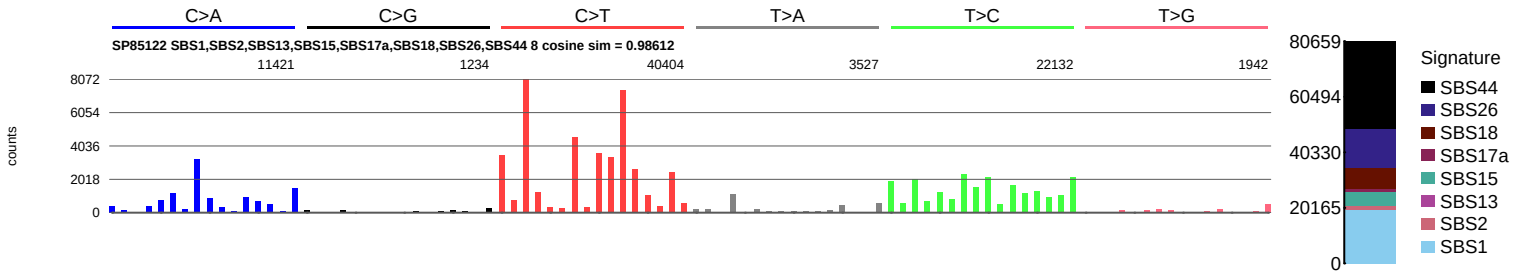
SBS5
SBS1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SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS44 8 cosine sim = 0.98612
11421
1234
40404
3527
22132
1942
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
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SBS44
SBS26
SBS18
SBS17a
SBS15
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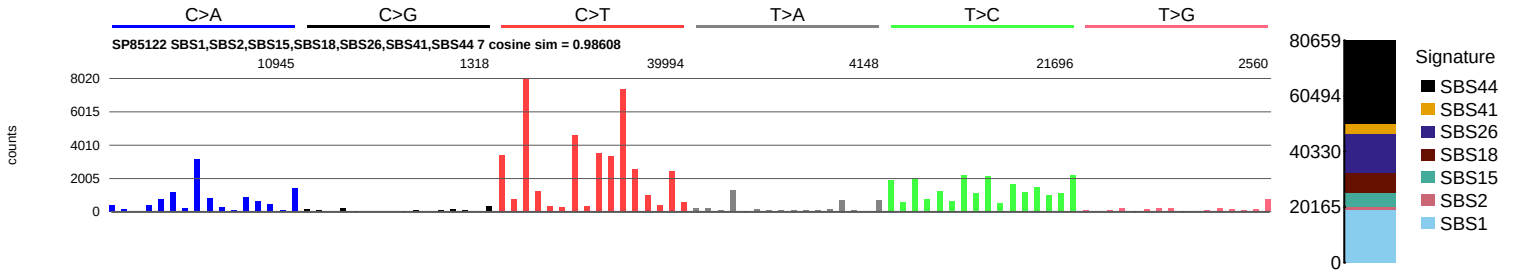
SBS2
SBS1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SP85122 SBS1,SBS2,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98608
10945
1318
39994
4148
21696
2560
counts
8020
6015
4010
2005
0
80659
60494
40330
20165
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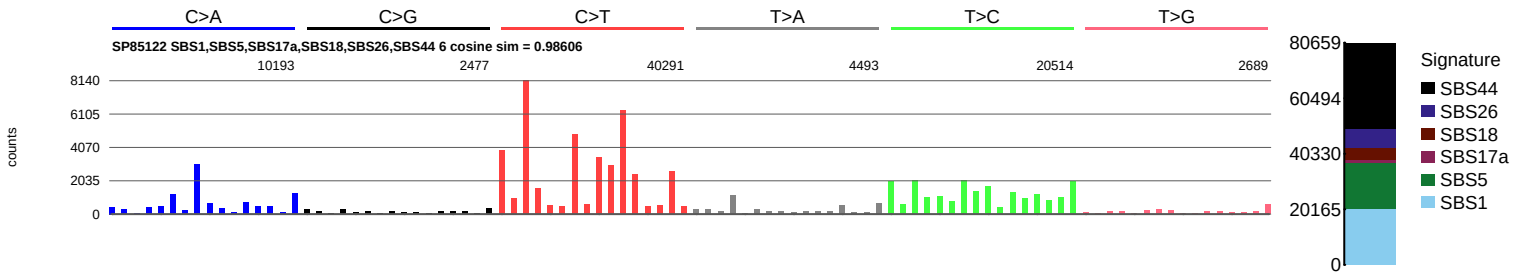
SBS1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SP85122 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98606
10193
2477
40291
4493
20514
2689
counts
8140
6105
4070
2035
0
80659
60494
40330
20165
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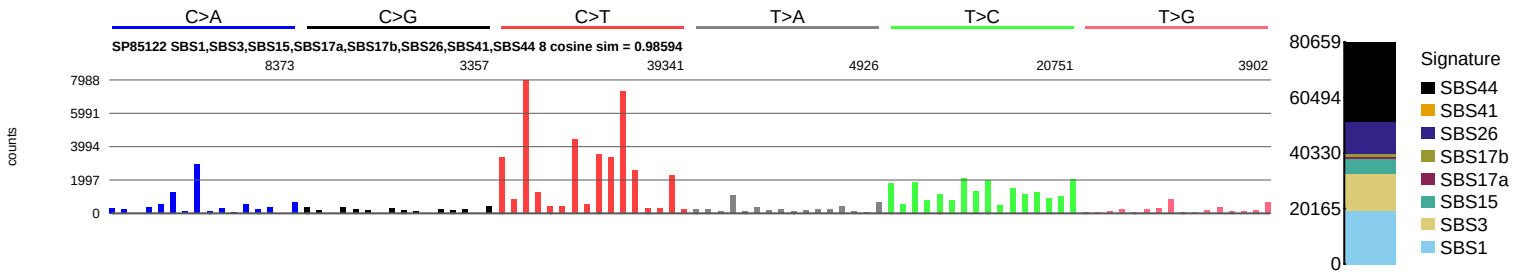
SBS26
SBS18
SBS17a
SBS5
SBS1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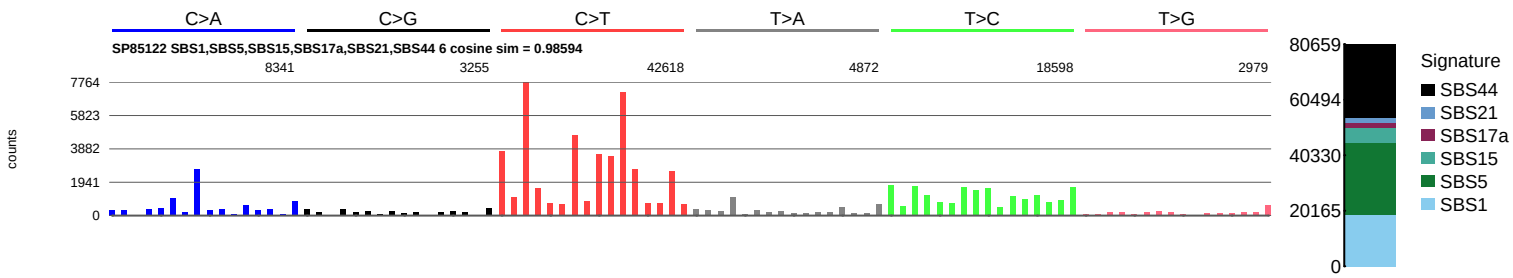
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98594
8373
3357
39341
4926
20751
3902
counts
7988
5991
3994
1997
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP85122 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS44 6 cosine sim = 0.98594
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3255
42618
4872
18598
2979
counts
7764
5823
3882
1941
0
80659
60494
40330
20165
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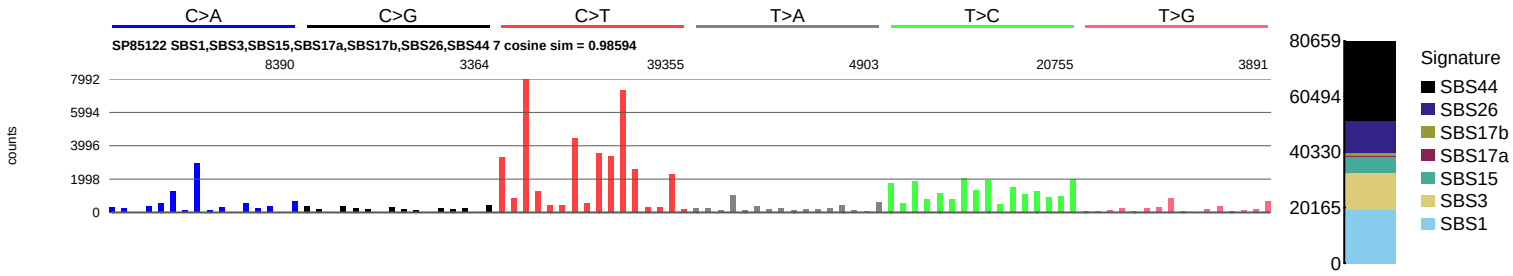
SBS44
SBS21
SBS17a
SBS15
SBS5
SBS1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SP85122 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98594
8390
3364
39355
4903
20755
3891
counts
7992
5994
3996
1998
0
80659
60494
40330
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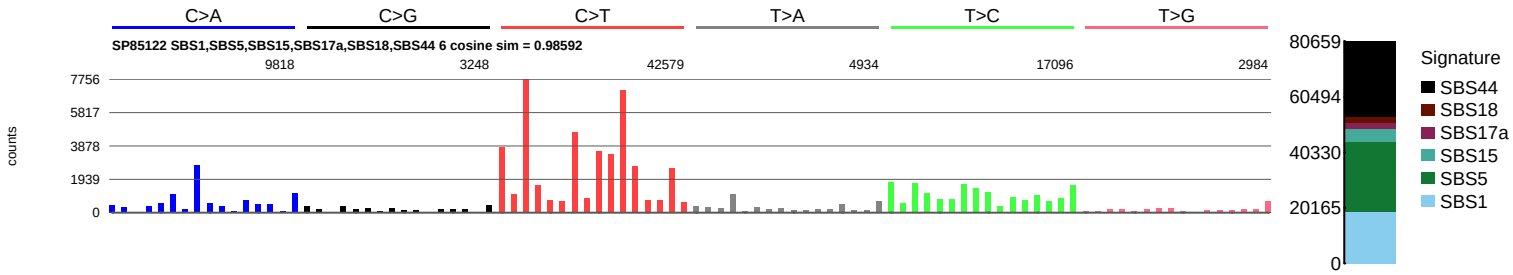
SBS17b
SBS17a
SBS15
SBS3
SBS1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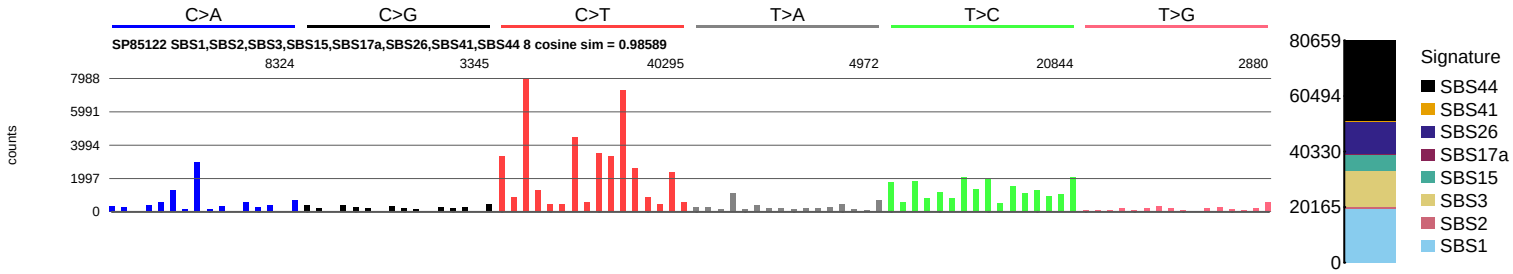
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS44 6 cosine sim = 0.98592
9818
3248
42579
4934
17096
2984
counts
7756
5817
3878
1939
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17a
SBS15
SBS5
SBS1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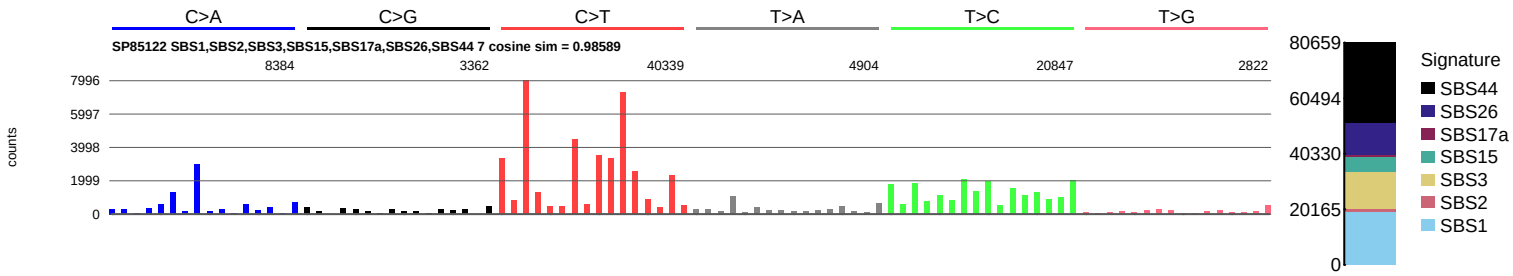
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98589
8324
3345
40295
4972
20844
2880
counts
7988
5991
3994
1997
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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T>C
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SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98589
8384
3362
40339
4904
20847
2822
counts
7996
5997
3998
1999
0
80659
60494
40330
20165
0
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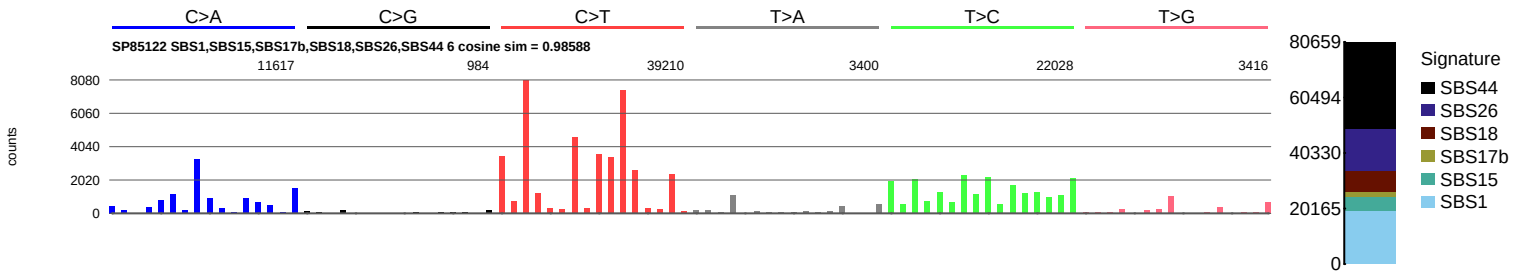
SBS26
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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SP85122 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98588
11617
984
39210
3400
22028
3416
counts
8080
6060
4040
2020
0
80659
60494
40330
20165
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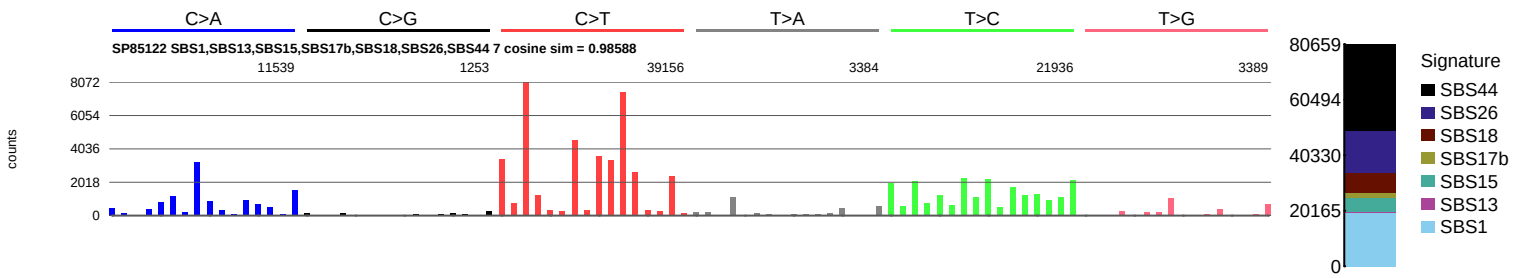
SBS26
SBS18
SBS17b
SBS15
SBS1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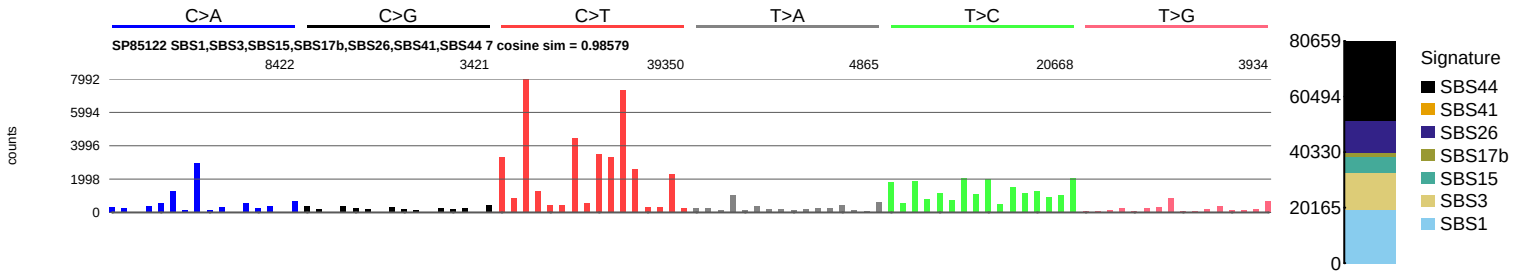
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98588
11539
1253
39156
3384
21936
3389
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS1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C>G
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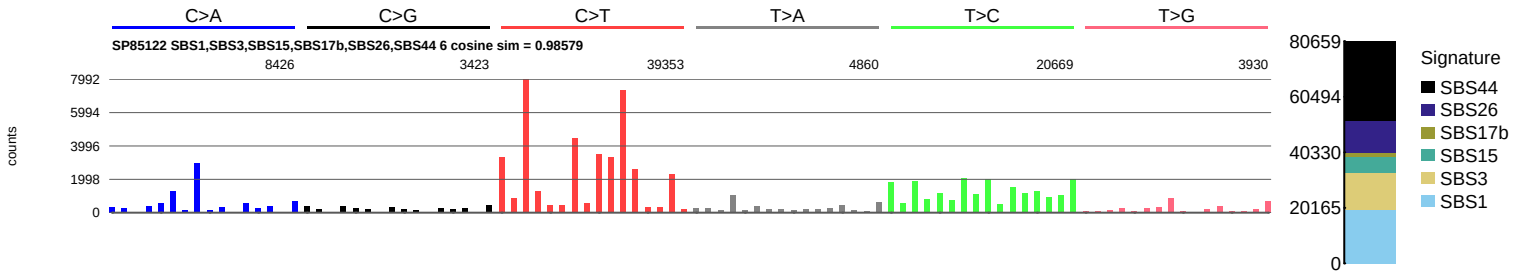
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98579
8422
3421
39350
4865
20668
3934
counts
7992
5994
3996
1998
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS3
SBS1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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98579
8426
3423
39353
4860
20669
3930
counts
7992
5994
3996
1998
0
80659
60494
40330
20165
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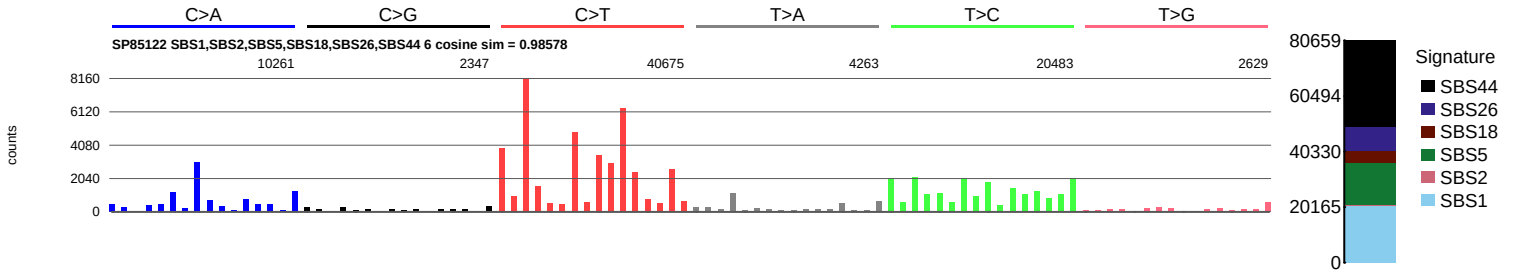
SBS26
SBS17b
SBS15
SBS3
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS18,SBS26,SBS44 6 cosine sim = 0.98578
10261
2347
40675
4263
20483
2629
counts
8160
6120
4080
2040
0
80659
60494
40330
20165
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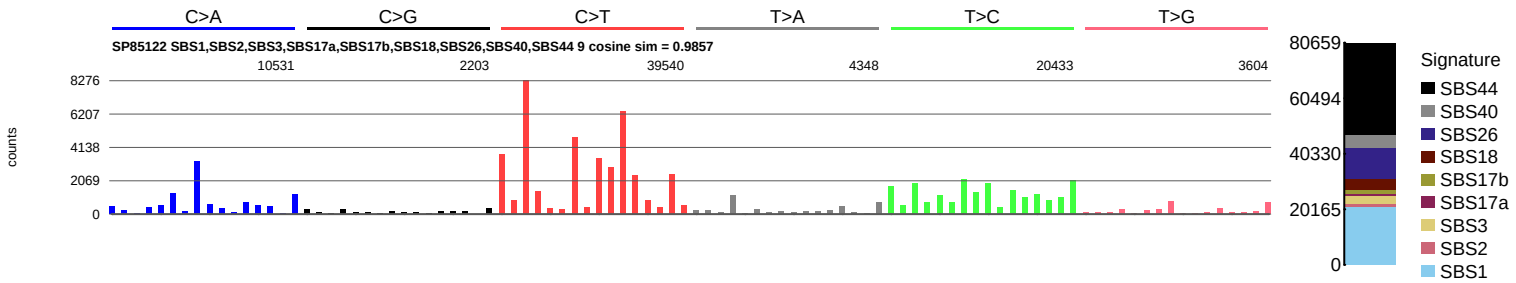
Signature
SBS44
SBS26
SBS18
SBS5
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9857
10531
2203
39540
4348
20433
3604
counts
8276
6207
4138
2069
0
80659
60494
40330
20165
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SBS26
SBS18
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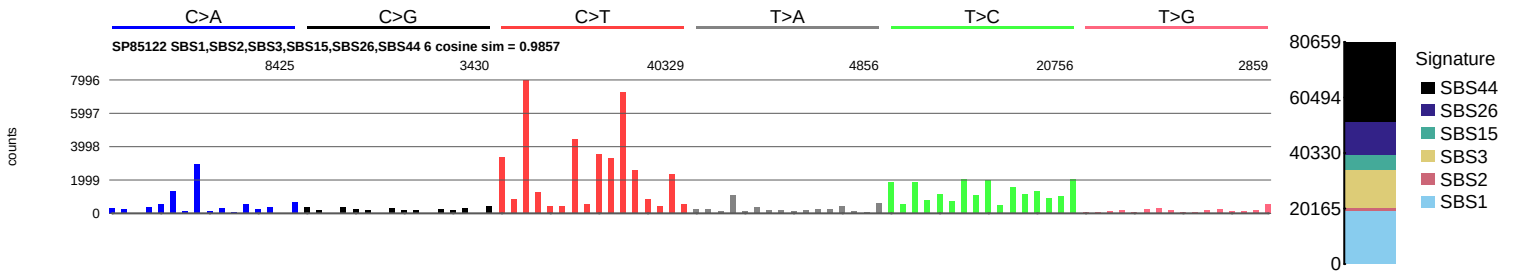
SBS3
SBS2
SBS1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T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS26,SBS44 6 cosine sim = 0.9857
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3430
40329
4856
20756
2859
counts
7996
5997
3998
1999
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80659
60494
40330
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SBS26
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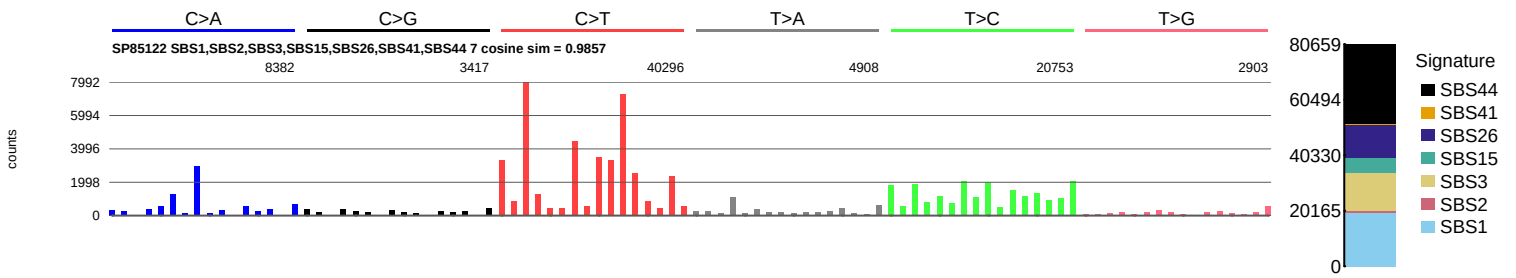
SBS1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C>G
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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.9857
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3417
40296
4908
20753
2903
counts
7992
5994
3996
1998
0
80659
60494
40330
20165
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SBS26
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SBS2
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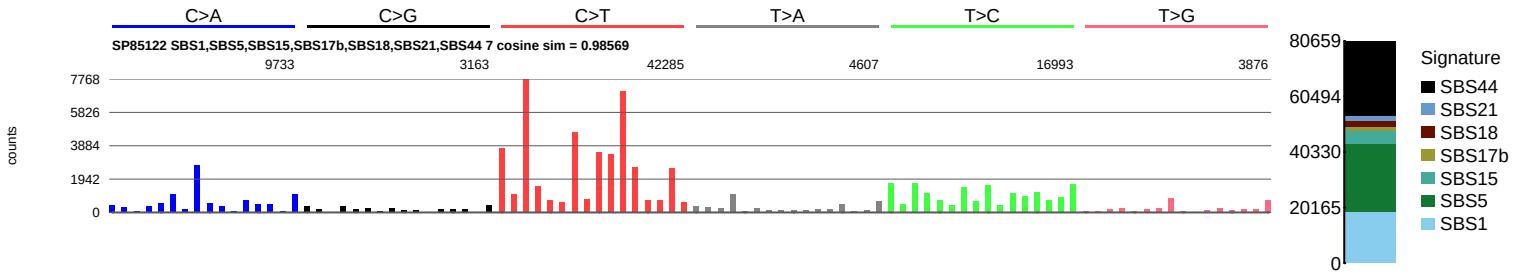
C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98569
9733
3163
42285
4607
16993
3876
counts
7768
5826
3884
1942
0
80659
60494
40330
20165
0
Signature
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SBS21
SBS18
SBS17b
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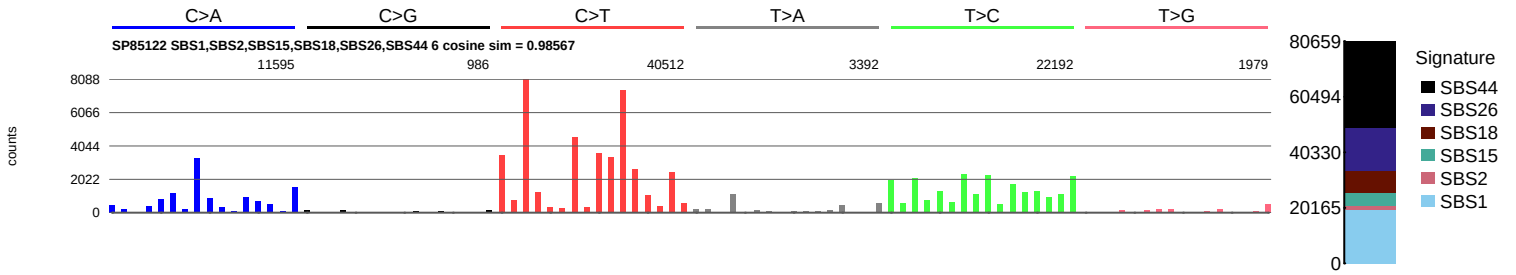
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98567
11595
986
40512
3392
22192
1979
counts
8088
6066
4044
2022
0
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60494
40330
20165
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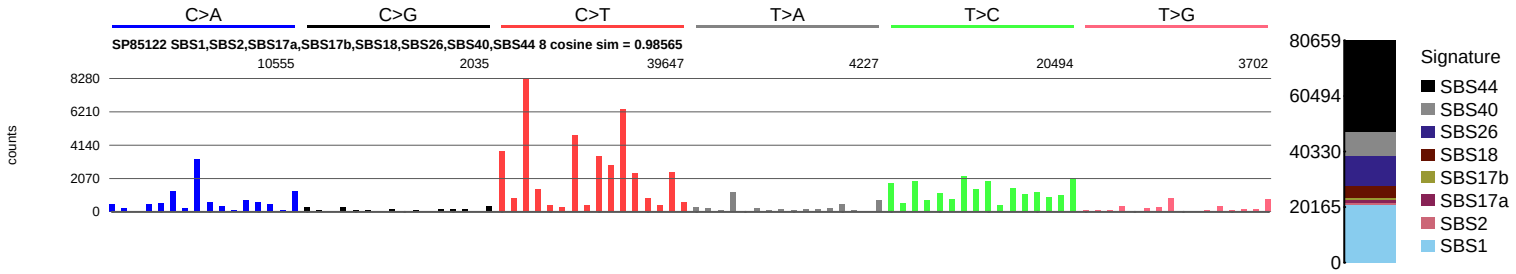
SBS44
SBS26
SBS18
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98565
10555
2035
39647
4227
20494
3702
counts
8280
6210
4140
2070
0
80659
60494
40330
20165
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SBS18
SBS17b
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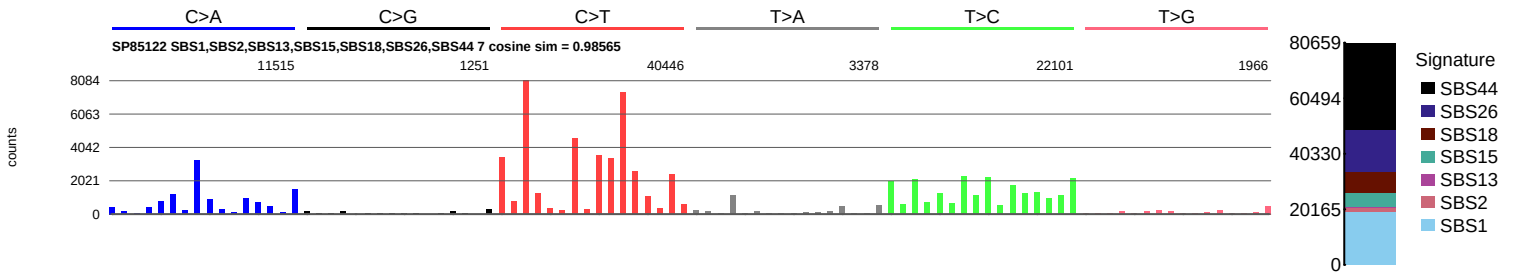
SBS2
SBS1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T>C
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SP85122 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98565
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1251
40446
3378
22101
1966
counts
8084
6063
4042
2021
0
80659
60494
40330
20165
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SBS18
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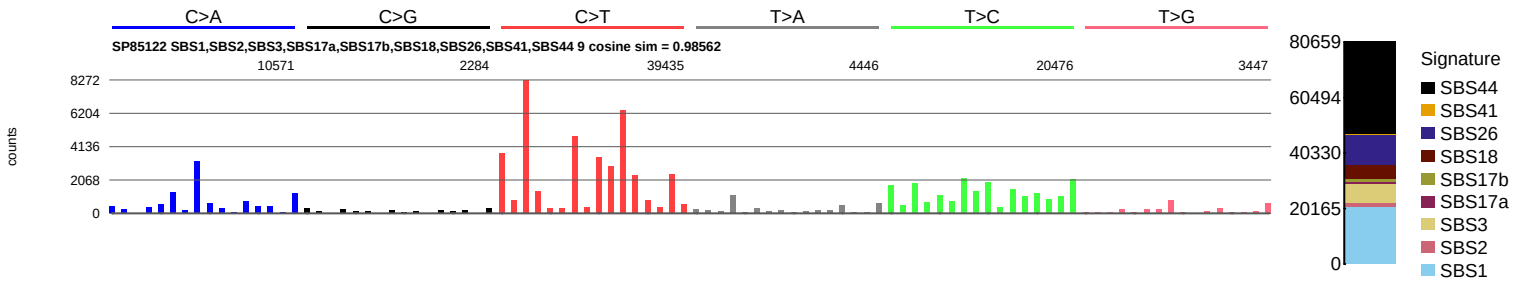
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98562
10571
2284
39435
4446
20476
3447
counts
8272
6204
4136
2068
0
80659
60494
40330
20165
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SBS41
SBS26
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SBS3
SBS2
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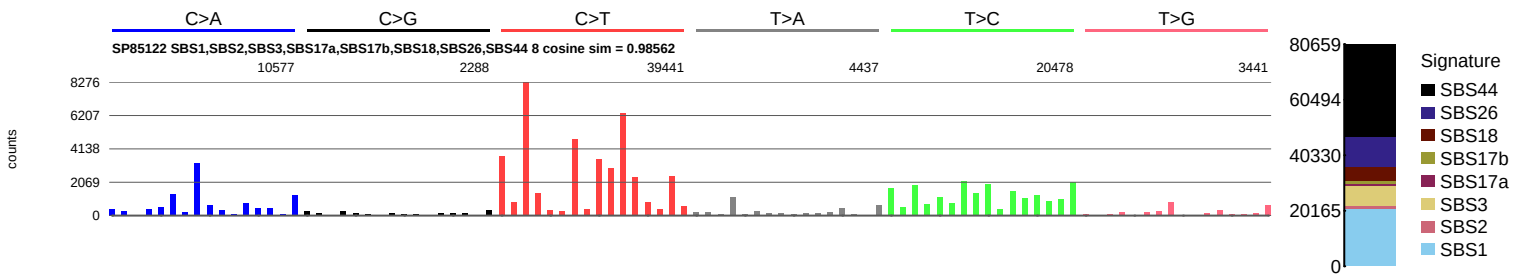
C>A
C>G
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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98562
10577
2288
39441
4437
20478
3441
counts
8276
6207
4138
2069
0
80659
60494
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SBS44
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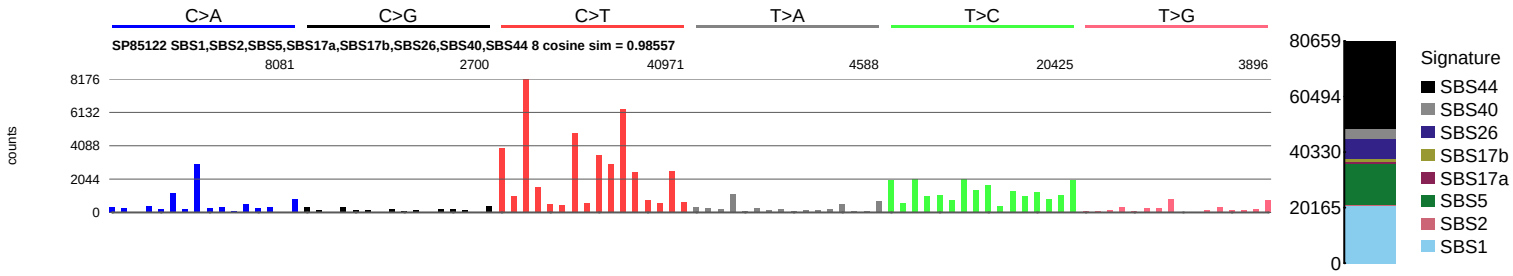
SBS2
SBS1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C>G
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T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98557
8081
2700
40971
4588
20425
3896
counts
8176
6132
4088
2044
0
80659
60494
40330
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SBS17b
SBS17a
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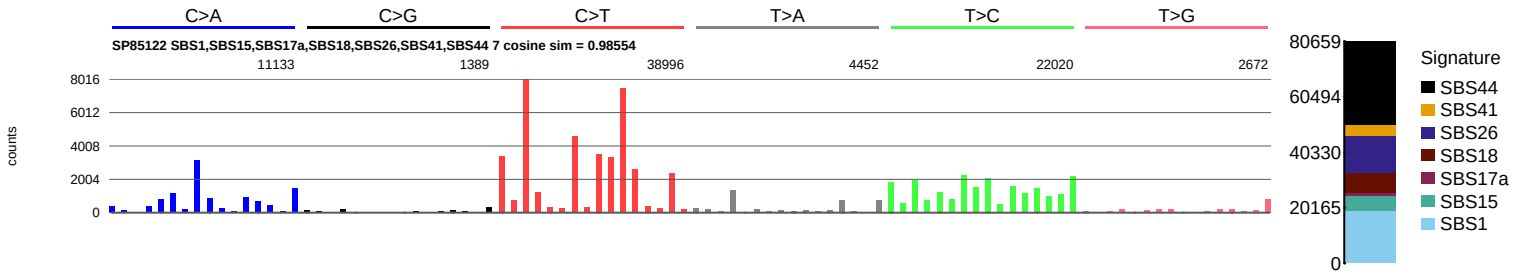
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98554
11133
1389
38996
4452
22020
2672
counts
8016
6012
4008
2004
0
80659
60494
40330
20165
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Signature
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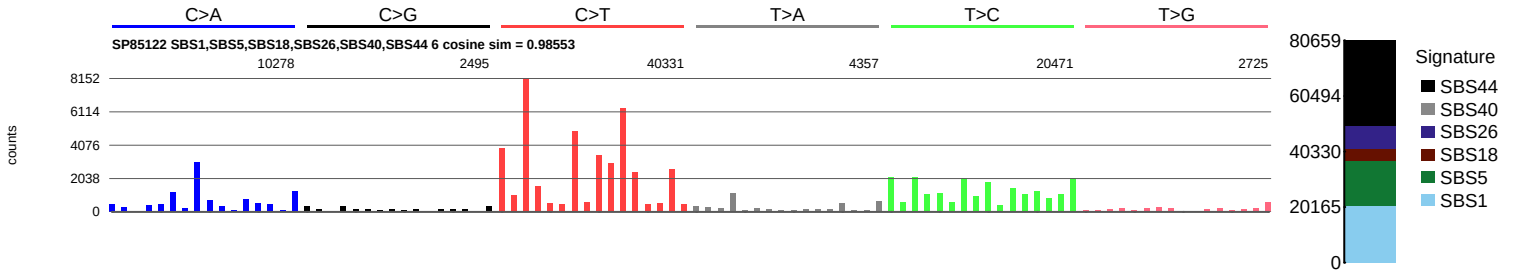
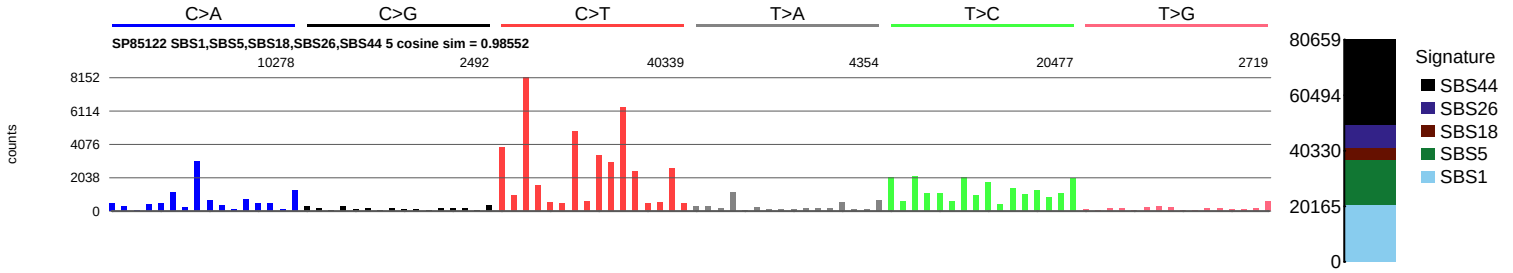
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98553
10278
2495
40331
4357
20471
2725
counts
8152
6114
4076
2038
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS5
SBS1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T>G
SP85122 SBS1,SBS5,SBS18,SBS26,SBS44 5 cosine sim = 0.98552
10278
2492
40339
4354
20477
2719
counts
8152
6114
4076
2038
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80659
60494
40330
20165
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SBS44
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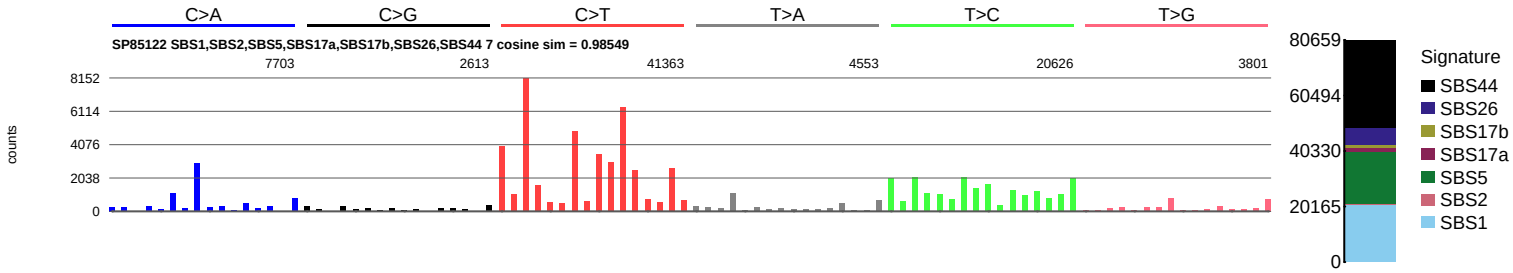
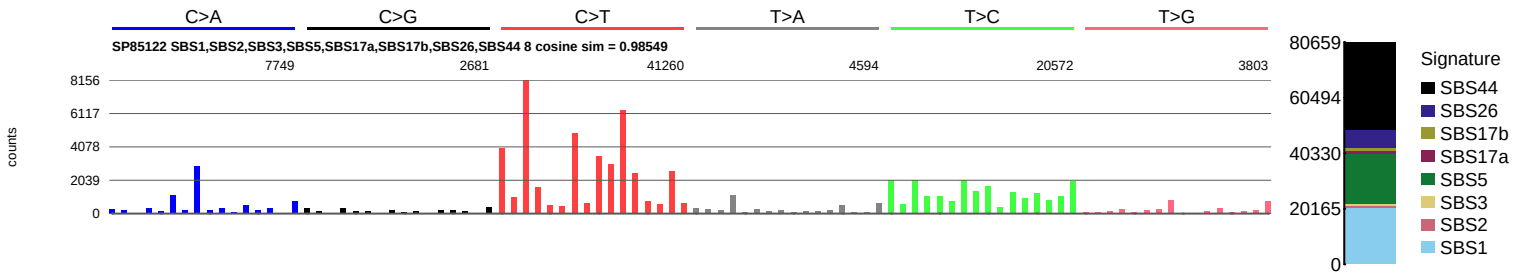
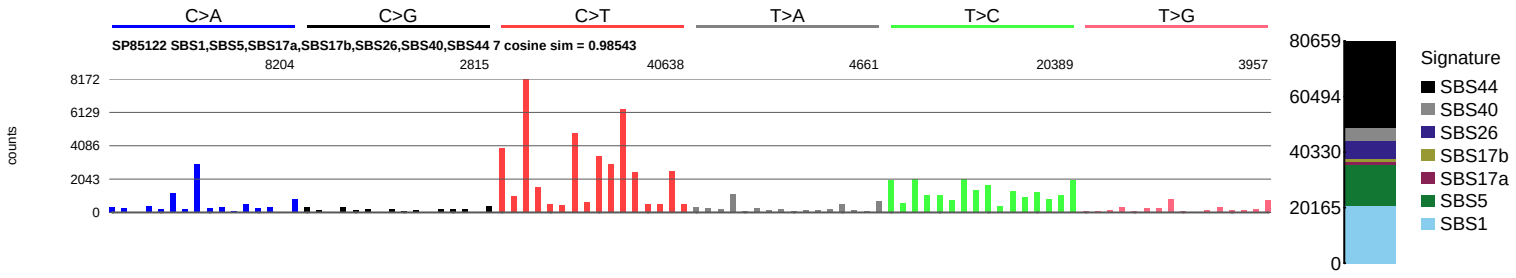
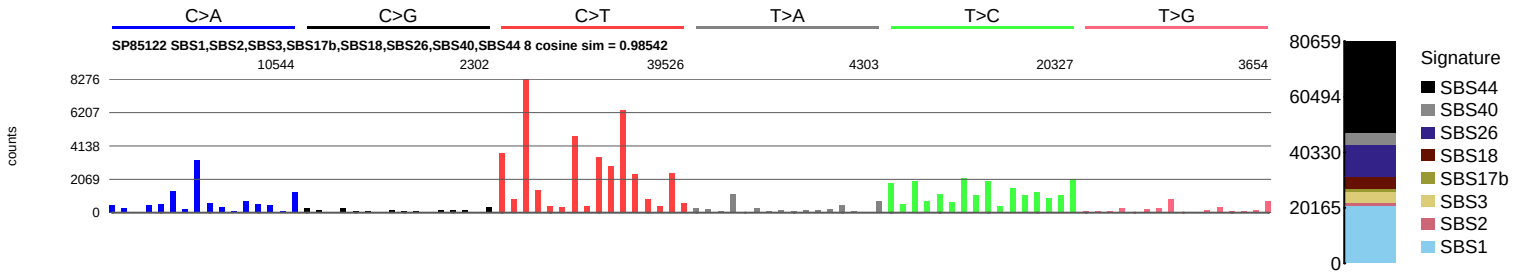
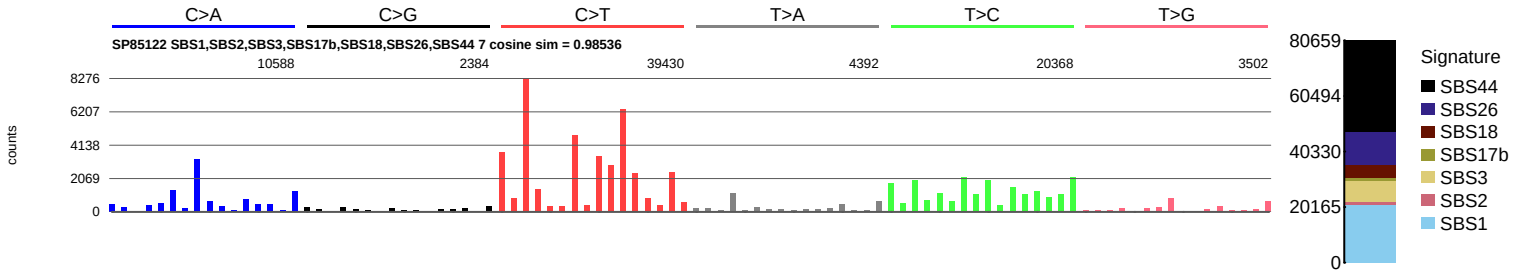
SBS5
SBS1

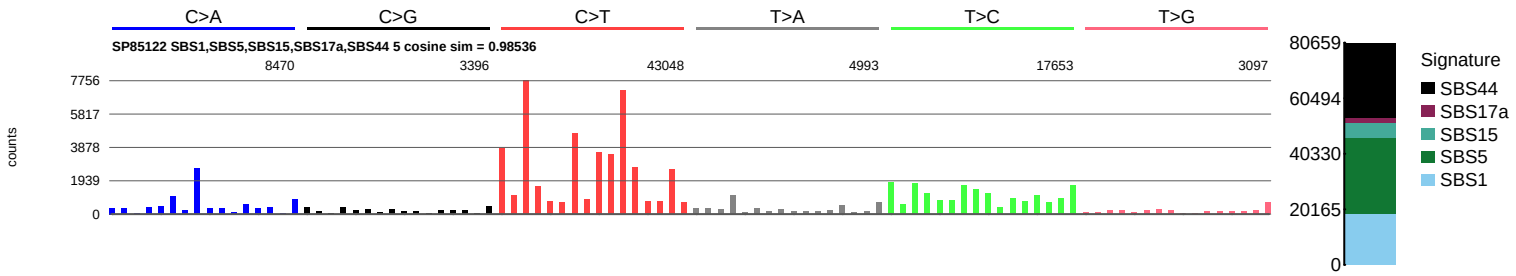

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98549
7703
2613
41363
4553
20626
3801
counts
8152
6114
4076
2038
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1

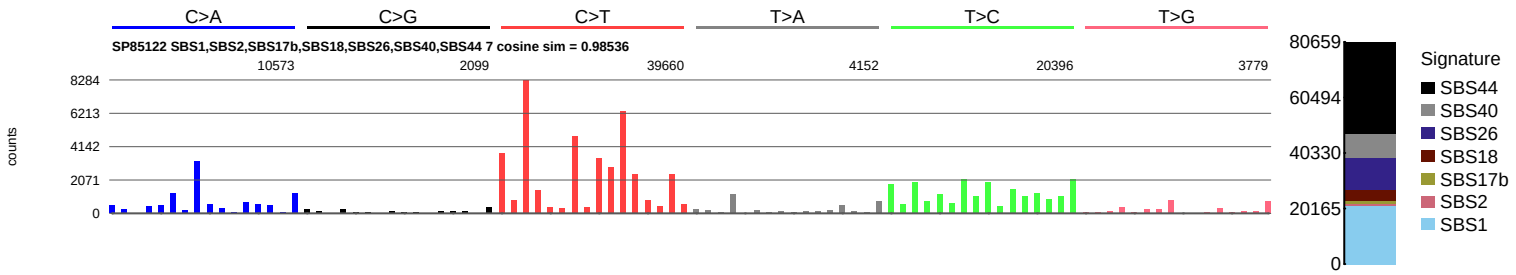

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98549
7749
2681
41260
4594
20572
3803
counts
8156
6117
4078
2039
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

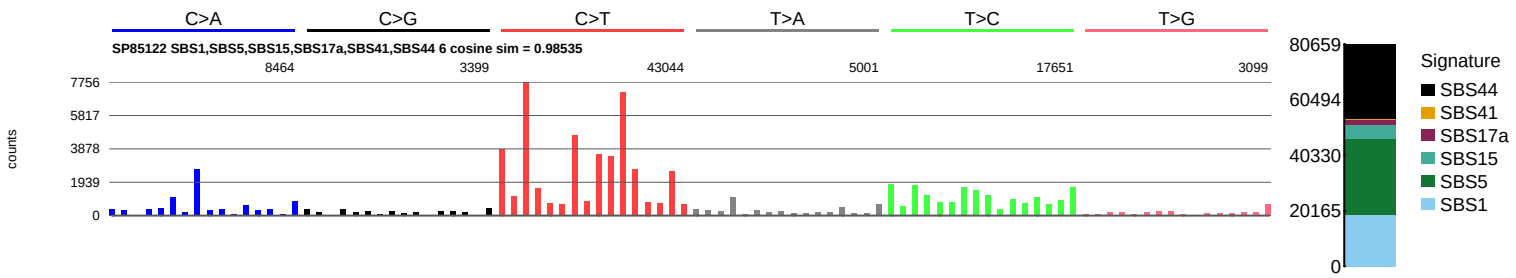

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98543
8204
2815
40638
4661
20389
3957
counts
8172
6129
4086
2043
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

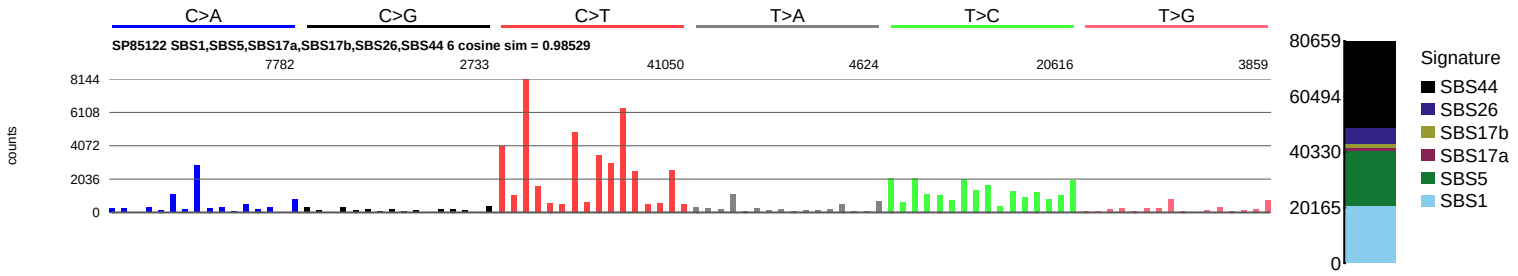

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98542
10544
2302
39526
4303
20327
3654
counts
8276
6207
4138
2069
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

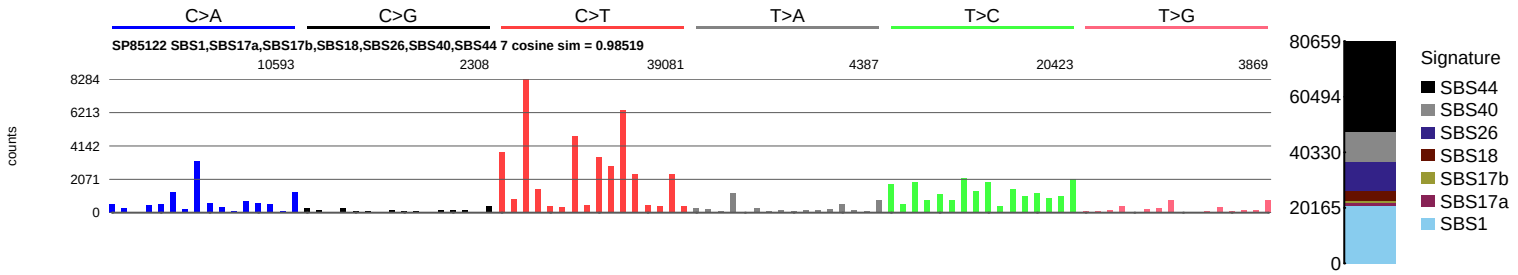

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98536
10588
2384
39430
4392
20368
3502
counts
8276
6207
4138
2069
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

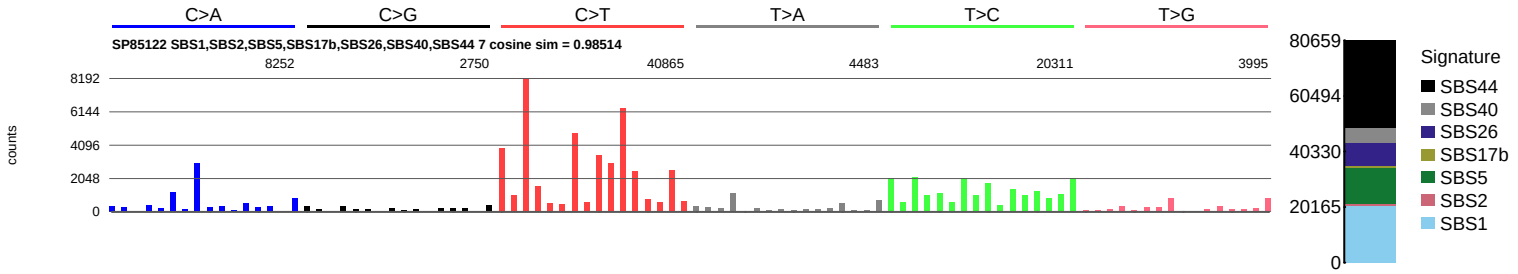

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS44 5 cosine sim = 0.98536
8470
3396
43048
4993
17653
3097
counts
7756
5817
3878
1939
0
80659
60494
40330
20165
0
Signature
SBS44
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98536
10573
2099
39660
4152
20396
3779
counts
8284
6213
4142
2071
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17a,SBS41,SBS44 6 cosine sim = 0.98535
8464
3399
43044
5001
17651
3099
counts
7756
5817
3878
1939
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98529
7782
2733
41050
4624
20616
3859
counts
8144
6108
4072
2036
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98519
10593
2308
39081
4387
20423
3869
counts
8284
6213
4142
2071
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98514
8252
2750
40865
4483
20311
3995
counts
8192
6144
4096
2048
0
80659
60494
40330
20165
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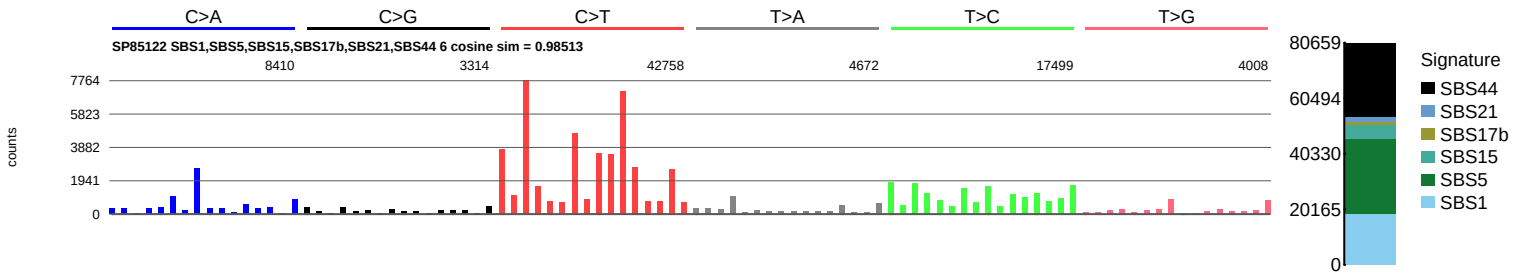
SBS26
SBS17b
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS44 6 cosine sim = 0.98513
8410
3314
42758
4672
17499
4008
counts
7764
5823
3882
1941
0
80659
60494
40330
20165
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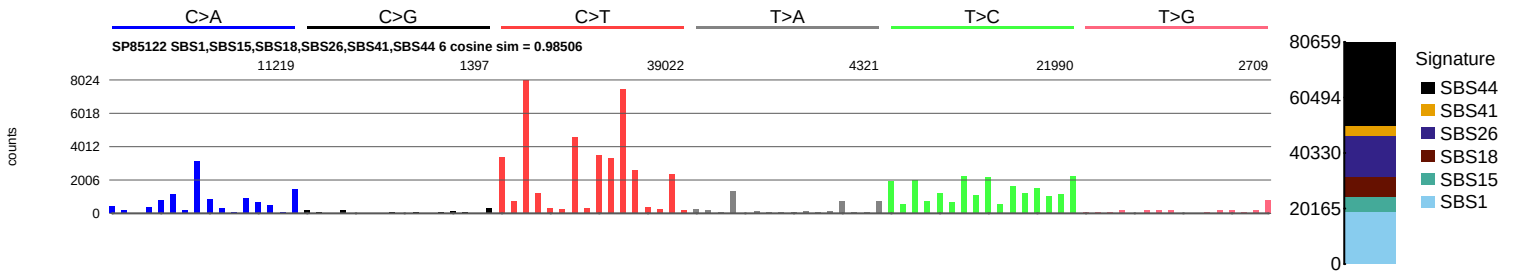
SBS21
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98506
11219
1397
39022
4321
21990
2709
counts
8024
6018
4012
2006
0
80659
60494
40330
20165
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Signature
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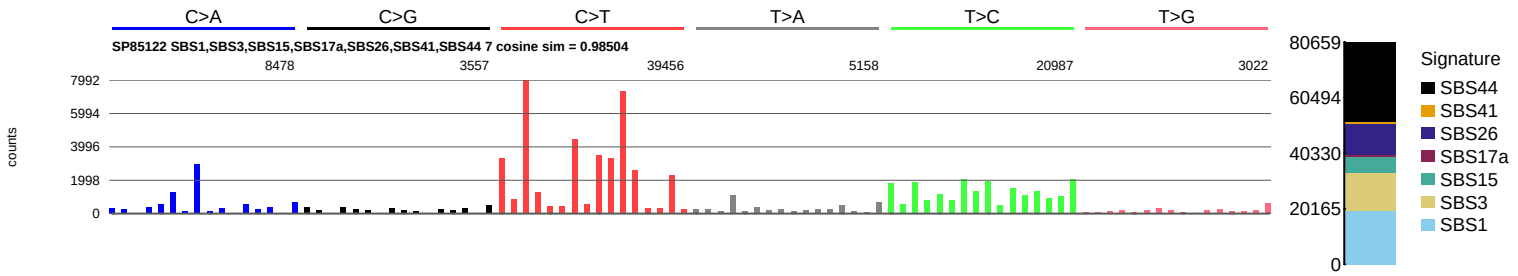
SBS41
SBS26
SBS18
SBS15
SBS1


C>A
C>G
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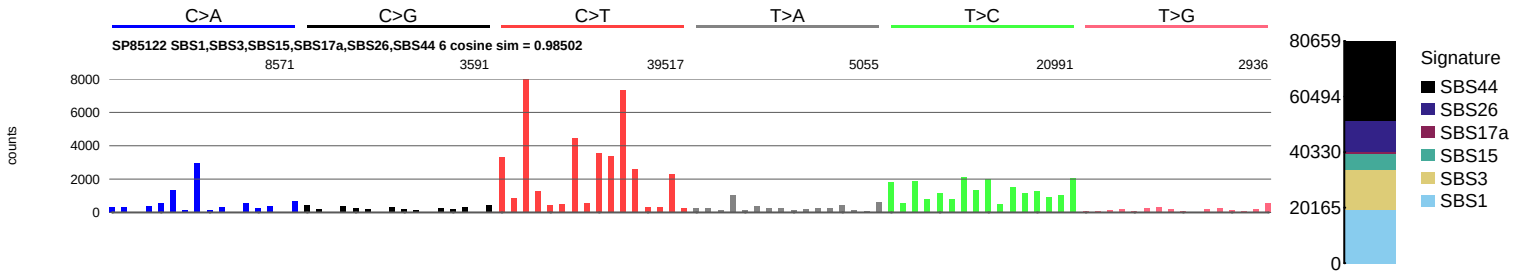
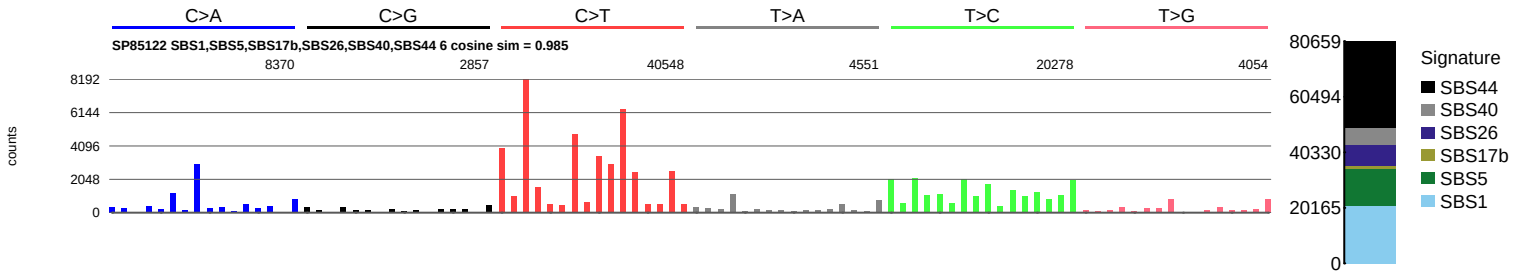
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98504
8478
3557
39456
5158
20987
3022
counts
7992
5994
3996
1998
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98502
8571
3591
39517
5055
20991
2936
counts
8000
6000
4000
2000
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS3
SBS1


C>A
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T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.985
8370
2857
40548
4551
20278
4054
counts
8192
6144
4096
2048
0
80659
60494
40330
20165
0
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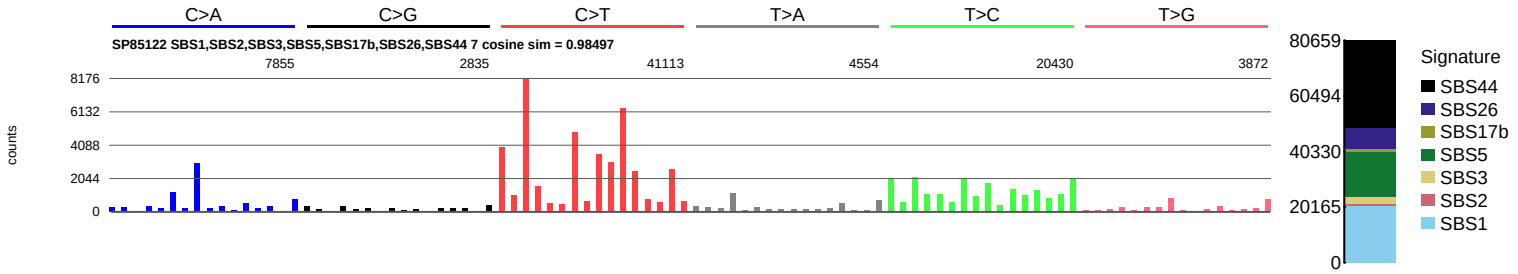
SBS44
SBS40
SBS26
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS26,SBS44 7 cosine sim = 0.98497
7855
2835
41113
4554
20430
3872
counts
8176
6132
4088
2044
0
80659
60494
40330
20165
0
Signature
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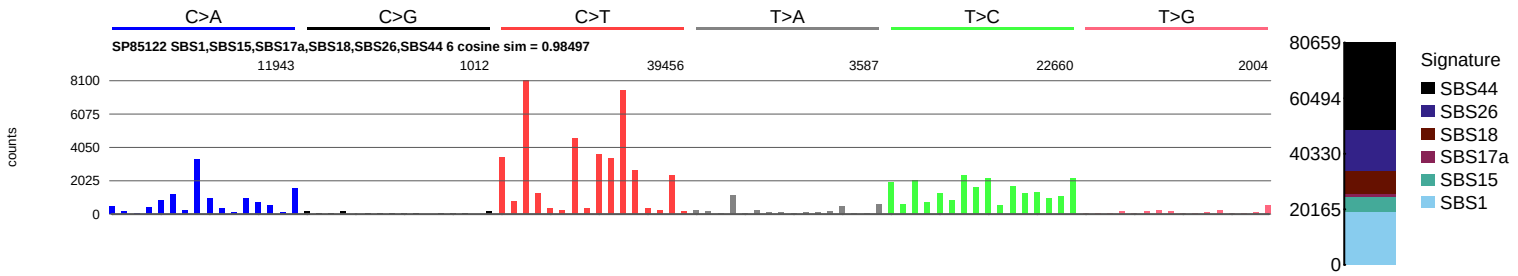
SBS26
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98497
11943
1012
39456
3587
22660
2004
counts
8100
6075
4050
2025
0
80659
60494
40330
20165
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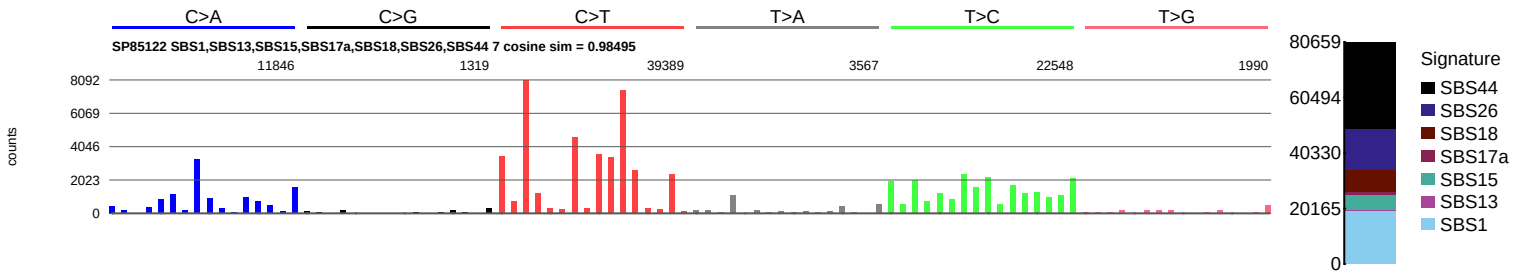
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98495
11846
1319
39389
3567
22548
1990
counts
8092
6069
4046
2023
0
80659
60494
40330
20165
0
Signature
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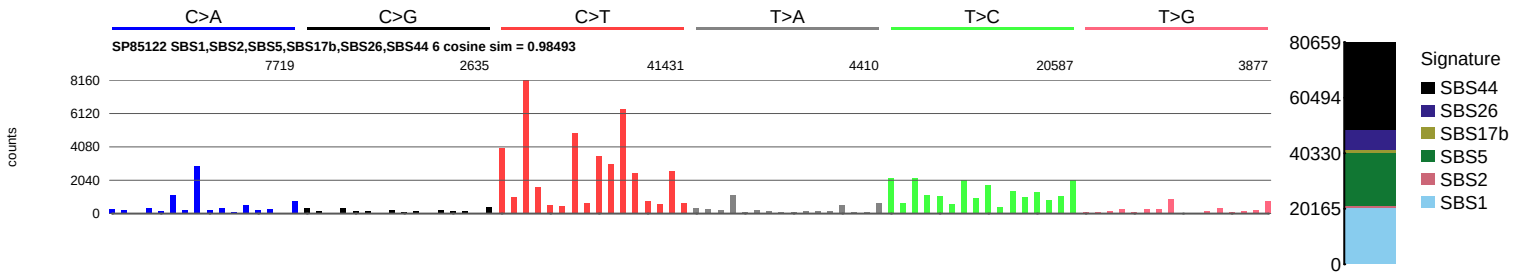
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS44 6 cosine sim = 0.98493
7719
2635
41431
4410
20587
3877
counts
8160
6120
4080
2040
0
80659
60494
40330
20165
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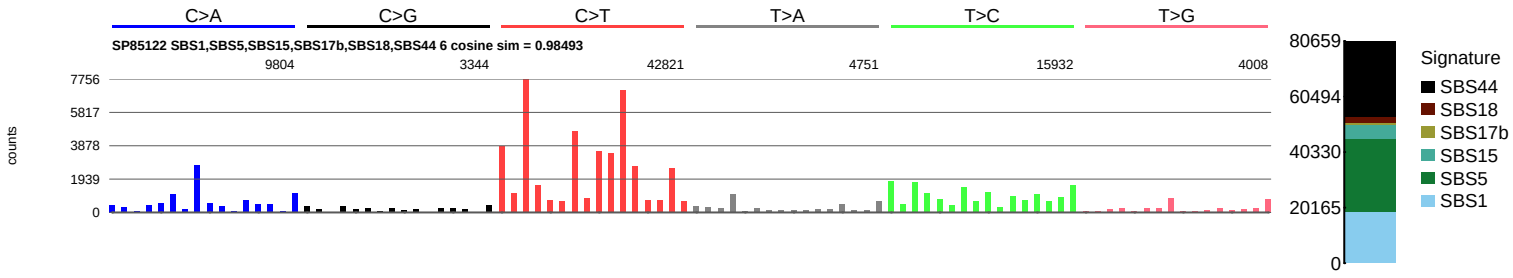
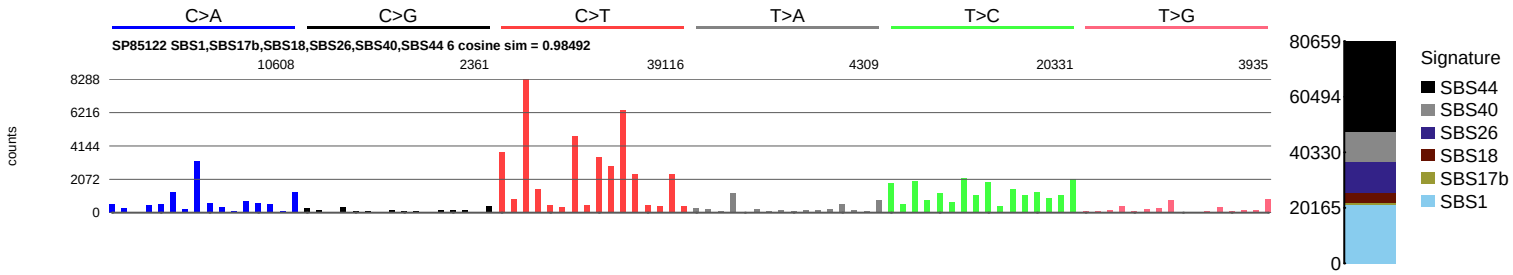
SBS44
SBS26
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS44 6 cosine sim = 0.98493
9804
3344
42821
4751
15932
4008
counts
7756
5817
3878
1939
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS5
SBS1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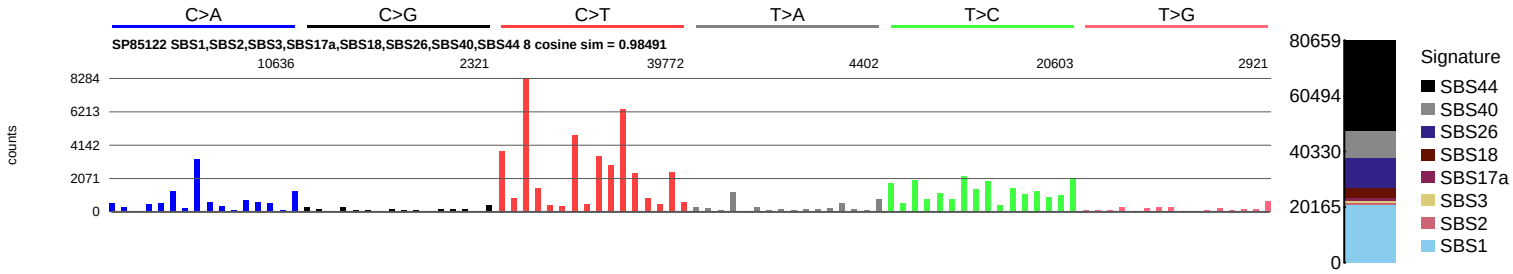
SP85122 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98492
10608
2361
39116
4309
20331
3935
counts
8288
6216
4144
2072
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS1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C>T
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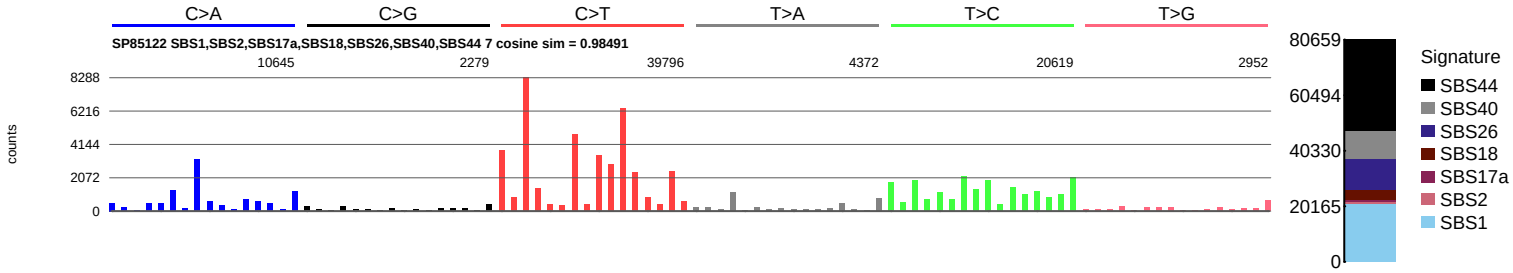
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98491
10636
2321
39772
4402
20603
2921
counts
8284
6213
4142
2071
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1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C>G
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SP85122 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98491
10645
2279
39796
4372
20619
2952
counts
8288
6216
4144
2072
0
80659
60494
40330
20165
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SBS44
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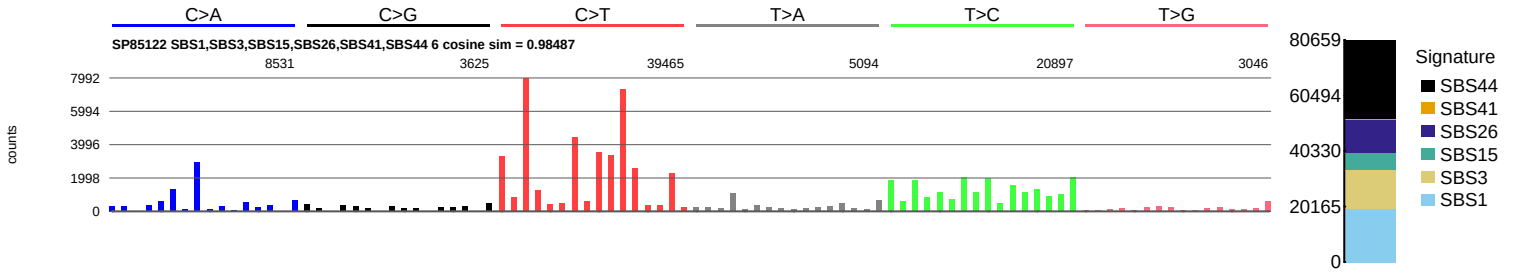
SBS26
SBS18
SBS17a
SBS2
SBS1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T>G
SP85122 SBS1,SBS3,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98487
8531
3625
39465
5094
20897
3046
counts
7992
5994
3996
1998
0
80659
60494
40330
20165
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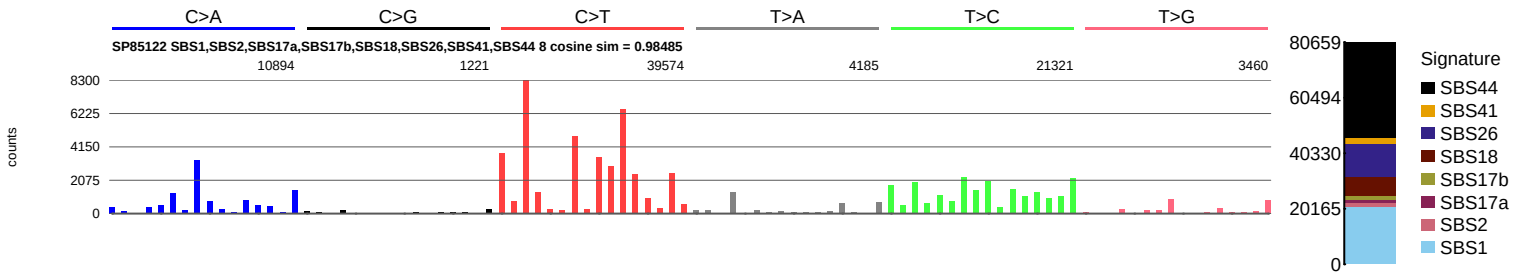
SBS44
SBS41
SBS26
SBS15
SBS3
SBS1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SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98485
10894
1221
39574
4185
21321
3460
counts
8300
6225
4150
2075
0
80659
60494
40330
20165
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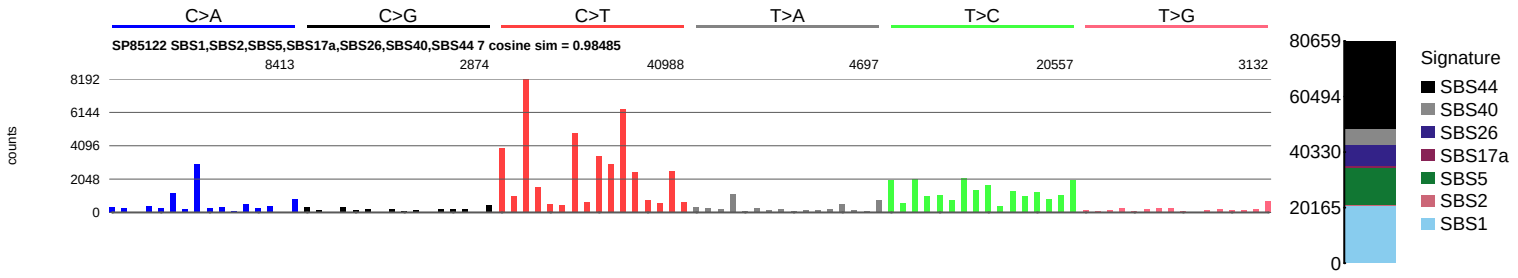
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS26,SBS40,SBS44 7 cosine sim = 0.98485
8413
2874
40988
4697
20557
3132
counts
8192
6144
4096
2048
0
80659
60494
40330
20165
0
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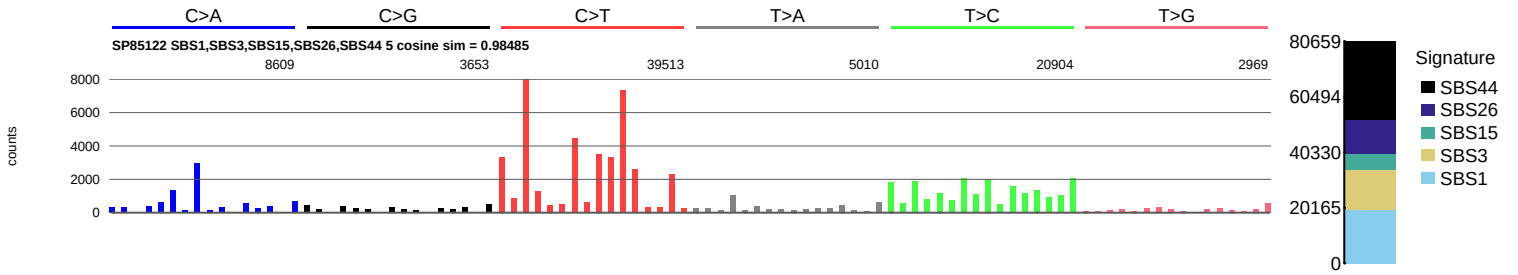
SBS44
SBS40
SBS26
SBS17a
SBS5
SBS2
SBS1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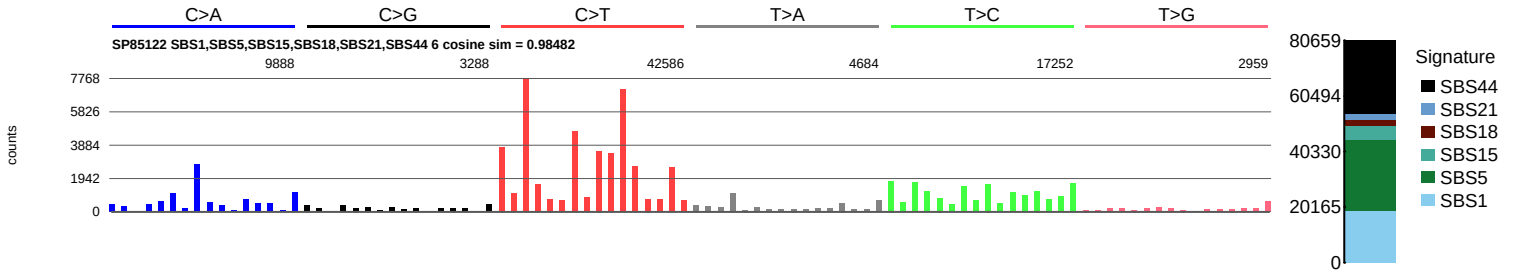
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS15,SBS26,SBS44 5 cosine sim = 0.98485
8609
3653
39513
5010
20904
2969
counts
8000
6000
4000
2000
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS15
SBS3
SBS1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C>G
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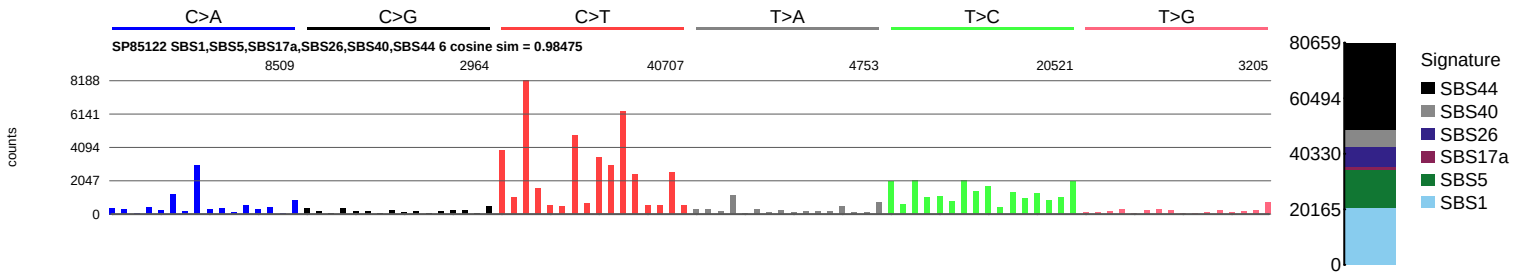
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS18,SBS21,SBS44 6 cosine sim = 0.98482
9888
3288
42586
4684
17252
2959
counts
7768
5826
3884
1942
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS15
SBS5
SBS1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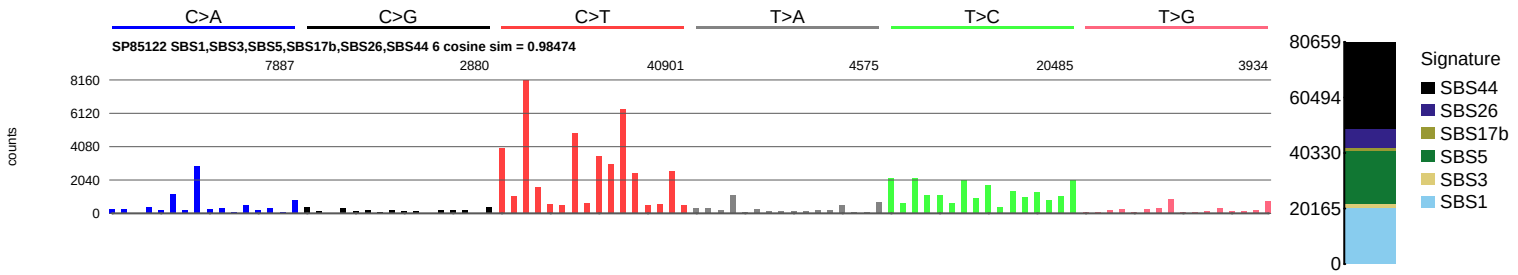
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS26,SBS40,SBS44 6 cosine sim = 0.98475
8509
2964
40707
4753
20521
3205
counts
8188
6141
4094
2047
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17a
SBS5
SBS1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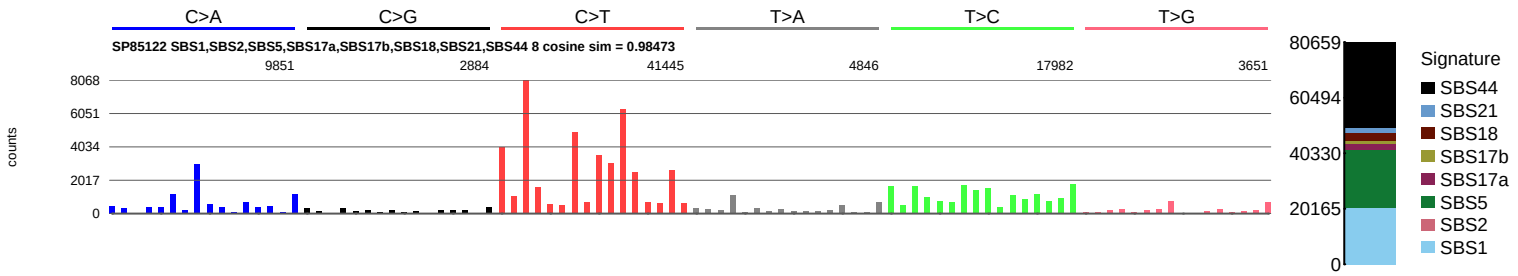
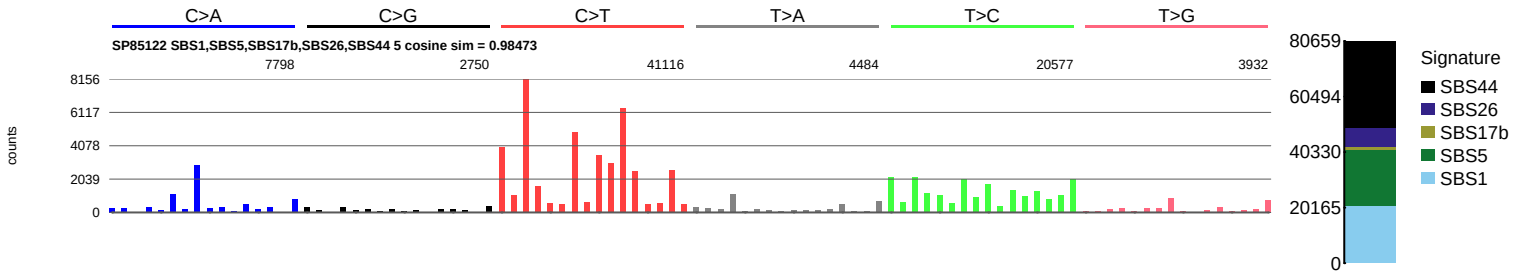
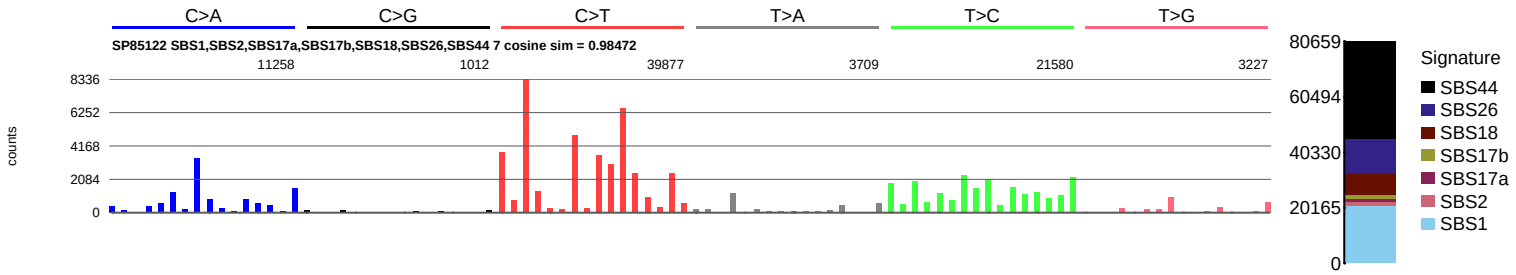
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS5,SBS17b,SBS26,SBS44 6 cosine sim = 0.98474
7887
2880
40901
4575
20485
3934
counts
8160
6120
4080
2040
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98473
9851
2884
41445
4846
17982
3651
counts
8068
6051
4034
2017
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17b,SBS26,SBS44 5 cosine sim = 0.98473
7798
2750
41116
4484
20577
3932
counts
8156
6117
4078
2039
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS5
SBS1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C>G
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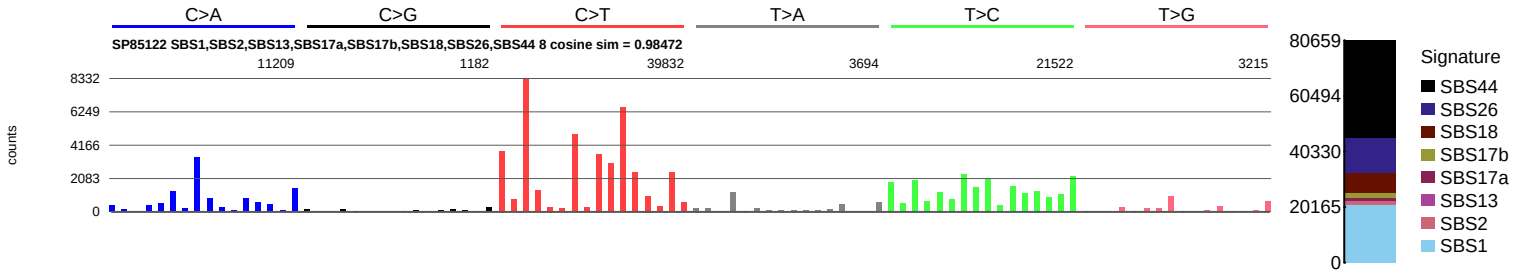
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98472
11258
1012
39877
3709
21580
3227
counts
8336
6252
4168
2084
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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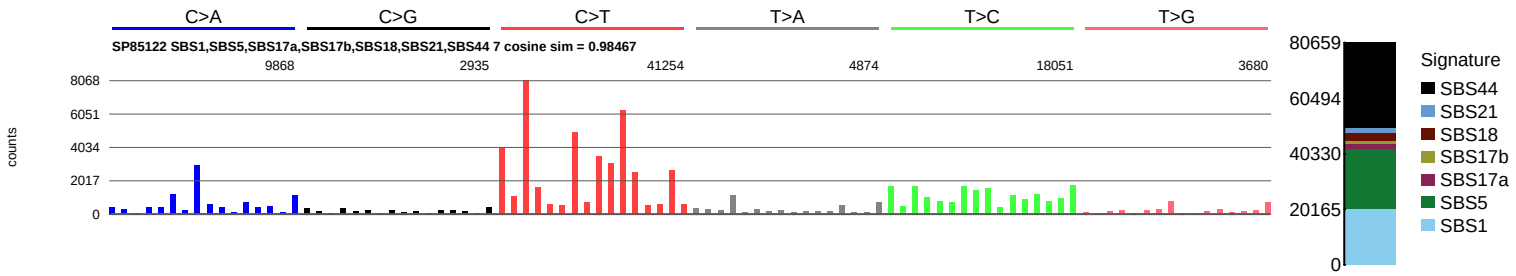
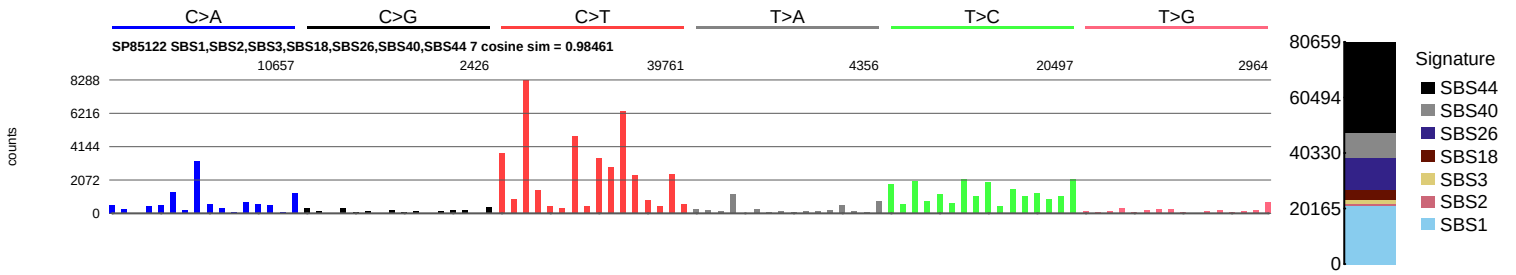
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98472
11209
1182
39832
3694
21522
3215
counts
8332
6249
4166
2083
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98467
9868
2935
41254
4874
18051
3680
counts
8068
6051
4034
2017
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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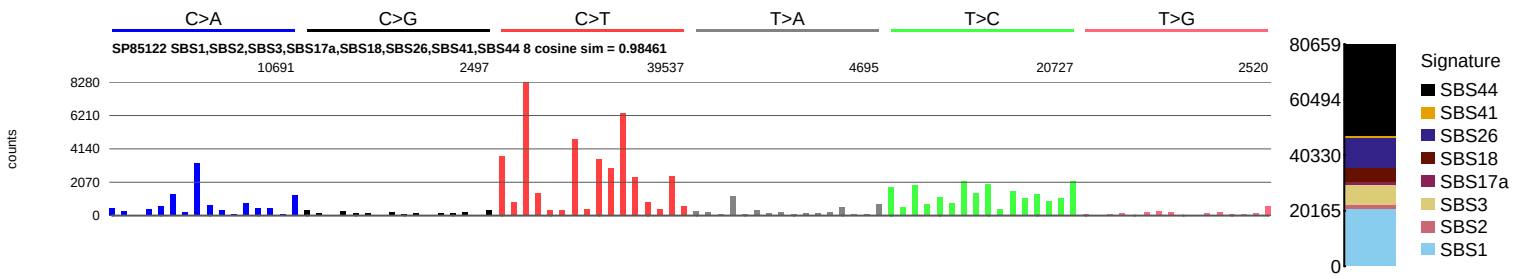
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98461
10657
2426
39761
4356
20497
2964
counts
8288
6216
4144
2072
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS3
SBS2
SBS1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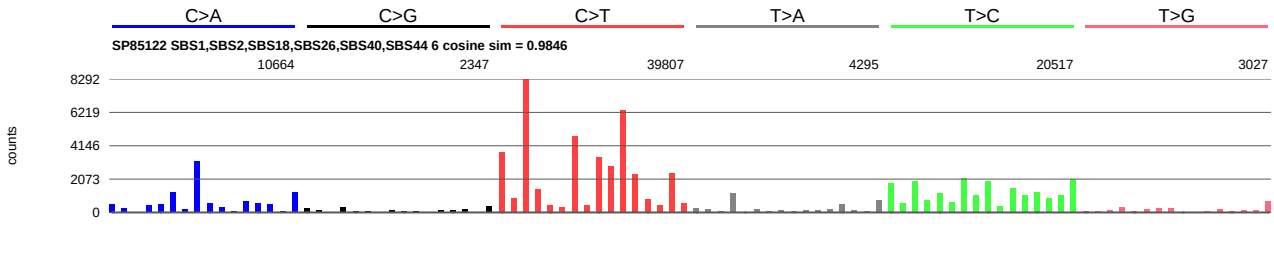
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98461
10691
2497
39537
4695
20727
2520
counts
8280
6210
4140
2070
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1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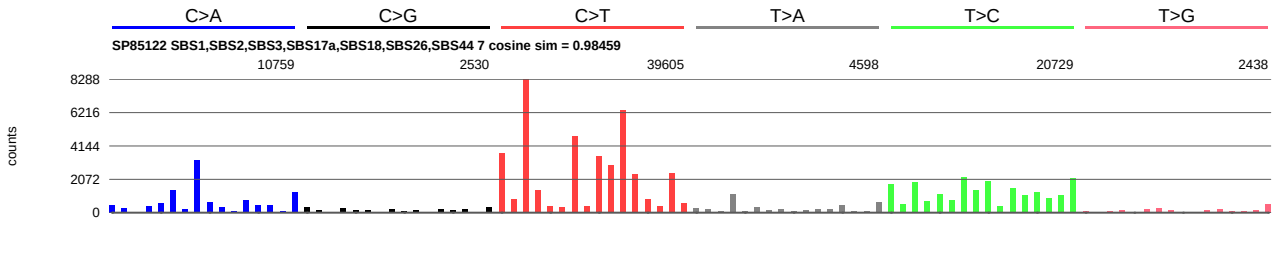
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.9846
10664
2347
39807
4295
20517
3027
8292
6219
4146
2073
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS2
SBS1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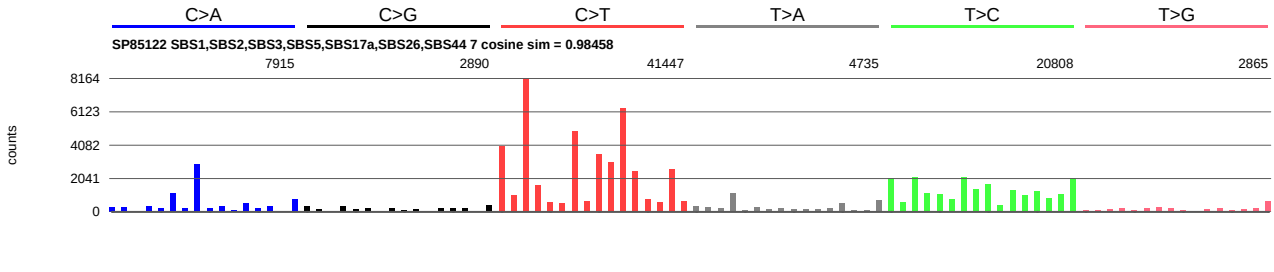
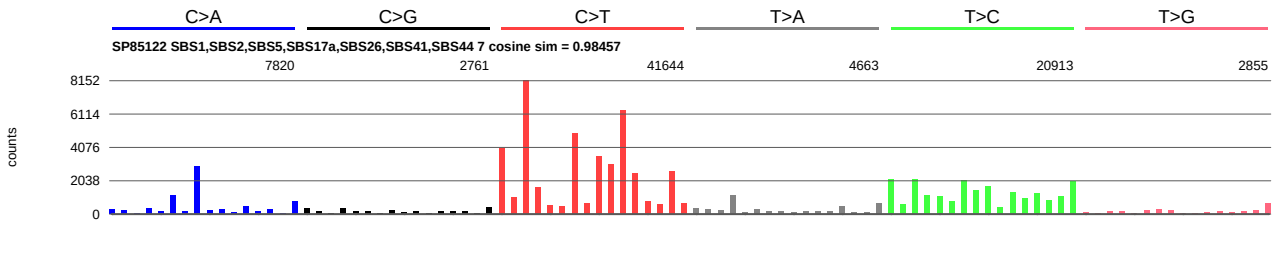
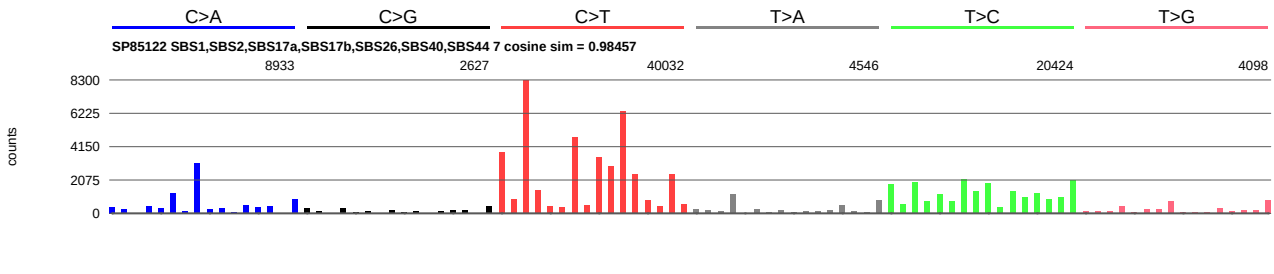
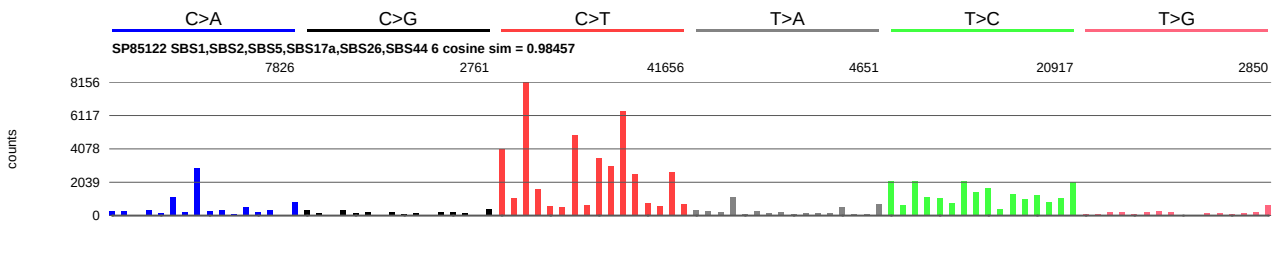
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98459
10759
2530
39605
4598
20729
2438
8288
6216
4144
2072
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS26,SBS44 7 cosine sim = 0.98458
7915
2890
41447
4735
20808
2865
8164
6123
4082
2041
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98457
7820
2761
41644
4663
20913
2855
8152
6114
4076
2038
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98457
8933
2627
40032
4546
20424
4098
8300
6225
4150
2075
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS26,SBS44 6 cosine sim = 0.98457
7826
2761
41656
4651
20917
2850
8156
6117
4078
2039
0
counts
80659
60494
40330
20165
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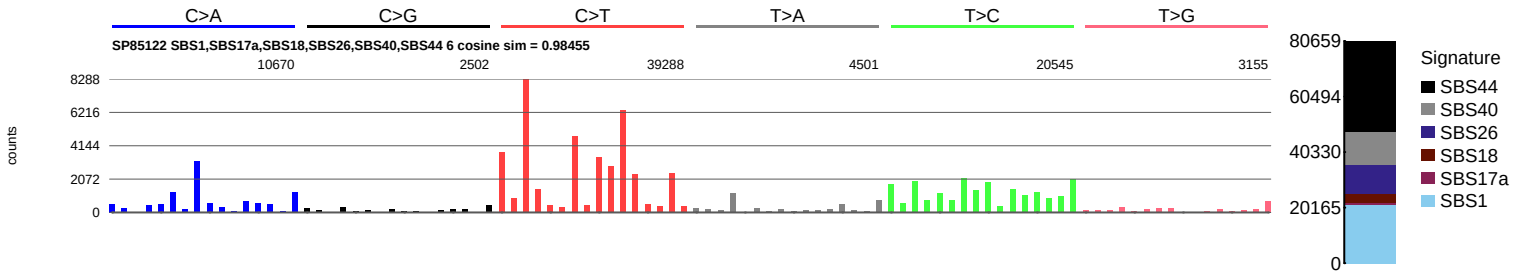
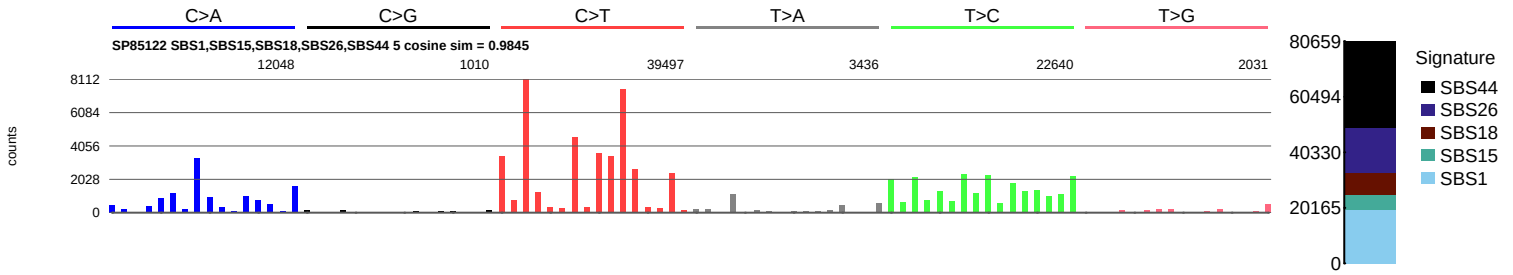
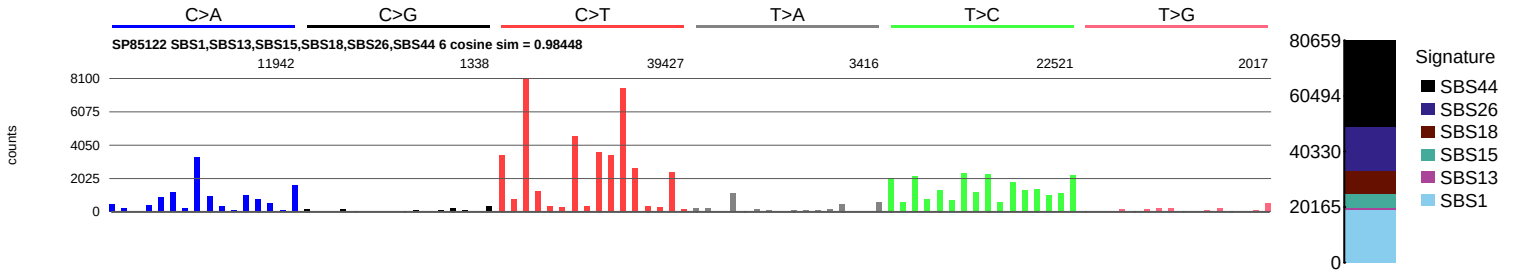
Signature
SBS44
SBS26
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98455
10670
2502
39288
4501
20545
3155
counts
8288
6216
4144
2072
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS18,SBS26,SBS44 5 cosine sim = 0.9845
12048
1010
39497
3436
22640
2031
counts
8112
6084
4056
2028
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98448
11942
1338
39427
3416
22521
2017
counts
8100
6075
4050
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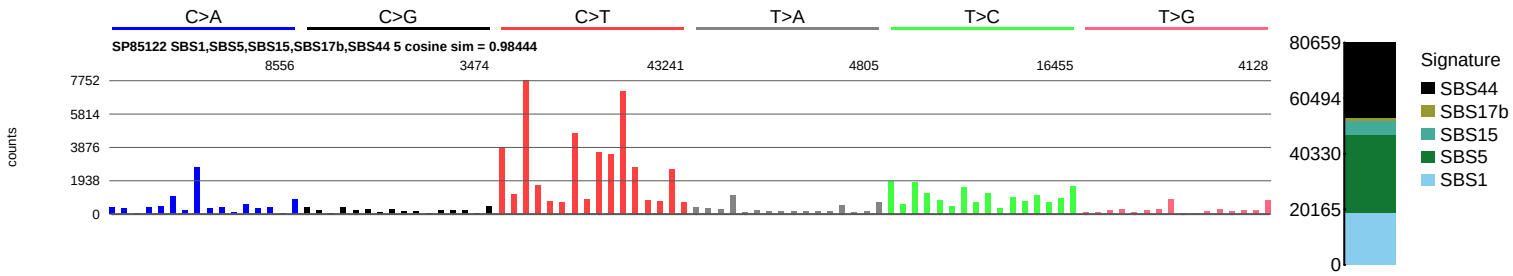
SBS1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C>G
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SP85122 SBS1,SBS5,SBS15,SBS17b,SBS44 5 cosine sim = 0.98444
8556
3474
43241
4805
16455
4128
counts
7752
5814
3876
1938
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80659
60494
40330
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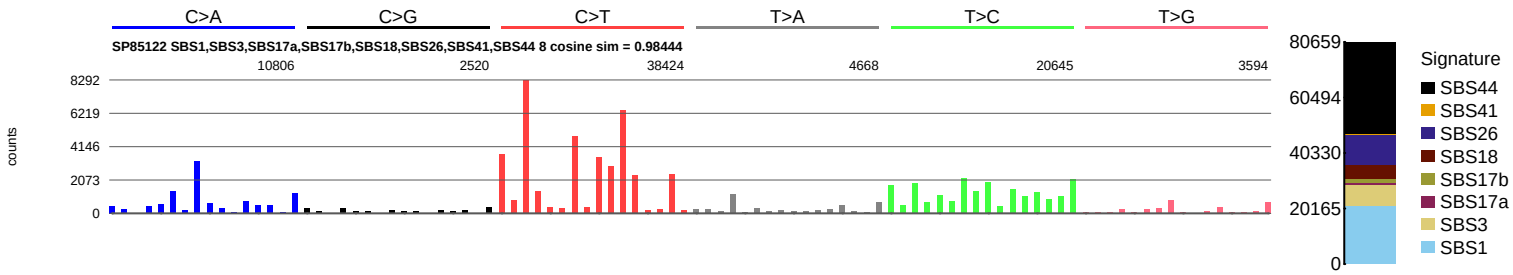
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SP85122 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98444
10806
2520
38424
4668
20645
3594
counts
8292
6219
4146
2073
0
80659
60494
40330
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SBS41
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SBS17b
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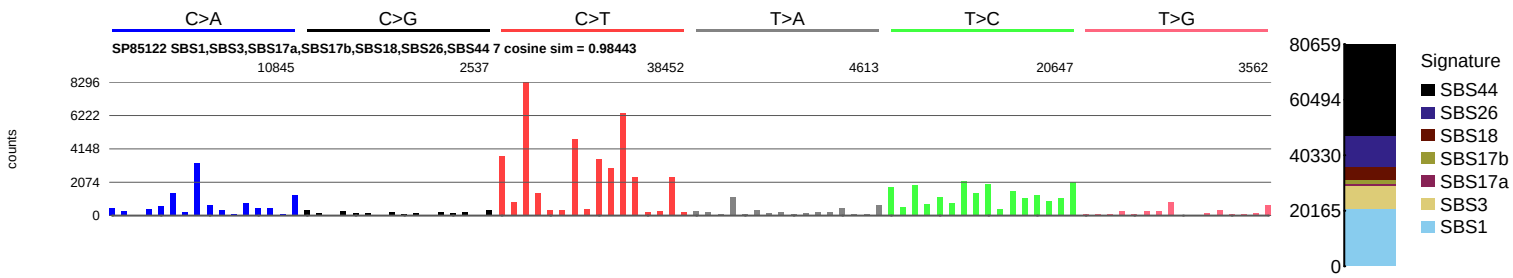
SBS3
SBS1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SP85122 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98443
10845
2537
38452
4613
20647
3562
counts
8296
6222
4148
2074
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40330
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SBS17b
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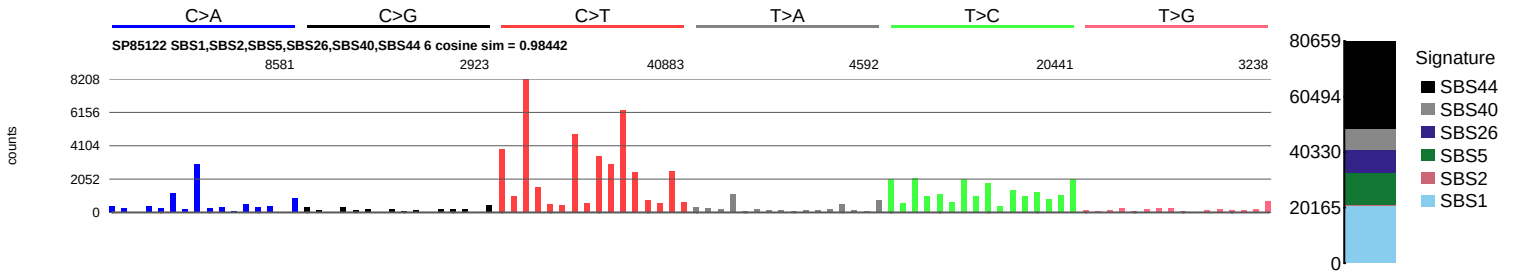
SBS3
SBS1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SP85122 SBS1,SBS2,SBS5,SBS26,SBS40,SBS44 6 cosine sim = 0.98442
8581
2923
40883
4592
20441
3238
counts
8208
6156
4104
2052
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60494
40330
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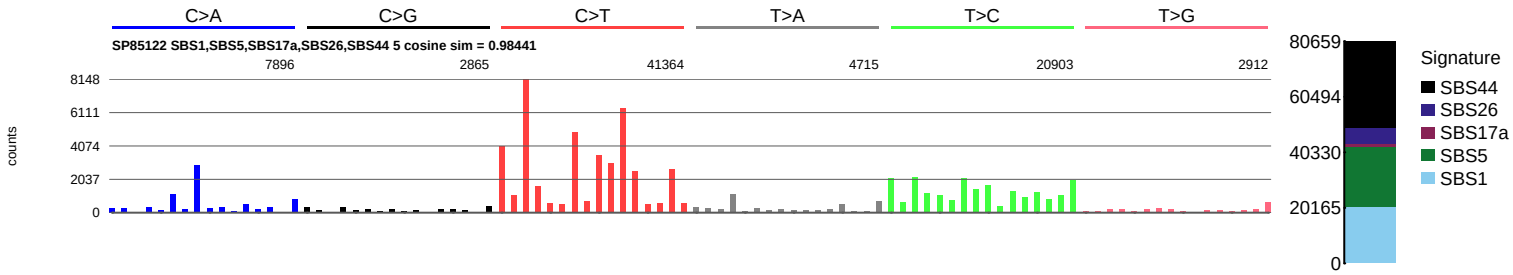
SBS1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SP85122 SBS1,SBS5,SBS17a,SBS26,SBS44 5 cosine sim = 0.98441
7896
2865
41364
4715
20903
2912
counts
8148
6111
4074
2037
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80659
60494
40330
20165
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SBS44
SBS26
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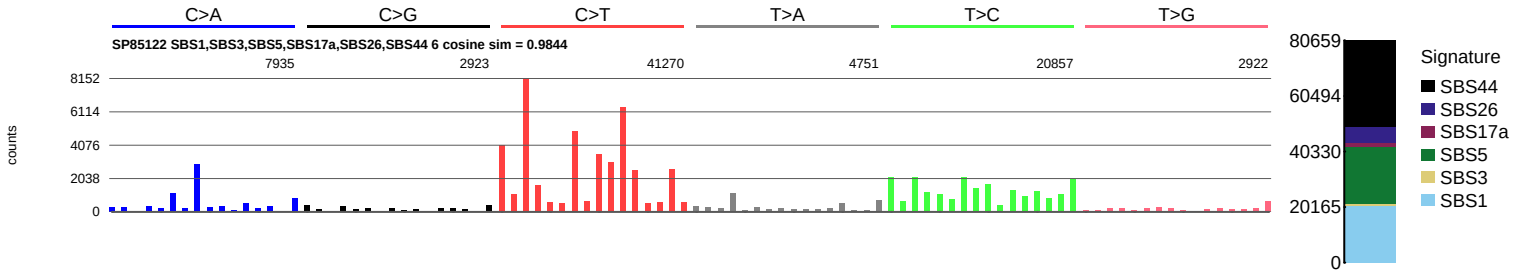
SBS5
SBS1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C>G
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T>G
SP85122 SBS1,SBS3,SBS5,SBS17a,SBS26,SBS44 6 cosine sim = 0.9844
7935
2923
41270
4751
20857
2922
counts
8152
6114
4076
2038
0
80659
60494
40330
20165
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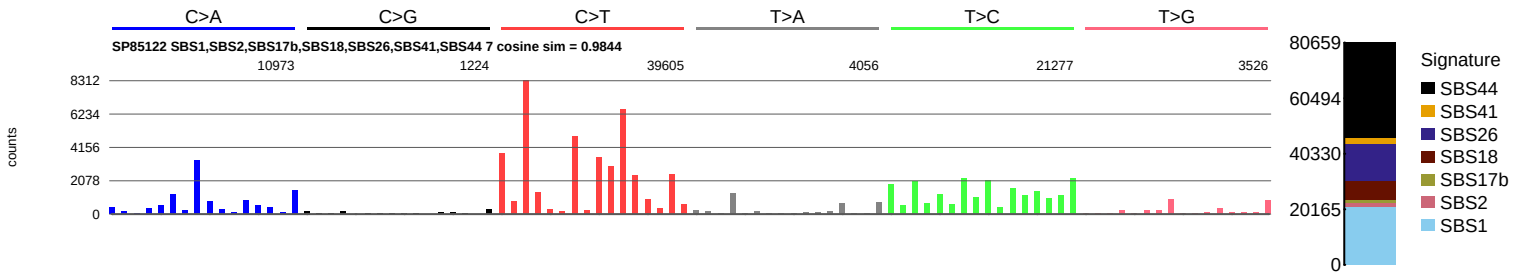
SBS5
SBS3
SBS1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SP85122 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.9844
10973
1224
39605
4056
21277
3526
counts
8312
6234
4156
2078
0
80659
60494
40330
20165
0
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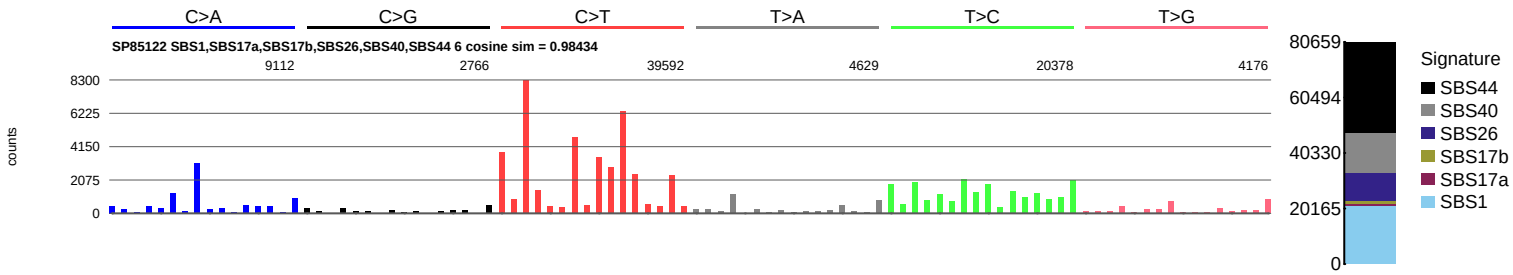
SBS41
SBS26
SBS18
SBS17b
SBS2
SBS1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T>C
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SP85122 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98434
9112
2766
39592
4629
20378
4176
counts
8300
6225
4150
2075
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60494
40330
20165
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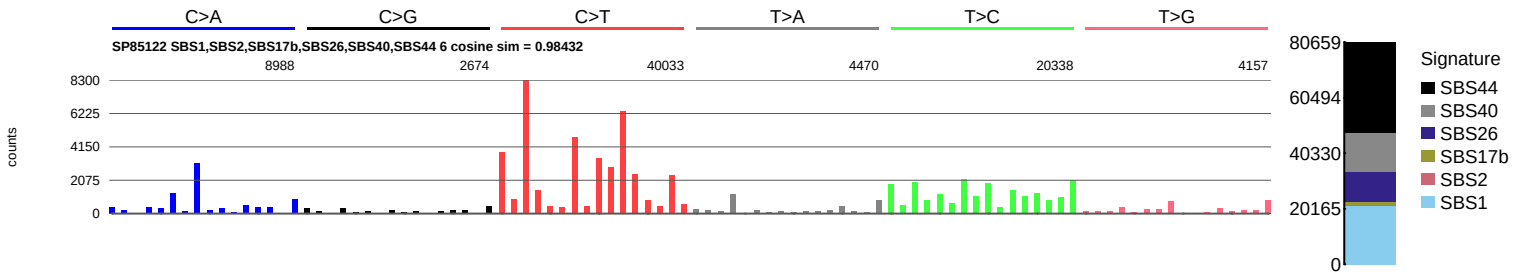
SBS26
SBS17b
SBS17a
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98432
8988
2674
40033
4470
20338
4157
counts
8300
6225
4150
2075
0
80659
60494
40330
20165
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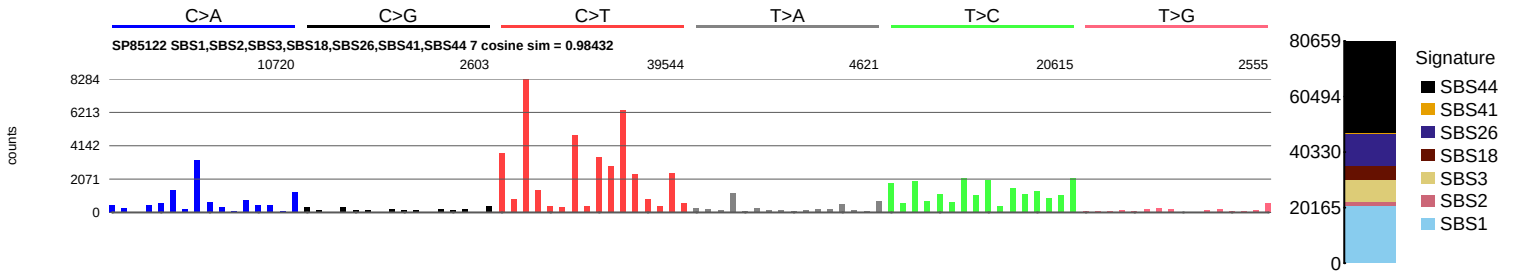
SBS44
SBS40
SBS26
SBS17b
SBS2
SBS1


C>A
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T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98432
10720
2603
39544
4621
20615
2555
8284
6213
4142
2071
0
counts
80659
60494
40330
20165
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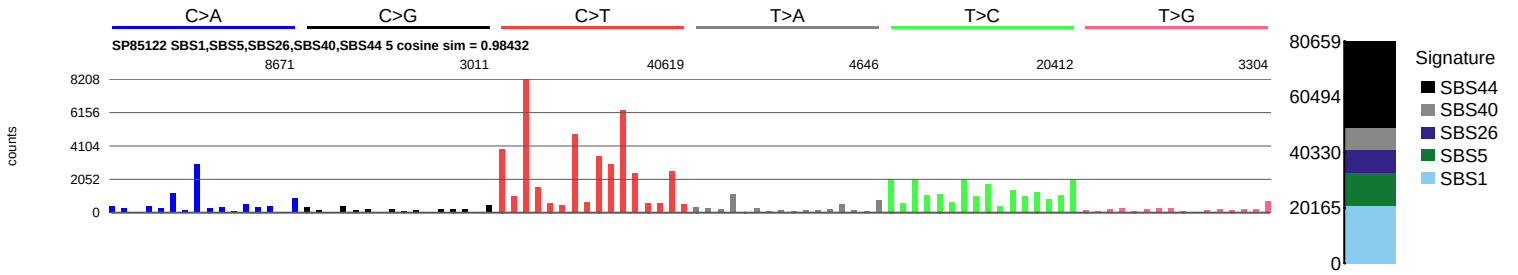
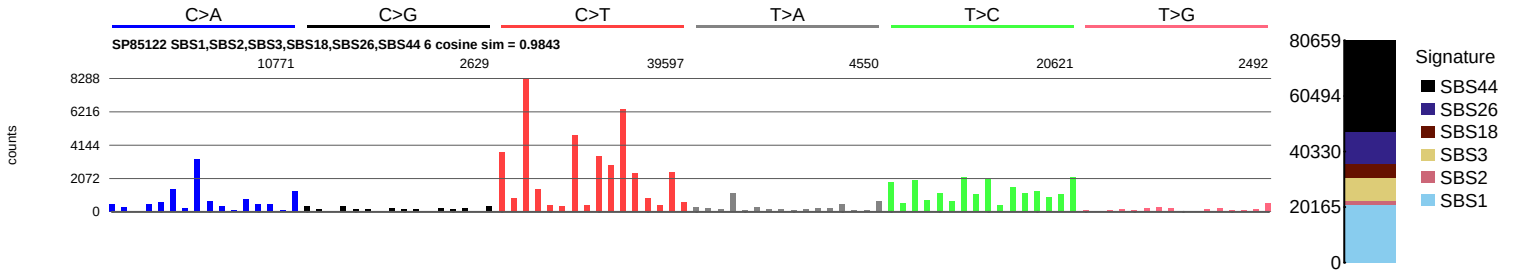
SBS41
SBS26
SBS18
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS26,SBS40,SBS44 5 cosine sim = 0.98432
8671
3011
40619
4646
20412
3304
8208
6156
4104
2052
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS5
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS18,SBS26,SBS44 6 cosine sim = 0.9843
10771
2629
39597
4550
20621
2492
8288
6216
4144
2072
0
counts
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60494
40330
20165
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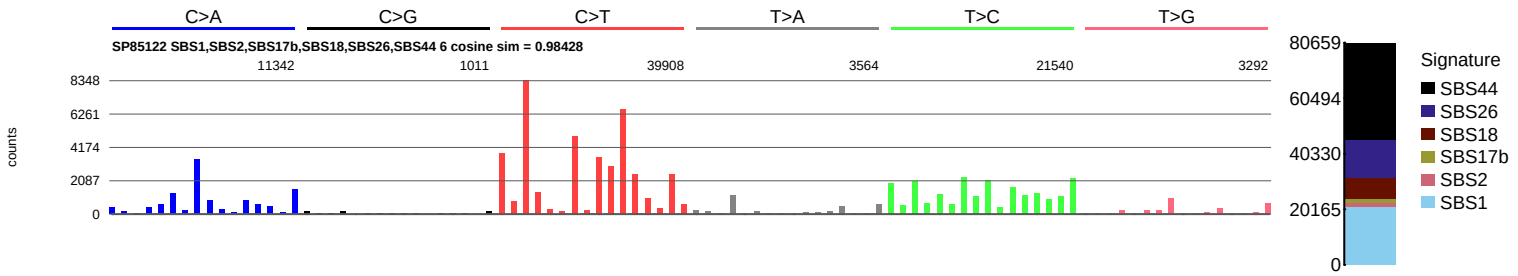
SBS26
SBS18
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98428
11342
1011
39908
3564
21540
3292
8348
6261
4174
2087
0
counts
80659
60494
40330
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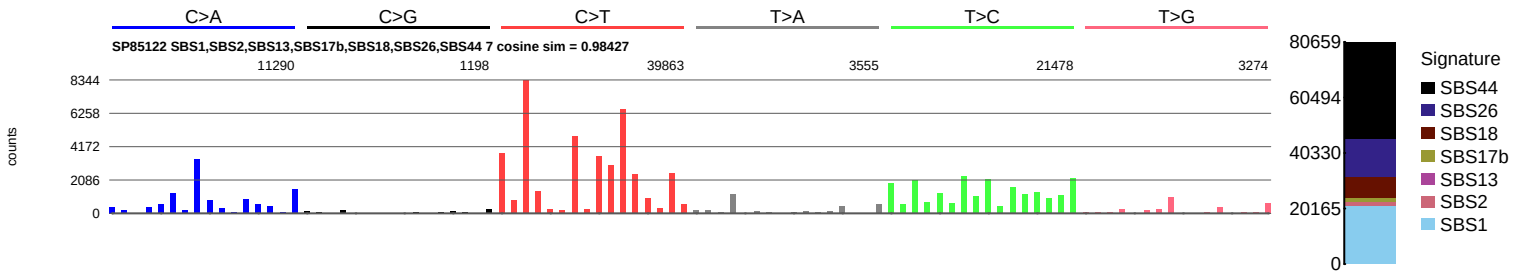
SBS17b
SBS2
SBS1


C>A
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T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98427
11290
1198
39863
3555
21478
3274
8344
6258
4172
2086
0
counts
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20165
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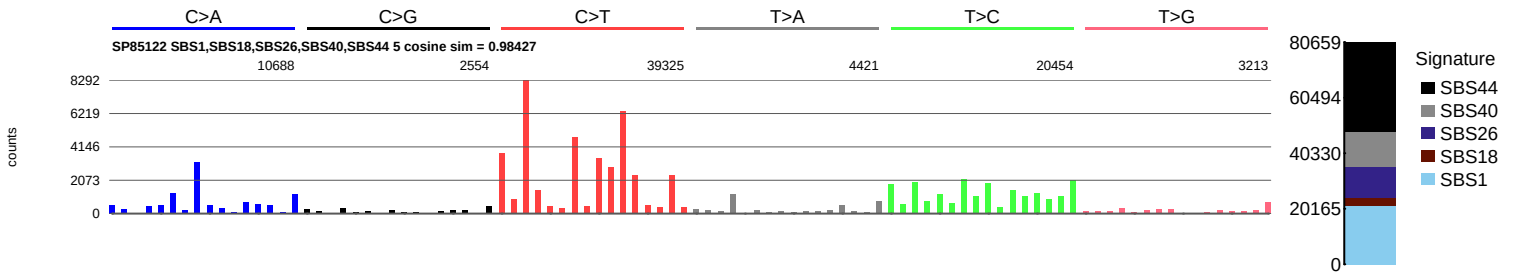
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS18,SBS26,SBS40,SBS44 5 cosine sim = 0.98427
10688
2554
39325
4421
20454
3213
8292
6219
4146
2073
0
counts
80659
60494
40330
20165
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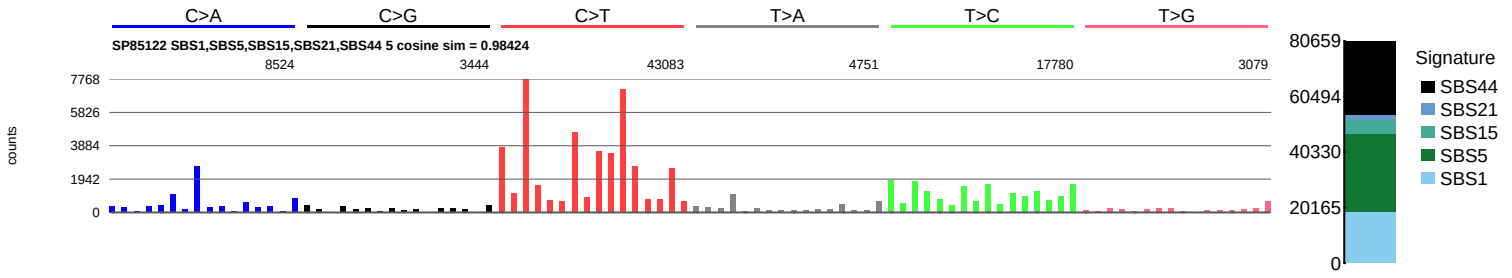
SBS40
SBS26
SBS18
SBS1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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS21,SBS44 5 cosine sim = 0.98424
8524
3444
43083
4751
17780
3079
counts
7768
5826
3884
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60494
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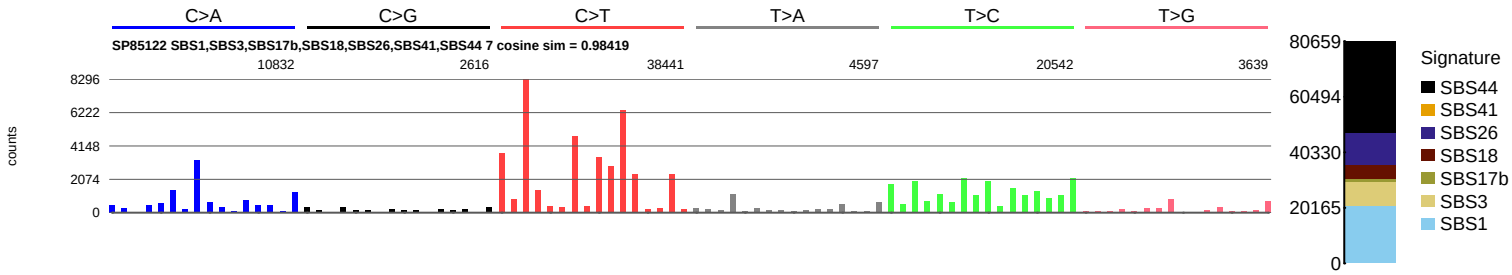
SBS5
SBS1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T>C
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SP85122 SBS1,SBS3,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98419
10832
2616
38441
4597
20542
3639
counts
8296
6222
4148
2074
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40330
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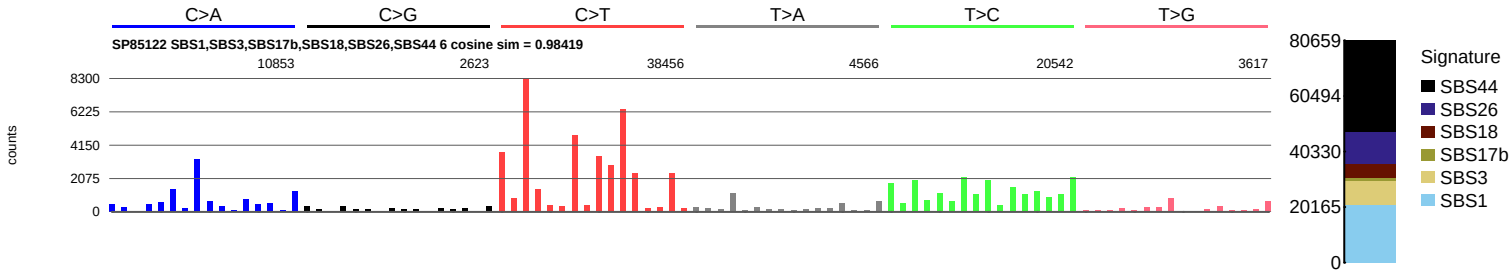
SBS3
SBS1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T>C
T>G
SP85122 SBS1,SBS3,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98419
10853
2623
38456
4566
20542
3617
counts
8300
6225
4150
2075
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60494
40330
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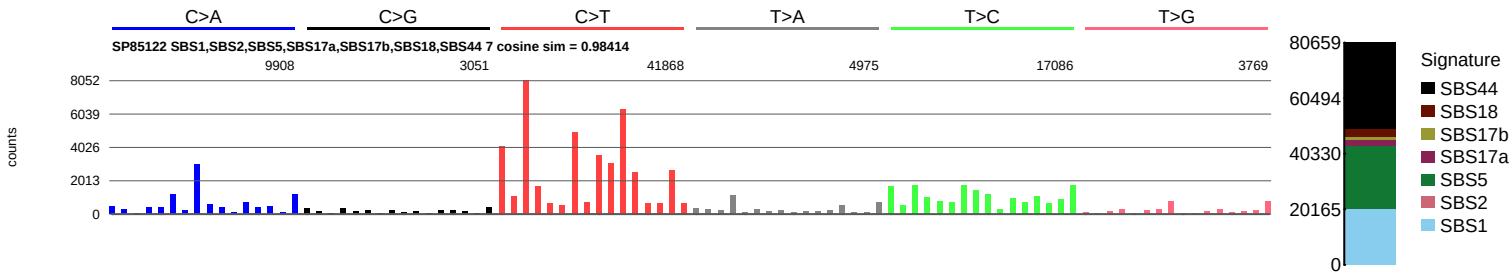
SBS1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T>C
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SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98414
9908
3051
41868
4975
17086
3769
counts
8052
6039
4026
2013
0
80659
60494
40330
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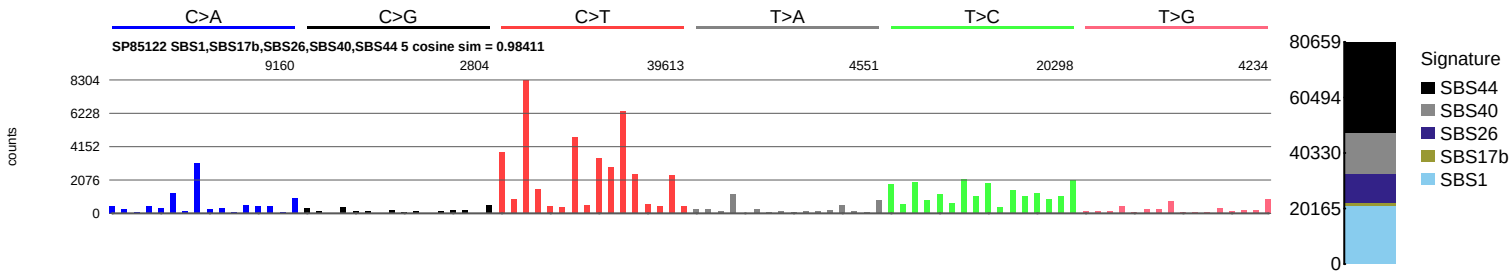
SBS2
SBS1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SP85122 SBS1,SBS17b,SBS26,SBS40,SBS44 5 cosine sim = 0.98411
9160
2804
39613
4551
20298
4234
counts
8304
6228
4152
2076
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80659
60494
40330
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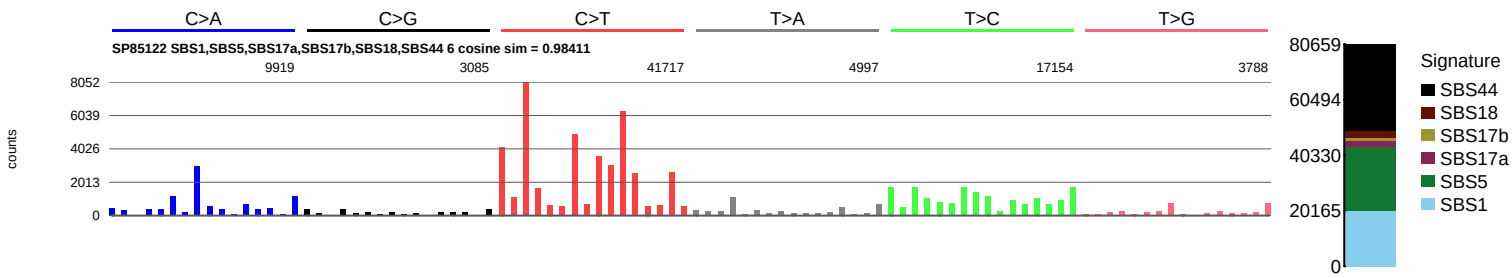
SBS26
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
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SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98411
9919
3085
41717
4997
17154
3788
counts
8052
6039
4026
2013
0
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SBS17b
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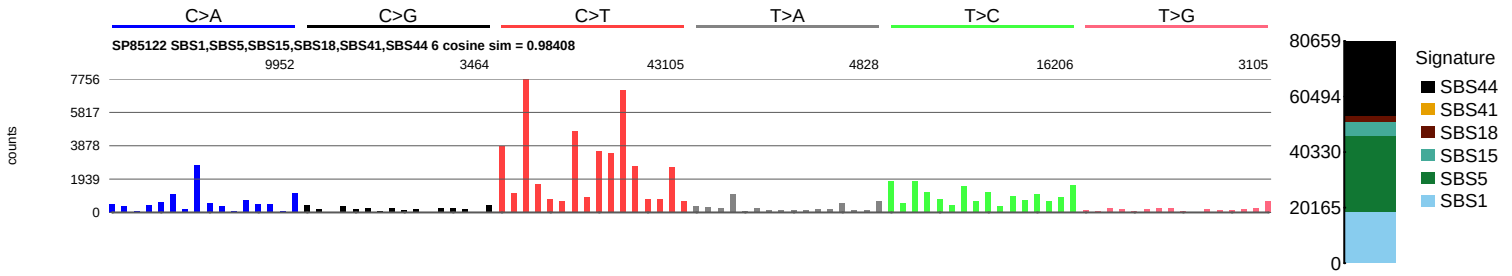
SBS5
SBS1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C>G
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T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS18,SBS41,SBS44 6 cosine sim = 0.98408
9952
3464
43105
4828
16206
3105
counts
7756
5817
3878
1939
0
80659
60494
40330
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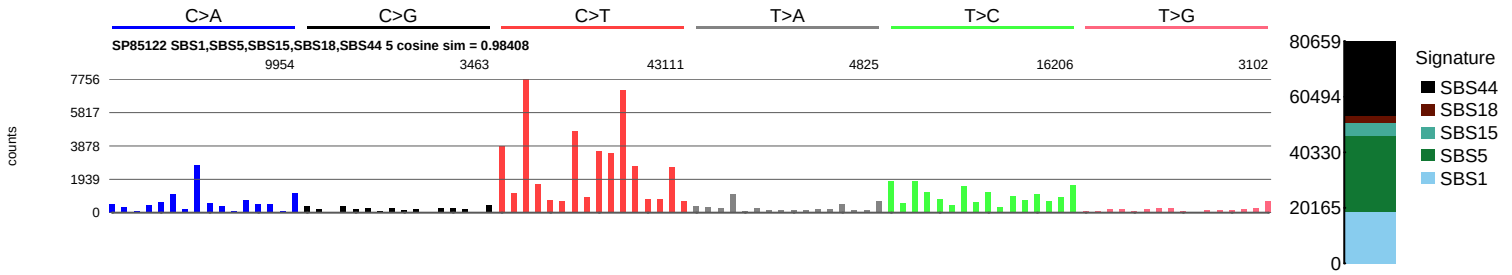
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS18,SBS44 5 cosine sim = 0.98408
9954
3463
43111
4825
16206
3102
counts
7756
5817
3878
1939
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80659
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40330
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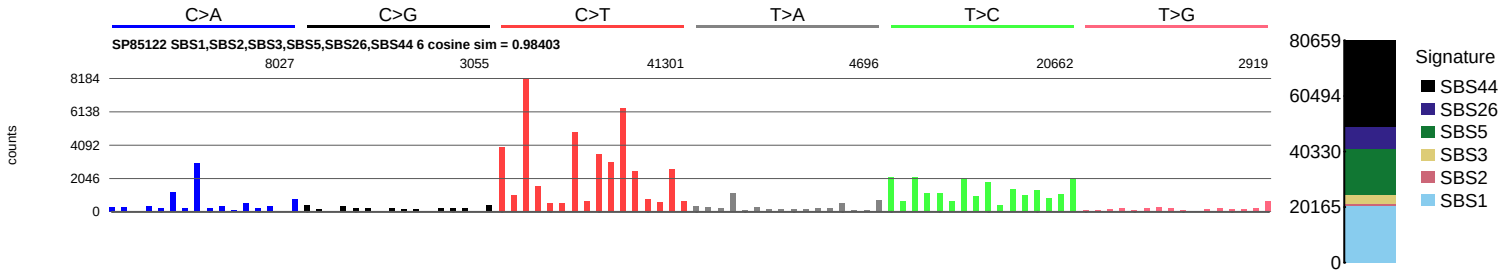
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SP85122 SBS1,SBS2,SBS3,SBS5,SBS26,SBS44 6 cosine sim = 0.98403
8027
3055
41301
4696
20662
2919
counts
8184
6138
4092
2046
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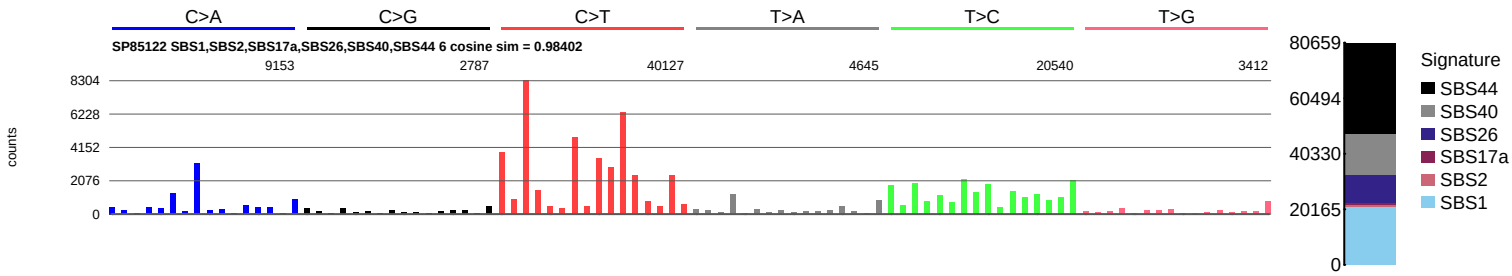
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T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS26,SBS40,SBS44 6 cosine sim = 0.98402
9153
2787
40127
4645
20540
3412
counts
8304
6228
4152
2076
0
80659
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40330
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Signature
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SBS40
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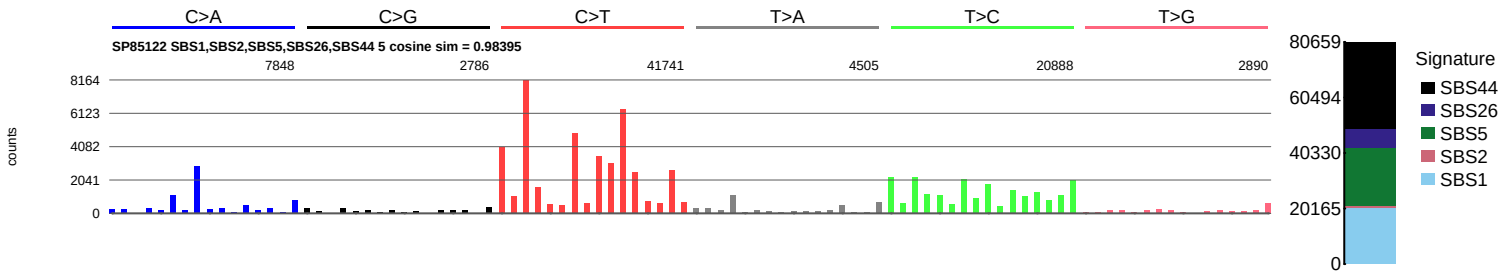
SBS17a
SBS2
SBS1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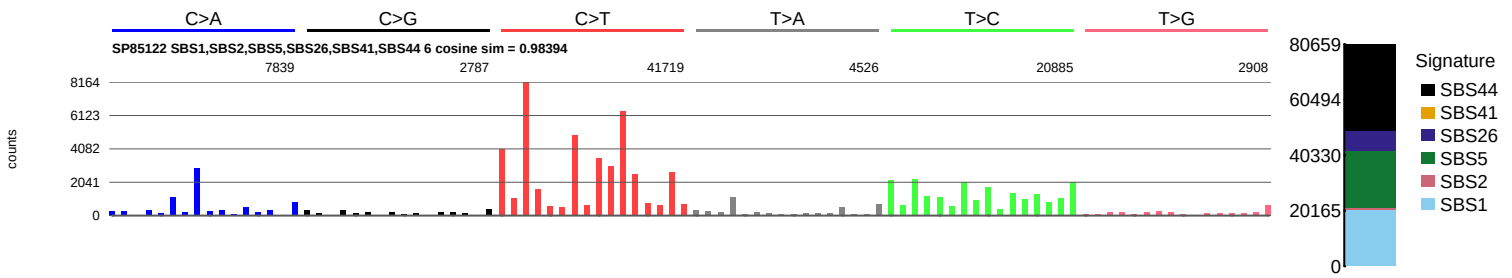
SP85122 SBS1,SBS2,SBS5,SBS26,SBS44 5 cosine sim = 0.98395
7848
2786
41741
4505
20888
2890
counts
8164
6123
4082
2041
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS5
SBS2
SBS1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SP85122 SBS1,SBS2,SBS5,SBS26,SBS41,SBS44 6 cosine sim = 0.98394
7839
2787
41719
4526
20885
2908
counts
8164
6123
4082
2041
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
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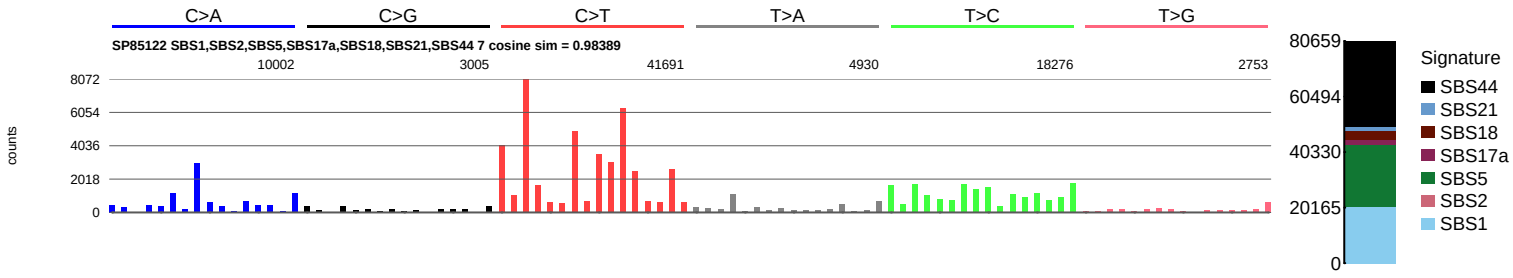
SBS2
SBS1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SP85122 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS44 7 cosine sim = 0.98389
10002
3005
41691
4930
18276
2753
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
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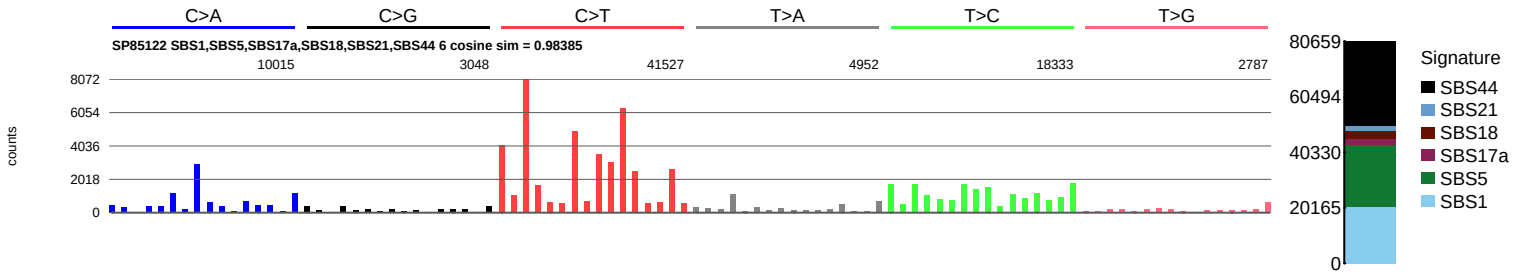
Signature
SBS44
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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SP85122 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS44 6 cosine sim = 0.98385
10015
3048
41527
4952
18333
2787
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
0
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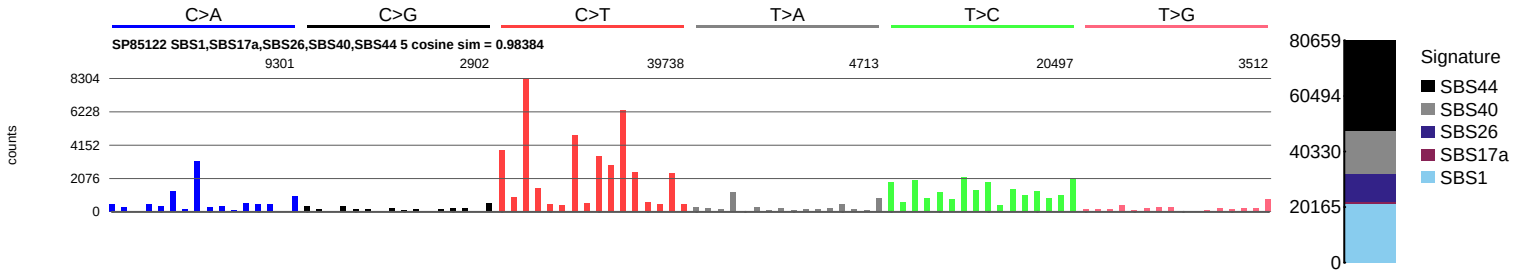
SBS44
SBS21
SBS18
SBS17a
SBS5
SBS1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SP85122 SBS1,SBS17a,SBS26,SBS40,SBS44 5 cosine sim = 0.98384
9301
2902
39738
4713
20497
3512
counts
8304
6228
4152
2076
0
80659
60494
40330
20165
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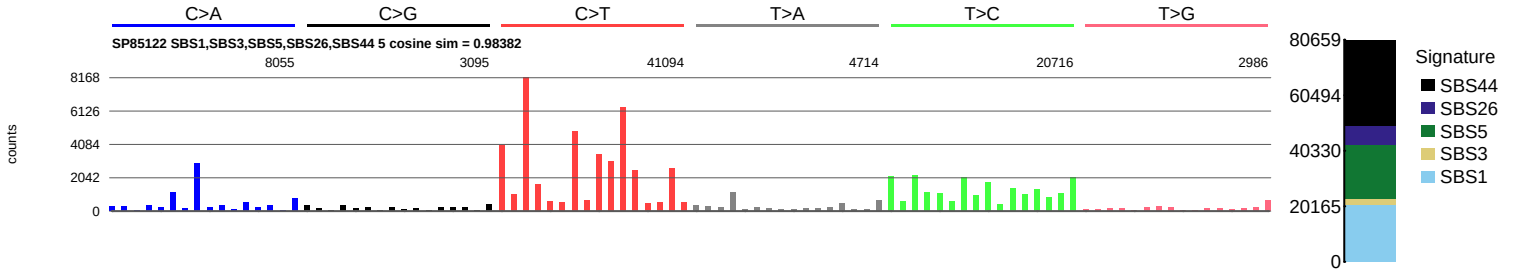
Signature
SBS44
SBS40
SBS26
SBS17a
SBS1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SP85122 SBS1,SBS3,SBS5,SBS26,SBS44 5 cosine sim = 0.98382
8055
3095
41094
4714
20716
2986
counts
8168
6126
4084
2042
0
80659
60494
40330
20165
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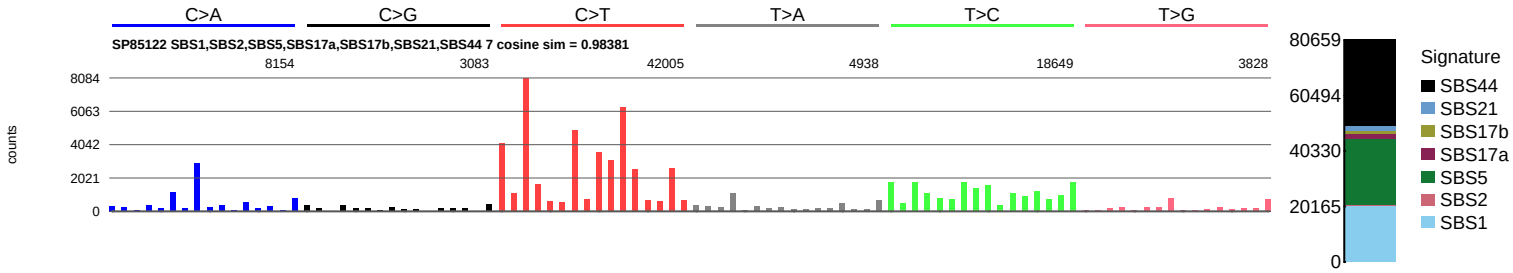
Signature
SBS44
SBS26
SBS5
SBS3
SBS1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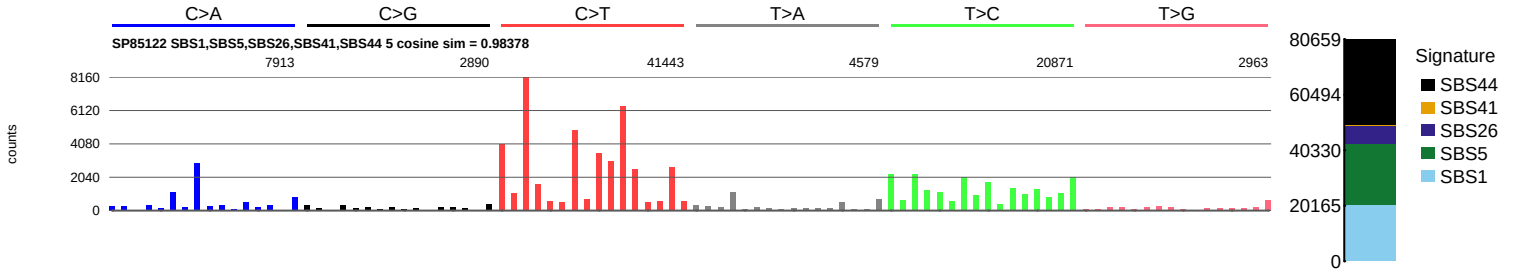
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98381
8154
3083
42005
4938
18649
3828
counts
8084
6063
4042
2021
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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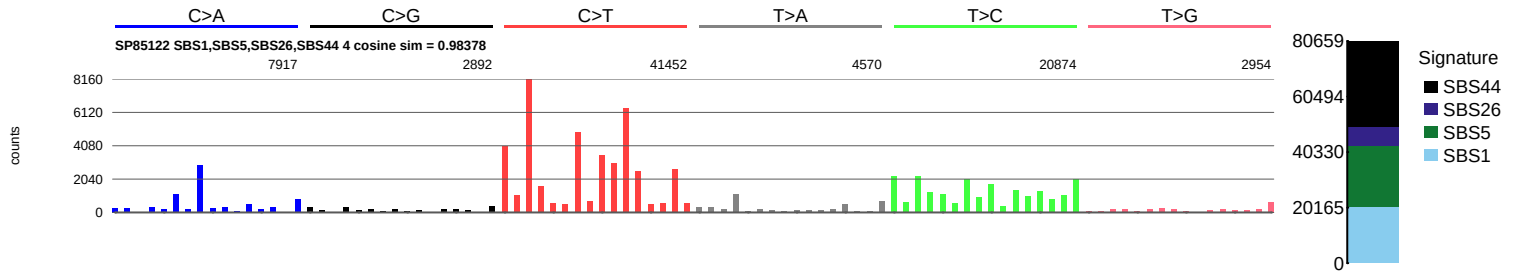
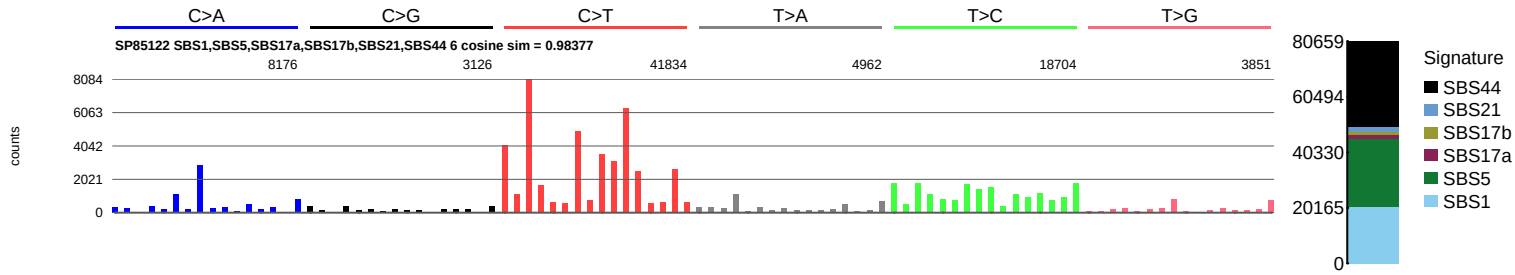
SP85122 SBS1,SBS5,SBS26,SBS41,SBS44 5 cosine sim = 0.98378
7913
2890
41443
4579
20871
2963
counts
8160
6120
4080
2040
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS26,SBS44 4 cosine sim = 0.98378
7917
2892
41452
4570
20874
2954
8160
6120
4080
2040
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS5
SBS1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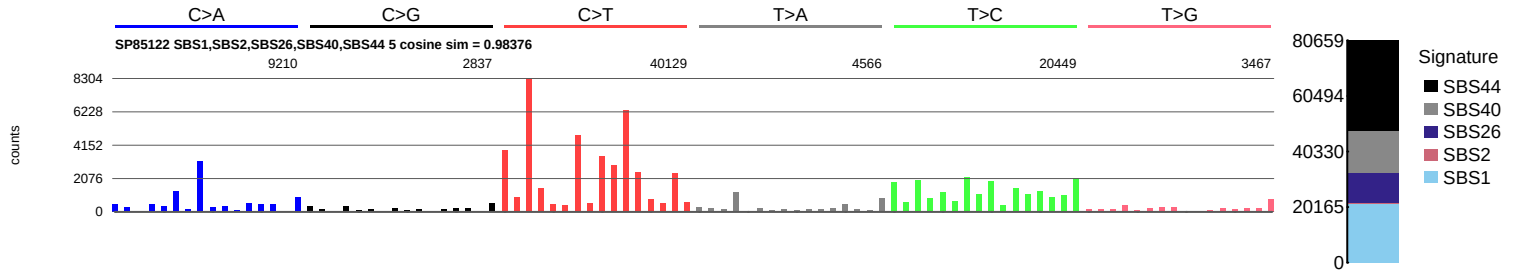
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98377
8176
3126
41834
4962
18704
3851
8084
6063
4042
2021
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS1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T>A
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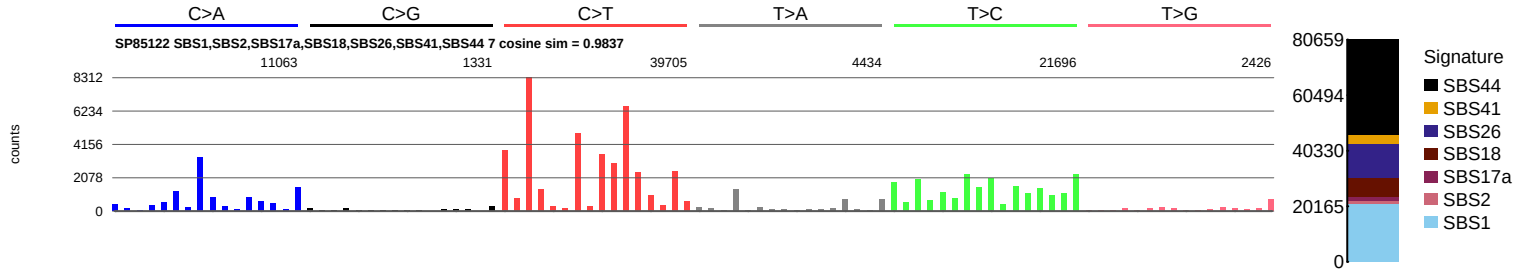
T>G
SP85122 SBS1,SBS2,SBS26,SBS40,SBS44 5 cosine sim = 0.98376
9210
2837
40129
4566
20449
3467
8304
6228
4152
2076
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS2
SBS1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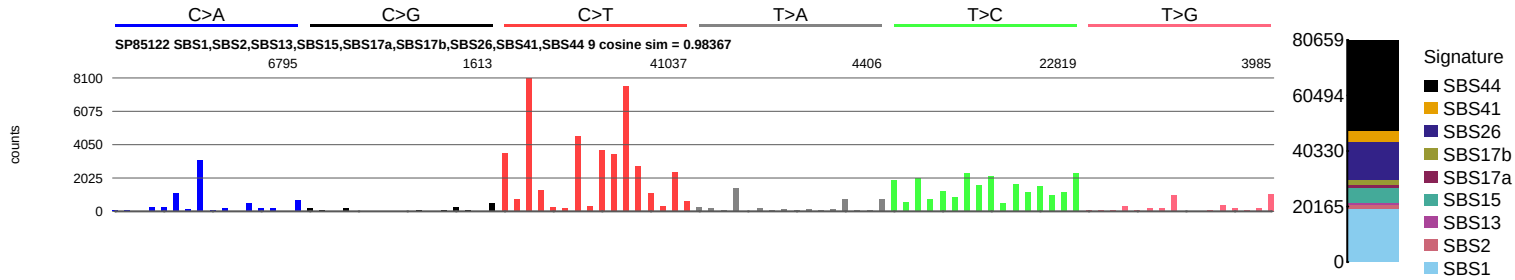
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.9837
11063
1331
39705
4434
21696
2426
8312
6234
4156
2078
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS2
SBS1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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98367
6795
1613
41037
4406
22819
3985
8100
6075
4050
2025
0
counts
80659
60494
40330
20165
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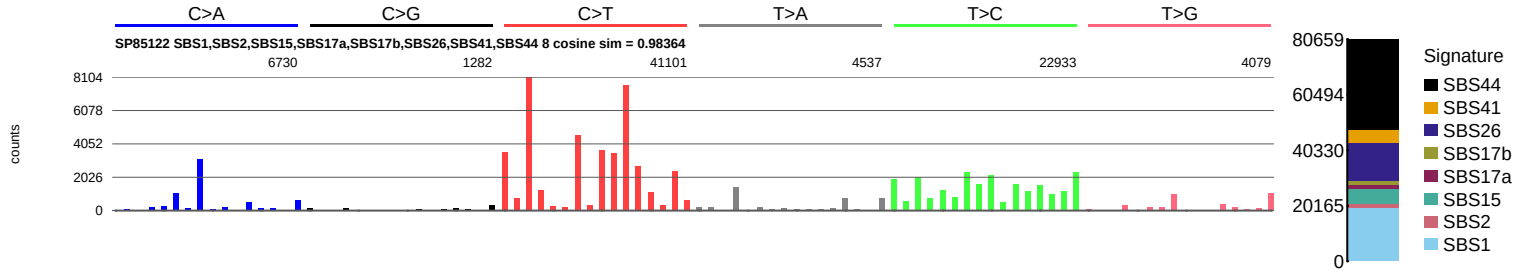
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98364
6730
1282
41101
4537
22933
4079
8104
6078
4052
2026
0
counts
80659
60494
40330
20165
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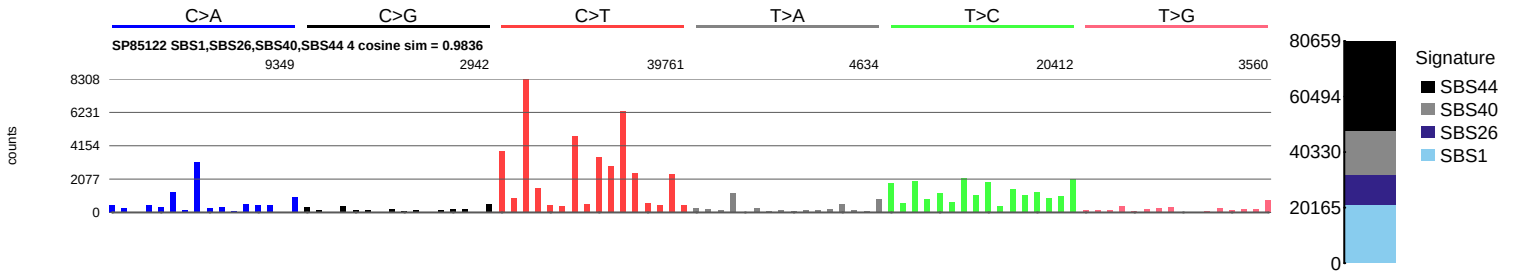
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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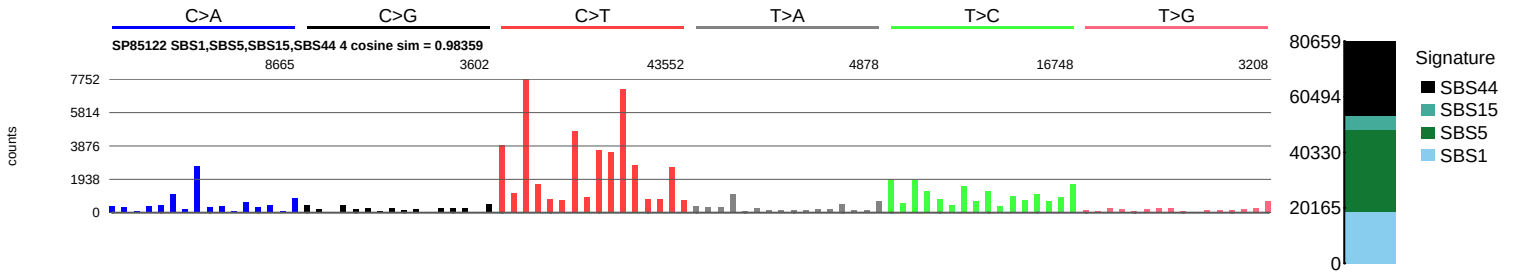
T>C
T>G
SP85122 SBS1,SBS26,SBS40,SBS44 4 cosine sim = 0.9836
9349
2942
39761
4634
20412
3560
8308
6231
4154
2077
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS26
SBS1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C>T
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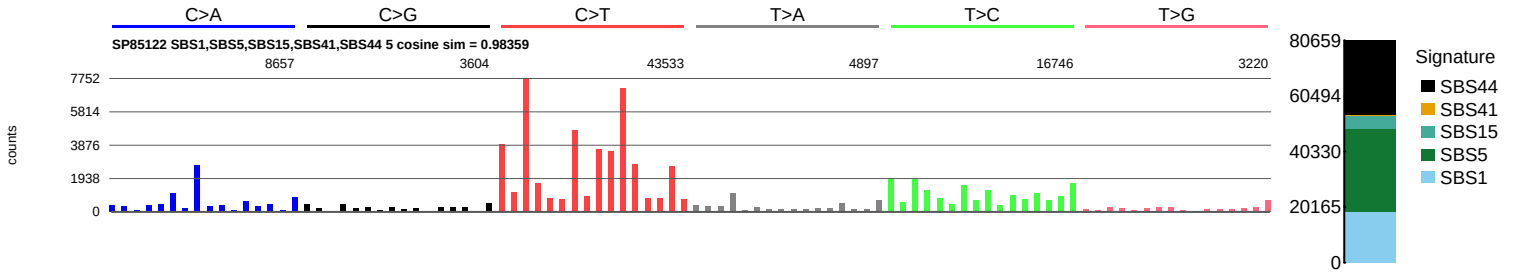
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS44 4 cosine sim = 0.98359
8665
3602
43552
4878
16748
3208
7752
5814
3876
1938
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS15,SBS41,SBS44 5 cosine sim = 0.98359
8657
3604
43533
4897
16746
3220
7752
5814
3876
1938
0
counts
80659
60494
40330
20165
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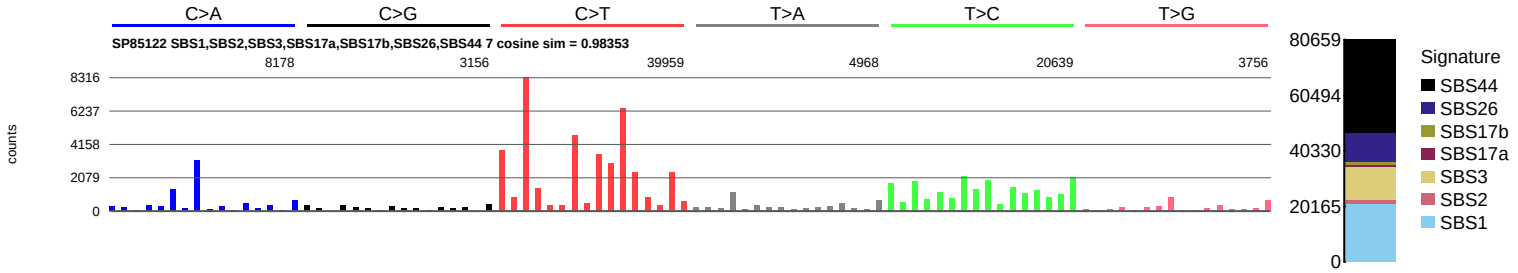
SBS41
SBS15
SBS5
SBS1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C>G
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T>C
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SP85122 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98353
8178
3156
39959
4968
20639
3756
8316
6237
4158
2079
0
counts
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60494
40330
20165
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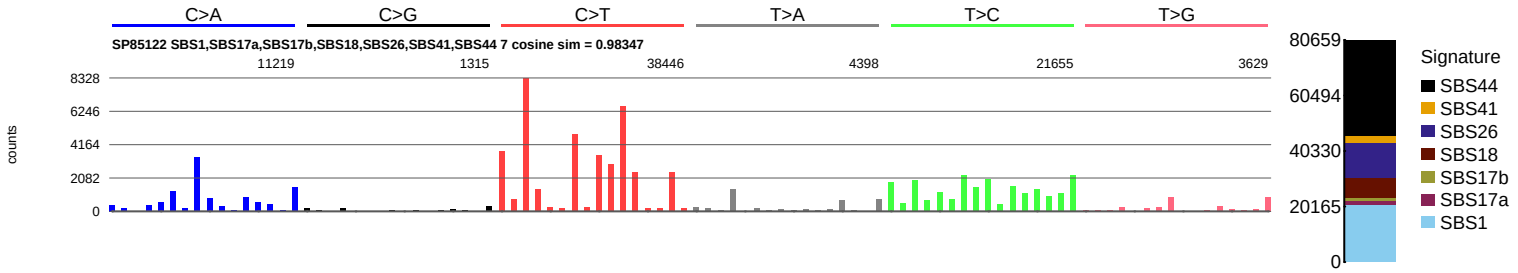
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98347
11219
1315
38446
4398
21655
3629
8328
6246
4164
2082
0
counts
80659
60494
40330
20165
0
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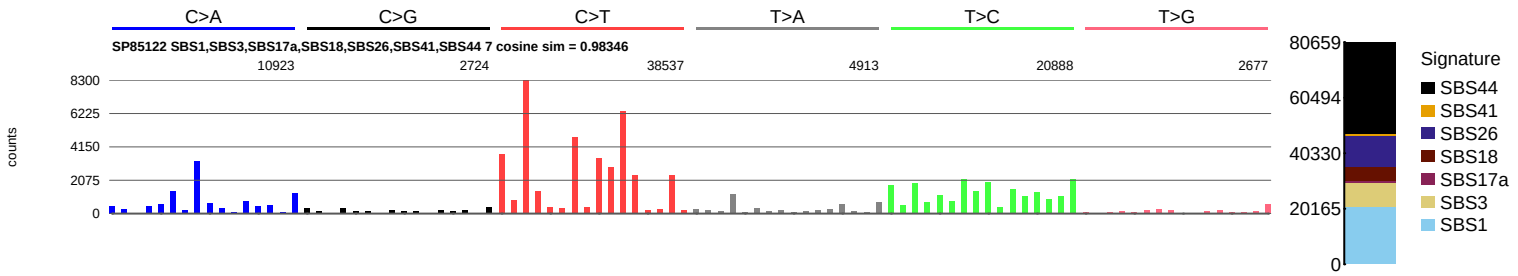
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
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T>G
SP85122 SBS1,SBS3,SBS17a,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98346
10923
2724
38537
4913
20888
2677
8300
6225
4150
2075
0
counts
80659
60494
40330
20165
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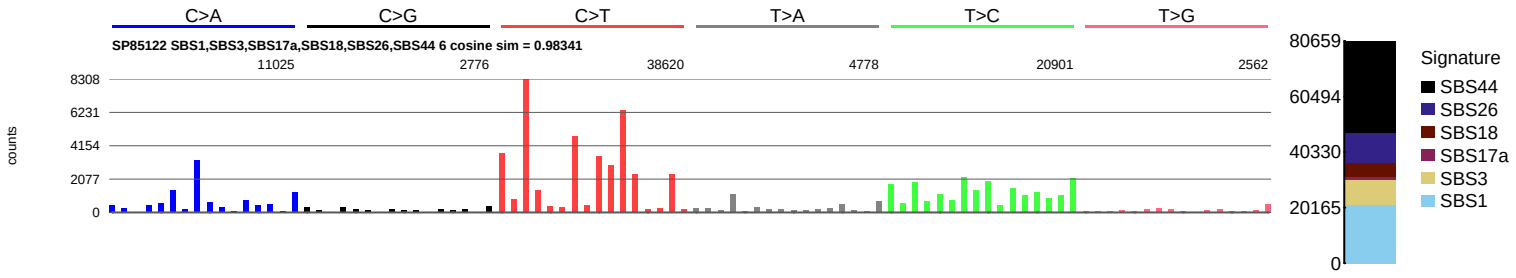
SBS41
SBS26
SBS18
SBS17a
SBS3
SBS1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T>C
T>G
SP85122 SBS1,SBS3,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98341
11025
2776
38620
4778
20901
2562
counts
8308
6231
4154
2077
0
80659
60494
40330
20165
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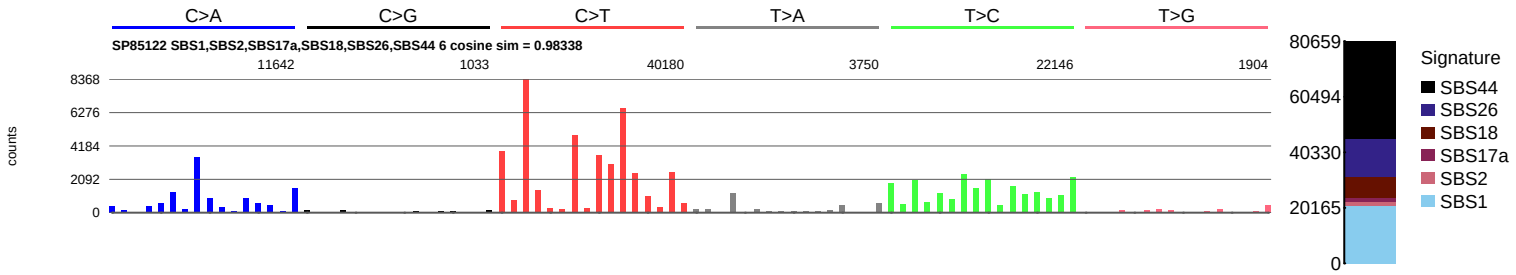
SBS44
SBS26
SBS18
SBS17a
SBS3
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98338
11642
1033
40180
3750
22146
1904
counts
8368
6276
4184
2092
0
80659
60494
40330
20165
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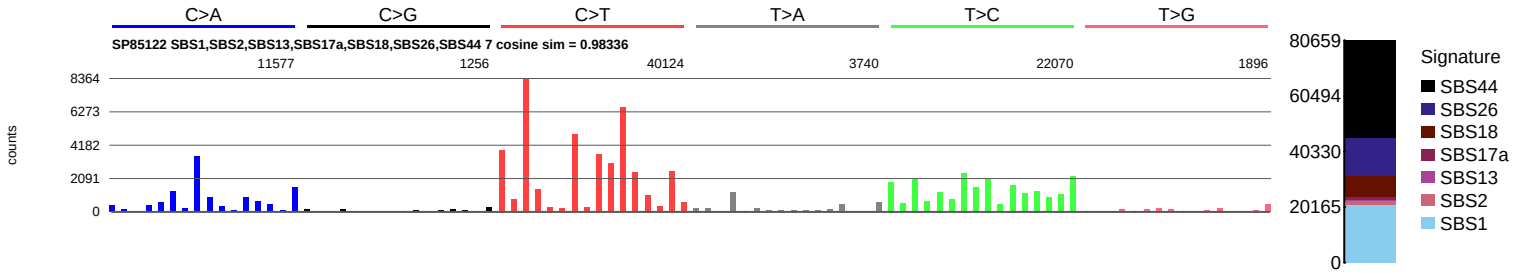
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98336
11577
1256
40124
3740
22070
1896
counts
8364
6273
4182
2091
0
80659
60494
40330
20165
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Signature
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SBS26
SBS18
SBS17a
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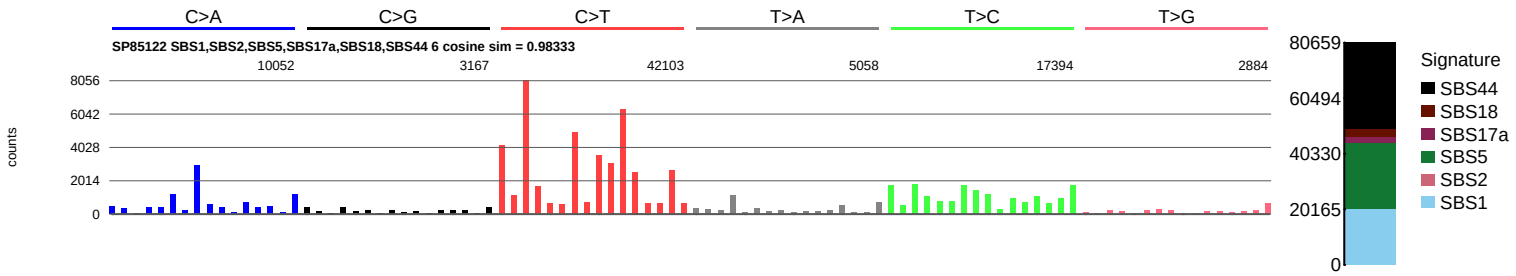
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS44 6 cosine sim = 0.98333
10052
3167
42103
5058
17394
2884
counts
8056
6042
4028
2014
0
80659
60494
40330
20165
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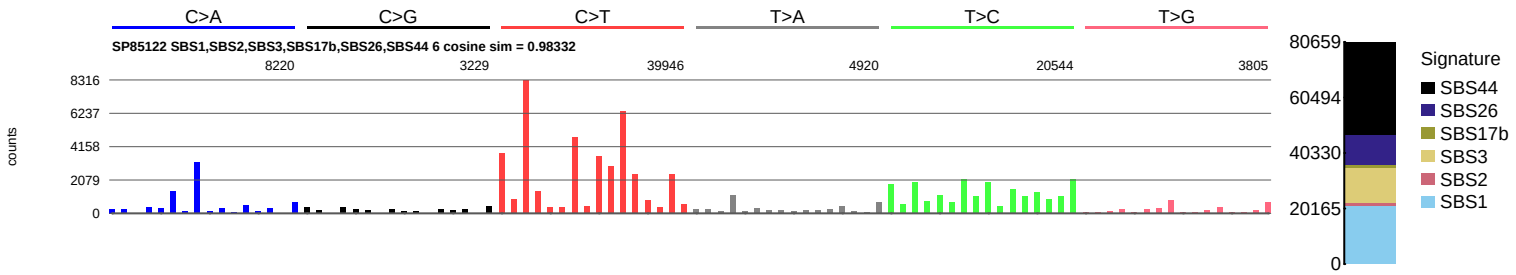
SBS44
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
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C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17b,SBS26,SBS44 6 cosine sim = 0.98332
8220
3229
39946
4920
20544
3805
counts
8316
6237
4158
2079
0
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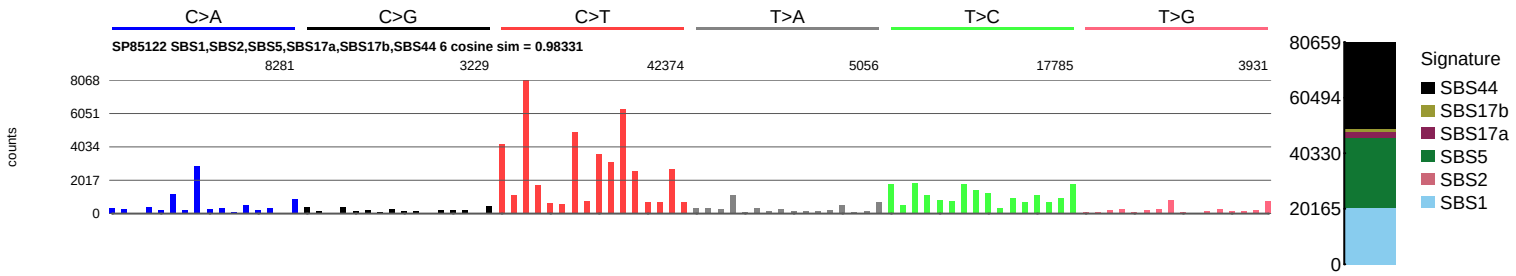
C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98331
8281
3229
42374
5056
17785
3931
counts
8068
6051
4034
2017
0
80659
60494
40330
20165
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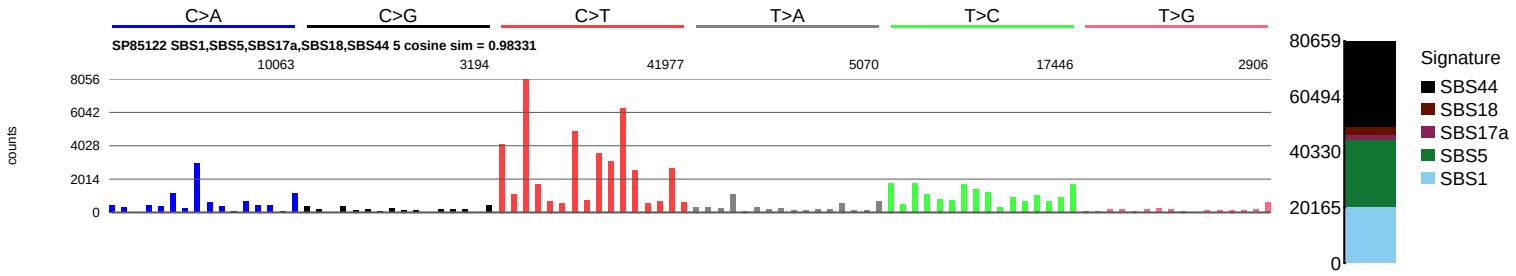
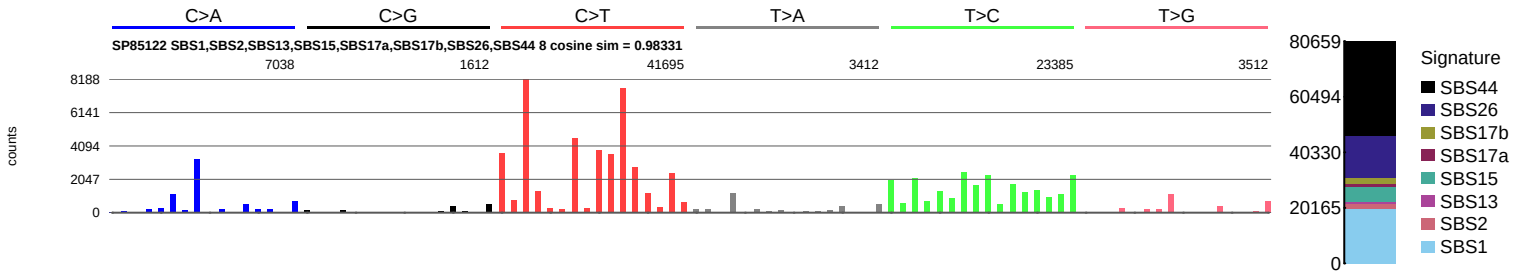
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS18,SBS44 5 cosine sim = 0.98331
10063
3194
41977
5070
17446
2906
counts
8056
6042
4028
2014
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17a
SBS5
SBS1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SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98331
7038
1612
41695
3412
23385
3512
counts
8188
6141
4094
2047
0
80659
60494
40330
20165
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Signature
SBS44
SBS26
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SBS17a
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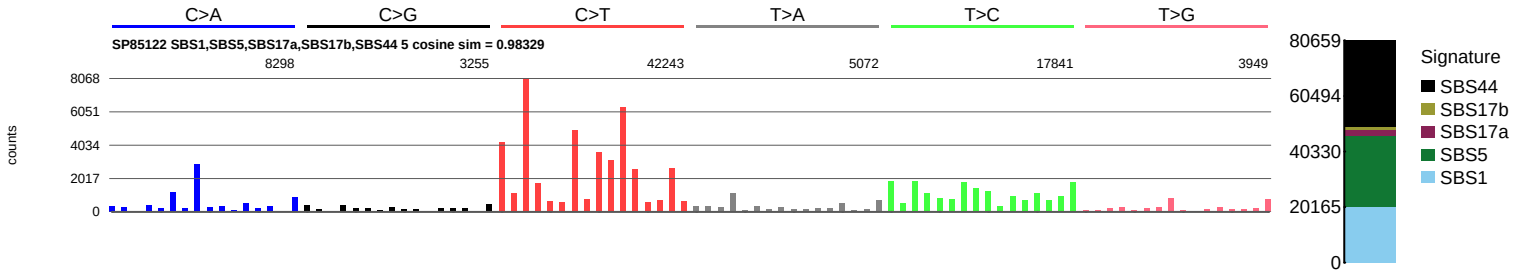
SBS2
SBS1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SP85122 SBS1,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98329
8298
3255
42243
5072
17841
3949
counts
8068
6051
4034
2017
0
80659
60494
40330
20165
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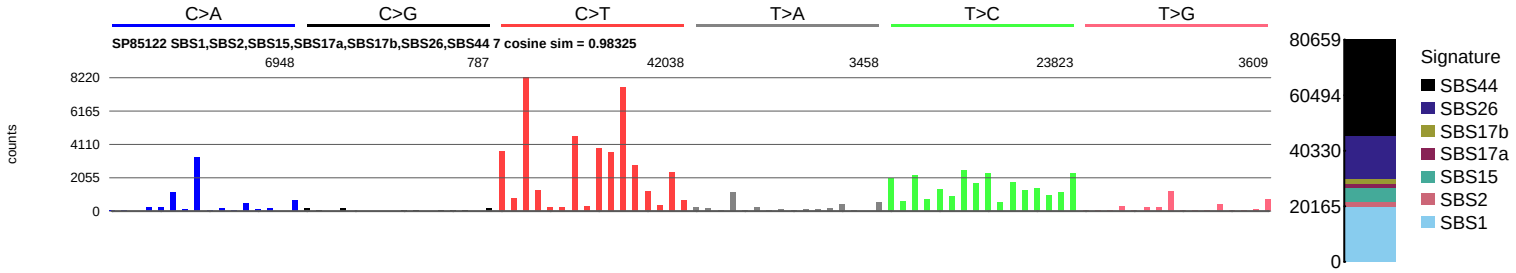
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS1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SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98325
6948
787
42038
3458
23823
3609
counts
8220
6165
4110
2055
0
80659
60494
40330
20165
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SBS26
SBS17b
SBS17a
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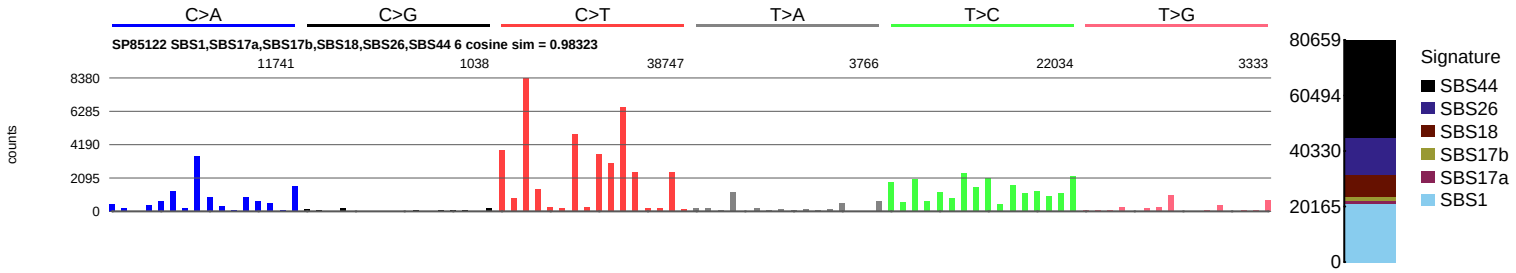
SBS2
SBS1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SP85122 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98323
11741
1038
38747
3766
22034
3333
counts
8380
6285
4190
2095
0
80659
60494
40330
20165
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SBS44
SBS26
SBS18
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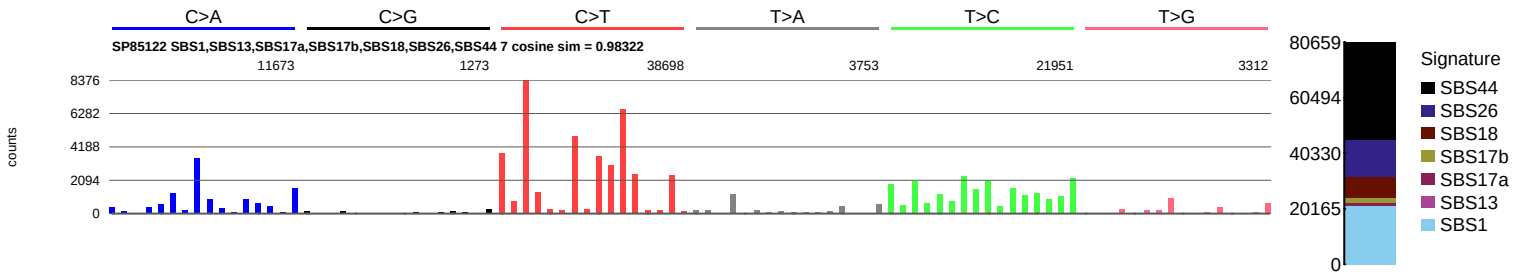
SBS17a
SBS1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SP85122 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98322
11673
1273
38698
3753
21951
3312
counts
8376
6282
4188
2094
0
80659
60494
40330
20165
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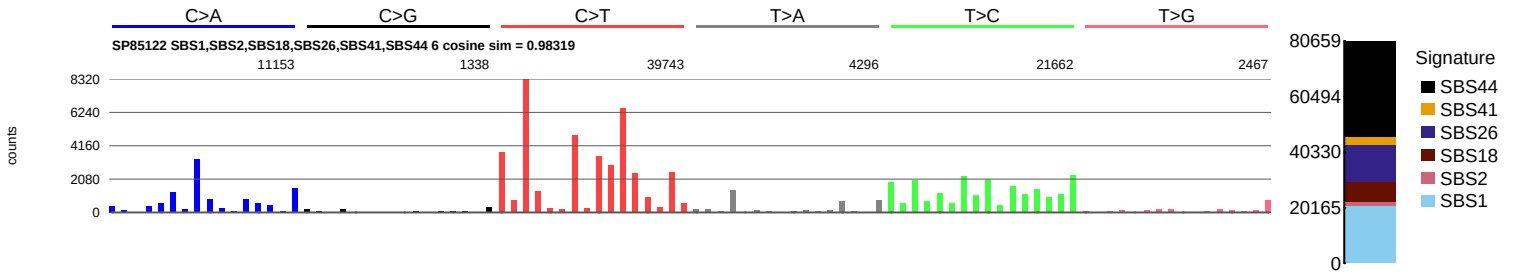
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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SP85122 SBS1,SBS2,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98319
11153
1338
39743
4296
21662
2467
8320
6240
4160
2080
0
counts
80659
60494
40330
20165
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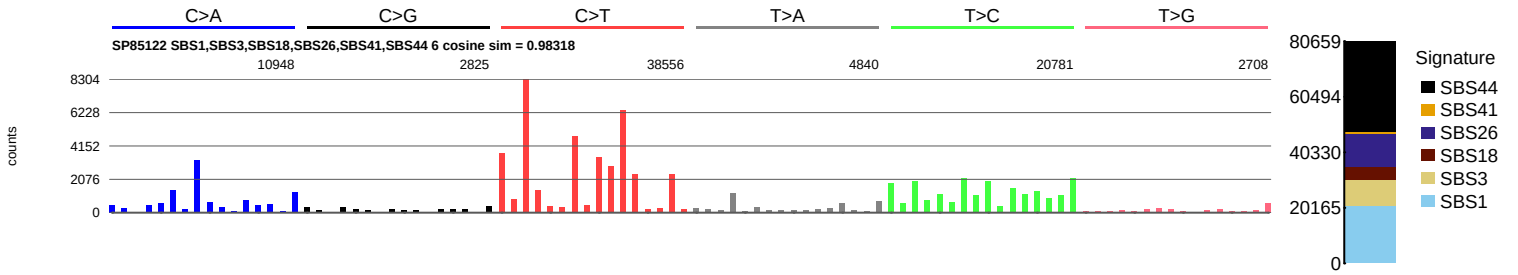
Signature
SBS44
SBS41
SBS26
SBS18
SBS2
SBS1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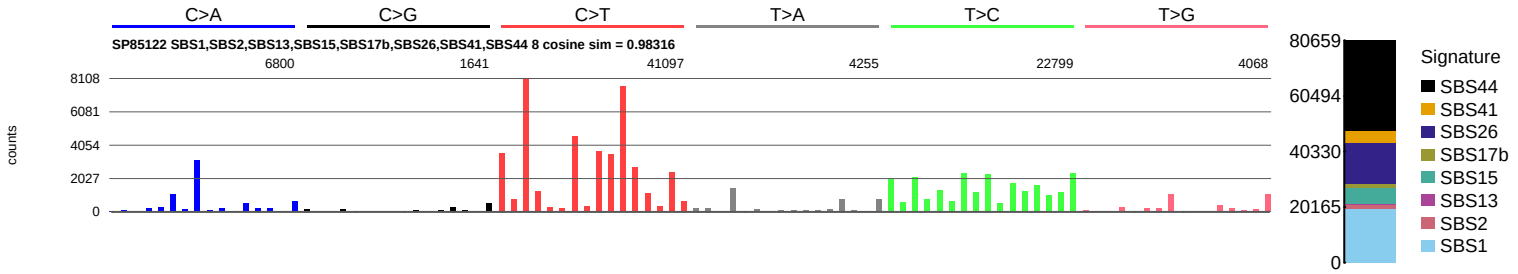
SP85122 SBS1,SBS3,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98318
10948
2825
38556
4840
20781
2708
8304
6228
4152
2076
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS3
SBS1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C>T
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T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98316
6800
1641
41097
4255
22799
4068
8108
6081
4054
2027
0
counts
80659
60494
40330
20165
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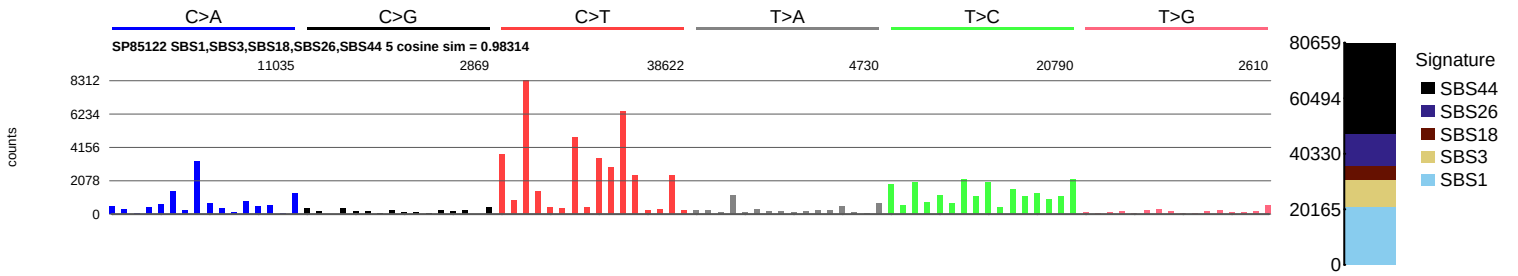
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1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SP85122 SBS1,SBS3,SBS18,SBS26,SBS44 5 cosine sim = 0.98314
11035
2869
38622
4730
20790
2610
8312
6234
4156
2078
0
counts
80659
60494
40330
20165
0
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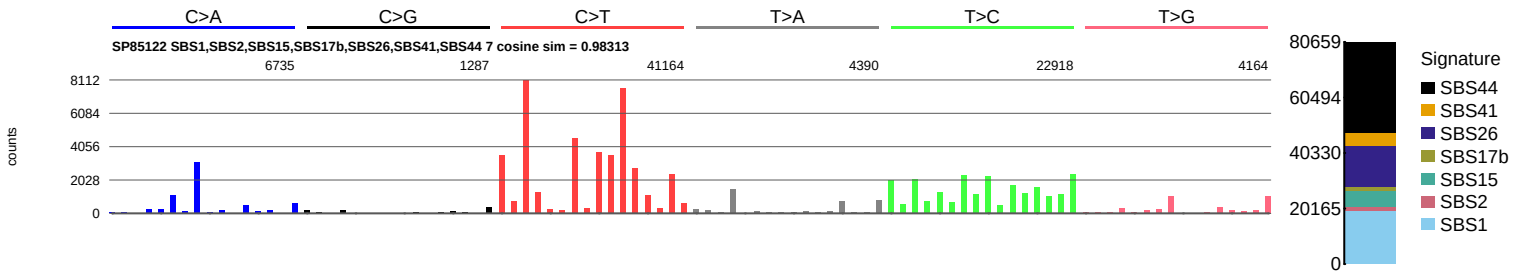
SBS44
SBS26
SBS18
SBS3
SBS1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C>G
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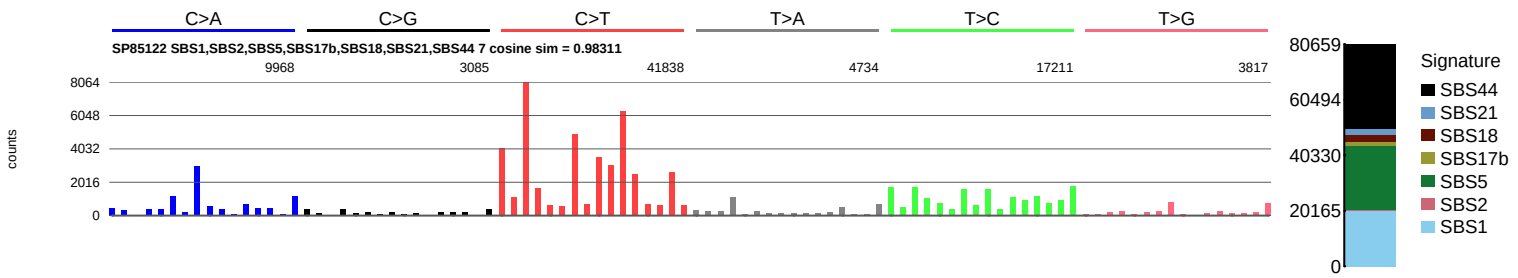
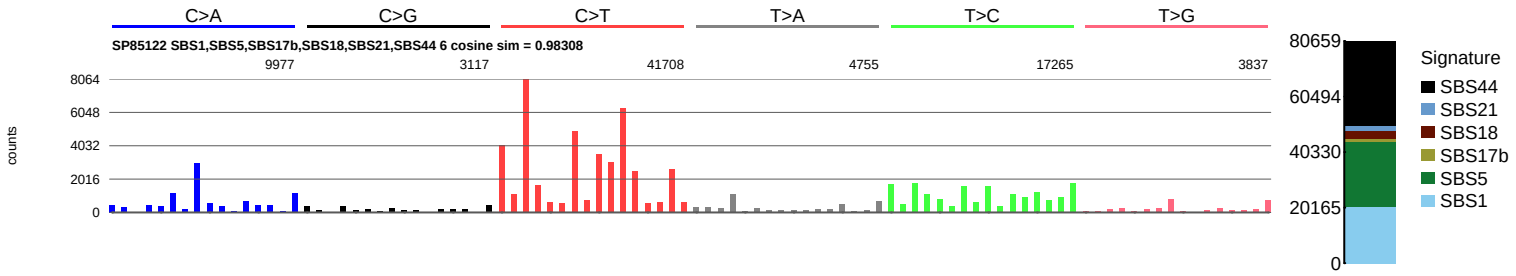
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98313
6735
1287
41164
4390
22918
4164
8112
6084
4056
2028
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98311
9968
3085
41838
4734
17211
3817
8064
6048
4032
2016
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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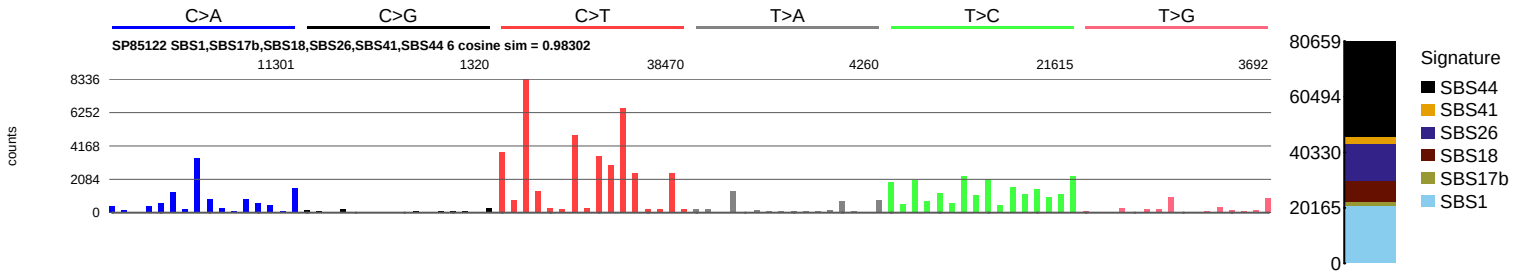
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.98308
9977
3117
41708
4755
17265
3837
counts
8064
6048
4032
2016
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS1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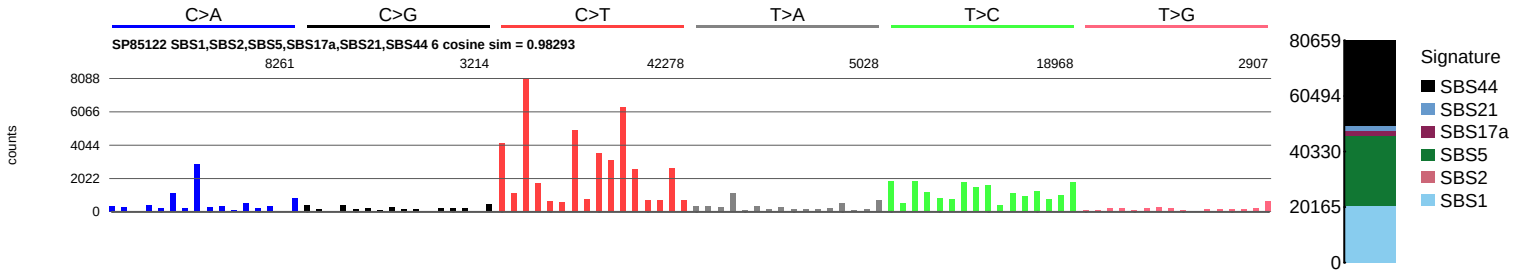
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS17b,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98302
11301
1320
38470
4260
21615
3692
counts
8336
6252
4168
2084
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS1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C>G
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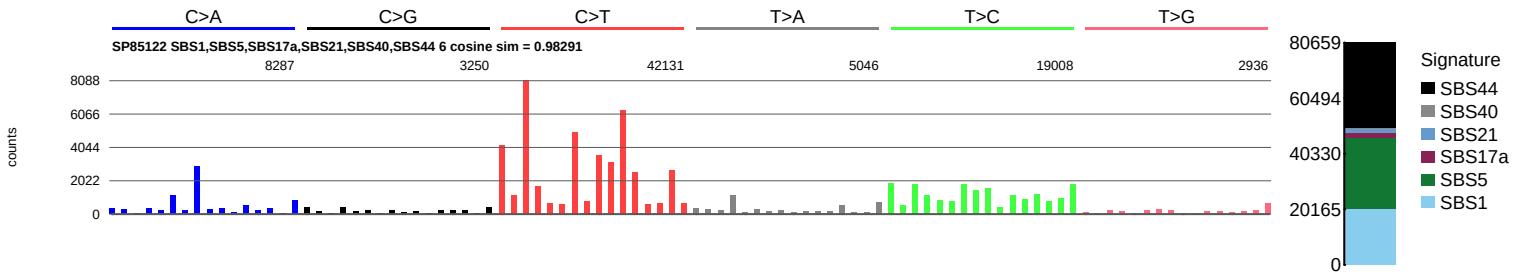
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS21,SBS44 6 cosine sim = 0.98293
8261
3214
42278
5028
18968
2907
counts
8088
6066
4044
2022
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17a
SBS5
SBS2
SBS1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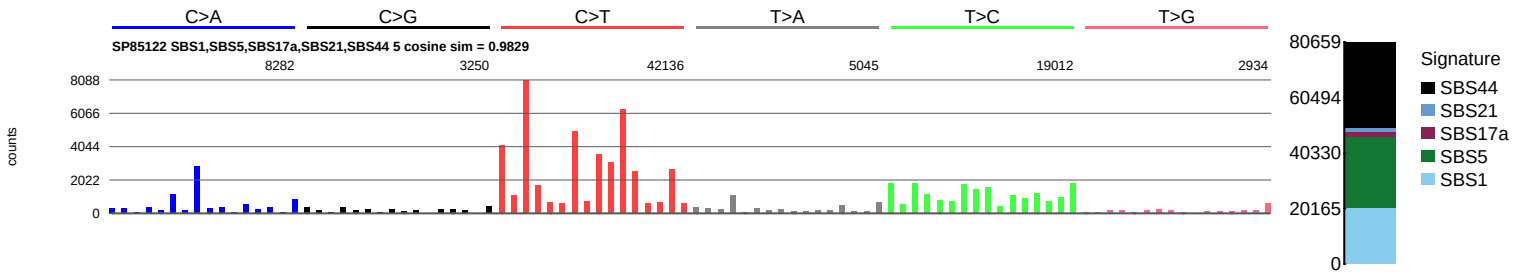
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS21,SBS40,SBS44 6 cosine sim = 0.98291
8287
3250
42131
5046
19008
2936
counts
8088
6066
4044
2022
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS5
SBS1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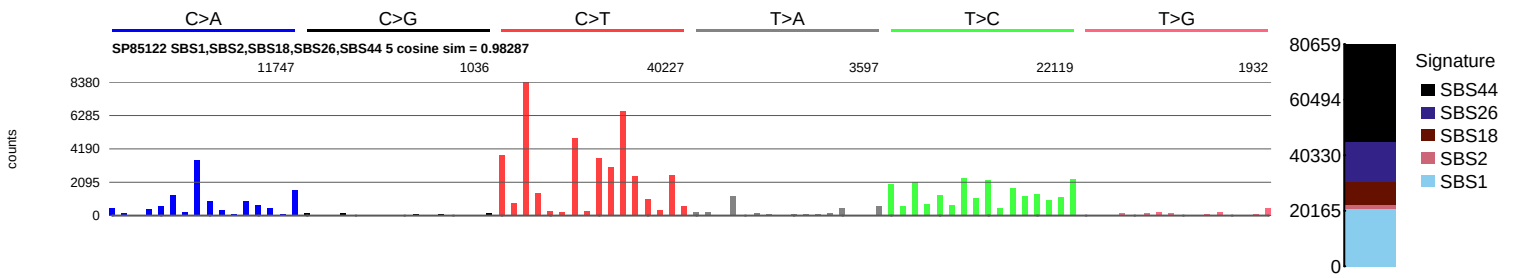
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS21,SBS44 5 cosine sim = 0.9829
8282
3250
42136
5045
19012
2934
counts
8088
6066
4044
2022
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17a
SBS5
SBS1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SP85122 SBS1,SBS2,SBS18,SBS26,SBS44 5 cosine sim = 0.98287
11747
1036
40227
3597
22119
1932
counts
8380
6285
4190
2095
0
80659
60494
40330
20165
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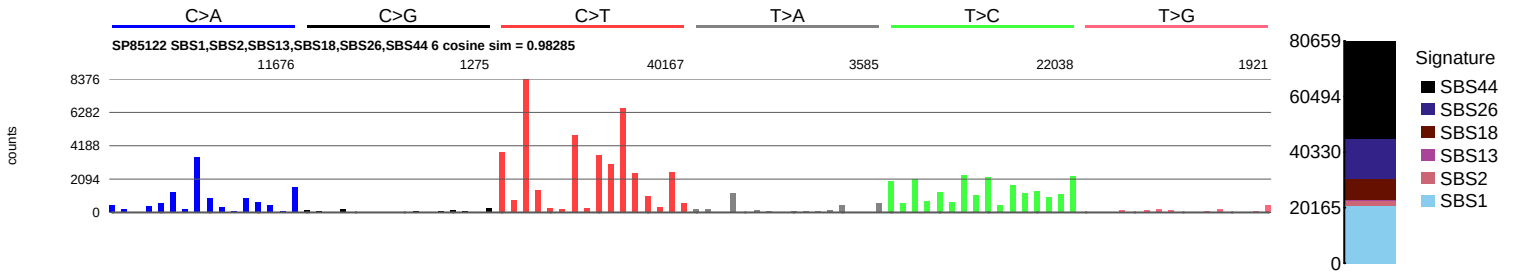
SBS44
SBS26
SBS18
SBS2
SBS1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T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS18,SBS26,SBS44 6 cosine sim = 0.98285
11676
1275
40167
3585
22038
1921
counts
8376
6282
4188
2094
0
80659
60494
40330
20165
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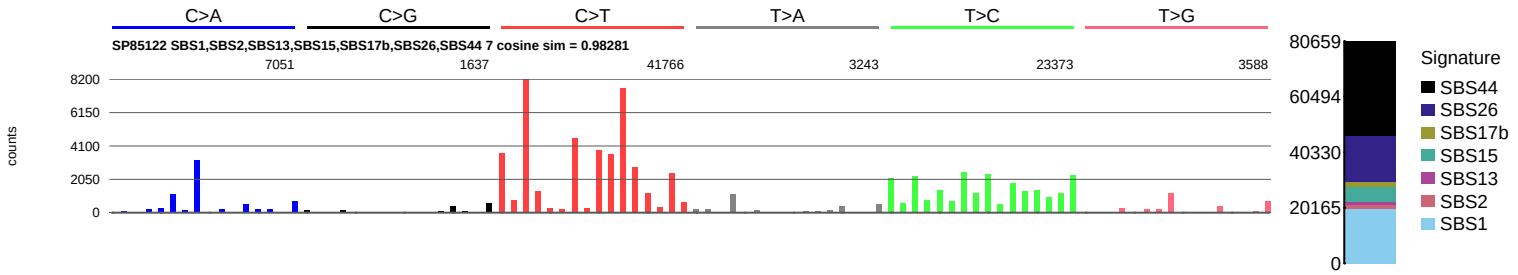
Signature
SBS44
SBS26
SBS18
SBS13
SBS2
SBS1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SP85122 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98281
7051
1637
41766
3243
23373
3588
counts
8200
6150
4100
2050
0
80659
60494
40330
20165
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SBS26
SBS17b
SBS15
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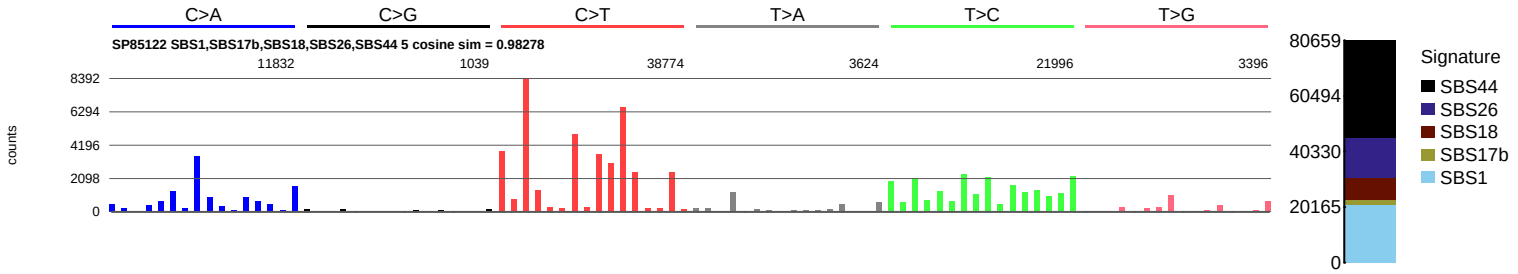
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP85122 SBS1,SBS17b,SBS18,SBS26,SBS44 5 cosine sim = 0.98278
11832
1039
38774
3624
21996
3396
counts
8392
6294
4196
2098
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80659
60494
40330
20165
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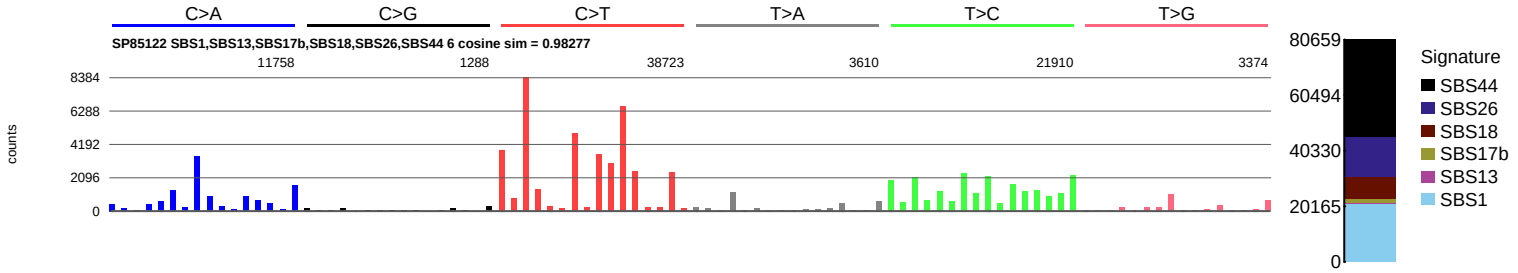
Signature
SBS44
SBS26
SBS18
SBS17b
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98277
11758
1288
38723
3610
21910
3374
counts
8384
6288
4192
2096
0
80659
60494
40330
20165
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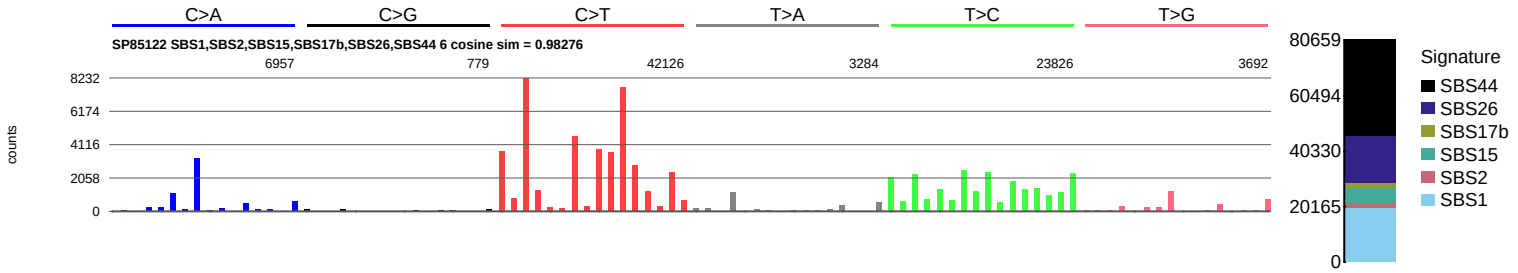
SBS18
SBS17b
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98276
6957
779
42126
3284
23826
3692
counts
8232
6174
4116
2058
0
80659
60494
40330
20165
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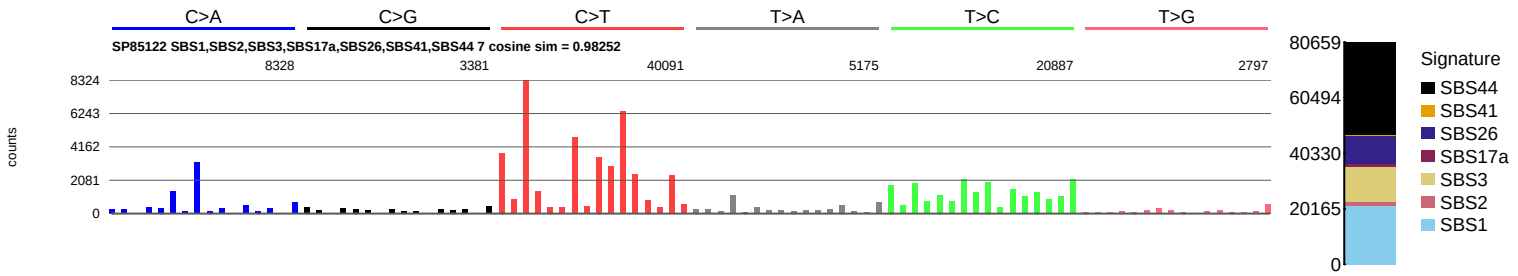
Signature
SBS44
SBS26
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98252
8328
3381
40091
5175
20887
2797
counts
8324
6243
4162
2081
0
80659
60494
40330
20165
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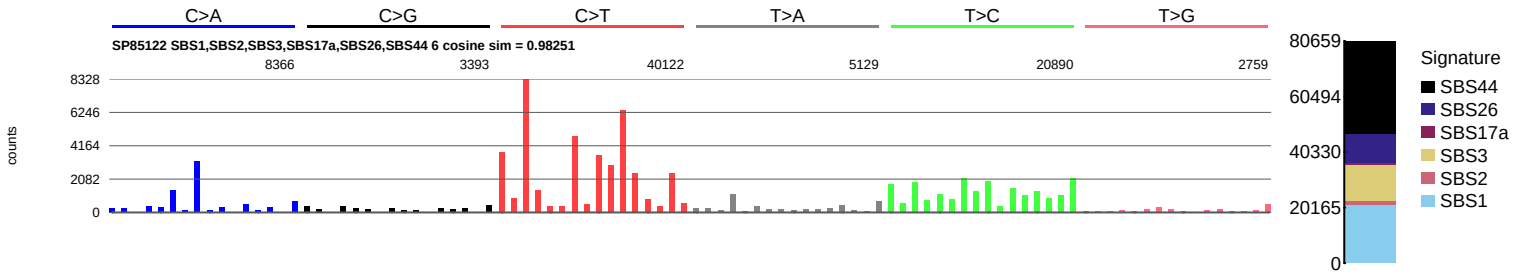
SBS41
SBS26
SBS17a
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS17a,SBS26,SBS44 6 cosine sim = 0.98251
8366
3393
40122
5129
20890
2759
counts
8328
6246
4164
2082
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40330
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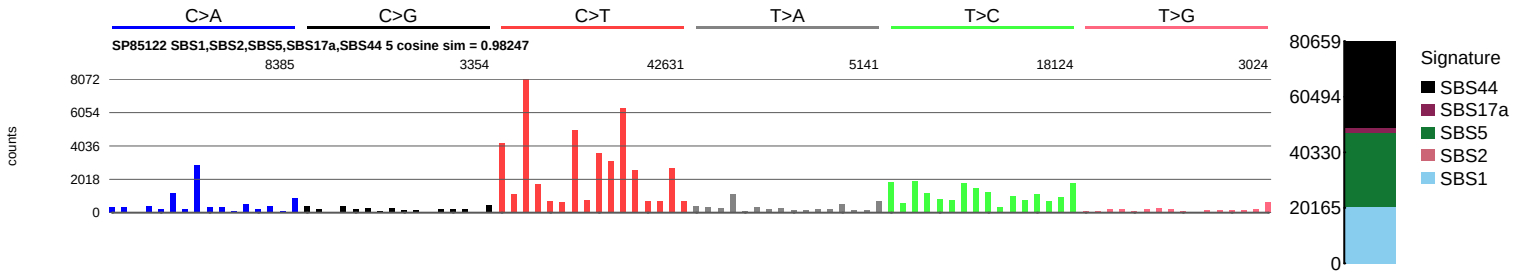
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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17a,SBS44 5 cosine sim = 0.98247
8385
3354
42631
5141
18124
3024
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
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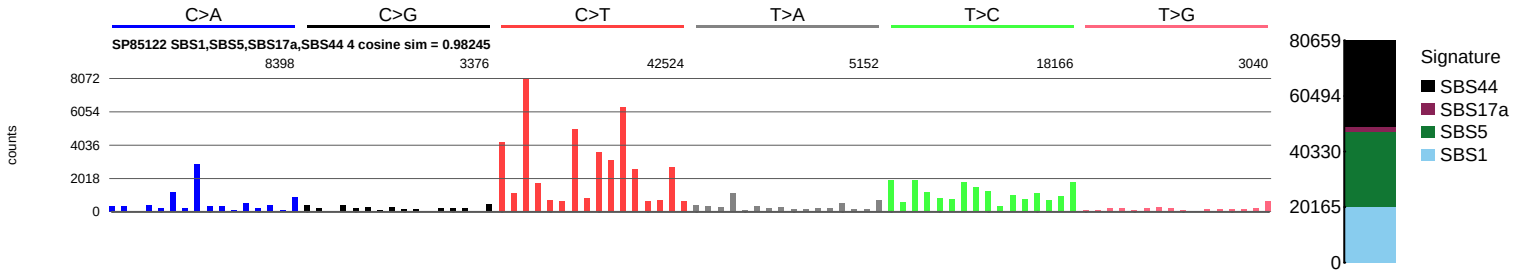
SBS17a
SBS5
SBS2
SBS1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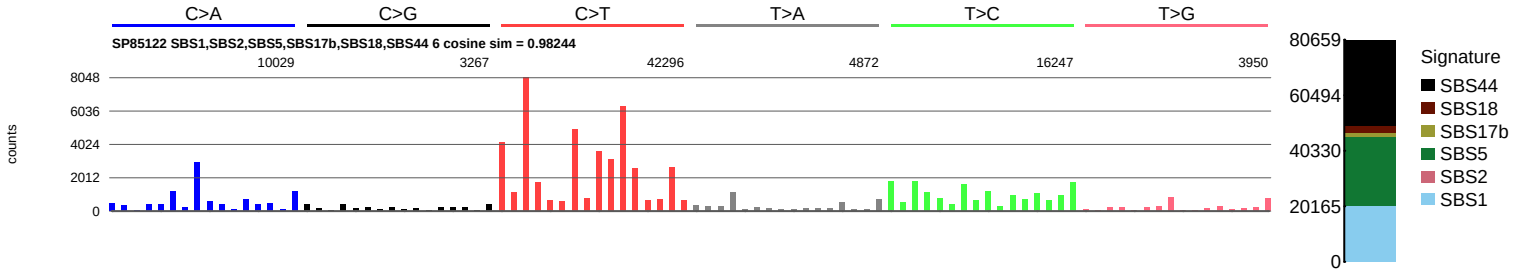
T>C
T>G
SP85122 SBS1,SBS5,SBS17a,SBS44 4 cosine sim = 0.98245
8398
3376
42524
5152
18166
3040
counts
8072
6054
4036
2018
0
80659
60494
40330
20165
0
Signature
SBS44
SBS17a
SBS5
SBS1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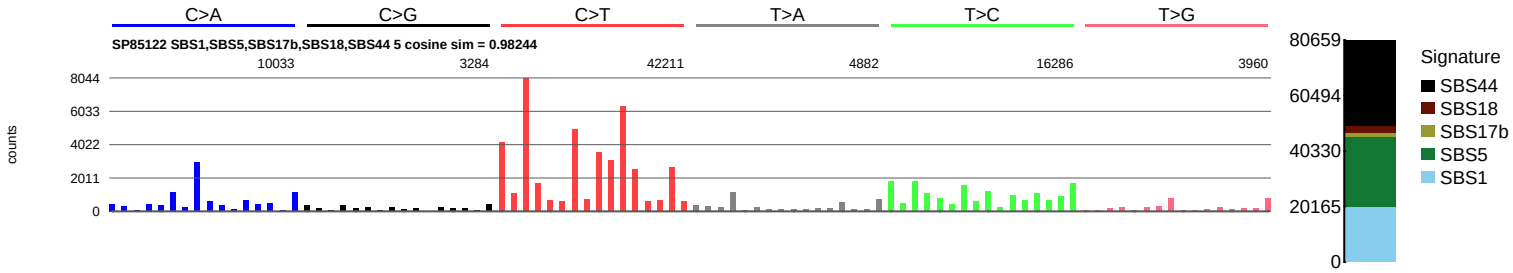
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS44 6 cosine sim = 0.98244
10029
3267
42296
4872
16247
3950
counts
8048
6036
4024
2012
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS2
SBS1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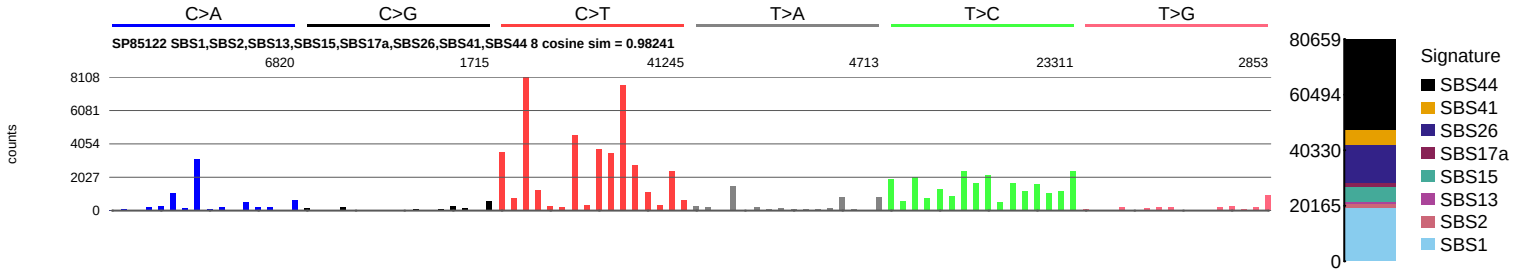
SP85122 SBS1,SBS5,SBS17b,SBS18,SBS44 5 cosine sim = 0.98244
10033
3284
42211
4882
16286
3960
counts
8044
6033
4022
2011
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS1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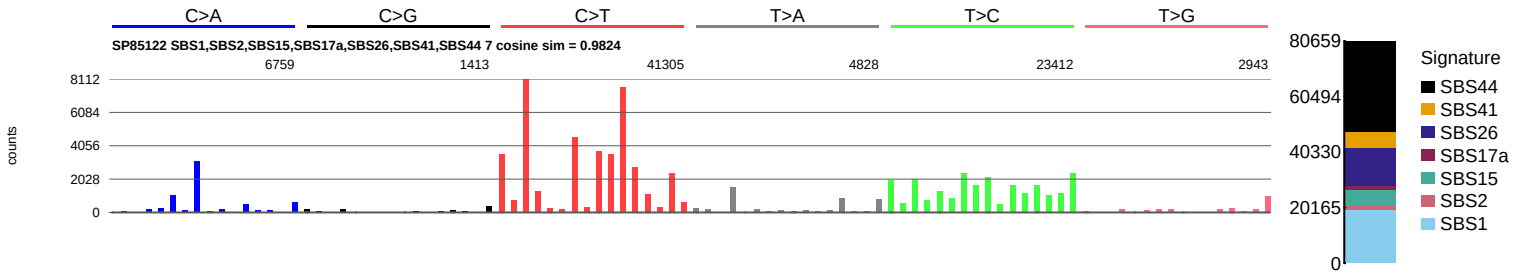
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98241
6820
1715
41245
4713
23311
2853
counts
8108
6081
4054
2027
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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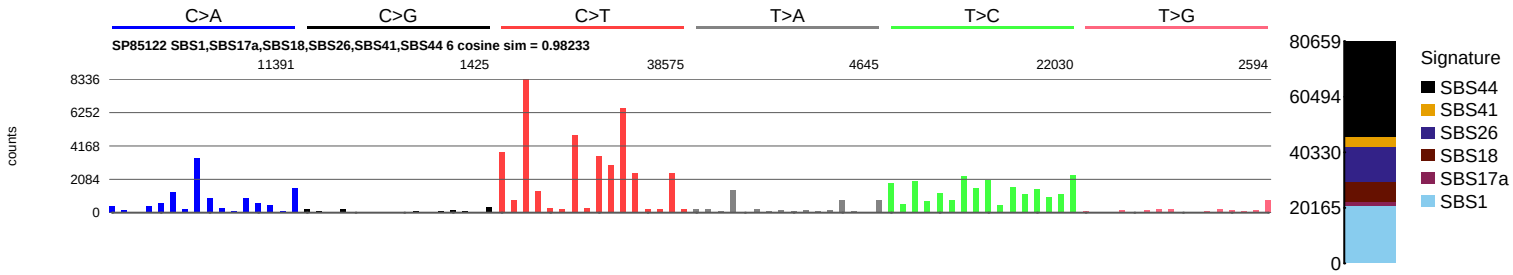
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.9824
6759
1413
41305
4828
23412
2943
counts
8112
6084
4056
2028
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS2
SBS1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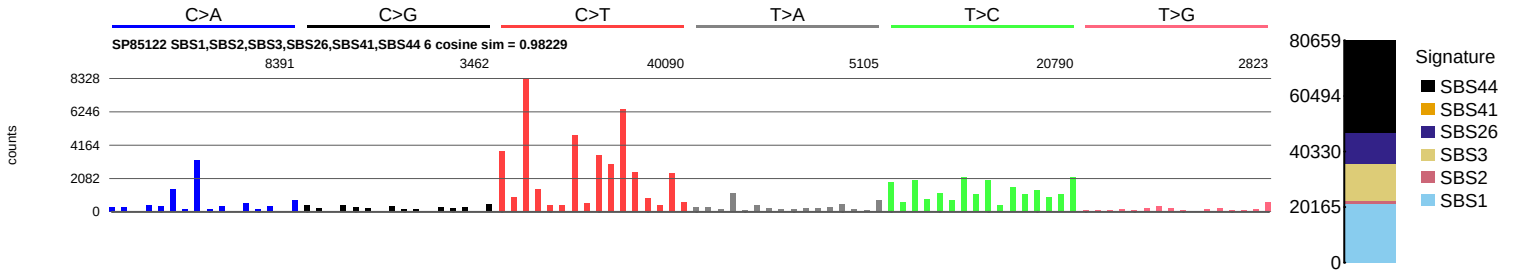
SP85122 SBS1,SBS17a,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98233
11391
1425
38575
4645
22030
2594
counts
8336
6252
4168
2084
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS1


C>A
C>G
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T>A
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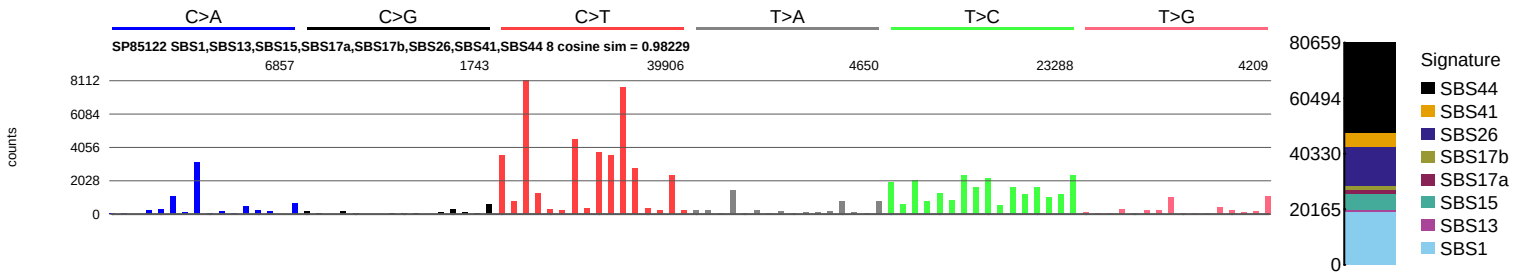
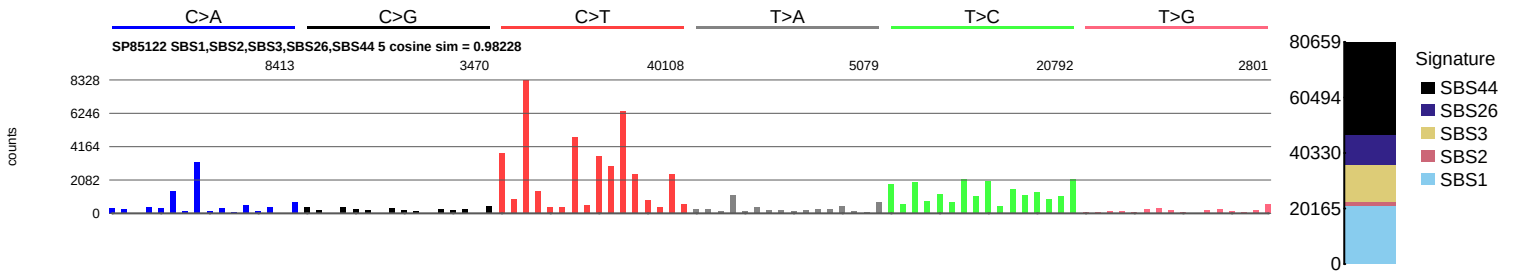
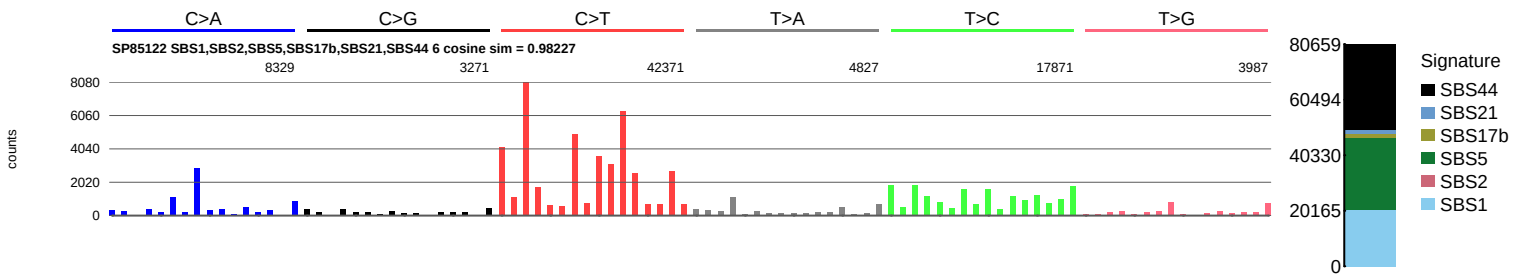
SP85122 SBS1,SBS2,SBS3,SBS26,SBS41,SBS44 6 cosine sim = 0.98229
8391
3462
40090
5105
20790
2823
counts
8328
6246
4164
2082
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98229
6857
1743
39906
4650
23288
4209
counts
8112
6084
4056
2028
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS26,SBS44 5 cosine sim = 0.98228
8413
3470
40108
5079
20792
2801
counts
8328
6246
4164
2082
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS3
SBS2
SBS1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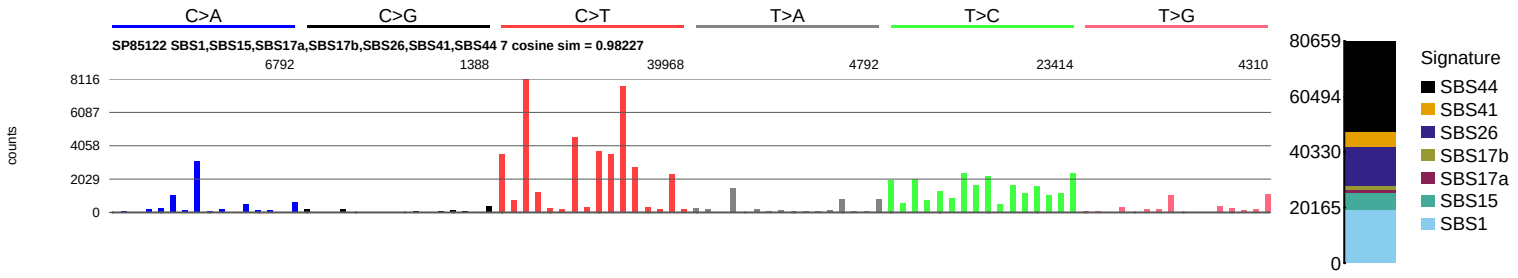
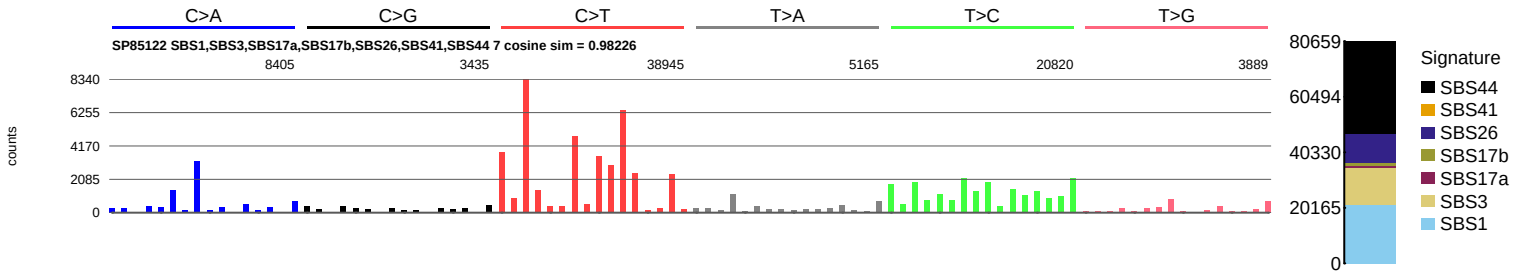
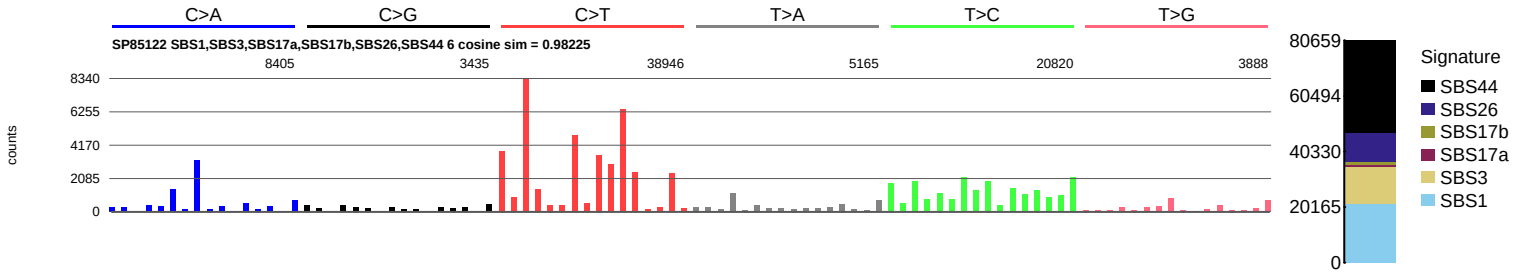
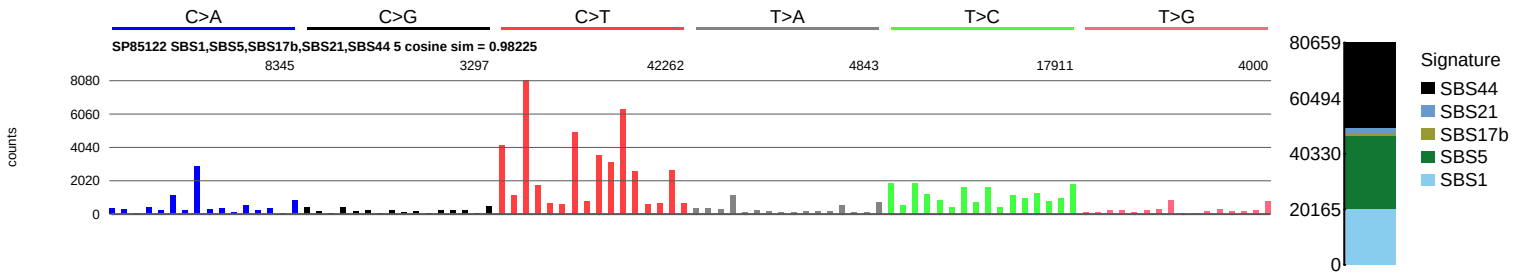
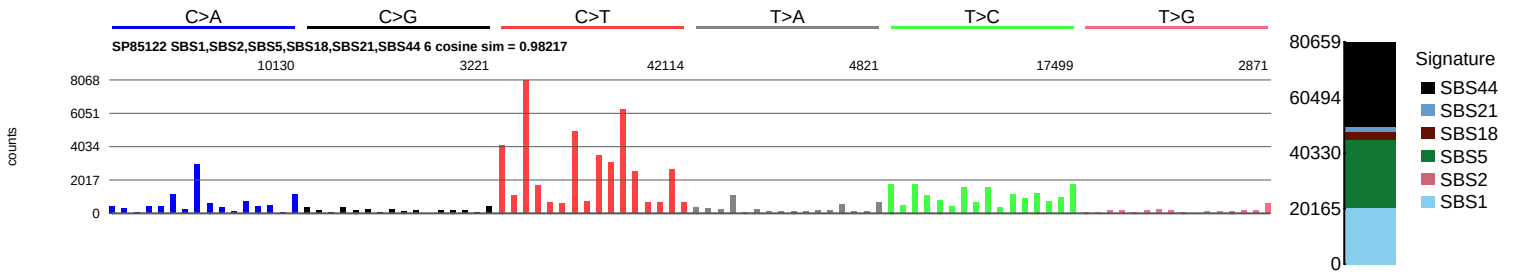
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS44 6 cosine sim = 0.98227
8329
3271
42371
4827
17871
3987
counts
8080
6060
4040
2020
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17b
SBS5
SBS2
SBS1

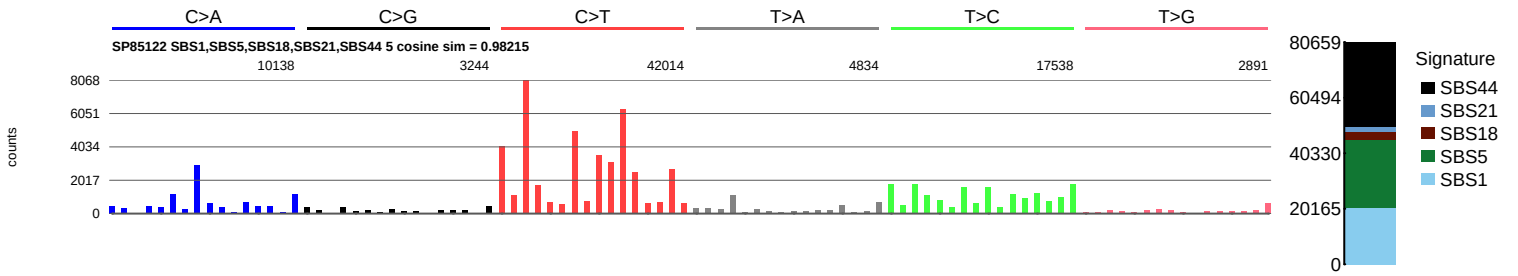

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98227
6792
1388
39968
4792
23414
4310
8116
6087
4058
2029
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98226
8405
3435
38945
5165
20820
3889
8340
6255
4170
2085
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98225
8405
3435
38946
5165
20820
3888
8340
6255
4170
2085
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS17b,SBS21,SBS44 5 cosine sim = 0.98225
8345
3297
42262
4843
17911
4000
8080
6060
4040
2020
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS18,SBS21,SBS44 6 cosine sim = 0.98217
10130
3221
42114
4821
17499
2871
8068
6051
4034
2017
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS5
SBS2
SBS1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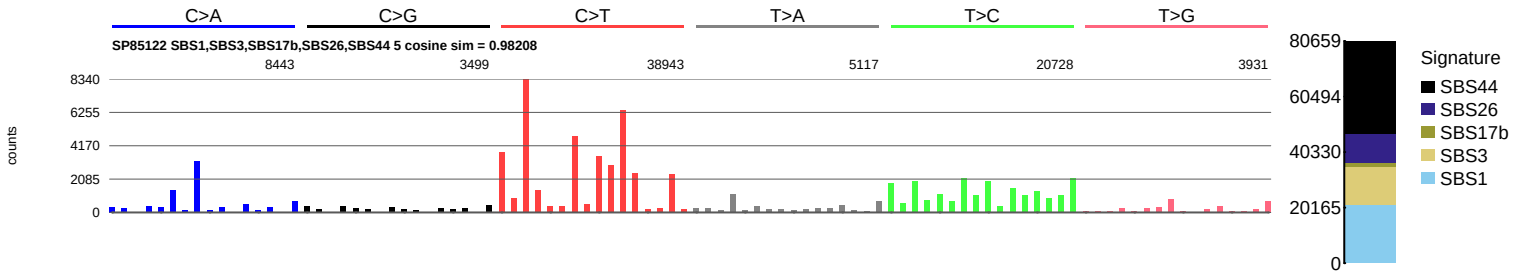
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS18,SBS21,SBS44 5 cosine sim = 0.98215
10138
3244
42014
4834
17538
2891
8068
6051
4034
2017
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS18
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS17b,SBS26,SBS44 5 cosine sim = 0.98208
8443
3499
38943
5117
20728
3931
counts
8340
6255
4170
2085
0
80659
60494
40330
20165
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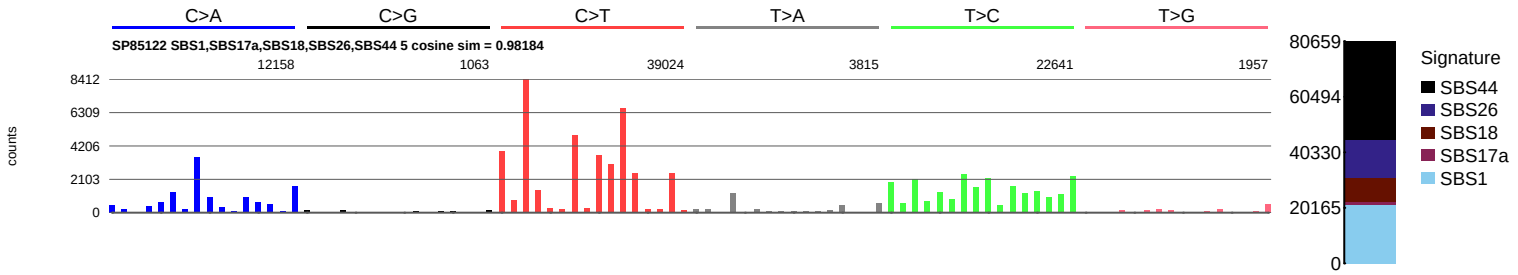
Signature
SBS44
SBS26
SBS17b
SBS3
SBS1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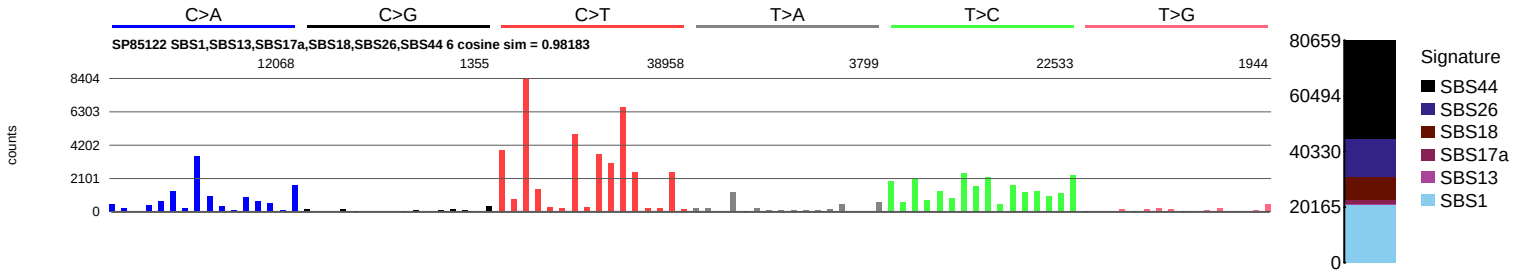
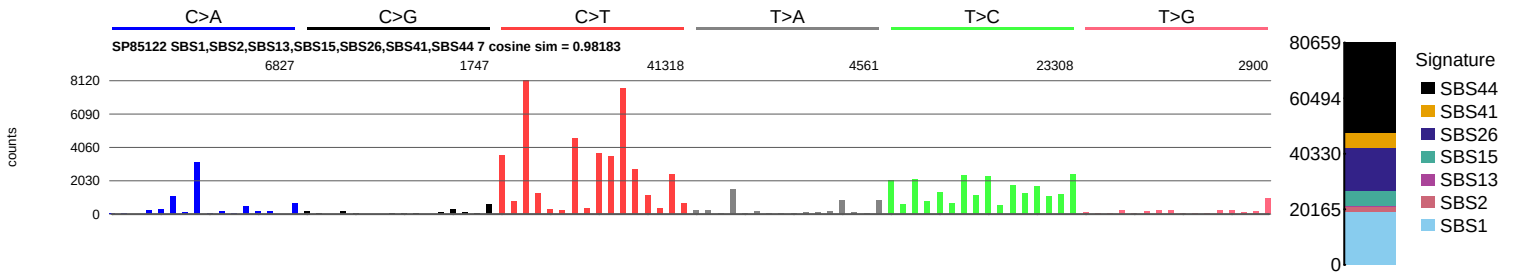
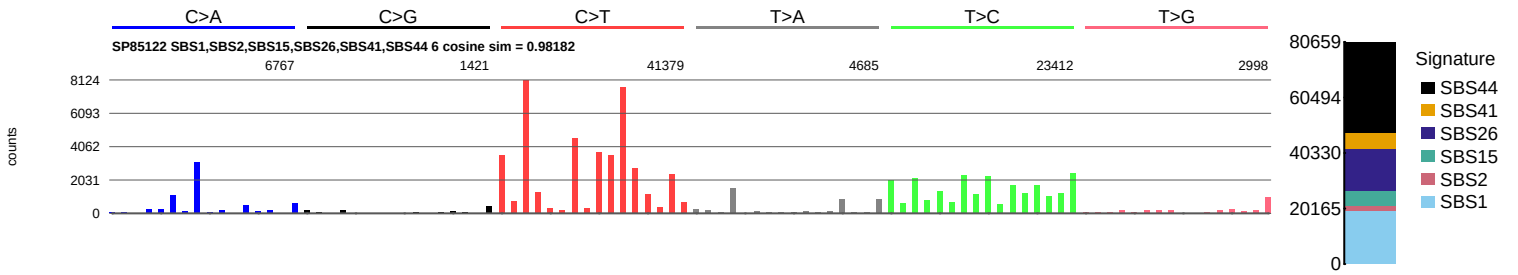
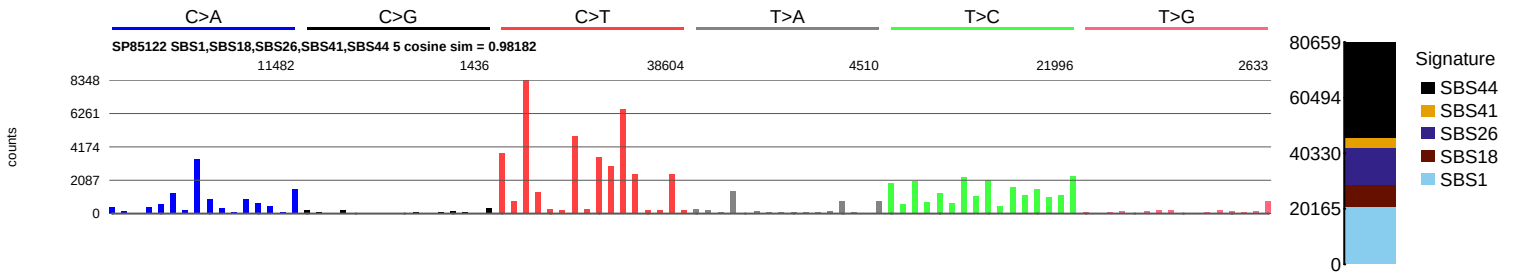
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS17a,SBS18,SBS26,SBS44 5 cosine sim = 0.98184
12158
1063
39024
3815
22641
1957
counts
8412
6309
4206
2103
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98183
12068
1355
38958
3799
22533
1944
counts
8404
6303
4202
2101
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.98183
6827
1747
41318
4561
23308
2900
counts
8120
6090
4060
2030
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98182
6767
1421
41379
4685
23412
2998
counts
8124
6093
4062
2031
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS18,SBS26,SBS41,SBS44 5 cosine sim = 0.98182
11482
1436
38604
4510
21996
2633
counts
8348
6261
4174
2087
0
80659
60494
40330
20165
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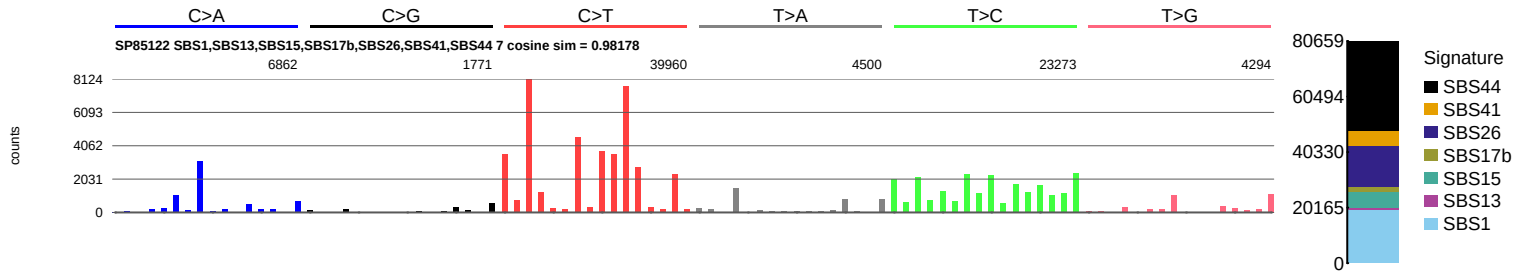
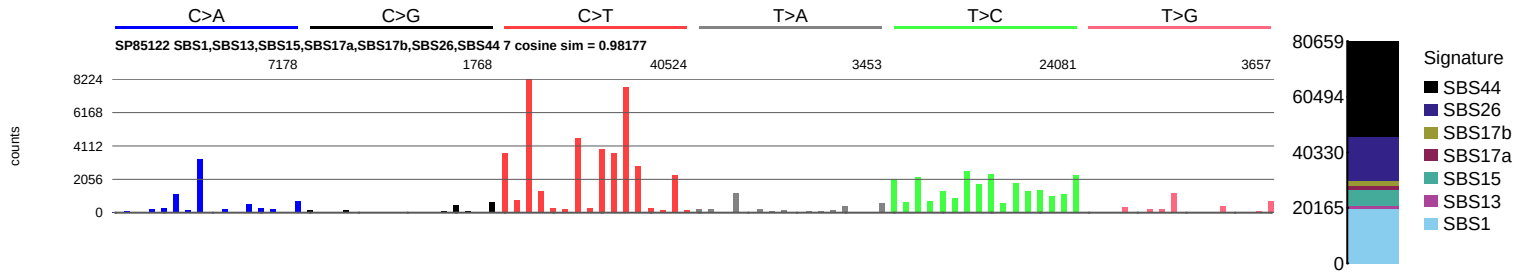
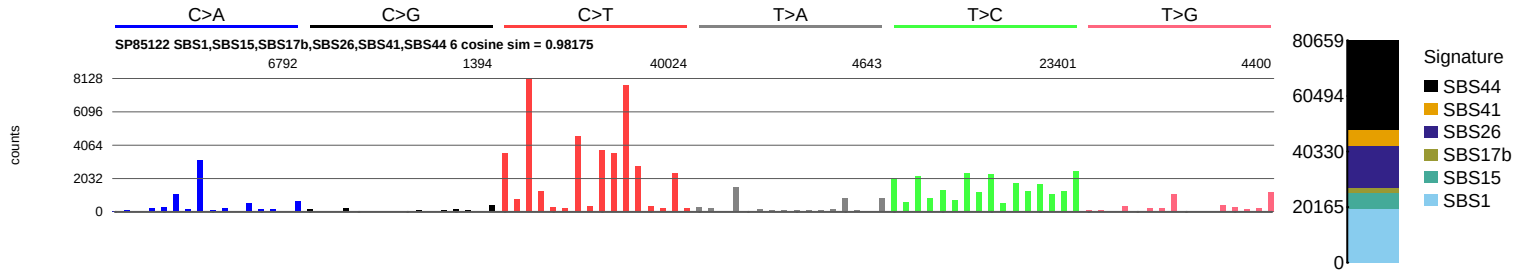
SBS44
SBS41
SBS26
SBS18
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98178
6862
1771
39960
4500
23273
4294
8124
6093
4062
2031
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98177
7178
1768
40524
3453
24081
3657
8224
6168
4112
2056
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98175
6792
1394
40024
4643
23401
4400
8128
6096
4064
2032
0
counts
80659
60494
40330
20165
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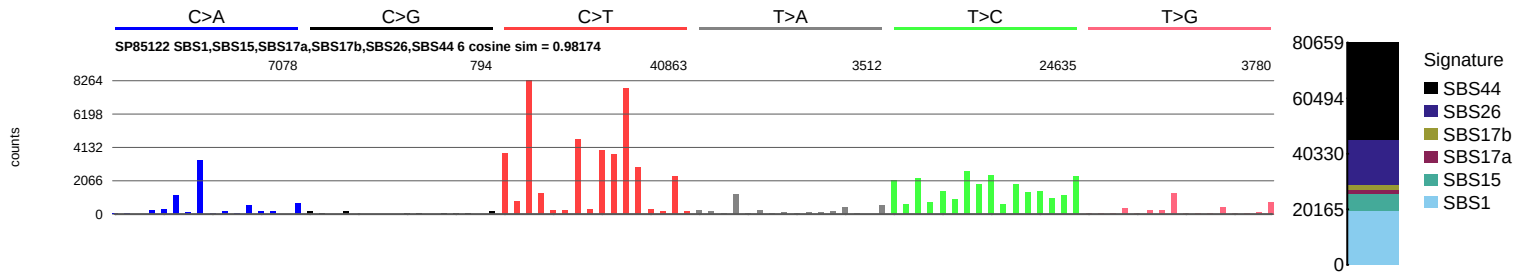
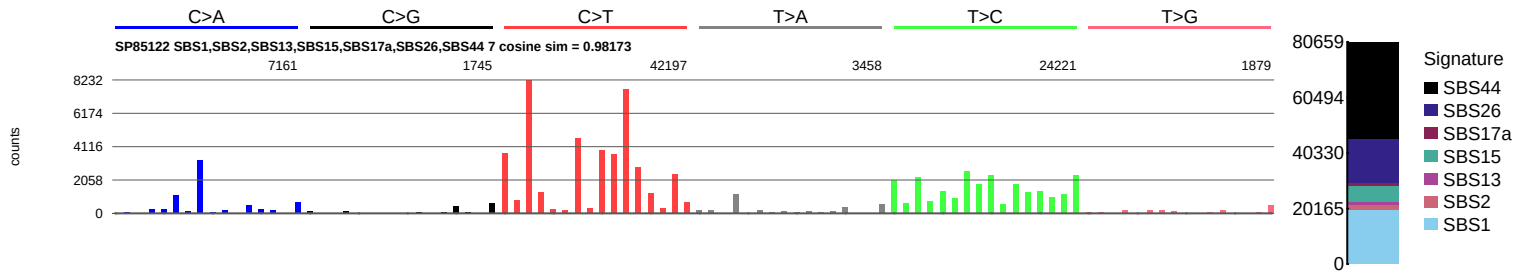
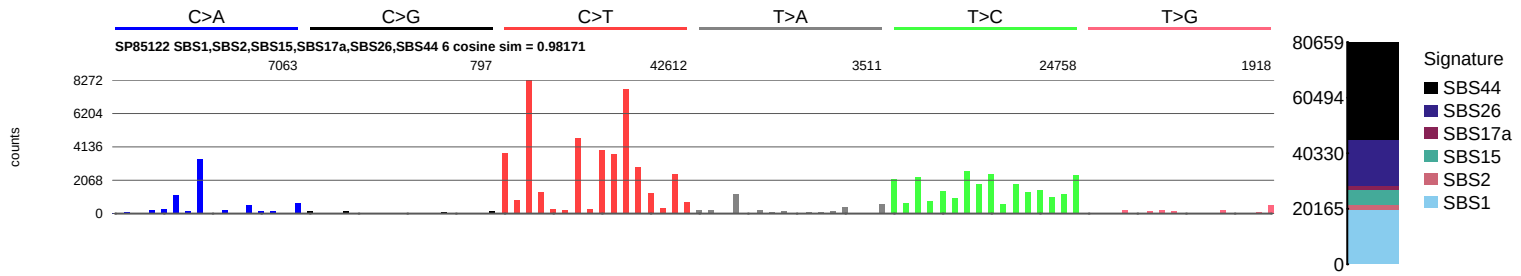
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98174
7078
794
40863
3512
24635
3780
8264
6198
4132
2066
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98173
7161
1745
42197
3458
24221
1879
8232
6174
4116
2058
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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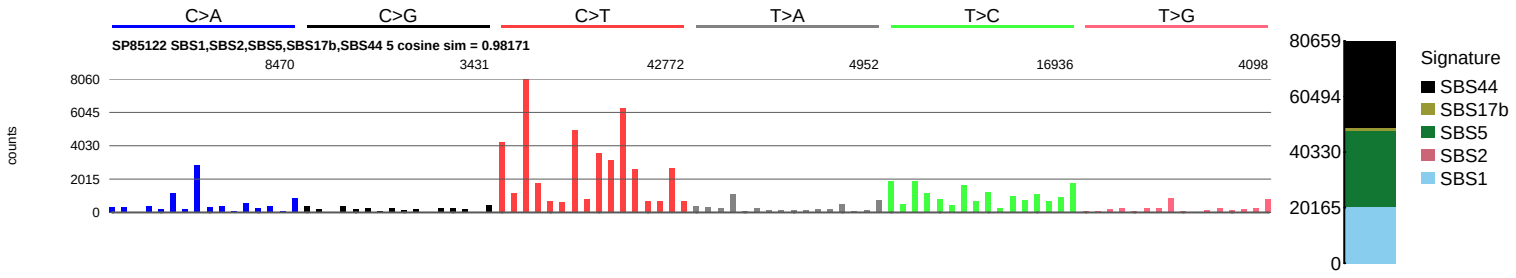
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98171
7063
797
42612
3511
24758
1918
8272
6204
4136
2068
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS2
SBS1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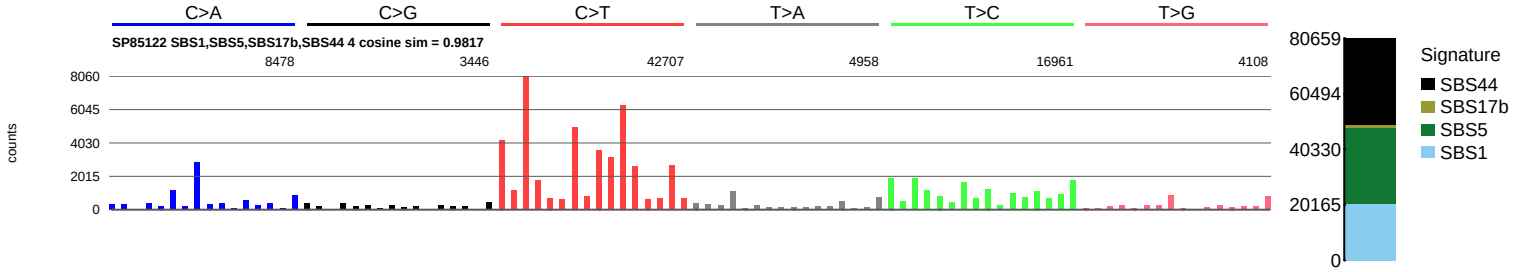
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS17b,SBS44 5 cosine sim = 0.98171
8470
3431
42772
4952
16936
4098
counts
8060
6045
4030
2015
0
80659
60494
40330
20165
0
Signature
SBS44
SBS17b
SBS5
SBS2
SBS1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T>C
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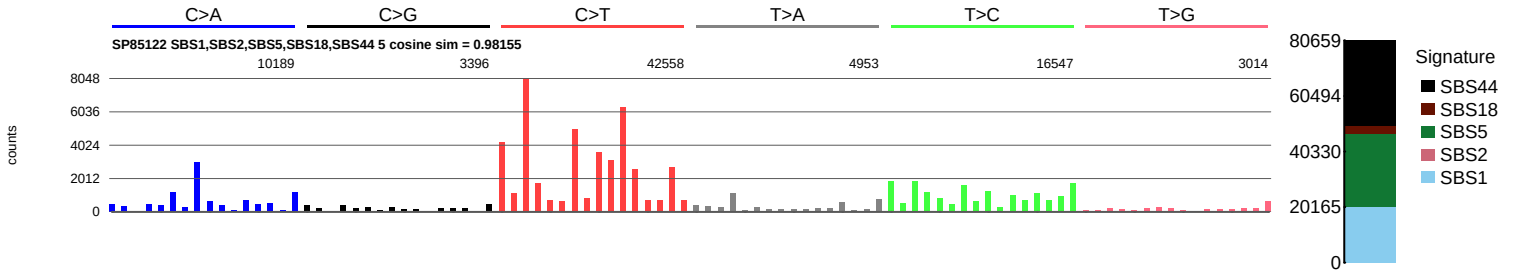
SP85122 SBS1,SBS5,SBS17b,SBS44 4 cosine sim = 0.9817
8478
3446
42707
4958
16961
4108
counts
8060
6045
4030
2015
0
80659
60494
40330
20165
0
Signature
SBS44
SBS17b
SBS5
SBS1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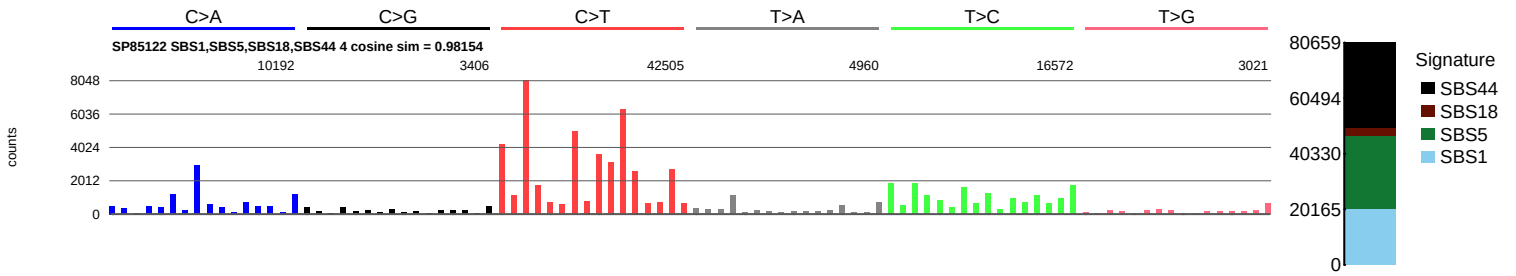
SP85122 SBS1,SBS2,SBS5,SBS18,SBS44 5 cosine sim = 0.98155
10189
3396
42558
4953
16547
3014
counts
8048
6036
4024
2012
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS5
SBS2
SBS1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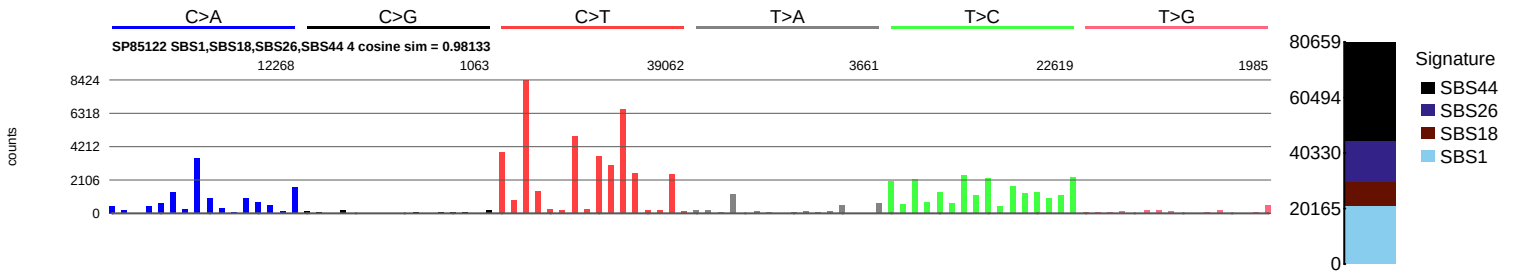
SP85122 SBS1,SBS5,SBS18,SBS44 4 cosine sim = 0.98154
10192
3406
42505
4960
16572
3021
counts
8048
6036
4024
2012
0
80659
60494
40330
20165
0
Signature
SBS44
SBS18
SBS5
SBS1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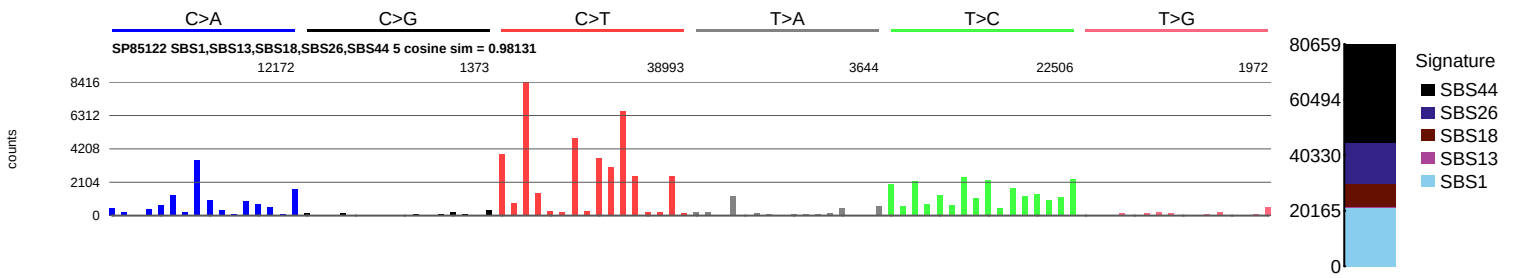
SP85122 SBS1,SBS18,SBS26,SBS44 4 cosine sim = 0.98133
12268
1063
39062
3661
22619
1985
counts
8424
6318
4212
2106
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS1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C>T
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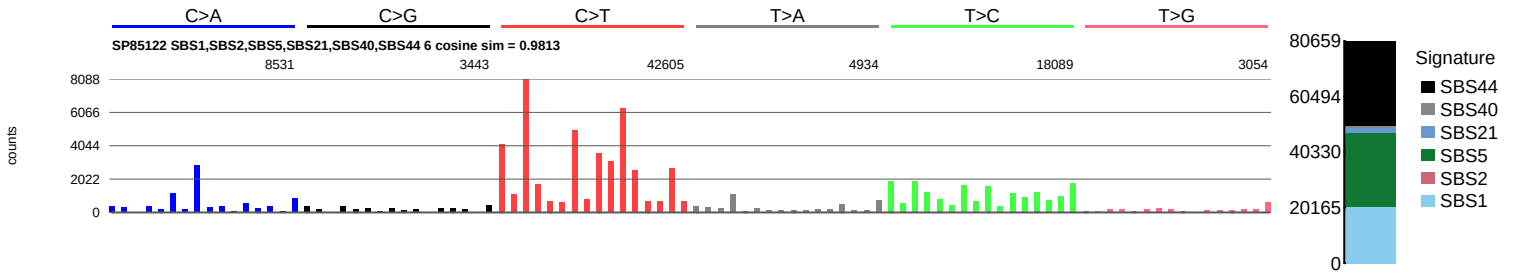
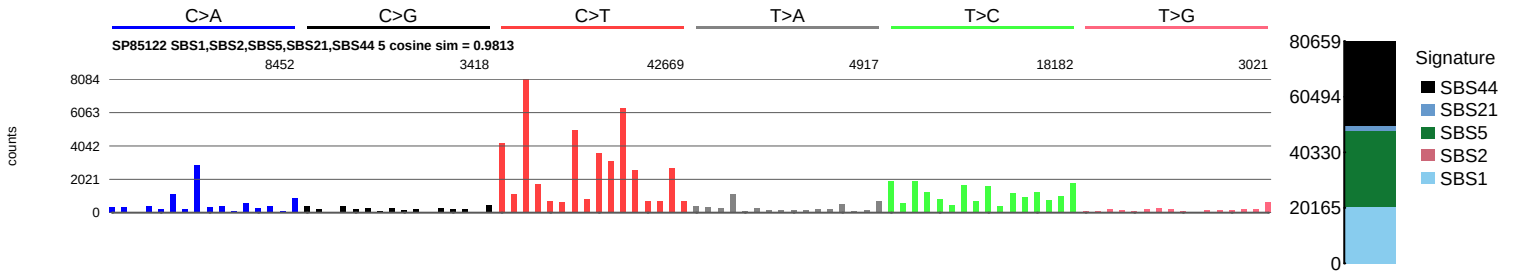
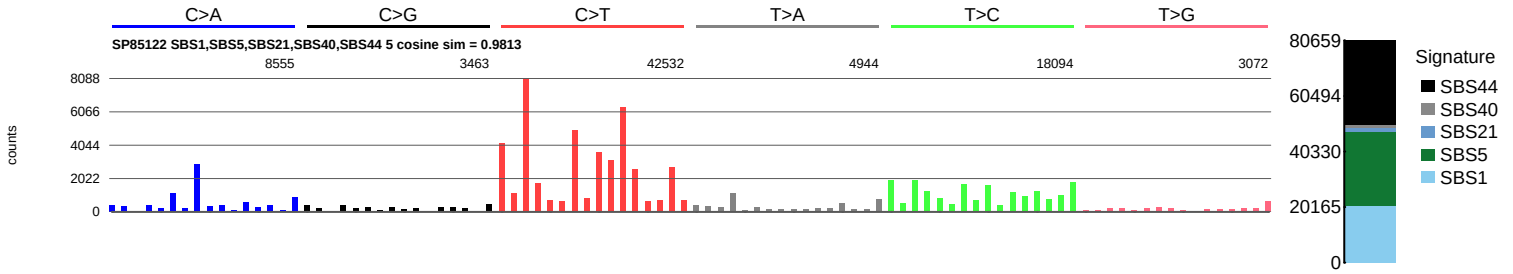
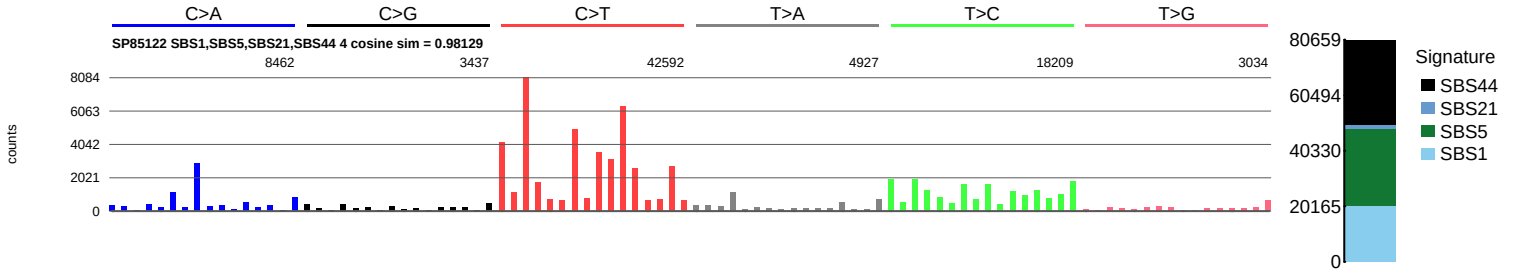
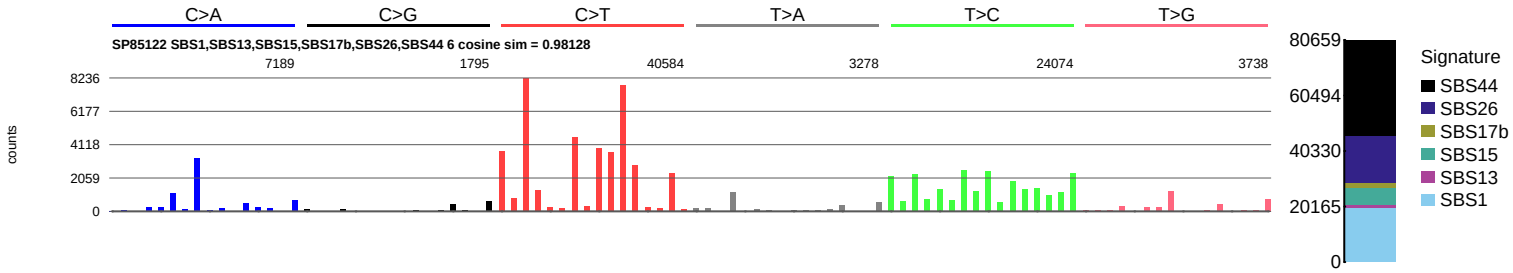
T>C
T>G
SP85122 SBS1,SBS13,SBS18,SBS26,SBS44 5 cosine sim = 0.98131
12172
1373
38993
3644
22506
1972
counts
8416
6312
4208
2104
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS18
SBS13
SBS1

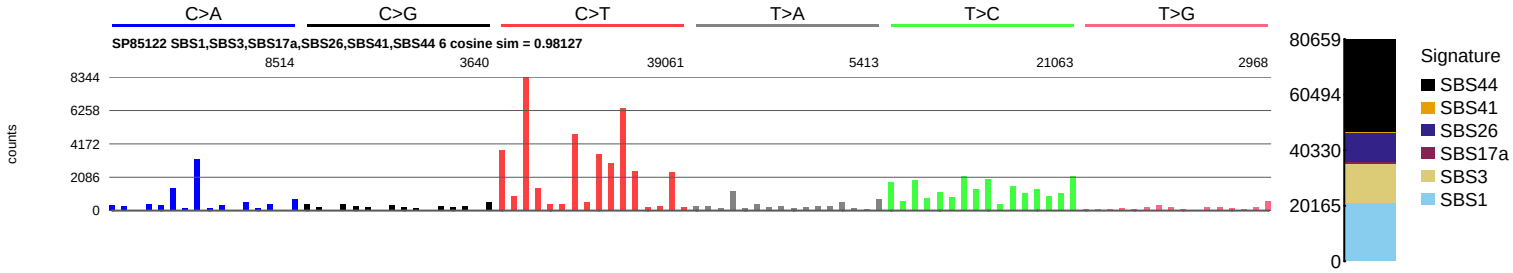

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS21,SBS40,SBS44 6 cosine sim = 0.9813
8531
3443
42605
4934
18089
3054
counts
8088
6066
4044
2022
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS5
SBS2
SBS1

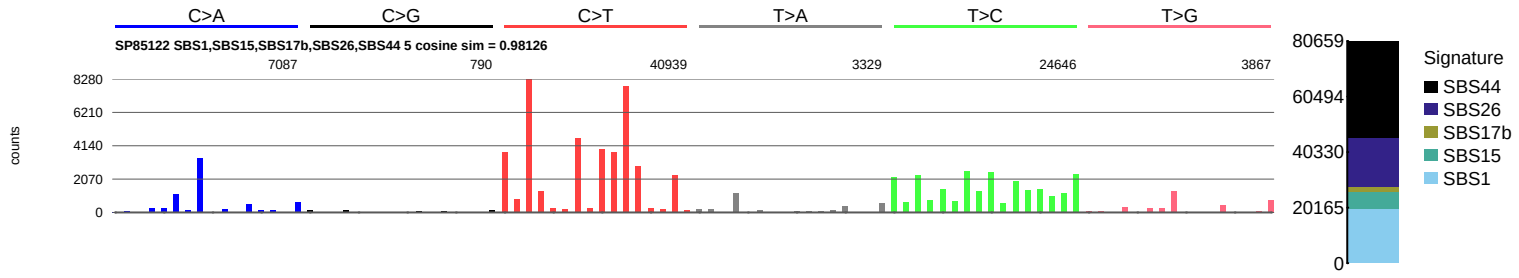

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS21,SBS44 5 cosine sim = 0.9813
8452
3418
42669
4917
18182
3021
counts
8084
6063
4042
2021
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS5
SBS2
SBS1

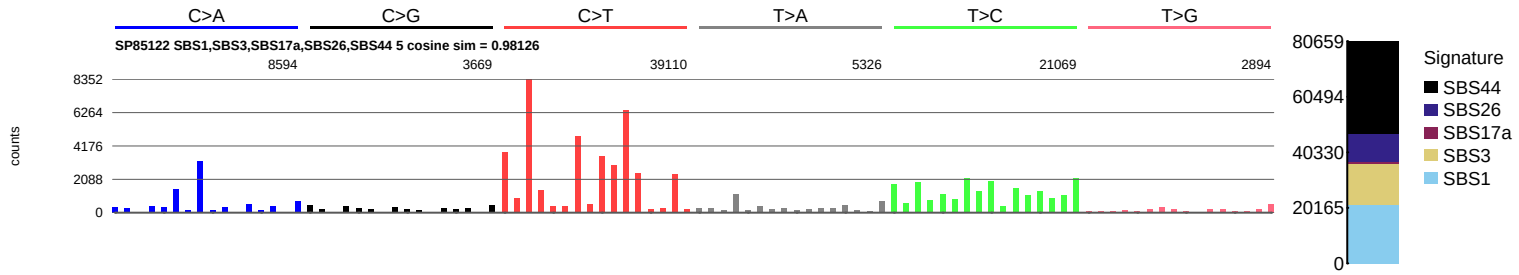

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS21,SBS40,SBS44 5 cosine sim = 0.9813
8555
3463
42532
4944
18094
3072
counts
8088
6066
4044
2022
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS5
SBS1

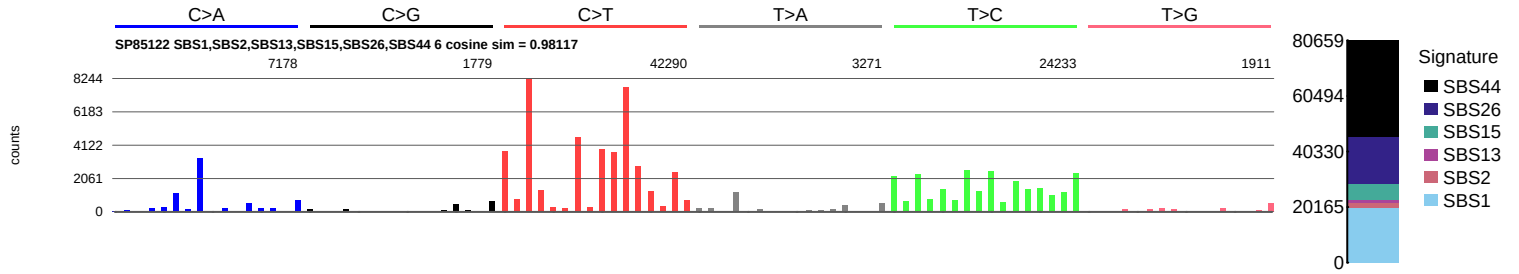

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS21,SBS44 4 cosine sim = 0.98129
8462
3437
42592
4927
18209
3034
counts
8084
6063
4042
2021
0
80659
60494
40330
20165
0
Signature
SBS44
SBS21
SBS5
SBS1

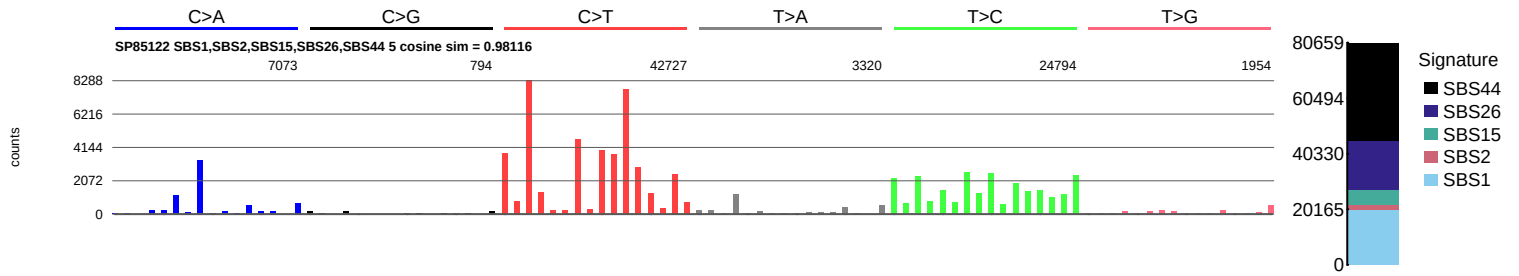

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98128
7189
1795
40584
3278
24074
3738
counts
8236
6177
4118
2059
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS17a,SBS26,SBS41,SBS44 6 cosine sim = 0.98127
8514
3640
39061
5413
21063
2968
counts
8344
6258
4172
2086
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17b,SBS26,SBS44 5 cosine sim = 0.98126
7087
790
40939
3329
24646
3867
8280
6210
4140
2070
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS17a,SBS26,SBS44 5 cosine sim = 0.98126
8594
3669
39110
5326
21069
2894
8352
6264
4176
2088
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS15,SBS26,SBS44 6 cosine sim = 0.98117
7178
1779
42290
3271
24233
1911
8244
6183
4122
2061
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS26,SBS44 5 cosine sim = 0.98116
7073
794
42727
3320
24794
1954
8288
6216
4144
2072
0
counts
80659
60494
40330
20165
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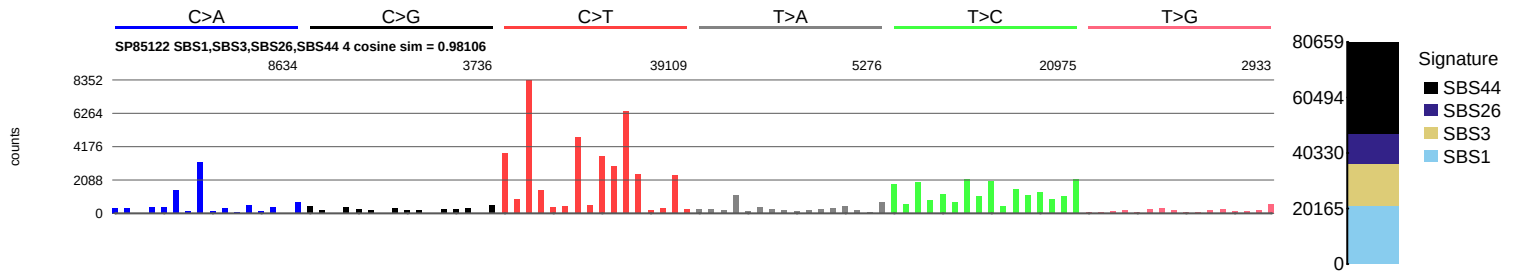
Signature
SBS44
SBS26
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS26,SBS44 4 cosine sim = 0.98106
8634
3736
39109
5276
20975
2933
8352
6264
4176
2088
0
counts
80659
60494
40330
20165
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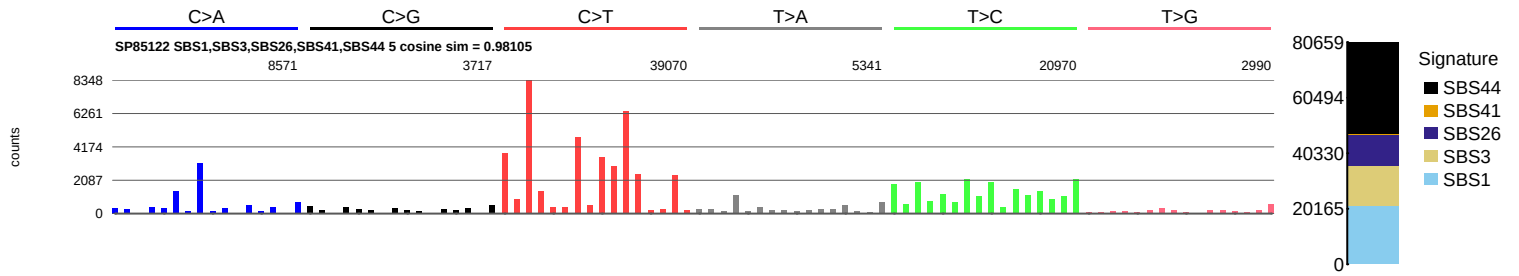
Signature
SBS44
SBS26
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS3,SBS26,SBS41,SBS44 5 cosine sim = 0.98105
8571
3717
39070
5341
20970
2990
8348
6261
4174
2087
0
counts
80659
60494
40330
20165
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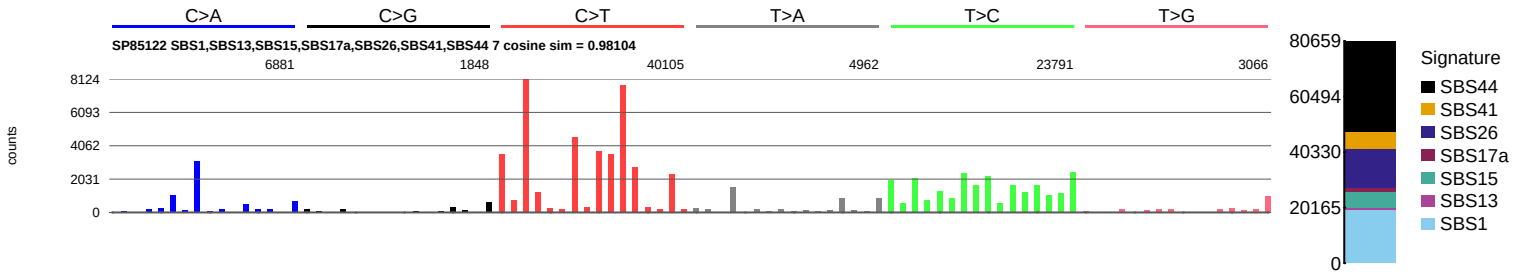
SBS44
SBS41
SBS26
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98104
6881
1848
40105
4962
23791
3066
8124
6093
4062
2031
0
counts
80659
60494
40330
20165
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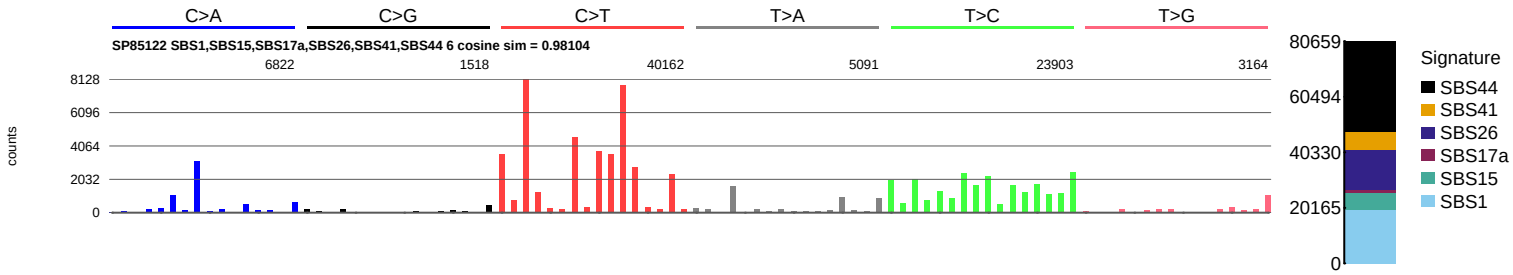
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS26,SBS41,SBS44 6 cosine sim = 0.98104
6822
1518
40162
5091
23903
3164
8128
6096
4064
2032
0
counts
80659
60494
40330
20165
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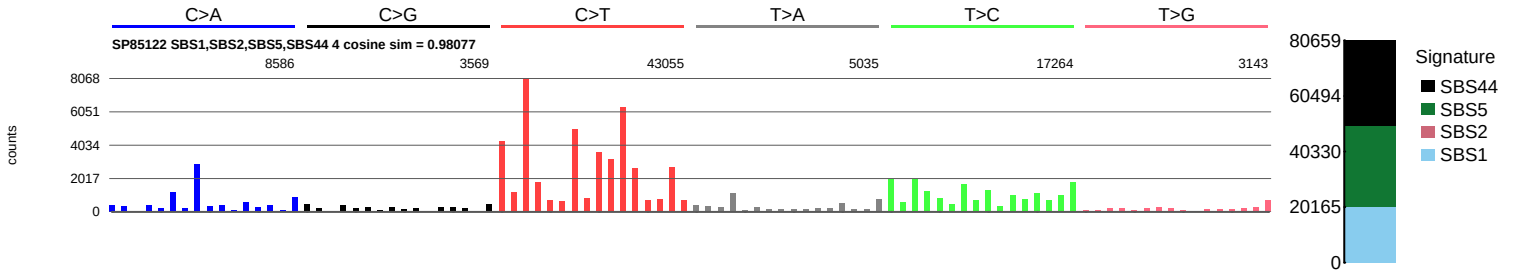
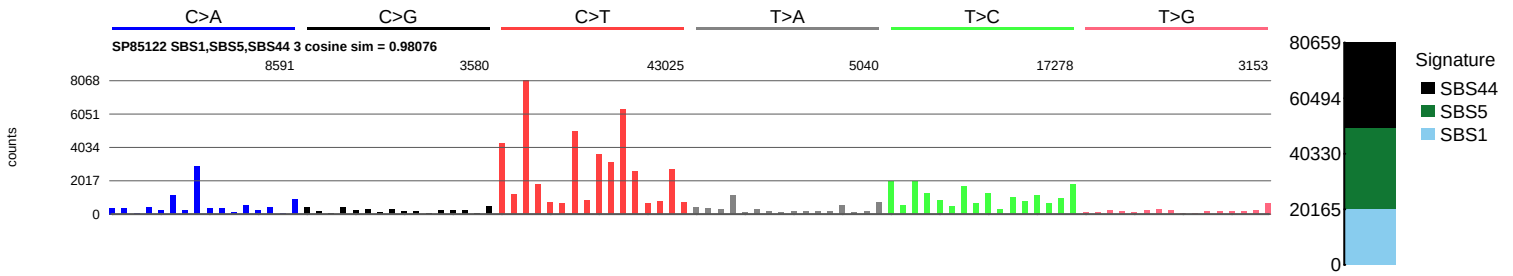
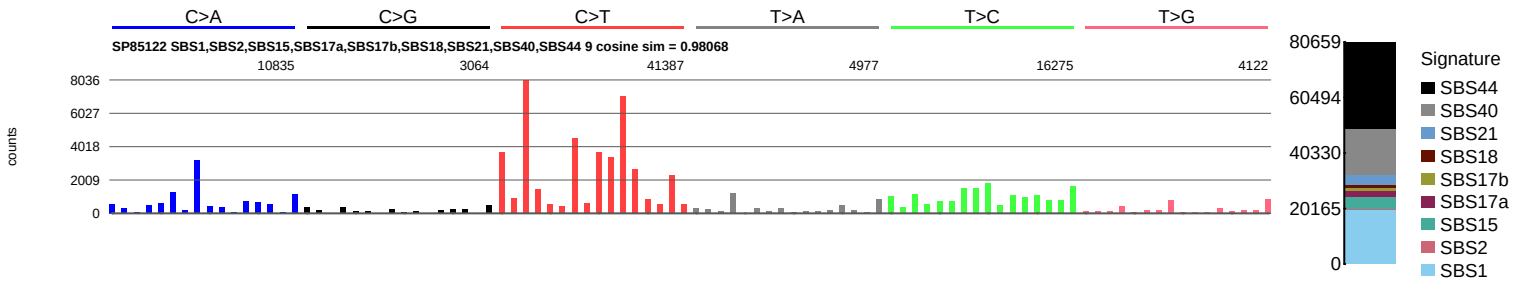
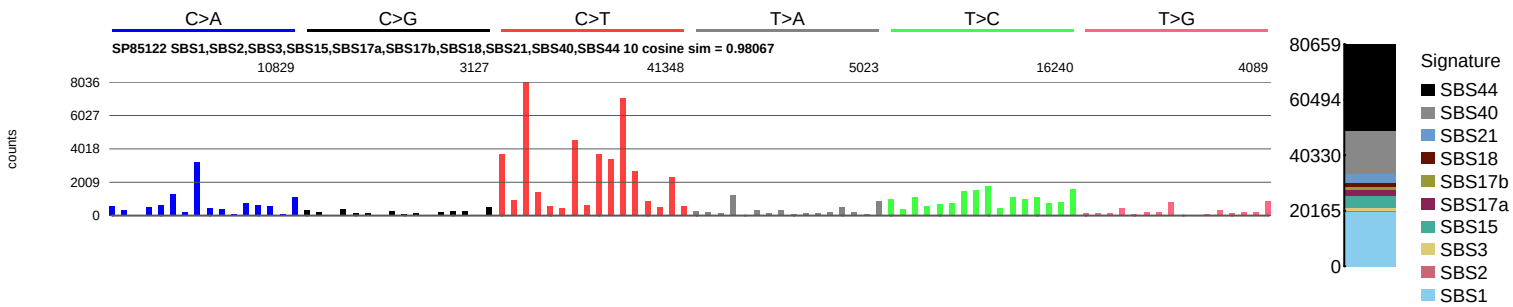
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS5,SBS44 4 cosine sim = 0.98077
8586
3569
43055
5035
17264
3143
8068
6051
4034
2017
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS5,SBS44 3 cosine sim = 0.98076
8591
3580
43025
5040
17278
3153
8068
6051
4034
2017
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98068
10835
3064
41387
4977
16275
4122
8036
6027
4018
2009
0
counts
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98067
10829
3127
41348
5023
16240
4089
8036
6027
4018
2009
0
counts
80659
60494
40330
20165
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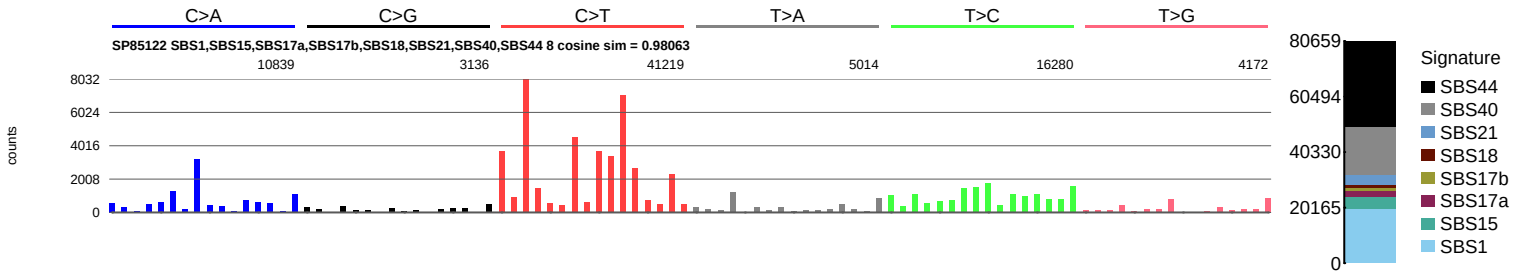
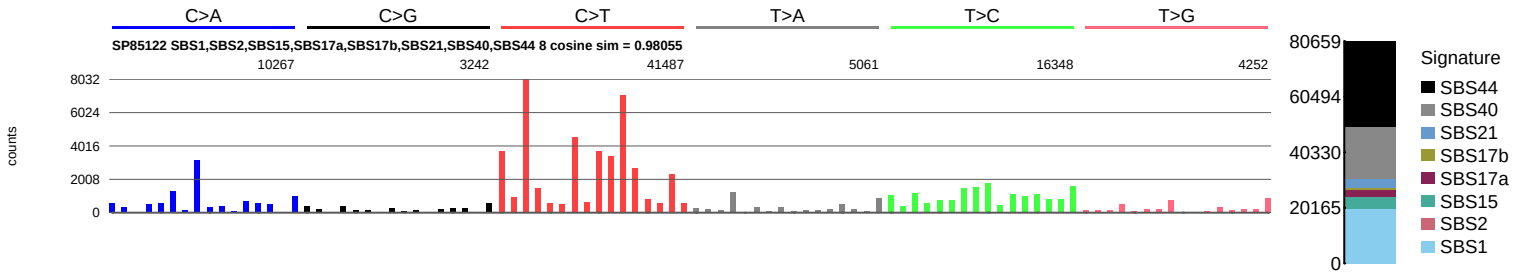
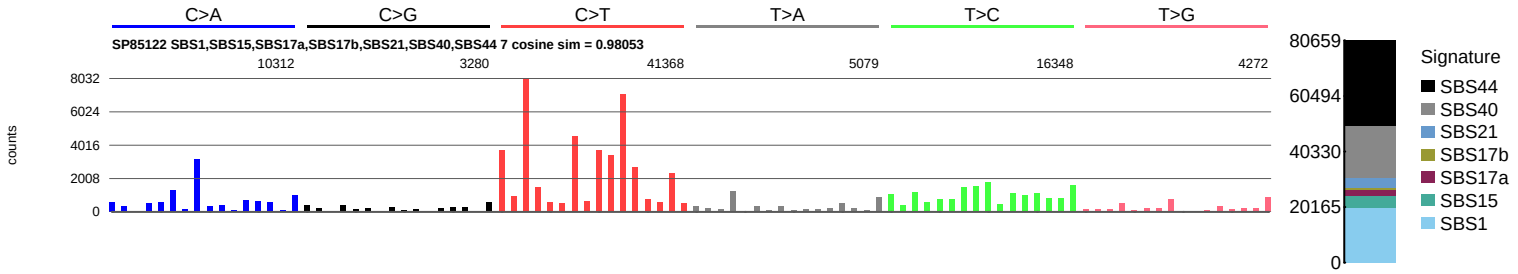
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98063
10839
3136
41219
5014
16280
4172
counts
8032
6024
4016
2008
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98055
10267
3242
41487
5061
16348
4252
counts
8032
6024
4016
2008
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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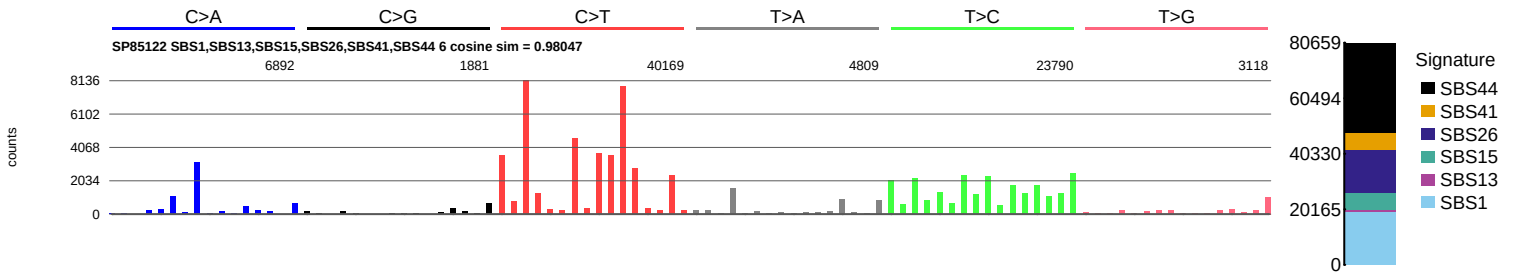
T>G
SP85122 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98053
10312
3280
41368
5079
16348
4272
counts
8032
6024
4016
2008
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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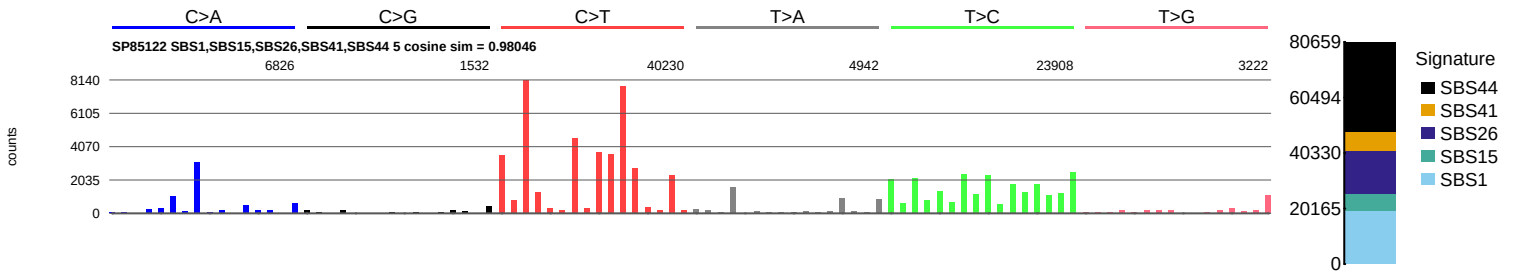
SP85122 SBS1,SBS13,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98047
6892
1881
40169
4809
23790
3118
counts
8136
6102
4068
2034
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS13
SBS1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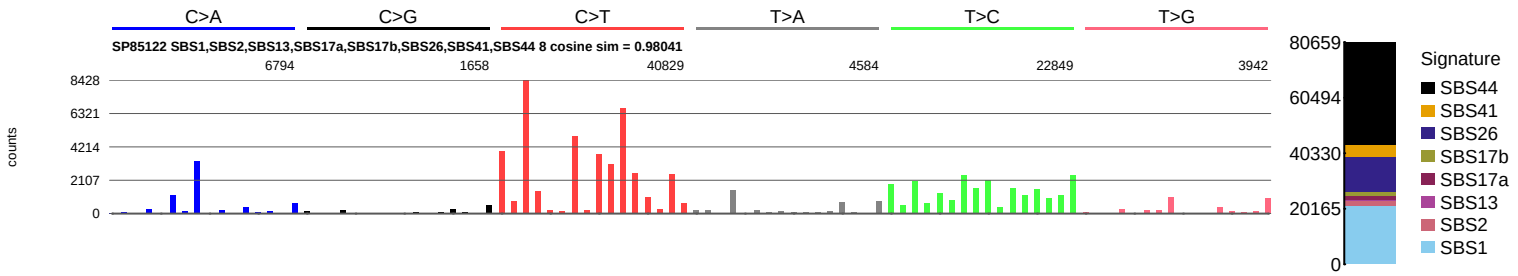
SP85122 SBS1,SBS15,SBS26,SBS41,SBS44 5 cosine sim = 0.98046
6826
1532
40230
4942
23908
3222
counts
8140
6105
4070
2035
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS1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C>G
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T>C
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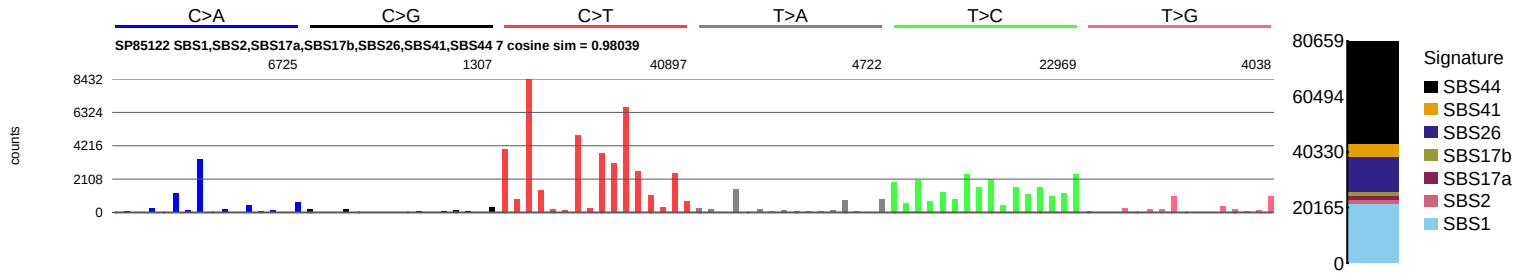
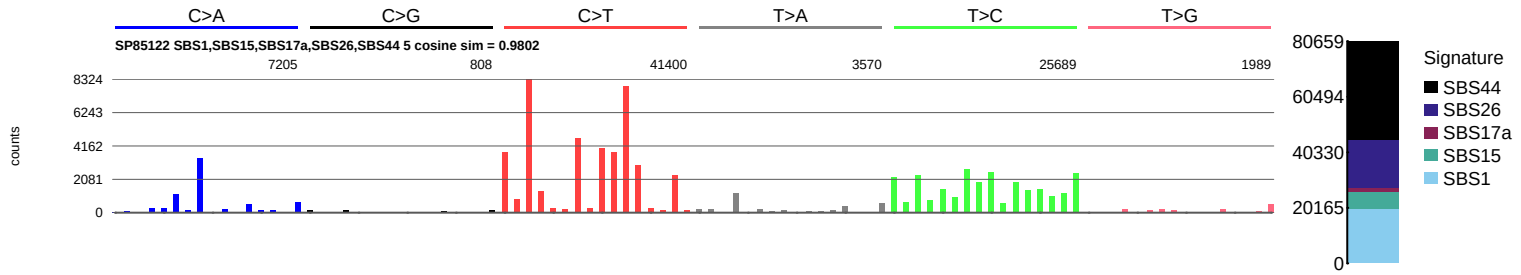
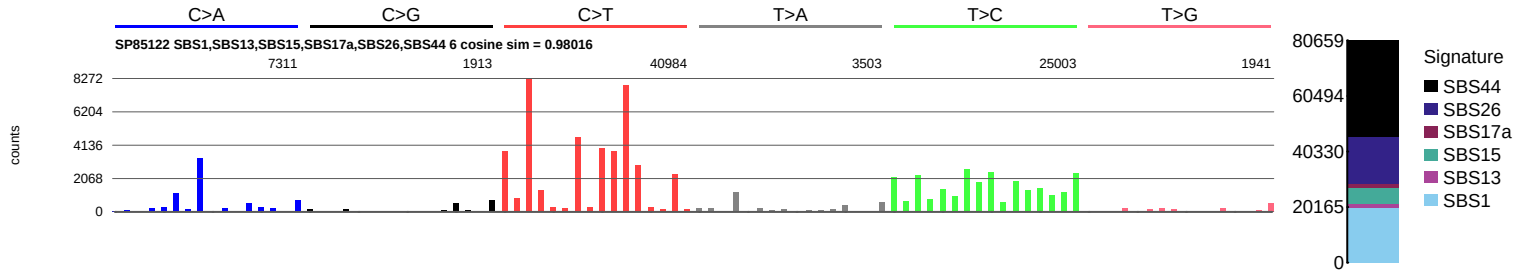
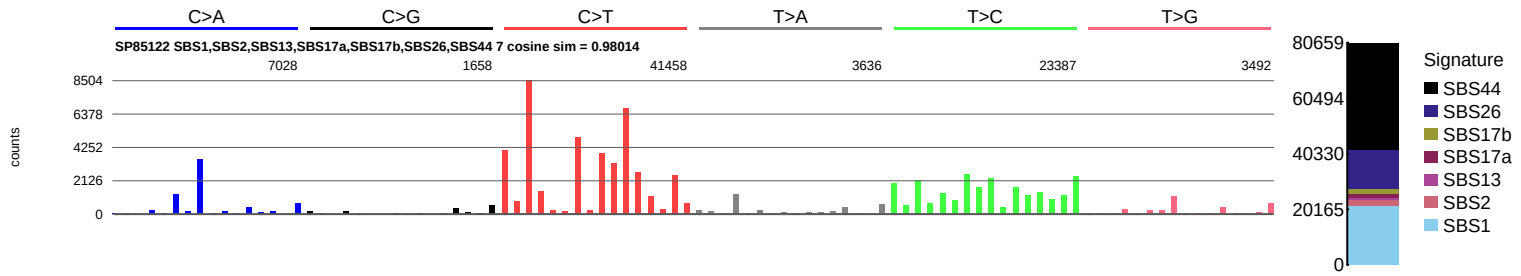
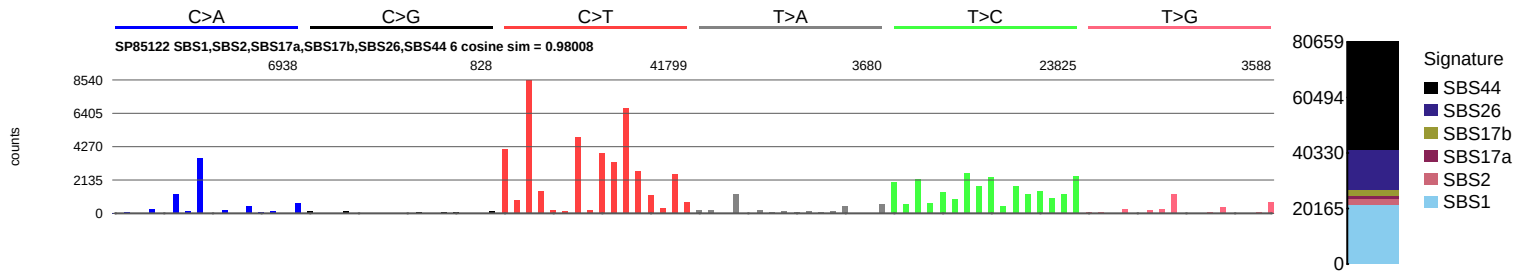
SP85122 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98041
6794
1658
40829
4584
22849
3942
counts
8428
6321
4214
2107
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

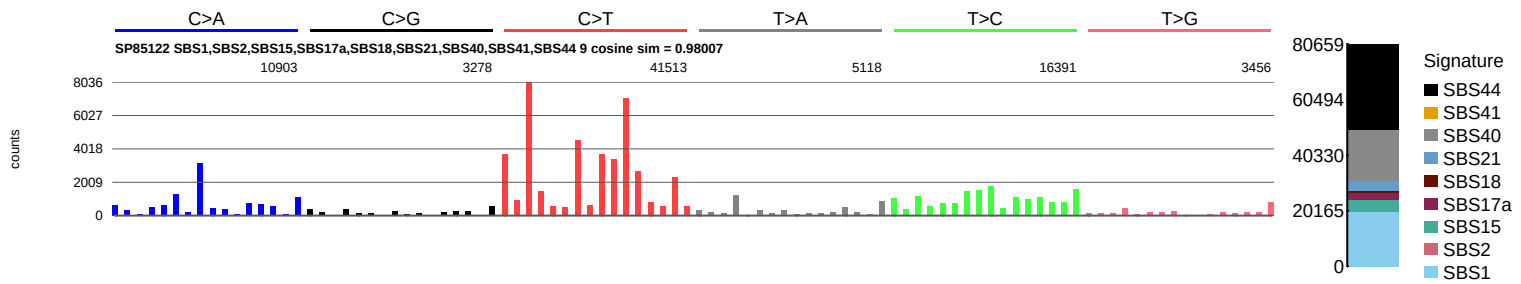

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98039
6725
1307
40897
4722
22969
4038
counts
8432
6324
4216
2108
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1

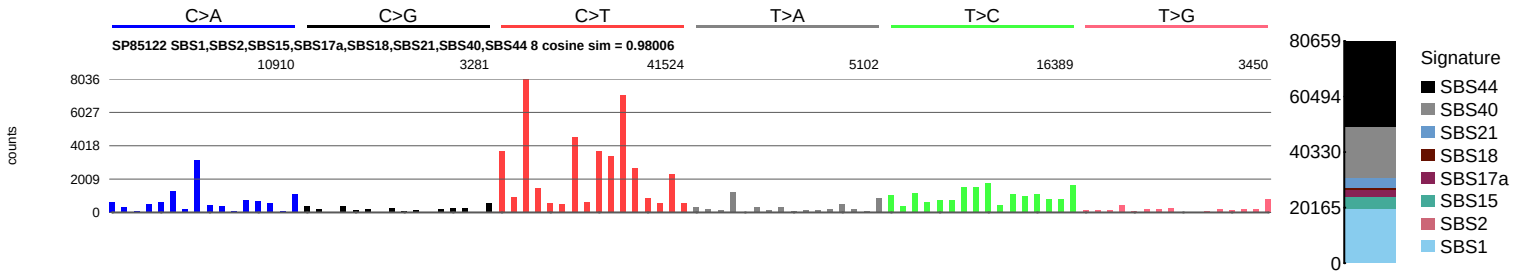

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS26,SBS44 5 cosine sim = 0.9802
7205
808
41400
3570
25689
1989
counts
8324
6243
4162
2081
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS1

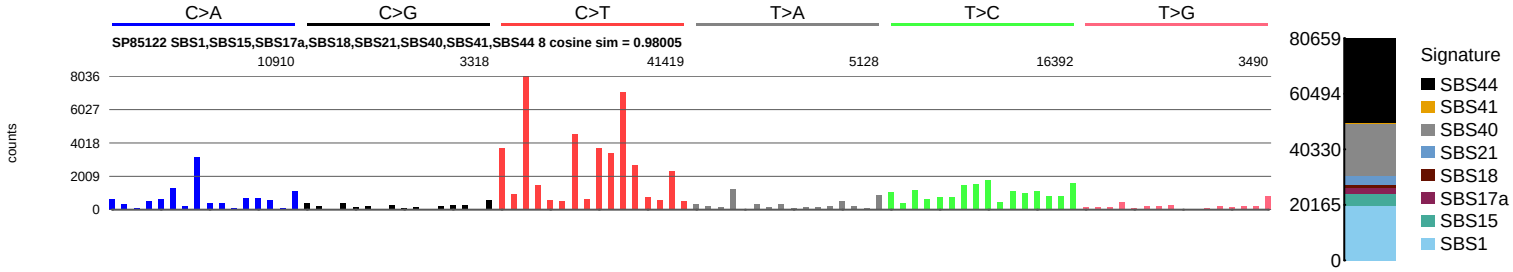

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS13,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98016
7311
1913
40984
3503
25003
1941
counts
8272
6204
4136
2068
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS13
SBS1

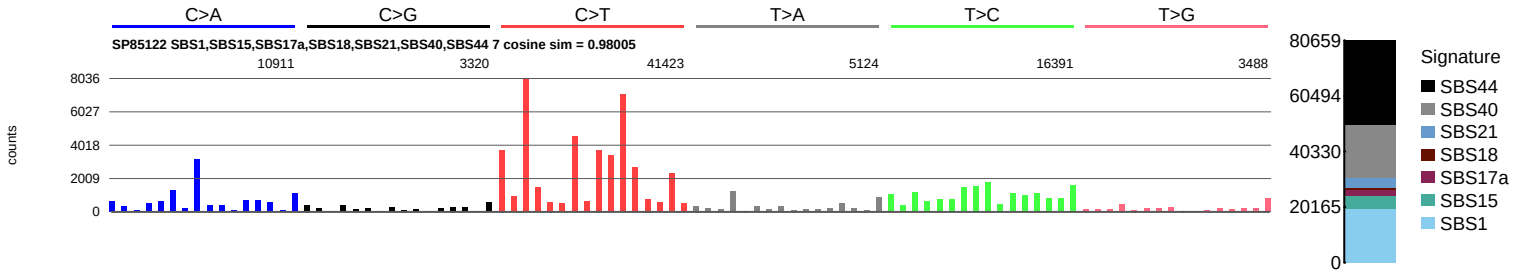

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98014
7028
1658
41458
3636
23387
3492
counts
8504
6378
4252
2126
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

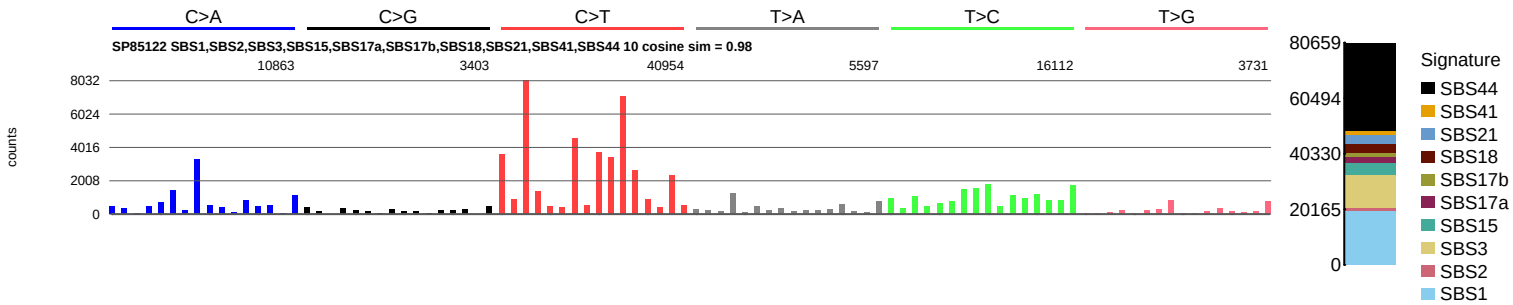

C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98008
6938
828
41799
3680
23825
3588
counts
8540
6405
4270
2135
0
80659
60494
40330
20165
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98007
10903
3278
41513
5118
16391
3456
counts
8036
6027
4018
2009
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98006
10910
3281
41524
5102
16389
3450
counts
8036
6027
4018
2009
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98005
10910
3318
41419
5128
16392
3490
counts
8036
6027
4018
2009
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98005
10911
3320
41423
5124
16391
3488
counts
8036
6027
4018
2009
0
80659
60494
40330
20165
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP85122 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98
10863
3403
40954
5597
16112
3731
counts
8032
6024
4016
2008
0
80659
60494
40330
20165
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1